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Multi-Criteria Decision Making Using Bayesian Networks

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This thesis is dedicated to my father who gave me wings to *fly*

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Abstract

Utility is a theoretical entity that helps us explain decision making under uncertainty. In classic decision making (e.g., the idealized agents of much of economic theory) rational decisions are understood as efforts to maximize the expected utility of actions – i.e., the probability-weighted value of the action’s outcomes. This often works well as a normative standard. At the margins, however, the analysis encounters difficulties.

In this thesis, we examine a different set of issues: how to represent and combine the utilities elicited from multiple experts across all the utility dimensions. When experts conflict in their utility estimates one approach is to “force” a consensus between them, for example, using Delphi methods. Another is to somehow weight the experts and provide a weighted average.

We look at these methods, but also consider new approaches. Specifically, it is proposed that some utility dimensions are related to others, without being readily inter-convertible with each other. For example, the quality of engineering is related to the quality of manufacture of high tech products, but they are not the same, nor are they a simple function of one another. While we may rate these two qualities independently of each other, the results are likely to be (probabilistically) dependent.

We learn and formally represent those interdependencies using Bayesian network models and then incorporate the resultant probabilistic representations of utility in larger scale Bayesian decision networks. This approach is experimentally compared with the objectives above (and others). To the best of our knowledge, treating utility dependencies has never been reported in conjunction to real-life multi-criteria decision-making problems.

This thesis proposes the development of a new methodology to handle dependent uncertainties in Multi-Criteria Decision Analysis (MCDA) problems and to compare decision options based on criteria distributions and weight distributions representing the unknown, partly known, and potentially conflicting preferences of decision makers. Additionally, we develop a tool and software for eliciting multi-criteria utilities, criteria weighting and utility dependencies in the form of Bayesian Network (BN) structures from multiple experts. After eliciting the preferences from the experts, we use causal discovery algorithms to incorporate them into the BN.

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Introduction

Real world domains are characterized by incomplete knowledge, inconsistency and change over time. Both theoretical ignorance, the limitation of our knowledge by unknown facts, and practical ignorance, insufficient or unrevealed details, force us to deal with uncertain hypotheses [6]. Before choosing some actions, we cannot know each and every thing related to the problem because things can change and life is unpredictable. People are uncertain about the results or possible outcomes of their actions because it may depend on various circumstances like environments which are non-deterministic, partial observations, noisy sensors and other complex features involved.

Let us suppose we have to take a flight but we do not know that we will be on time to catch that flight. The uncertainty about the answer involves different situations, e.g., car failure, any natural disaster, accident, or road condition can be related to partial observations, noisy sensors can provide unreliable weather news or updates about traffic and lastly the complex modelling features may include the traffic on the route. Purely logical agents can not take strong decisions in such types of uncertain situations, as the decisions are not always binary (true/false, yes/no, etc.); also they may not know what course of action to make. In such cases, the agents need to make use of whatever information they have in order to make a rational decision amongst available options. In addition, a difficulty often exists when the relative utility of each alternative action is not precisely known, i.e., identified on a numerical scale, which is necessary to specify that how good the alternatives are in a specific decision situation.

In various disciplines, people have to make complex decisions and those decisions may usually involve multiple criteria. Those criteria may be:

1. Very large in number and conflicting. For example, cost/price of a car is one main criterion and some measure of quality is typically another criterion, but often in conflict with cost.
2. Interdependent upon each other. In purchasing a car; cost, comfort, safety, and fuel economy may be some of the main criteria we consider - it is unusual that the cheapest

car is the most comfortable and the safest one.

3. Uncertain because criteria evaluation and preference weighting between criteria often involve uncertainty.
4. Subjective preferences - values/preferences are subjective assessments from experts which can differ or partially overlap [7].

The scope of complex (multi-criteria) decision making problems is enormous — e.g., government policies, medical surgeries, etc. — but here we are just looking at the car buying and defence system acquisition problems as examples of the broader range of multi-criteria decision making.

1.1 Research Issues

Multi-Criteria Decision Analysis (MCDA) is the decision making process of selecting the best alternative among a set of actions, involving multiple criteria which may be interdependent, often in conflict and may be uncertain. In our work here, we learn and formally represent those interdependencies using Bayesian network (BN) models — probabilistic graphical models that consist of nodes and directed edges representing conditionally dependent and conditionally independent relationships between random variables [2, 8]. More specifically, in this thesis, we use Bayesian decision networks (BDNs), a variant of ordinary BNs extended with decision and utility nodes, for decision making based on expected utility. We then incorporate the resultant probabilistic representations of utility in a larger scale Bayesian decision network, as they allow qualitative and quantitative representations of the domain and specifically support decision analysis. A detailed comparative analysis of the existing MCDA techniques using some real life example is given in chapter 2. While utility dependencies have been treated in various ways, there has been limited application to real-life multi-criteria decision-making problems, and to our knowledge BNs have not previously been used for this purpose.

The main research issues that this PhD research addresses:

1.1.1 Large Number of Criteria

In various disciplines people have to make complex decisions and those decisions usually involve multiple criteria. Those criteria may be very large in number and often in conflict with each other. Our research handles those criteria and represent them in an informative way to make inference easy.

1.1.2 Dependencies among Criteria

The criteria involved in a decision problem may be interdependent with each other. Let suppose we want to purchase a car; cost, comfort, safety, and fuel economy may be some of the main criteria we consider — it is unusual that the cheapest car is the most comfortable and the safest one. We address the trade-off among the criteria using BN models in conjunction with MCDA techniques.

1.1.3 Uncertainty

There may be uncertainty involved in criteria evaluation and preference weighting. We develop a preference elicitation method to elicit utilities from the experts and incorporate those utilities into Bayesian Decision networks to assist in multi-criteria decision making problems.

1.1.4 Over-Confidence

The criteria weights are subjective preferences — values or preferences are subjective assessments from experts which can differ or partially overlap [7]. There can be bias/overconfidence in weighting the criteria such that a person can be over-confident in the assessment of her/his own preference. Our research focuses on preference elicitation from multiple experts and aggregating those preferences to make better decisions.

1.1.5 Causal Discovery with Utilities

When a decision problem involves multiple experts / decision makers to make a single best decision, the preferences from those experts are required to be elicited and combined. Utilities elicited from different experts may have dependencies among each other such that different experts may prefer different options resulting in different decisions, so we aim to find those dependencies. This can be achieved by applying causal discovery algorithms, specifically CaMML to learn the causal structure from preference information, a technique not previously utilized.

1.2 Objectives and Motivation

Primarily, this PhD research develops Bayesian Network models for Multi-Criteria Decision Analysis (MCDA) problems. The following steps are involved in the development of these models.

- The development of a methodology to handle dependent uncertainties in MCDA problems. Most of the existing techniques - while modelling uncertainties, assume that there is no dependency among the criteria (that is, the external objective random variables) involved in the decision problem.
- Comparing decision options based on criteria distributions and weight distributions representing the unknown, partly known, and potentially conflicting preferences of decision makers. Modelling of large multivariate distribution for interdependencies among the criteria is a major challenge because it is often very difficult to achieve. Most existing MCDA approaches do not model interdependencies among the utilities associated with criteria (that is, the values that people attach to each of the criteria), but rather they assume that the utilities involved in a decision making problem are independent. Hence, another objective is using BDNs to capture and represent utility dependencies, to reflect the dependencies and interactions, including trade-offs, that are common in real-life multi-criteria decision problems.
- Developing a method for eliciting multi-criteria utilities from multiple experts, criteria weighting and utility dependencies that can be converted into the form of BDN structures, and development of a method based on the BN-based machine learning technique of causal discovery where large preference data is available. These can be applicable to various problems such as equipment procurement, fleet purchases, consumer preference modelling in e-commerce settings, etc.

Current BN software tools like Netica¹, for example, treats the utilities represented in different utility nodes as independent and additive. Let suppose we have an example BDN of a land combat vehicle selection problem in Figure 1.1 where the given criteria are *Mobility*, *Cost*, *Protection* and *Reliability* associated with their respective utility nodes whom are the parents of *Overall Score* utility node. These criteria are interactive and dependent upon each other — if we need more protection we lose mobility of the vehicle and so on. The dependencies between the variables (the oval nodes) are straightforward to represent by the directed arcs in the BN, as shown in the figure. However there is no explicit structural representation of the dependencies between the associated utilities.² We aim to use BDNs to represent dependencies among the expert elicited utilities by applying causal discovery algorithms.

In Figure 1.2 *CV* is a utility possibility curve which represents the various levels of utilities of a person i.e., *Money* and *Health*. According to Pareto criterion, a movement from *Q* to *R*,

¹This tool is available at <https://www.norsys.com/netica.html>

²These dependencies can be *implicitly* represented in the numerical parameters of a utility node, capturing preferences that reflect interactions between the factors influencing the decision making. Though we also note that such utility functions are generally deterministic due to limitations in the major inference engines.

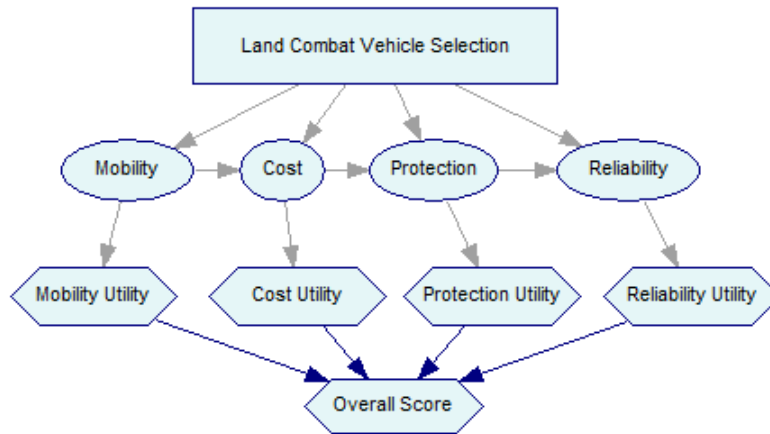


Figure 1.1: Simple BDN example

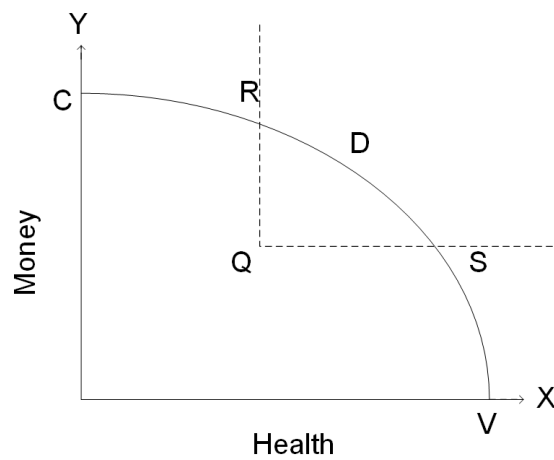


Figure 1.2: Pareto Criterion

or Q to D , or Q to S indicates the increase in social welfare because in such movements the utility of either *Money* or *Health* or both increases. A movement from Q to R indicates that the utility *Money* increases, while that of *Health* remains the same.

On the other hand, a movement from Q to S implies that while utility of *Health* has become better off, *Money* is no worse off. And a movement from Q to D or any other point on the segment between R and S will mean increase in utility of both options. Thus points R , D and S are preferable to Q from the point of view of social welfare.

We aim to represent and combine the utilities elicited from multiple experts across all the utility dimensions. Bayesian network models are probabilistic graphical models, a rich framework to encode multivariate distribution over many interacting random variables while allowing users to do reasoning efficiently. They directly enable the user to model the uncertainties and interdependencies among the criteria involved in a decision making problem.

1.3 Research Goal

Our main research goal is to propose a new methodology for eliciting preferences for complex multi-criteria decision problems from experts and representing these within a BDN. Such BDN structures might be developed based on expert knowledge alone, or combined with large data sets (such as simulation generated data) and causal discovery. In parallel, we propose a new method for accounting for preference interdependencies by applying causal discovery algorithms where larger amounts of preference data can be collected.

1.3.1 Preference Elicitation

We propose using a variant of the IDEA method (Investigate, Discuss, Estimate, Aggregate Judgments) for preference elicitation, taking it beyond its prior use in probability elicitation [9]. Due to time and resource constraints, we truncate the full IDEA method into what we call the 4 Point Elicitation method, taking two of the steps from IDEA that are easily implemented online. We compare this method with an alternative straightforward "Baseline" method. These methods both produce Bayesian decision networks as a final step for assessing decisions.

1.3.2 Causal Discovery

In many cases, large sets of preference data may be available or can be collected. For example, surveys can be distributed to large audiences who directly possess the preferences we are trying to capture (for example, this is very feasible in online e-commerce environments), or who are otherwise similar to or can provide insight into the preferences of our target individuals of interest. In other cases, we may be able to capture preference data via observed choices, in the spirit of revealed preference theory [10].

When such data sets are available, we can apply causal discovery algorithms to learn causal structures from the preference data, allowing us to capture both dependencies and uncertainties amongst the preferences. We evaluate the learned causal models in comparison with the above BN models.

1.4 Thesis Organisation

This chapter introduces the thesis, the motivation for the research, and the main contribution of the project as well as providing a brief reference to the related literature. The rest of the thesis is organized as follows:

- **Chapter 2** presents terminologies and the relevant research literature, in the areas of

Bayesian decision networks, including causal discovery, expert elicitation and the existing techniques used for Multi-Criteria Decision Analysis (MCDA) problems. This provides the fuller context for our research problem as well as the current limitations in Multi-Criteria Decision Making (MCDM).

- **Chapter 3** presents a method for learning BNs from simulation data and developing BDNs based on those learned BNs. The method is illustrated with a case study for learning BN meta-models from the unclassified land combat simulation data. It uses existing well known causal discovery algorithms for learning the BN structures, applies existing modelling techniques to convert these into BDNs and demonstrates some validation steps that can be applied to the meta-models.
- **Chapter 4** proposes the methods used for eliciting preferences from the experts and calculating the utility functions based on two new methods, the Baseline and 4 Point Elicitation methods. It also introduces two novel methods for creating BDNs in cases where it is possible to collect larger sets of preference information, one method based on Analytic Hierarchical Process (AHP) and another method based on causal discovery.
- **Chapter 5** presents a case study used to demonstrate the application and usefulness of the proposed methods, based on the fleet vehicle purchase problem for any large organization. It provides details of the elicitation survey, preprocessing of the elicited data, and how this is subsequently used to parameterize the expert-elicited BDN structure. To support the large data-dependent methods, the chapter also introduces a set of artificial preference models that are used to generate data for testing these methods. The chapter ends with a description of the evaluation metrics that can be applied to the models produced by each method.
- **Chapter 6** presents the experimental results and an empirical analysis from the fleet vehicle purchase case study, using the methods presented in Chapter 4 on the elicited survey data and the artificially generated data. The chapter contains two major parts: Section 6.1, which evaluates the Baseline and 4 Point Elicitation methods on the collected survey data; and Section 6.2, which starts with the data generated from the preference models introduced in the previous chapter and uses these data sets to evaluate all 4 methods, including the large data-dependent methods (the data-based AHP and causal discovery methods) and the survey-based methods.
- Finally, the thesis concludes in **Chapter 7** with a summary of the research contributions along with suggesting future directions to enhance and extend this field of research.

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- Glossary and Symbol tables are presented in **Appendices 7.2.4.**

Background

In Chapter 1 we briefly discussed the research problem we aim to address, and related research issues and challenges. We also discussed our motivation for this research followed by our research goals and our proposed solution. In this chapter we explore the existing research methods used for reasoning with Bayesian networks under uncertainty and multi-criteria decision making analysis. It also covers the background of research and the problems arising in this field in order to get a better foundation for understanding our approach.

Section 2.4 explains the MCDA approaches currently used to assist Decision Makers (DMs) in multi-criteria decision making problems and also covers the limitations and drawbacks of the existing methods. We describe preference elicitation from experts in Section 2.9 followed by causal discovery using preference data in section 2.10. In the last Section 2.11, we discuss the validation and evaluation techniques we use in this thesis for the empirical analysis of our methods.

2.1 Reasoning under Uncertainty

In real life, decision making problems are often very complex to handle, usually very large in scale and may involve many uncertainties. We normally need to reason under uncertainty, with and about propositions with unknown truth values, as in gambling. Indeed, most real-life decision problems can be viewed as gambles. For instance, people needing to reach the airport by a certain time will not know with certainty what the traffic conditions are (even with Google Maps), so how can they plan rationally in such situations?

If they must arrive at the airport by 9 pm on a week night, perhaps they can “safely” leave for the airport an hour before. There is some probability that the trip may take longer, but that probability may be low. They may need more time to get to the airport, if it’s on a Friday at 4:30 pm and here the probability could be relatively high that it takes over 1.5 hours. To act rationally under uncertainty one must be able to evaluate how likely these kinds of events are. By assessing the probabilities of relevant events, one can develop techniques for acting

rationally under uncertainty [11, 12].

Uncertainty can also exist over preferences, and this can take many forms. Examples include not knowing one's own preferences clearly (introspective psychology is the weakest approach to psychology, because our insight into our own minds is demonstrably weak); changes in preferences over time; and preferences may be organizational, or otherwise other-oriented, and so a matter of uncertain inference. Techniques for handling these kinds of uncertainty are much the same if we treat utilities as just another type of uncertain but measurable data.

2.2 Probability

Probability is used to measure the uncertainty of an event's occurrence. For an instance, let event A be that "Tomorrow it will rain in Melbourne." It is problematic to predict that A will happen with certainty; we all know that the weather forecasts are not fully reliable. Therefore, we should prefer to quantify the uncertainty with $P(A)$, known as probability of A . Probability is defined to be a non-negative number between 0 and 1 and should not exceed 1. The two extreme cases can be represented by the statements; the probability that event A is impossible to happen is $P(A) = 0$ and the probability of A to happen for sure is $P(A) = 1$. So the $P(A) = 0$ indicates that A will not occur and $P(A) = 1$ indicates that A will occur for sure [13]. An English Statistician *Thomas Bayes* introduced a theorem called "Bayes' Theorem" in 1763, which is being used to model probabilities in various real life problems. Bayesian Networks (BN) are based on this theorem.

$$P(h|e) = (P(e|h)P(h))/(P(e))$$

Where $P(h|e)$ is the posterior probability of h given the evidence e , $P(e|h)$ is the likelihood of evidence e if hypothesis h is true, $P(h)$ is the prior probability and $P(e)$ is the prior probability that the evidence e itself is true.

2.2.1 Probabilistic Models

Conditional dependencies among random variables can be modelled using graphical structures — Probabilistic Graphical Models (PGMs) — which allow us to infer unknown quantities, make predictions and learn from data [14]. Probabilistic models began to be widely adopted in the late 1980s. PGMs are being used in many diverse domains such as medical diagnosis, machine learning, marketing, genetic data analysis and many others. The most visible and influential application of PGMs is associated with the development of Bayesian networks which provide support for any type of reasoning in a diagnostic and predictive manner. The graphical structure of probabilistic models helps researchers visualize and intuitively understand the

underlying relationships and dependencies.

2.2.2 Bayesian Networks

There are many kinds of PGMs; one of the most widely used are Bayesian Networks or Belief Networks (these models are based on directed graphs and show the asymmetric effects and dependencies, e.g., causal relationships, among the variables) and the second is Markov Models (based on undirected graph structures and represent symmetric effects and dependencies, e.g., non-causal relationships, among the variables).

A Bayesian Network is a graphical structure representing the directed dependencies among a set of random variables, with the dependencies quantified by the set of Conditional Probability Tables (CPTs) [2, 15]. Bayesian Networks have gained a tremendous popularity in the representation of encoding uncertain expert information in expert systems over the last decade [16]. A Bayesian Network must have the following properties:¹

- A set of random variables and directed arcs connecting those variables.
- Every node/variable must have finite set of mutually exclusive states.
- The set of variables and the directed arcs among them must create a Directed Acyclic Graph (DAG). E.g., let suppose there is a path from $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_n$ but in an acyclic graph there is no path from A_n to A_1 and no only arc.
- For each node B having its parents $A_1, A_2, \dots, A_n$, there must be a CPT for $P(B|A_1, A_2, \dots, A_n)$.
- For each node B without parents, the CPT reduces to unconditional probability $P(B)$ [17].

To illustrate this Bayesian Network definition, Figure 2.1 shows a very basic BN example, the well-known ‘metastatic cancer’ BN [1–3], consisting of 5 nodes representing variables in the Bayesian Network and 5 directed arcs representing the dependency relationships among them. Associated with each Boolean node (two states, true (T) and false (F)) is a set of conditional probability distributions; a subset of these is given in Figure 2.1.

For the above example we have the factorization

$$P(M, S, B, C, H) = P(M) \cdot P(S|M) \cdot P(B|M) \cdot P(C|S, B) \cdot P(H|B)$$

¹We restrict ourselves here to discrete Bayesian networks, which are the most common in applied modelling. Continuous BNs introduce complications which would only distract us here.

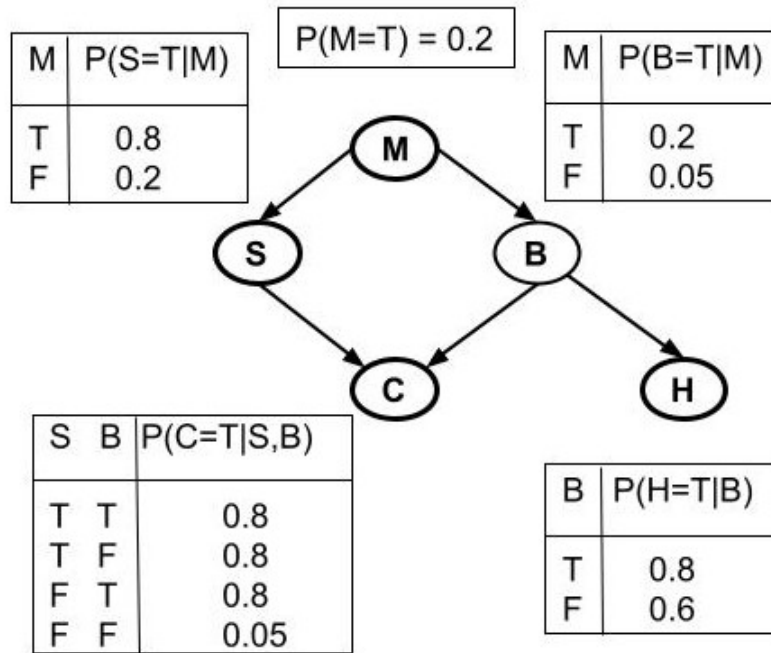


Figure 2.1: Bayesian Network Example for Metastatic Cancer [1–3]

where M = *Metastatic Cancer* can cause of B = *Brain Tumour* and can be also responsible for S = *Increased total serum calcium*, either of these explaining a patient in C = *Coma*. However H = *Severe Headache* is also associated with *Brain Tumour*.

2.2.2.1 Bayesian Decision Networks

Bayesian Decision Networks (BDNs)² [18] are extended versions of Bayesian Networks which support decision making, by providing two additional kinds of nodes: decision nodes to represent the alternative decisions under consideration, and utility nodes, which can capture the costs of decisions, including the benefits or value or preferences associated with different outcomes.

Utility Theory

Utility theory provides for assessing individual preferences or values and utilizing them in decision making. It is assumed each person knows about her/his likes and dislikes, at least to the extent that an individual can rank their choices according to their preferences. Utility is the measure of the relative desirability and satisfaction of an individual from various goods and services. Following Frank Ramsey, we can always generate a utility function over outcomes that yields an actor's consistent preference ordering [19].

Agents tend to take a course of action among multiple alternatives which has the highest expected gain or profit, which is known as maximizing expected utility. Given the utility

²Also called influence diagrams by Howard and Matheson [18]

function $U(O_i/A)$ for every possible outcome O_i that can occur as a result of a particular action A and the probability $P(O_i/A)$ of every such outcome that may happen, then we will be able to calculate the expected utility for that action by using below equation [8].

$$EU(A) = \sum_i U(O_i/A) \times P(O_i/A)$$

What will be the appropriate action to take when the future state of the world is not known? The answer is to find out the utility of each possible outcome of an event and select an action which has the highest expected utility.

There are three types of state nodes shown in the Table 2.1 which need to be considered in the formation of decision making process.

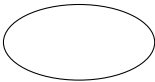
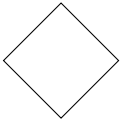
		
Chance nodes are represented by oval shape. They represent random variables/nodes within a decision network and a CPT is associated with each chance node to maintain its probability.	Decision nodes are shown by a rectangle. It indicates the action to be taken by a decision maker among multiple alternatives. These nodes are also called action nodes and may have other decision nodes as parents.	Diamond shape shows the utility node also known as value node. Each utility node maintains a utility table and its parent nodes describe the outcome state which directly affects the utility node.

Table 2.1: Nodes in Bayesian Decision Network

In a Bayesian Decision Network, we can identify the best action to take in the following way: identify all evidences, set the action nodes in each possible way, calculate posterior probabilities for all parents of utility node given the evidence, then calculate expected utility for each possible action and lastly choose the action with maximized utility.

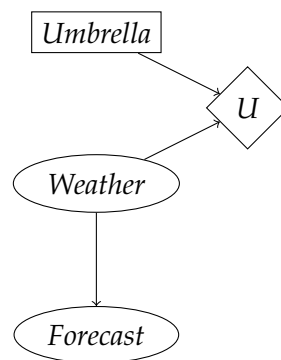


Figure 2.2: Bayesian Decision Network Example

Figure 2.2 represents a simple Bayesian Decision Network which has two chance nodes;

Weather and *Forecast* (showing the weather information) and a decision node *Umbrella* (whether to take or not). There is also a utility node U which measures the level of satisfaction of the Decision Maker (DM). The *Weather* may be sunny or rainy and the arcs from *Weather* and *Umbrella* to U show the satisfaction level of the person will be dependent upon taking the weather condition and the decision of taking or not the *umbrella*.

When there are multiple decision makers involved in making a single best decision based on some given criteria, the preferences from all the participants need to somehow be combined to get a collective response. Consider an example of a fleet vehicle purchase problem in an organization; where the purchase decision involves different types of vehicles to choose from and multiple users have different preferences. The fleet purchase decision has to be made based on a set of given criteria considering the preferences of multiple individuals. There may be uncertainties involved in any preference elicitation such that a participant may be unaware of her/his desires, or may use a different range of values to express her/his satisfaction level as compared to other participants. There is also uncertainty in criteria weighting as the criteria of different individuals may be dependent upon each other. For example, in the above mentioned problem, any two criteria such as purchase cost and safety may be dependent upon each other — if you prefer safety of the vehicle you will have to compromise on purchase cost. Similarly, in a collective single decision making problem, the utilities of different decision makers may also be dependent upon each other, which may in turn be reflected in utility dependencies. Therefore, dependencies between the elicited utilities are better modeled than ignored.

A decision is said to be rational if it remains consistent with respect to values, alternatives and information quantified by a single person or group who make it [20]. There are two perspectives, namely substantive rational and procedural rational which contribute in making a good decision. Substantive rationality indicates that a decision is said to be good if its outcome is good while in procedural rationality, a decision is considered good if the procedure adopted to make the decision was right [21]. The quality of decision processes can be judged by transparency, understanding, efficiency and participation of broad stakeholders [22].

The main purpose of decision support models is to help individuals make decisions efficiently. There are different kinds of techniques and methods used to support human decisions; among them one is Multi-Criteria Decision Analysis (MCDA).

2.3 Multi-Criteria Decision Analysis

Multi-Criteria Decision Making Analysis (MCDA) has become a very popular methodology to tackle the problems and improve decisions-making process when there are multiple objectives involved. Numerous studies show MCDA being widely used tool in various areas, especially

in environmental decision-making where dealing with multiple objectives is the norm, using different kinds of data as well as multiple stakeholders (e.g., [23–25]). The methodology of MCDA provides a systematic approach to analysing complex decision making with conflicting goals. It provides an insight to the nature of those conflicts among multiple objectives and helps to tackle them in an appropriate manner [26]. The framework commonly adopted is a continuous learning process consists of several stages to formally represent a decision. Figure 2.3 illustrates the basic key steps involved in the process of Multi-Criteria Decision Making Analysis.

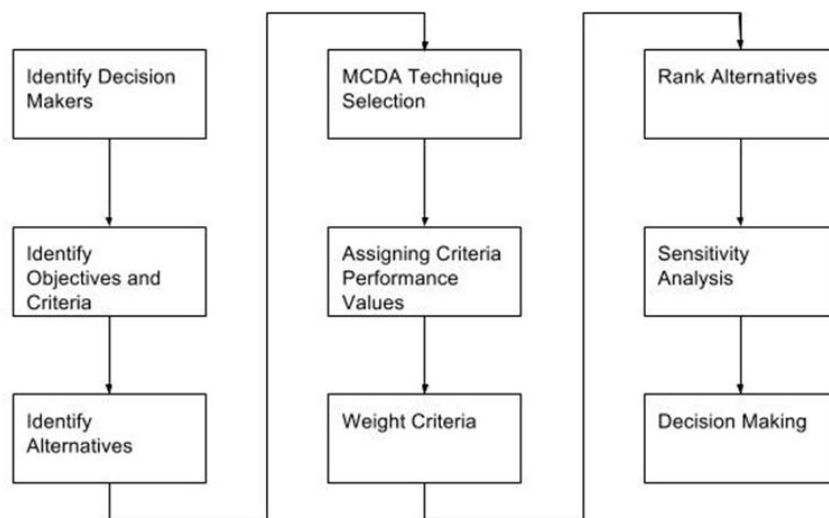


Figure 2.3: MCDA Process [4]

2.4 MCDA Techniques

There are different kind of techniques used to provide an insight into the nature of conflicting criteria in Multi-Criteria decision problems and also help us to improve the decision making process. Fundamentally, we have three different approaches used in various disciplines for MCDA.

1. Utility Based Approaches
2. Pairwise Comparison Approaches
3. Outranking Approaches

2.5 Utility-Based Approaches

In this kind of approaches are based on utility functions to evaluate the user's preferences and perform the decision making process according to that. Some of the most commonly used techniques are explained below.

2.5.1 Multi-Attribute Utility Theory (MAUT)

Multi-Attribute Utility Theory (MAUT) is a rank-based method which uses the utility functions to evaluate the suitability of preferences regarding the given criteria. Following are the key steps of MAUT [27].

- First of all, identify the alternatives and their relative attributes.
- Assign weights to each attribute.
- Sum up the weights and evaluate the attributes in order to get an overall value for every alternative using value function given below.
- Select the alternative highest overall utility after performing the sensitivity analysis.

$$U(v) = \sum_{i=1}^n a(i)u(i)$$

Where $U(v)$ is the overall utility of an outcome, $a(i)$ is magnitude of the option on i^{th} attribute, $u(i)$ is the utility of i^{th} attribute and n is the number of attributes.

To illustrate this technique, lets take an example of choosing a car among four available options (*Toyota, Honda, BMW, Mercedes*) where criteria are safety (*Safety*), Luxury (*Luxury*), cost (*Cost*), usability (*Usability*) and engine type (*Engine*). The highlighted values in the following Table 2.2 are the weights of the criteria; these values are multiplied by the utilities $u(i)$ of each option w.r.t the given criteria. The resultant values are then summed up to get an overall utility $U(v)$ for each alternative as shown below. The alternatives with the highest overall utility is then selected as the best option which this example is *BMW* having the highest overall utility of 68.

* is the highest overall utility among all options

2.5.1.1 Limitations of MAUT

MAUT is a very simple and understandable technique but it does have the following limitations.

		Attributes				$U(v)$ or overall utility of each option
		<i>Safety</i>	<i>Luxury</i>	<i>Cost</i>	<i>Usability</i>	
$weights(a) =$		4	5	2	1	
		$u \times a$	...	...	...	
Options	<i>Toyota</i>	$3 \times 4 = 12$	$3 \times 5 = 15$	$9 \times 2 = 18$	$3 \times 1 = 3$	48
	<i>Honda</i>	$3 \times 4 = 12$	$3 \times 5 = 15$	$3 \times 2 = 6$	$3 \times 1 = 3$	36
	<i>BMW</i>	$9 \times 4 = 36$	$5 \times 5 = 25$	$3 \times 2 = 6$	$1 \times 1 = 1$	68*
	<i>Mercedes</i>	$1 \times 4 = 4$	$3 \times 5 = 15$	$9 \times 2 = 18$	$9 \times 1 = 9$	46

Table 2.2: Multi-Attribute Utility Theory Example

- It does not deal with any dependencies and feedback during the decision making process.
- Does not treat causal relationships and uncertainties among distinct and conflicting criteria involved in the decision problem.
- Does not handle multi dimensions of a utility.

2.5.2 Stochastic Multi-Accessibility Analysis (SMAA)

This approach was introduced to deal with uncertain, missing and inaccurate information represented as probability distributions. SMAA considers a discrete decision making problem with multiple decision makers and allows the decision makers not to specify their preferences explicitly. It provides a methodology to explore the weight space based on an assumed utility function. Stochastic multi-objective Acceptability Analysis (SMAA) technique has been presented in two steps. First phase is deterministic case, where the exact criterion values are known for each alternative. In the next step they generalised the process for the case in which the criterion values are stochastic variables represented by probability distributions.

2.5.2.1 Deterministic Case

In this case, value g_{ij} for each alternative i and criterion j are given and an additive utility function is used $u_{ij} = u_j(g_{ij})$ where u_{ij} is utility of criterion j for alternative i and $u_j(g_{ij})$ is utility function of criterion j in the range of $[0, 1]$. For each alternative, the overall utility u_i is then represented as a convex combination of the criteria utilities using some unknown normalised weights w_j .

$$u_i = \sum_{j=1}^n w_j u_{ij} \quad w \in W$$

The weights need to be considered as non-negative and normalized having a weight space.

$$W = \{w \in R^n : w \geq 0 \text{ and } \sum_{j=1}^n w_j = 1\}$$

The set of feasible weight vectors W which is an $(n - 1)$ -dimensional simplex in the n -dimensional weight space. In the three-criteria case, the feasible weight space W is a two-dimensional area as shown in Figure 2.4.

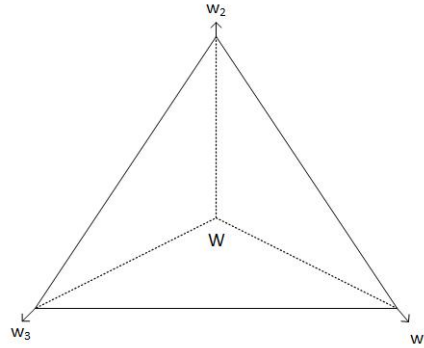


Figure 2.4: Feasible Weight Space

This feasible weight space represents various possible valuations from the decision makers. If the weight vector is already known in MCDA problem then u_i can easily be computed and the highest valued alternative can be chosen but for some reasons the weights are not always known. It is very unlikely that every decision maker may be willing to specify the weights. In this situation, SMAA uses a different approach where a set of weights W_i are determined for each alternative i to make the overall utility of that alternative greater than or equal to the utility of other alternatives. W_i is known as set of *favourable weights vectors* for alternative i because it comes to the best choice among others. W_i is the subset of feasible weight vector space and satisfying the constraint $u_i \geq u_k$ where $k = 1, \dots, m; k \neq i$. SMAA argues that without having prior knowledge of the decision maker's valuations, *central weight vector* is the best option to use as a single vector to represent the valuations of any typical decision maker who prefers alternative i . The *central weight vector* w_i^c for alternative i can be computed using the following formula.

$$w_i^c = \frac{\int_{w_i} w dw}{\int_{w_i} dw}$$

w_i^c is an unbiased estimate for weight vector w in a case where the alternative i is preferred. An acceptability index a_i is the measurement of different valuations (denoted by feasible weight vectors) allowing that particular alternative to be chosen. For acceptability index for alternative i is defined by the ratio between the volumes of W_i and feasible weight space W [28].

$$a_i = \frac{vol(W_i)}{vol(W)}$$

a_i can also be interpreted as the probability of an alternative being the best option to choose,

assuming the weight distribution used in calculations. If a_i is zero, it means that the polytope's dimension is lower than $n - 1$, so in order to make that alternative the best option, at least two of the weights must depend linearly from the other weights. A decision maker selecting an alternative with $a_i = 0$ should be able to justify why the weights must satisfy such additional dependencies. Moreover, there must be at least one other alternative option having $a_i \neq 0$ and whose favourable weight vector is the same, so the DM should also justify why she/he didn't select an equally good but more acceptable alternative instead. Without referring to criteria external to the model, choosing an alternative option with $a_i = 0$ can not be encouraged in SMAA.

2.5.2.2 Stochastic Case

In this case the value γ_{ij} for alternative i and criterion j are given as stochastic variables [29] [30]. A density function $f(\gamma)$ specifies the joint probability distribution of γ_{ij} , which can be computed as

$$f(\gamma) = \prod_{i,j} f_{ij}(\gamma_{ij})$$

The utility function is used for mapping the criterion distributions into utility distributions and the expected utilities are evaluated by the given function.

$$u_{ij} = \int_{\gamma} f(\gamma) u_j(\gamma_{ij}) d\gamma$$

The rest of the process is continued in same manner as in deterministic case. Here the overall utility $u_i(\gamma_i, w)$ is expressed as the convex combination of the criteria distributions.

$$u_i(\gamma_i, w) = \sum_{j=1}^n w_j u_j(\gamma_{ij}) \quad w \in W$$

Just like the deterministic case, here the *favourable weight vector* $W_i(\gamma)$ are computed as

$$W_i(\gamma) = \{w \in W : u_i(\gamma_i, w) \geq u_k(\gamma_k, w), \quad k = 1, \dots, m; k \neq i\}$$

The expected value of volume $W_i(\gamma)$ can be calculated

$$E(\text{vol}(W_i(\gamma))) = \int_{\gamma} f(\gamma) \int_{w_i \gamma} dw d\gamma$$

The *central weight vector* w_i^c for alternative i is expressed as

$$w_i^c = \int_{\gamma} f(\gamma) \left(\int_{w_i\gamma} w dw \int_{w_i\gamma} dw \right) d\gamma$$

For alternative i , the ratio between the expected volume of $W_i(\gamma)$ and the volume of feasible weight space W is evaluated to get *stochastic acceptability index* a_i .

$$a_i = \frac{E(\text{vol}(W_i(\gamma)))}{\text{vol}(W)}$$

In the deterministic case, the *central weight vector* makes an alternative i with $a_i \neq 0$ better than others with certainty but in the stochastic case, a *confidence factor* p_i^c for each alternative is used to measure the probability of that alternative to be preferred one if the w_i^c is chosen.

$$p_i^c = \int_{\gamma: u_i(\gamma_i, w_i^c) \geq u_k(\gamma_k, w_i^c)} f(\gamma) d\gamma$$

The *confidence factor* can be calculated for any given weight vector and alternative. It enables us to measure whether criteria assessment is accurate enough to discern among the alternatives when the w_i^c is given [31].

SMAA is not a single technique; it is a family of different versions developed to support different kinds of decision making problems. SMAA-2 is used for discrete stochastic multiple criteria decision making analysis involving multiple DMs. A ranking function is used to measure the preferences to make the alternative acceptable for each particular rank.

A new method SMAA-D is the combination of Data Envelopment Analysis (DEA) (used for classifying alternatives into efficient and non-efficient ones) and SMAA-2 which is used to handle the uncertain or imprecise criteria measurements and partial preference information by using stochastic distributions [32]. It is useful to implement where the criteria can be classified into inputs and outputs conveniently, and the participants/DMs agree to use DEA to represent the structure of their preference information. SMAA only addresses the issue of such uncertain, imprecise, missing or partially missing preference information. However, the uncertainties and dependencies among the utilities still need to be tackled.

2.6 Pairwise-Comparison Approaches

This approach is based on the comparison of entities in pairs to analyse which entity is the most preferred one, or has more weight or identical or not. This approach was first introduced by Psychometrician L.L. Thurstone in 1927 [33] who called it the law of comparative judgement; a model used to get measurements from any method of pairwise comparisons.

2.6.1 Analytic Hierarchy Process (AHP)

Thurstone's law of comparative judgement was later adopted by Thomas L. Saaty in the 1970s [34] in multi-criteria decision making analysis called Analytic Hierarchy Process (AHP) which is now widely used for the study of preferences, social choices, attribute and many other purposes. AHP is a technique which constructs a multiple criteria decision making problem in a hierarchical manner such as goal, criteria, sub-criteria and alternative decisions shown in Figure 2.5 below.

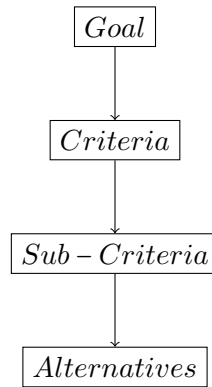


Figure 2.5: Analytic Hierarchy Process

It allows pairwise comparisons to measure the relative importance of the elements at each level in order to make best decisions [35]. The number of comparisons depend on the number of criteria involved in the problem shown in the Table 2.3 below.

The interdependencies among the elements of different hierarchies not being handled in the AHP, if there are some dependencies among the elements they must be put in the same level of hierarchy. The dependencies among the criteria in the AHP works like XOR (exclusive or) where you have to select one or the other but not both at one time i.e., you can prefer either safety or cost in a pairwise comparison or let suppose we have a family dinner but we can have my family invited or my husband's family but not both at the same dinner.

No. of criteria	1	2	3	...	n
No. of comparisons	0	1	3	...	$n = \frac{n(n-1)}{2}$

Table 2.3: Number of Comparisons w.r.t Criteria

A judgement scale from 1 to 9 is used for preferences of the users on each level, where 1 is of equal importance and 9 for extreme importance. Let suppose a car buying problem where we have 3 different criteria to consider i.e., safety, cost and comfort. Cost is more important than safety (judgement scale = 3) while safety is more preferable than comfort to user (judgement scale = 5) whereas cost has dominance over comfort (judgement scale = 7). The number of

comparisons here will be 3 and the comparison matrix A for these criteria based on the above judgement scales will be evaluated as

$$A = \begin{bmatrix} & \text{Safety} & \text{Cost} & \text{Comfort} \\ \text{Safety} & 1 & 0.33 & 5 \\ \text{Cost} & 3 & 1 & 7 \\ \text{Comfort} & 0.2 & 0.14 & 1 \end{bmatrix}$$

The lower diagonal values in the matrix are filled using the formula $a_{ji} = \frac{1}{a_{ij}}$ where a_{ij} is the element of row i and column j . After having the comparisons, a priority vector is calculated which is the normalized Eigenvector of the matrix. Eigenvectors and Eigenvalues are being calculated accordingly and the largest Eigenvalue called the principle Eigenvalue λ_{max} is obtained. The priority ranking based on the following formula

$$AX = \lambda_{max}X$$

where A is the comparison matrix of size $n \times n$ for n number of criteria, $\lambda_{max} \in R > n$ and X is the Eigenvector of size $n \times 1$. Lastly, consistency ratio CR is calculated to measure the consistency of the judgements i.e., if $A > B$ and $B > C$ then logically $A > C$. Consistency ratio can be measured using the formula $CR = \frac{CI}{RI}$ where CI is the Consistency Index and can be calculated as $CI = \frac{\lambda_{max} - n}{n - 1}$, RI is the Random Consistency Index which is an appropriate Consistency Index. If the CR is less than or equal to 0.1 then inconsistency is acceptable if it is greater than 0.1 then subjective judgements must be revised.

The AHP has been used to produce priorities that can be used as prior probabilities in Bayes' formula where the prior probabilities cannot be generated using the statistical approaches [36]. It is widely used in many areas to deal with MCDA problems however it has limitations when applied to complex real life problems that involve interdependence and feedback.

2.6.1.1 Limitations of AHP

AHP is very simple to understand and apply, in the way that it supports breaking down a complex problem into a hierarchy of components, but this hierarchy also introduces limitations, as follows.

- It requires a large number of comparisons among the criteria and is very difficult to use when there are more than 7 criteria.

- Its hard to add new criteria/alternative or remove existing ones because all the evaluations will need to be revised.
- It does not take the utility dependencies and uncertainties into account.

A generalized form of the AHP called Analytical Network Process (ANP) was introduced to address some of these limitations.

2.6.2 Analytic Network Process (ANP)

All decision problems cannot be structured hierarchically because in most of the problems the elements have to interact and may be dependent upon each other. Analytical Network Process (ANP) [37] considers those dependencies and interaction among the elements, that's why it is represented as a network rather than hierarchy. A network consists of clusters and there are two types of influences/dependencies given below:

- The influence/dependency of the elements in one cluster upon the elements of other cluster is known as Outer dependency/influence.
- While inner dependency/influence is the dependency among the elements of same cluster [38].

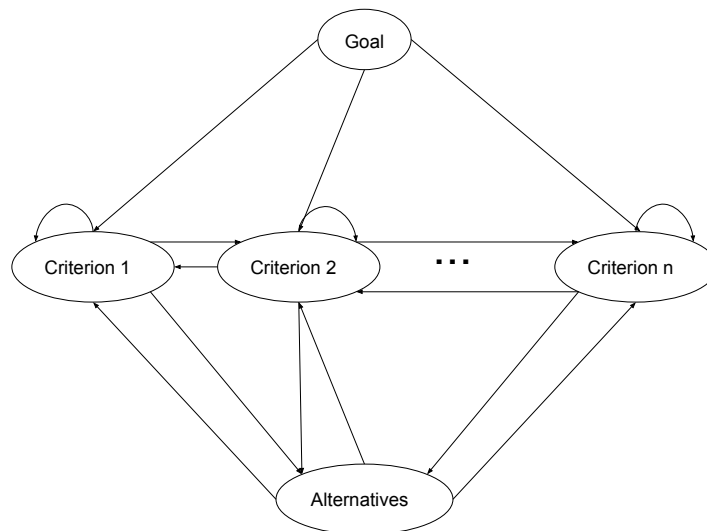


Figure 2.6: Analytic Network Process [5]

In ANP, the goal, criteria and alternatives are represented as clusters in a network and arrows show the outer and inner dependencies among those clusters as shown in the Figure 2.6. However, same judgement scale as the AHP from 1 - 9 is used here for the purpose of pairwise comparisons among the elements. While making pairwise comparisons, the user must take

care of any inconsistencies i.e., if *safety* of a car is preferable to *cost* and *cost* is preferable to *comfort* then logically *safety* is preferable to *comfort*. Lets take an example of choosing best scientist among three scientists to explain the process of ANP in detail. Here our Goal is choosing best scientist among *John*, *Harry* and *Peter*, the selection criteria are Course Development (*CD*), Popularity (*PO*), Research Papers (*RP*), Citations (*CI*) and Science Projects (*SP*) while Alternative outcomes are *John*, *Harry* and *Peter*. For optimizing our problem, lets make two clusters for criteria; *Teaching* (having Course Development (*CD*) and Popularity (*PO*) and *Research* (Research Papers (*RP*), Citations (*CI*) and Science Projects (*SP*) shown in the Figure 2.8 below.

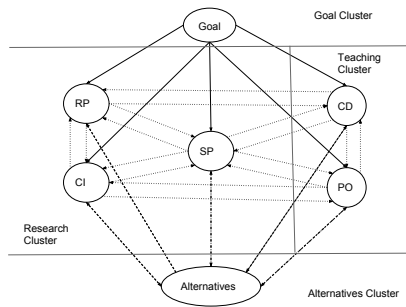


Figure 2.7: ANP Problem Breakdown

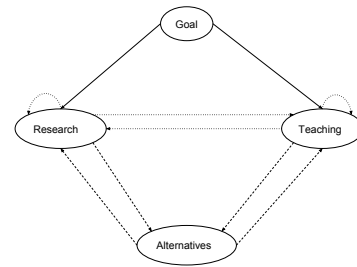


Figure 2.8: Selecting best Scientist

Dotted arcs show the dependencies among the criteria and the dashed arcs represent the dependencies among the alternatives upon criteria. Some of the arcs are missing due to the complexity of diagram but they do exist such as the arc between Research Papers (*RP*) and Popularity (*PO*).

Pairwise Comparison

After getting the network based structure of the decision making problem, an unweighted super matrix is created. That matrix contains all the nodes of the network and their associated local priorities. The judgement scale for the priorities is same as mentioned by Saaty in the AHP and there is a consistency ratio which measures the decision maker's inconsistency about preferences during pairwise comparisons and that ratio must be under 0.1 or 10 percent. In the above example, pairwise comparisons on nodes level with respect to the goal will be evaluated i.e., criteria in *Teaching* will be evaluated w.r.t goal and will be placed at rows *CD* and *PO* and so on. The calculations for priorities regarding *CD* and *PO* are as follows:

Goal	<i>CD</i>	<i>PO</i>
<i>CD</i>	1	1/2
<i>PO</i>	2	1
Sum	3	1.5

After inputting the judgement scales and user preferences we obtain the unweighted super

Goal	CD	PO	Average
CD	$1/3 = 0.33$	$0.5/1.5 = 0.33$	0.33
PO	$2/3 = 0.67$	$1/1.5 = 0.67$	0.67

matrix 2.4 [39,40]. For pairwise comparisons on cluster level, the unweighted super matrix is converted into weighted super matrix. Here we need to compare criteria based clusters w.r.t goal. Let suppose *Research* criterion is more important than *Teaching* criterion weighing 0.75 and 0.25 respectively, then 0.75 will be multiplied to all the elements of *Research* criterion (RP , CI , SP) and 0.25 to the elements in *Teaching* cluster (CD , PO). Compare all the values in cluster criteria w.r.t alternatives and vice versa. The resultant values are shown in the weighted super matrix 2.5.

	Goal	CD	PO	RP	CI	SP	<i>John</i>	<i>Harry</i>	<i>Peter</i>
Goal	0	0	0	0	0	0	0	0	0
CD	0.33	0	1	0	0	1	0.2	0.4	0.6
PO	0.67	1	0	0	0	0	0.8	0.6	0.4
RP	0.25	1	0	0	0.4	0.6	0.2	0.4	0.6
CI	0.25	0	0	0	0	0.4	0.2	0.5	0.3
SP	0.5	0	1	1	0.6	0	0.6	0.1	0.1
<i>John</i>	0	0.5	0.5	0.2	0.4	0.6	0	0	0
<i>Harry</i>	0	0.3	0.2	0.2	0.5	0.3	0	0	0
<i>Peter</i>	0	0.2	0.3	0.6	0.1	0.1	0	0	0

Table 2.4: Unweighted Super Matrix

	Goal	CD	PO	RP	CI	SP	<i>John</i>	<i>Harry</i>	<i>Peter</i>
Goal	0	0	0	0	0	0	0	0	0
CD	0.8	0	0.2	0	0	0.2	0.1	0.2	0.3
PO	0.17	0.2	0	0	0	0	0.4	0.3	0.2
RP	0.186	0.2	0	0	0.10	0.12	0.1	0.2	0.3
CI	0.186	0	0	0	0	0.08	0.1	0.25	0.15
SP	0.383	0	0.2	0.25	0.15	0	0.3	0.15	0.05
<i>John</i>	0	0.3	0.3	0.15	0.30	0.36	0	0	0
<i>Harry</i>	0	0.18	0.12	0.15	0.375	0.18	0	0	0
<i>Peter</i>	0	0.12	0.18	0.45	0.075	0.06	0	0	0

Table 2.5: Weighted Super Matrix

The next step is calculating Limit matrix; here we normalize the priorities of the elements in each cluster equal to one by multiplying weighted super matrix to itself until all the columns become identical. to get final priorities. These values are then listed as a limit matrix in order to get the desired priorities/weights of the elements with respect to the goal. The alternative with the highest value is then chosen as the best outcome of this process [41,42].

2.6.2.1 Limitations of ANP

ANP gained popularity and is being used widely for solving complex decision problems both real and hypothetical (e.g. [35]) but there are some limitations:

- The process of explaining the concept of ANP is extremely challenging.
- It requires a lot of calculations and formation of pairwise comparison matrices as compared to AHP method, and hence the user must be very careful in order to keep track of all the comparisons [43].
- It is very complex to use as a standard software tool for practical decision making in an organization.
- It is not suitable for multiple criteria decision making problems when there are uncertainties among the utilities.

2.7 Outranking Approaches

Outranking is one of the approaches used to help in making complex decision making problems involving multiple choices and multiple agents. It is basically dealing with the dominance of one alternative over the other and hence it focuses on whether there is available information to select the preferred choice over other. In this approach, there is no notion of value, and hence it does not consider the issue of the existence of a utility function and its functional form. Therefore, it is not directly applicable or comparable to the utility-based approaches taken in this thesis, but we include a brief description of two variants of this approach for completeness.

2.7.1 ELECTRE

ELECTRE stands for ELimination Et Choix Traduisant la REalité (ELimination and Choice Expressing REality) was first presented in 1966 to work on real world problem at a European Consultancy Company SEMA [44]. ELECTRE has two main parts: construction followed by the exploitation procedure. The construction phase aims to compare each pair of actions then an exploitation procedure is being used to elaborate the recommendations obtained in earlier phase. The recommendations can be choosing, sorting or ranking based on the problem being addressed [45]. Preferences in ELECTRE are being modelled using binary outranking relations, S , which indicates that "*at least as good as*". Let suppose we have two actions x and y , so four possibilities may occur:

- xSy and not ySx , i.e., xPy (x is strictly preferred to y)

- ySx and not xSy , i.e., yPx (y is strictly preferred to x)
- xSy and ySx , i.e., xIy (x is indifferent to y)
- Not xSy and not ySx , i.e., xRy (x is incomparable to y)

Outranking relations uses a new preference relation, R (*in-comparability*) to model preferences. This relation helps in the situations where the decision maker is not able to compare a pair of actions [46].

2.7.2 PROMETHEE

Preference Ranking Organization METHod for Enrichment of Evaluations (PROMETHEE) is an outranking approach based on mathematics and sociology, developed in the beginning of 1980s. This method assists the Decision Makers (DMs) to find out the alternatives that are most suitable to their goal rather than finding out the right decision [47]. It helps the DMs to have a better understanding about the decision problem by providing a comprehensive and structural framework [48]. This method is widely used around the world to different decision scenarios e.g., healthcare, business, transportation, education and governmental institutions.

2.8 Capabilities and Limitations of Traditional MCDA Techniques

The utility based approaches such as MAUT and SMAA use utility functions in order to evaluate users' preferences to assist the decision making process as discussed in 2.5. These approaches elicit weights from different experts to assign to different attributes in a decision, evaluate and sum up the attributes using a utility function and select the alternative with highest utility. These methods are limited to handling independent utilities elicited from different participants and are not capable of addressing uncertainty among the criteria weighting. Pairwise approaches such as AHP and ANP are discussed in Section 2.6 which are commonly used in various domains. These methods compare a pair of criteria involved in a multi-criteria decision making problem where the participants are asked to rate a criterion as being more important than another using a judgment scale. The elicited responses are then used to calculate the eigenvector and eigenvalues to select an alternative with the highest value. AHP considers dependencies among the criteria, but only pairwise and hierarchical while ANP handles graphical dependencies (including feedback) among the criteria, but only pairwise. These approaches are limited to problems with a small number of criteria and are not practical for complex problems. Neither they use utility functions nor consider uncertainties. Another type of MCDA techniques are outranking approaches such as ELECTRE and PROMETHEE

discussed in Section 2.7. These methods are used to assist in decision making problems by comparing a pair of actions followed by an exploitation procedure to elaborate the recommendations obtained in an earlier phase. These techniques are not based on utilities and values.

Capabilities/Limitations	MAUT	SMAA	AHP	ANP	ELECTRE	PROMETHEE	Our Survey Approach	Our CaMML Approach
Uses utility function	yes	yes	no	no	no	no	yes	yes
Hierarchical utility dependencies	no	no	yes	yes	no	no	yes	yes
Graphical utility dependencies	no	no	no	yes	no	no	no	yes
Pairwise relationships	no	no	yes	yes	no	no	yes	yes
Higher order relationships	no	no	no	no	no	no	no	yes
Uncertainty in preferences	no	yes	no	no	no	no	yes	yes
Handles over-confidence	no	no	no	no	no	no	yes	yes

Table 2.6: Capabilities/Limitations of MCDA Techniques vs Our Approach

Table 2.6 shows a summary of the capabilities and limitations of the existing MCDA techniques in comparison to our approach proposed in this thesis. Here we can see that neither the traditional methods deal with the arbitrary dependencies among the criteria nor do they consider the over-confidence and uncertainty in preference weighting. On the other hand, our approach is a utility based approach to develop a decision problem using a BDN with utilities (Section 5.2) and find out the dependencies among those utilities (Section 4.5.2). The preference elicitation method we propose in this thesis uses a pairwise comparison approach to compare different criteria involved in the decision making problem (Section 4.2), and addresses uncertainty and over-confidence in weighting those criteria (Section 4.3). The machine learning approach in thesis is capable of handling arbitrary dependencies (i.e. graphical, including feedback and representations of relationships of any order), however, the survey approach is currently limited to pairwise, but could be extended.

2.9 Eliciting Preferences

Decision making is one of the most crucial parts of almost every domain in our life. Decisions often need to be made with incomplete and imperfect data, involving uncertainty and risk. Therefore, experts' judgments and knowledge are required to make well-informed decisions [49]. Expert judgments are being elicited in a variety of different applications such as environmental impact assessment [50], predictive models [51], structured decision making [52] etc. Unstructured and ill-informed elicitation techniques are not only unreliable and unproducible, but often even disastrous. Well structured and systematic elicitation approaches can enhance the quality of expert judgments to make efficient decisions [53]. One of the recent structured protocols named IDEA protocol (Investigate, Discuss, Estimate, Aggregate judgments) distills the most useful elements from existing approaches, and combines them into one novel and practical protocol.

2.9.1 Traditional IDEA Protocol

The IDEA protocol [9, 53, 54] is a structured elicitation protocol used to elicit judgments from domain experts, especially judgments of probabilities and other real-valued physical parameters. IDEA stands for “Investigate”, “Discuss”, “Estimate” and “Aggregate”, which form the basic steps of this elicitation method. IDEA recruits a diverse group of participants/experts who are asked to provide probabilistic or quantitative answers to a set of questions. The steps in the IDEA protocol are as follows.

1. *Investigate*: In this first step, experts investigate the questions and understand the context of questions. In this round they provide their estimates privately: their lowest plausible value, highest plausible value and best single estimate, and (in one version, see below) their confidence.
2. *Discuss*: Next, they exchange feedback with each other on their given answers and discuss the results in the presence of a facilitator. This step helps in promoting critical thinking and understanding ambiguity, but is not meant necessarily to reach consensus.
3. *Estimate*: After discussing with each other, each expert provides a revised final round set of estimates privately.
4. *Aggregate*: In this last step, the elicited individual estimates from the experts are combined using a statistical aggregation.

IDEA protocol emphasizes both the wording (asking experts to consider plausible and realistic scenarios) and the questions order (lowest estimate, highest estimate, best guess, confidence). They intentionally use the words ‘plausible’ and ‘realistic’ to refrain the participants from specifying uninformative bounds (such as 0 and 1 on probability estimates). The order of questions (asking lowest and highest bound first) avoids anchoring on best estimates, and encourages to provide an interval and the evidence for relatively extreme values. The lowest estimate and highest estimate provide uncertainty bounds, best guess is interpreted as a best estimate (not as a mean, median, or mode) [53].

2.9.1.1 Elicitation Method of IDEA Protocol

Currently, the research discussion is restricted to using IDEA protocol to elicit probabilities of events to occur. However, the elicitation step in IDEA is based on earlier work on eliciting either probabilities or quantities, in either a 3 point or 4 point elicitation.

In 3 step elicitation [9], the participants are asked to provide lower and upper bounds to draw an uncertainty interval followed by asking participant’s “best guess” which is treated as

the best estimate of a particular event to occur. In the 4 point elicitation method [54], participants are asked to provide the same estimates as in the 3 point method – the lowest plausible value of an event to occur, the highest plausible value, their best estimate – with the fourth additional estimate being their confidence that the interval they have created from lowest to highest value actually holds the true value.

The 4 point elicitation method helps in reducing the over confidence issue by letting the participants or experts specify their own level of confidence [55]. We use the 4 point elicitation method for a quite different purpose in Chapter 4, namely to elicit preferences in the form of continuous values, instead of probabilities.

3 Point Elicitation	4 Point Elicitation
Lowest plausible value	Lowest plausible value
Highest plausible value	Highest plausible value
Best estimate	Best estimate
	Confidence

Table 2.7: 3 Point vs 4 Point Elicitation by IDEA Protocol

2.10 Causal Discovery

Existing machine learning techniques can predict variables based on correlation, but correlation not being the same as causation [56]. Correlation only informs us about the association or relation if there exists among the variables, for example it tells us there is a relation between variable A and B but it does not necessarily mean that A can alter B. On the other hand, causation tells us the directional relationship between variables, for example there is a relationship between two variables A and B such that changes in A leads to changes in B [57]. Causal relations are crucial because the understanding of underlying causal relations assist in predicting the behaviour of a system when it is subjected to interventions [58].

The main difference between correlation and causation lies in the potential of common cause, for example there is no direct causal relationship between two variables A and B but a third variable C causes both A and B. In such case, there is a strong correlation between A and B but changes in A would not alter B and the common cause C is called a co-founder which can be observed or latent. Causal structure discovery can be categorized as the problem of learning the causal structure from the observational data [57].

There are two types of causal structure discovery algorithms called constraint-based algorithms and score-based algorithms. Constraint-based algorithms aim to generate the causal structure by identifying the conditional independence constraints with statistical test, and connect the nodes that are not found independent. Score-based algorithms construct a number of

candidate causal graphs, each candidate graph is assigned a network reflecting its goodness of fit, which the algorithms then attempt to maximise.

2.10.1 PC Algorithm

PC algorithm is a well-known constraint-based algorithm that was named after its authors Peter and Clark. It is widely used for causal discovery because it is easily explained and runs quickly. It was developed by [59] and uses classical significance testing for dependencies to identify local structural constraints. PC starts with a fully connected *skeleton* network, where every node is connected to every other, recursively removing edges according to any significance tests reporting conditional independence. For example, an edge between X and Y is removed if we find a set Z that does not include X and Y , and when conditioning on Z , X and Y are independent. PC has been implemented in various open-source software and has become a very reliable tool to learn the causal structures from the data [60].

2.10.2 K2 Algorithm

This structure learning algorithm is one of the earliest BN learning algorithms, that was developed by [61]. It is a score-based algorithm which uses greedy search approach in an iterative process to obtain the best DAG structure per a Bayesian metric. This algorithm requires a variable order and a maximal number of parents for each node before the search process can start, but is often combined with a minimum weighted spanning tree algorithm to provide an ordering. Given an initial node ordering reduces the search space significantly which makes K2 algorithm computationally efficient [62]. But an improper initial node ordering may result in a very poor performance, so it is very important to provide it with an effective initial node ordering.

2.10.3 CaMML

CaMML or Causal model using the Minimum Message Length (MML) is one of the causal discovery algorithms used to learn the causal structure from the data using the MML method [8]. MML principle is used for inductive inference (generalized conclusions are drawn from specific observations) where the authors assume that given data D is a binary-coded message which consists of two parts; hypothesis message and data message, and the MML principle supports the inference that minimizes the length of this binary-coded message. For CaMML, we measure the MML cost of models using efficient encoding methods and sample data, and search through the space of possible models. This search and score method uses a greedy search approach starting with an automatically generated seed model and running through

each pair of nodes in an arbitrary order. Then it attempts to add an arc if there is no arc between those two nodes, delete or change the direction of the arc if there is one. The change is effected if the current model's MML cost minimizes, this loop continues for all the node and the whole cycle is repeated until no change is effective. Causal models learned from generated data using CaMML are considered to be very accurate reflections of the original models.

2.11 Validation and Evaluation of our Models

Validation and Evaluation of models is a very important step throughout the development of the model. Validation is the process of comparing the model predictions with a real-world data set, in order to assess its accuracy. Without having a proper validation, it is impossible to do a comparative analysis of the model with other models. Evaluation of the model is performed to assess the adequacy of the needs of the concerned users and in the remainder of this section we describe the main validation and evaluation used in this thesis.

2.11.1 Evaluation Approaches

We require model evaluation approaches to quantify the model performance. In this section, we review some of the most commonly used approaches for evaluating our learned models.

2.11.2 Expert Opinion and Internal Validity

Expert opinions are the subjective assessments obtained from the experts to evaluate the predictive model's structure and behaviour. We use the experts at fleet procurement department at Monash University to assess the results of each technique based on case analysis and internal validity. Case analysis takes specific instances and examines how the model deals with them, which may and should include both normal and extreme cases. Internal validity determines the causal relationship between the dependent and independent variables. It specifically checks the internal behaviour of the model regardless of any external changes [63]. We use each method to generate option rankings and ask the experts to judge which technique is better in terms of option rankings.

2.11.3 Predictive Accuracy

Predictive accuracy one of the most commonly used technique to validate the predictive models such a BNs, regression models, classification trees etc. Predictive accuracy is used to measure the fit to data of each model via the following algorithm [63]:

1. Set a counter i to 0.

2. For each case in the test data set.
 - (a) Add any available evidence for the attribute nodes.
 - (b) Update the network.
 - (c) Take the predicted value for target variable to be the one that has the highest posterior probability (the modal probability).
 - (d) If this value is equal to the actual value for that variable in the sample, increment i .
3. Return i divided by the sample size.

2.11.4 F Score Measure

F-measure is also known as F-score which is a measure of a test's accuracy. F-measure considers both the precision and the recall of the test to calculate the score. The result is a value between 0.0 for the worst F-measure and 1.0 for a perfect F-measure. Precision is the number of true positive results divided by the number of all positive results including the incorrectly identified. However, Recall is the number of true positive results divided by the number of all relevant samples (total actual positives). F1 score or balance score is calculated using the formula 2.1 given below:

$$F1 \text{ Score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (2.1)$$

2.11.5 Kullback Leibler Divergence

Kullback-Leibler divergence or KL divergence is a method of comparing two probability distributions over the same variable z . In probability and statistics, we often tend to replace the observed data with approximating distribution, so KL divergence assist us to measure how much information we lose when we choose an approximation. It is a non-symmetric measure between two probability distribution $p(z)$ and $q(z)$. We denote the KL divergence of $q(z)$ from $p(z)$, by $D_{KL}(p(z), q(z))$, to measure the information we lose when we use $q(z)$ to approximate $p(z)$ [64].

Let say we have two probability distributions $p(z)$ and $q(z)$ of a discrete random variable z . Whereas, both $p(z)$ and $q(z)$ add up to 1, and $p(z) > 0$ and $q(z) > 0$ for any z in Z . Using the equation 2.2 we calculate the KL divergence $D_{KL}(p(z), q(z))$.

$$D_{KL}(p(z)||q(z)) = \sum_{z \in Z} p(z) \ln \frac{p(z)}{q(z)} \quad (2.2)$$

The operator “ $||$ ” denoted the divergence of distribution $p(z)$ from $q(z)$.

2.11.6 Log Score Function

Logarithmic score is the log predicted likelihood which was originally proposed by Good in 1952. It is one of the strictly proper scoring rules which assigns a harsh penalty to low probability cases and is highly sensitive to extreme cases. The forecaster can maximize the expected reward by forecasting the true distribution and be honest when reporting the distribution. Log score is calculated as the log of the probability estimate of the actual outcome, defined by

$$\log S(f, x_t) = \log f(x_t)$$

Where f is the distribution forecast for the random variable X_t evaluated at the outcome x_t . Log score is (much) better than predictive accuracy, because (as a proper scoring rule) it maximally rewards the true distribution.

It is one of the widely used scoring rule to evaluate the accuracy of the predictive models, so we use it to validate our models (see details in Section 5.5.4 in Chapter 5).

2.11.7 Bayesian Information Reward (BIR)

Bayesian Information Reward (BIR) is the generalized version of Good's information reward method which was originally proposed by I.J. Good in 1952 [65]. The original IR positively rewards binary classifications which are informed relative to an uninformed uniform prior, however, BIR generalizes the binary case to multinomial predictions, taking into account the non-uniform priors. It also appropriately punishes miscalibration; that is, it will penalize any *correct* prediction with a confidence different from its actual probability over the long term, meaning it is a strictly proper scoring function [66]. The reward increases asymptotically to one for extreme predictions close to one or zero, however, for incorrect predictions the negative reward increases without bound.

$$IR_B = \sum \frac{I_i}{n}$$

Where n is the possible target values, $I_i = I_i^+$ for true values and $I_i = I_i^-$ otherwise, and

$$I_i^+ = \log \frac{\hat{p}_i}{p_i}$$

$$I_i^- = \log \frac{1 - \hat{p}_i}{1 - p_i}$$

Where $\hat{p}_i$ is the posterior probability and p_i is the prior probability of the variable taking the value. This method is better than predictive accuracy as predictive accuracy fails to take uncertainty of predictions into account; for example, a prediction of a disease's probability of 0.52

counts the same as a prediction of a disease's probability of 1.0. BIR on the other hand, rewards for an informative prediction which is completely ignored by predictive accuracy and not adequately addressed by Good's Information Reward method. We use this validation method for our proposed elicitation models as it is very efficient and informative (see the detailed explanation and example in Section 5.5.5 of Chapter 5).

2.11.8 Expected Value (EV)

One of the evaluation techniques we use in our evaluation process is calculating Expected Value (EV) for each model and compare the results to see which model has the best performance. EV is the probability-weighted sum of the all possible values that a random variable can take on. It helps us analyze the situations where the Decision Maker (DM) must make a decision without any knowledge about the outcomes that may occur from that decision. Such situations are called decision making under uncertainty and the DM will tend to choose a decision / action that will results in maximum expected utility value. The origin of expected utility theory was first introduced by a mathematician named "Daniel Bernoulli" in 1738 to solve St. Petersburg paradox [67]. According to his concept, peoples' preferences reflect their expected benefit from a lottery instead of its expected monetary value. In 1940, Jhon Von Neumann and Oskar Morgenstern began to lay the axiomatic foundation of expected utility theory [68]. They presented the use of mathematical function i.e., "utility function" to calculate peoples' preferences towards lotteries. EV can be calculated by multiplying all the possible outcomes by the likelihood of each outcome that it will occur, and then sum up all those values to get an overall EV as shown by the formula below.

$$EV = \sum P(X_i) \times X_i$$

Where EV is the Expected Value, $P(X_i)$ is the probability of the event and X_i is the event.

2.12 Conclusion

In this chapter, we discussed reasoning under uncertainty and how to use probabilities to measure uncertainties. We introduced probabilistic graphical models to represent the dependencies among various random variables, specifically Bayesian networks which are the focus of the models developed in this thesis. We also presented Bayesian decision networks, an extension for supporting decision making. We discussed multi-criteria decision making and explained the existing MCDA methods that are used in various domains to assist in decision making. Traditional MCDA techniques have various limitations and issues which were

also explored in this chapter. In the next chapters, we use BNs to model multi criteria decision making problems. We surveyed various methods for eliciting preferences, including the IDEAL protocol which inspired our proposed preference elicitation methods. We also introduced causal discovery to learn the causal structure from the data, which we later apply with preference data. Lastly, we explored the validation and evaluation metrics used for empirical analysis of the predictive models, which we use to empirically evaluate and compare our various approaches.

Learning BNs from Simulation Data

This chapter is based on a published paper “Bayesian network meta-models from combat simulation for defence decision analysis”, with Steven Mascaro, Thang Cao, Kevin B. Korb. Thang Cao provided us with unclassified simulation data from DST group which was used for creating and validating BNs. I am the main author of this paper who was responsible for generating the BNs from the data and validating them in consultation with Steven Mascaro and Thang Cao whereas Kevin B. Korb was responsible for proof writing and expert advice.

In the previous chapter we discussed the traditional MCDA techniques and their limitations, described the structured IDEA approach to preference elicitation and introduced causal discovery. We also described some validation and evaluation techniques that are necessary for evaluating MCDA approaches. In this chapter we introduce a practical method for creating BDN models suitable for decision support based on multidimensional simulation data. Today, simulations can produce enormous quantities of data that are often difficult to summarise well, particularly in a format that provides decision makers with concise and relevant information that does not overwhelm. BDNs provide the ideal opportunity to transform simulation data into a form that can directly drive decision support tools. This chapter provides an end-to-end process for creating such BDNs from simulation data, applied to real simulations for Land Combat Vehicle (LCV) assessment. This process uses causal discovery algorithms to learn the BN structures from the simulation data and validate those models against the data to check their accuracy; performs sensitivity analyses and uses causal visualizations and what-if analyses to reveal key information; and extends these machine learned models with utilities to provide standalone models that summarise and support assessments of the various decision options.

In Section 3.1 we introduce the problem description, motivation and our proposed solution. Section 3.2 explains the background of the problem followed by the case study based on combat simulation. We develop a BN model for combat simulation study using causal discovery algorithms in Section 3.4 and validate those models in Section 3.5. We explore the causal visualization and what if analysis in Section 3.6 and 3.7 followed by the addition to utility

nodes to the BNs in Section 3.8. Lastly, we conclude this chapter in Section 3.9 by adding the research issues we are addressing in further chapters.

3.1 Land Combat Procurement Problem

Defence Science and Technology Group (DST) is investigating the characteristics of LCVs with a view to improving purchasing decisions between different LCV options. Using a combination of closed-loop combat simulation (i.e., one run without any human interventions) and subject matter experts (SMEs), these studies seek to understand the close combat capability factors for a future LCV. These factors are being investigated for different kinds of physical environment and different levels of enemy lethality, with each LCV's lethality, survivability, signature (the ease with which the enemy identifies friendly forces) and knowledge acquisition (friendly forces identifying enemies) being measured, yielding a multi-dimensional view of the different LCVs and their performance. Our study describes a new approach, developing a Bayesian network (BN) meta-model (i.e., a model based upon the combat simulation model; see Fig 3.1) that combines those multiple dimensions in a single multi-criterial decision model.

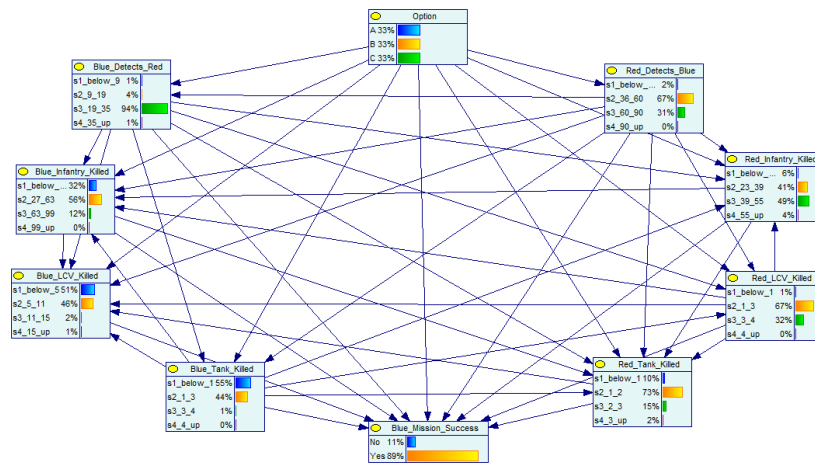


Figure 3.1: A Bayesian network meta-model (from the PC algorithm).

Both deterministic and stochastic methods have been developed for multi-criteria decision analysis (MCDA). Limitations to prior approaches include treating utilities as independent of each other and also very limited ability to deal with uncertainty about preferences. Bayesian networks offer advantages in these regards, so here we develop an alternative approach using a BN meta-model for decision making and evaluating the operational impact of different LCV choices. BNs support both the prediction of operational results and their causal explanation. These BNs were developed both from expert elicitation and causal discovery (data mining) from combat simulation data, and enable causal visualization, sensitivity analysis and opti-

mization. We evaluate the performance of some of these BNs against alternatives, using common accuracy-based evaluation metrics. Please note that the example and data presented in this chapter are not representative of the real LCV options and testing scenario, they are used for methodology illustration only.

3.2 Background

In multi-criteria decision analysis (MCDA), decision makers (DMs) choose from m alternatives using n criteria. Often criteria are themselves composed of weighted sub-criteria as discussed in Section 2.3 of Chapter 2. The preferences (weights, utilities) associated with the (sub)criteria may be determined by interviews, surveys, etc. with SMEs, direct measurements, or simulations. Whatever their sources, real world applications must cope with uncertain and incomplete data. The weights are also frequently uncertain and occasionally incomplete. In our study, the criteria utilities and weights are evaluated using both combat simulation and SMEs.

Deterministic methods of ranking and selecting the “best option” from simulation has been extensively studied by [69] for a single response metric, and there are some studies, e.g., [70] and [71], which apply different MCDA methods to multi-criterial problems. AHP weighting is the most common deterministic weighting approach in MCDA defence problems [70, 72] due to its capacity to establish the weights of attributes in a systematic and robust manner and allow the DMs to check the consistency and consensus between the rankings of the criteria in group decision making.

However, DST deals extensively with MCDA problems that require handling uncertainty about both attributes and utilities. Several other MCDA methods do support handling uncertain or inaccurate criterial information. For example, ELECTRE [73] and PROMETHEE [74] handle different kinds of uncertainty using threshold models. In multi-attribute utility theory (MAUT) the uncertainty is represented using probability distributions, and alternatives are evaluated based on their expected utility [75]. Of these, the MAUT-based methods are able to handle criteria uncertainties, but not preference uncertainties. As in MAUT, the Stochastic multi-criteria Acceptability Analysis (SMAA) methods (see e.g., [28], and [76]) represent uncertain or inaccurate criteria through probability distributions and are thus able to handle arbitrary dependencies between criteria uncertainties.

We here present an alternative approach using BN meta-models for decision making and evaluate their performance against alternatives, using common accuracy-based evaluation metrics.

3.3 Combat Simulation Study

The combat simulation used in these studies is based on a mission requiring the rapid seizure of an urban fringe area. The Blue force (the side for which LCV options were being assessed) was a Combat Team (CT) of three infantry platoons, a Tank troop and two direct fire support vehicles. The Red force was one third the size of Blue and its mission was to delay Blue by 24 hours. The CT assembly area was 10km from the objective in a heavily vegetated area, with one viable road to the objective. Red planned an ambush along the road. Red planned an ambush along the road.

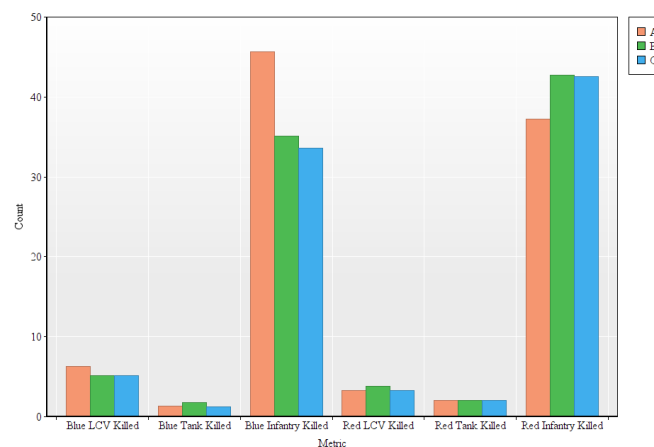


Figure 3.2: Simulation results (means and confidence intervals). The sample size is 10,000, so intervals are very small.

Three LCV options were evaluated: (A) Wheeled, Weapon E, Armour X; (B) Tracked, Weapon F, Armour Y; (C) Tracked, Weapon E, Armour Y. Armour protection was measured by comparing casualty rates and kinds of infantry and vehicle damage; lethality measures were the same but applied to enemy platforms and infantry. The signature metrics reflected the total number of Blue force entities detected and their range at detection. Knowledge acquisition captured the same information in relation to detecting Red. A binary mission success variable recorded whether Blue achieved its mission, the seizure of its objective within 24 hours before reaching attrition thresholds. SMEs were consulted in order to generate appropriate plans and tactics for both sides. These plans were then encoded in the COMBATXXI simulation, a stochastic, entity-based closed-loop combat simulation developed by the US Army [77]. The simulation was repeated 10,000 times for each LCV. The output variables measured for each side were: LCVs Killed; Tanks Killed; Infantry Killed; Detection of entities; Blue Mission Success. The mean and confidence interval of the metrics are shown in Figure 3.2.

3.4 Using Bayesian Network Models

We developed Bayesian networks (BNs) to model combat effectiveness, as they allow qualitative and quantitative representations of the domain and specifically support decision analysis [78]. A BN consists of a DAG over a set of random variables, each with a CPT representing conditional dependencies over its parents as explained in Section 2.2.2 of Chapter 2. The elements of this example domain are represented in a BDN (see Section 2.2.2.1 for details), where the decision node is *Option* with three different vehicle configuration options *A*, *B* and *C* to choose from as mentioned in Section 3.3. We have 9 chance nodes i.e., *Blue_Detects_Red*, *Red_Detects_Blue*, *Blue_Infantry_Killed*, *Red_Infantry_Killed*, *Blue_LCV_Killed*, *Red_LCV_Killed*, *Blue_Tank_Killed*, *Red_Tank_Killed* and *Blue_Mission_Success*. Any choice between *A*, *B* and *C* in *Option* is made based on the *Yes* or *No* state of *Blue_Mission_Success* node which shows the success rate of Blue force. The states of other nodes represents the number of kills for example, *s1_below_1* state means that the number of kills is 0 and so on.

Our chosen BN tool, GeNIe, provides a user friendly interface for developing BN models. It also includes the SMILE Engine [79] for programmatic access to Bayesian inference with the developed BN model. We discretized the simulation data prior to applying two of the causal discovery algorithms within GeNIe, PC [59] and a Bayesian search based on K2 [61] that uses BDeu (Bayesian Dirichlet equivalent uniform) as the metric [80] (see details about PC and K2 in Sections 2.10.1 and 2.10.2 of Chapter 2). We chose PC and K2 algorithms to apply on the simulation data to learn the causal structures; as they are pretty straight forward, useful and easily available to use in most of the BN software tools such as Netica and GeNIe.

3.5 Meta-Model Validation

For validating meta-models we used two common measures directly supported by GeNIe, namely predictive accuracy and the log score function, applying them in *five-fold cross validation*.

Figure 3.3 shows the accuracy results for each node in the model generated using the PC algorithm (i.e., the model in Figure 3.1). The primary output node of interest, *Mission Success*, reports a mean accuracy across all folds of about 0.965, which is statistically significantly higher than the Base Rate of 0.893 (i.e., the overall *Mission Success* rate across the simulations). The difference between the two is the “effect size” — the improvement obtained in predictive accuracy when using the observed values of the other variables to predict whether the mission will be successful. We also performed five-fold cross validation on the network generated by the K2 algorithm (not shown), resulting in Figure 3.4. The mean accuracy of about 0.96 for

	5 Folds					Accuracy	Base rate	Difference	Log Score
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5				
Mission Success	0.9755	0.9645	0.9666	0.9643	0.9545	0.96508	0.893	0.07	-0.108
Blue LCV Killed	0.6606	0.6608	0.6543	0.6648	0.6683	0.66176	0.512	0.15	-0.673
Blue Tank Killed	0.6116	0.599	0.6068	0.6205	0.594	0.60638	0.554	0.052	-0.694
Blue Infantry Killed	0.6268	0.6176	0.6176	0.625	0.6245	0.6223	0.555	0.067	-0.806
Red LCV Killed	0.6665	0.6725	0.662	0.6648	0.682	0.66956	0.675	0	-0.659
Red Tank Killed	0.7586	0.7538	0.7406	0.748	0.7591	0.75202	0.729	0.023	-0.711
Red Infantry Killed	0.5943	0.5825	0.583	0.5676	0.58	0.58148	0.487	0.094	-0.841
Red Detects Blue	0.7065	0.6923	0.6953	0.7086	0.6936	0.69926	0.667	0.0323	-0.63
Blue Detects Red	0.972	0.9543	0.9698	0.9693	0.972	0.96748	0.942	0.025	-0.083

Figure 3.3: Predictive Accuracy for the PC meta-model. Bold indicates a statistically significant improvement in the Overall result compared to the Base Rate.

	5 Folds					Accuracy	Base rate	Difference	Log Score
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5				
Mission Success	0.955	0.956	0.971	0.9696	0.954	0.96112	0.904	0.06	-0.133
Blue LCV Killed	0.6503	0.633	0.639	0.644	0.6355	0.64036	0.517	0.123	-0.664
Blue Tank Killed	0.606	0.607	0.582	0.595	0.588	0.5956	0.551	0.045	-0.699
Blue Infantry Killed	0.6306	0.624	0.622	0.6275	0.629	0.62662	0.557	0.07	-0.793
Red LCV Killed	0.672	0.677	0.6721	0.668	0.685	0.67482	0.675	0	-0.647
Red Tank Killed	0.757	0.7535	0.746	0.747	0.7605	0.7528	0.739	0.014	-0.719
Red Infantry Killed	0.565	0.569	0.594	0.555	0.5375	0.5641	0.486	0.08	-0.87
Red Detects Blue	0.697	0.686	0.694	0.701	0.681	0.6918	0.668	0.02	-0.626
Blue Detects Red	0.9705	0.965	0.9583	0.9603	0.975	0.96582	0.954	0.012	-0.0802

Figure 3.4: Predictive Accuracy for the K2 meta-model. Bold indicates a statistically significant improvement in the Overall result compared to the Base Rate.

Mission Success is again statistically significant, but slightly worse than that using PC. In both models, the other variables show some improvement in predictive accuracy as well, although not in all cases; the log scores are closely aligned with predictive accuracies. Since both models are similar, we limit subsequent discussion to the PC model.

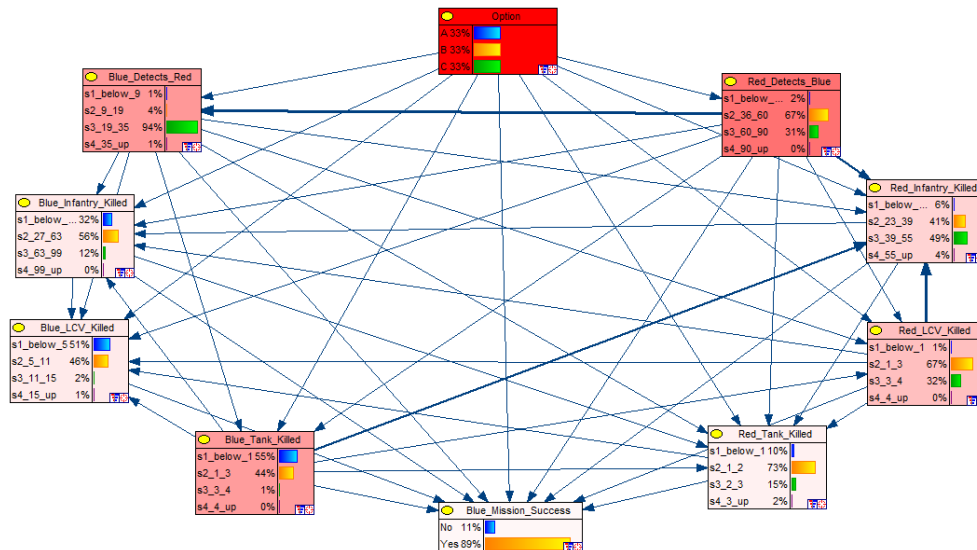


Figure 3.5: Sensitivity Analysis using causal visualization of the PC model.

3.6 Causal Visualization

Figure 3.5 illustrates one kind of visual sensitivity analysis that can be performed using BNs. The degree of saturation of red indicates a node’s relative importance to *Mission Success*, with greater saturation being more important, while the thickness of the arrows showing the paths of greatest influence. These have been assessed using the mutual information between *Mission Success* and the other nodes. Examination of the visualization can readily reveal interesting points from the simulation. For example, it shows that detection of enemy forces, together with *Blue Tank Killed*, are the most important factors for *Mission Success*. Furthermore, Blue detecting Red is strongly related to Red detecting Blue. The model asserts that Red detecting Blue *causes* Blue detecting Red, however, that needs to be taken with a “grain of salt”: since that causal arrow is “covered” (i.e., the two detection nodes have a common parent), causal discovery algorithms cannot determine which way the arrow should go. Here, it can simply be read as implying a strong association between the two detection nodes. (The causal discovery literature explains these issues in depth; see, e.g., Part II of [8].)

3.7 What-If Analysis

Causal BN models support arbitrary hypothetical or counterfactual analyses. That is, using them we can ask any kind of “What if” question we like. For example, what if we can protect all the Blue LCVs, Blue Tanks and Blue Infantry, what then will be the chance of Blue mission success?

If little to no harm comes to any of our Blue forces, i.e., *Blue LCV Killed*, *Blue Tank Killed* and

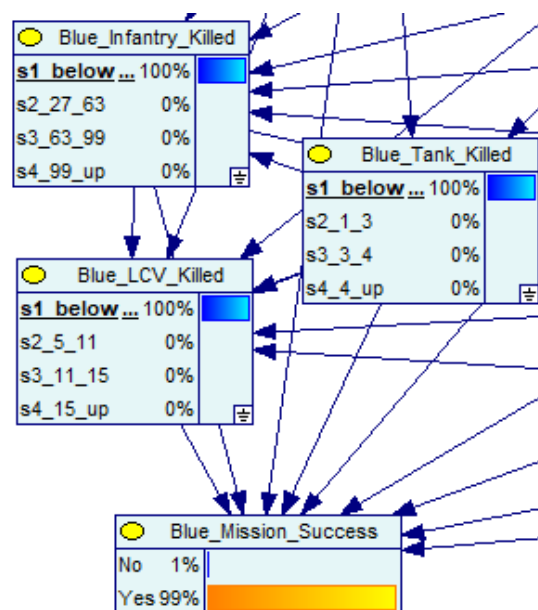


Figure 3.6: What-if Analysis

Blue Infantry Killed are all in their lowest possible states, then we can see in Figure 3.6 that the success rate is about 99 percent. Note that this is strictly an *observational* analysis: it reveals a probabilistic dependency between observing Blue forces being protected and *Mission Success*, without revealing that *acting* specifically to protect Blue forces would lead to *Mission Success*. This is an example of correlation not being the same as causation.

A true causal BN can also answer causal hypotheticals: by manually altering Figure 3.6 by cutting all in-bound (parental) arrows to the nodes *Blue LCV Killed*, *Blue Tank Killed* and *Blue Infantry Killed* and then setting them to the states of interest (Blue forces protected), you will discover that directly intervening to protect Blue forces (as if by an “act of God”) will indeed lead to greater *Mission Success* — increasing to about 96%, about 7% above the base rate. The observational and causal reasoning result in the same qualitative conclusion in this case, but in general observation does not substitute for causal intervention, any more than a Lecturer coloring her hair white would get her promoted to Professor. The ability to support both observational and causal reasoning is a particular advantage to using causal BNs for decision making that other approaches lack.

3.8 Extension to Utilities

Our BN can inform us about the chance of mission success given our three LCV options. If our sole concern was mission success, this would be sufficient but we are also concerned with other outcomes. We would like to minimize the loss of our own forces and equipment and

inflict the same on the enemy. Each of these outcomes will matter to the DMs to a different degree and should therefore be weighted in our decision appropriately.

Figure 3.7 represents the model after adding utility nodes to each outcome of interest. There are six utilities associated with the forces (infantry, LCVs and tanks) of both Blue and Red (*BlueInfantry*, *BlueLCV*, *BlueTank*, *RedInfantry*, *RedLCV*, *RedTank*) and one associated with *MissionSuccess*. A final utility node is for estimating the effectiveness of the mission in a broader context, beyond just the current mission (*MissionEffectiveness*), which here is the weighted sum of all other utility nodes.

Each force utility node is a simple numerical translation of its outcome. For example, the utility node *RedInfantry* maps each state of *Red Infantry Killed* to the mid-point value of the state's range (so 0–23 is mapped to 11.5, 23–39 is mapped to 31, and so forth). The same applies to Blue force utility nodes, except that the associated utility is negative. *Mission success* is simply assigned 1 for success, and 0 for failure.

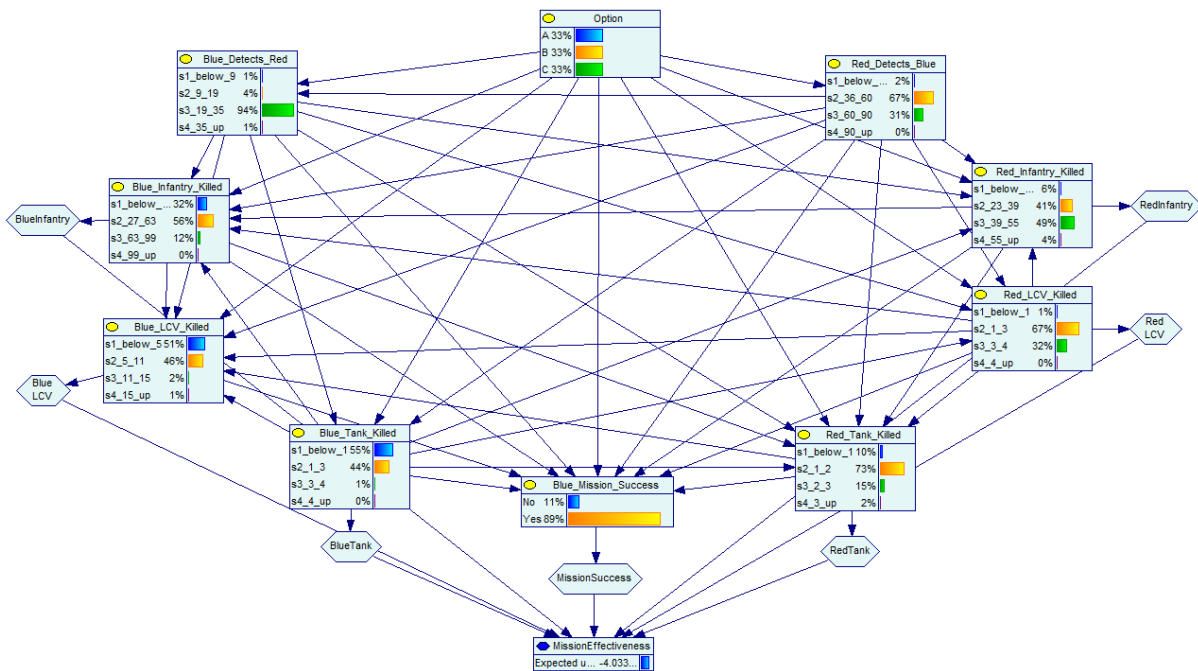


Figure 3.7: BN Meta Model with Utilities

Of course, each such utility is on an independent scale; we need to combine them in the final *MissionEffectiveness* utility node. The combining weights, then, are similar to regression coefficients, but dependencies are being handled further up in the network. Since there is no obvious scale for the final utility node, we simply measured in terms of number of Blue infantry; e.g., -3 for mission effectiveness would be equal in value to the loss of 3 infantry. We describe these units as “utiles”.

The combining weights need to reflect the values of the DMs. The approach we used to

estimate them was a mixture of the Trade-off Method [81,82] and Swing Weights [83], where the weights were estimated via ratio values between the base criteria value and other criteria values. The weights were chosen after consultation over equipment capabilities: Blue Infantry 1; Blue LCV 15; Blue Tank 8; Red Infantry 0.2; Red LCV 7.5; Red Tank 4; Mission Success 100. The Base Weights were chosen after a brief survey of potential equipment capabilities and are shown in Table 3.1. The critical issue of this decision problem is the trade-off between the competing objectives: reducing our force casualties, achieving mission success, protecting important assets such as LCV and tanks, and decreasing enemy capability. As the unit for the scale, Blue infantry are considered to have a utility of (positive) 1. Since it is only possible for Blue infantry to be killed during a mission (not added), the *Blue Infantry* utility node will always be negative or zero, as is also the case for other Blue forces. We give the corresponding Red utility nodes less weight. While there is an advantage to incurring losses on the side of the enemy, they are less likely to be as important as losses to our own side outside the context of the current mission. For infantry, enemy losses are expected to be of much less value than our own losses, and so count as a small fraction (0.2) of the Blue infantry equivalent.

For estimating the value of LCVs and tanks, we took into account their capacity to carry crew and infantry, and their much greater strategic value. LCVs are particularly valuable, and their loss would make missions substantially more difficult. Hence these have been estimated to contribute the highest utility: 15 utiles for Blue LCVs, and 7.5 for Red LCVs. Again, Blue losses are expected to be more significant to us than Red losses, but given the strategic value of LCVs, to a lesser degree than for infantry. The same considerations were applied to tanks. Finally, mission success is likely to be highly context dependent — here, we suppose we are interested in missions that have high strategic value, hence have set mission success as equivalent to 100 Blue infantry (or 100 utiles). Please note the utility weight values presented here may not reflect Australian Defence strategic guidance; the relevant point is that in any concrete analysis, some weighting must be adopted.

Outcome Utility	Weight
BlueInfantry	1
BlueLCV	15
BlueTank	8
RedInfantry	0.2
RedLCV	7.5
RedTank	4
MissionSuccess	100

Table 3.1: Combination weights for the utility nodes for assessing mission effectiveness

With the Base Weights in place, we can assess the LCV options. The Table 3.2 represents the overall scores for option *A*, *B*, and *C*, where we can clearly see that option *A* is inferior with

an expected utility of -30.14, approximately 40 utiles (or blue infantry equivalents) lower than options *B* and *C*. However, options *B* and *C* are very close; any choice between them should be based upon a wider context, possibly requiring additional simulation scenarios.

Option	Mission Effectiveness (Utiles)
A	-30.14
B	8.23
C	9.81

Table 3.2: Mission Effectiveness of the Network

3.9 Conclusion

In this chapter, we explored three Land Combat Vehicle (LCV) options, evaluating them using multiple dimensions of utility, such as Blue force survivability and lethality. Bayesian Network (BN) meta-models were learned from combat simulation data by applying two common causal discovery algorithms. The resultant causal BN meta-models were then validated on the data; they could also be validated using expert opinion or a combination of expert opinion and data (see, e.g., [63]). Once you have a validated causal BN meta-model, you can explore it for insights into the domain using sensitivity analysis, including “what if” hypothetical analysis. Our causal BN meta-models were then further extended using expert-elicited utilities, yielding a causal decision network. We then used the causal decision network to determine the expected utilities for each LCV option, finding that for the simulated environment used in this study, option *A* had the worst expected utility, while options *B* and *C* were of about equal best value. To make a decision between *B* and *C* additional simulation environments will likely be necessary. This study shows the method and value of using BN meta-models to interpret combat simulations, or other simulations, with multiple dimensions of utility.

We collected the preferences from a single expert to calculate the utilities of different outcomes but our aim is to involve multiple DMs by eliciting their preferences and aggregating those preferences in order to make optimal decision in a specific situation. In the Chapter 4, we present 4 different preference elicitation methods elicit preferences from different experts, combine them and use them to calculate the utility functions in a BDN. We also extend these methods by applying causal discovery algorithms to the data extended by elicited utilities. That is, experts’ elicited utilities are themselves measurements of random variables (reporting their subjectively assessed utilities associated with aspects of each scenario). The variations and co-variations between utilities have a causal explanation as much as other variables, so causal discovery can provide insights into how and why they are related to each other, as well as to other random variables. We anticipate that such research may reveal new insights into

multi-criteria decision making.

Multi-Criteria Preference Elicitation

The previous Chapter 3 we presented a method for building a Bayesian decision network through a combination of learning from simulation data, and relatively ad hoc elicitation of utilities from a single expert. In this chapter, we turn our attention to the specific problem of eliciting preferences from a group of people – who might be stakeholders, or experts – and aggregating those elicited preferences in a way that allows them to be incorporated into the utility component of a BDN. More specifically, we present a number of methods for eliciting preferences from survey participants, including weighing different utility dimensions, and aggregating these preferences.

We start with a “Baseline” preference elicitation method (Section 4.2), which is a straightforward combination of elements from the AHP and probability elicitation methods. While developing it to the point of having a clear algorithm, we encountered some serious limitations with it, especially a failure to account for overconfidence in self-assessments. Canvassing the literature revealed a superior preference elicitation method that has had success recently and which specifically address degrees of self-confidence. We describe this and its adaptation to the special circumstances of our survey problem in Section 4.3. We extend this preference elicitation method, which assumes the survey data will provide exhaustive coverage of pairwise preferences, to deal with small samples/subsets of data in Section 5.3.6, i.e. to ensure the method can be applied in practice when scaling up to real-world complex decision making. In Section 4.4 we also describe the use of the AHP method in a BDN; while this is not the focus of this thesis, this translation of the AHP method is used in the comparative analysis of our methods. Lastly, we explain the causal discovery with preference data to learn the interdependencies within the data in Section 4.5.

Here, we present the methods in a general form. In Chapters 5 and 6, we present a real-world case study, that demonstrates how the methods can be applied, and provides an empirical comparative analysis of the usefulness of the methods in practice.

4.1 Scaling Difficulties

During the development of the methods presented in this chapter, and prior to running any actual survey, we developed a set of artificial data to do qualitative analysis on how different methods for parameterizing BDN utility functions might work (see Chapter 6). Our exploration yielded a scaling problem. The number of utilities requiring elicitation is exponential in the number of variables that are inputs to the utility nodes. The required data set size for a reasonable estimate of all parameters is, therefore, the square of an exponential, meaning that the preference data collection problem is in principle non-feasible. Of course, exactly the same scaling problem occurs when eliciting the CPTs in Bayesian decision networks. As we developed an expert-elicited BDN for the experimental case study (see Section 5.2.1), it was clear that this wasn't just a theoretical complexity issue, but that it was also in practice non-feasible for real-world complex decision making. We decided to deal with this in a couple of ways. First, we could assume that the distribution of utilities is roughly linear across each dimension. In that case, we could use an interpolation tool, such as that developed to handle the scaling problem with CPT elicitation,¹ to estimate unsolicited values, while soliciting preferences only for extreme case utilities (and perhaps a few midpoints for sanity checks). We have used this interpolation method when we were parameterizing the chance nodes of our expert-elicited BN (see Section 5.2.2). Second, since this project is really an elaboration and test of a concept — a new approach to understanding multi-dimensional utilities — we do not need actual data. Real data for illustrating a technique is surely helpful, especially in persuading practitioners to adopt a new method. But we can use entirely artificial data if need be, in order to explore and illustrate a new method (we have used the artificial data generated from scenario based preference models in Section 5.4.1.2 of Chapter 5). Testing can still be done in a limited fashion, to show that the method is at least promising, without showing that it clearly works in real cases.

We propose to follow both of these approaches in dealing with sparse preference data.

4.2 The Baseline Method

Here we present our initial preference elicitation method, which we call the “Baseline method”, inspired in part by the Analytic Hierarchical Process (AHP) [84] and probability elicitation methods [85]. The Baseline method combines these with a Likert scale [86] for assessing confidence, using a weighted combination of scores. As we know, in any multi-criteria decision making problem, a decision maker has a set of alternatives to choose from and a set of cri-

¹This tool is available at <https://bayesian-intelligence.com/interpolator/>

teria to consider while making the decision. Each individual aims to select the best option among those alternatives based on their own needs or preferences. For our purposes here, the individuals are the survey participants. The Baseline method aims to elicit each participant's preferences by comparing and rating every pair of criteria in a decision making problem in order to select a best alternative among multiple options. The participants are also required to provide their confidence when weighing one criterion over other. The preferences elicited from the participants must then be aggregated in a way that specifies the utility functions in a BDN.

The following are the main steps of our Baseline preference elicitation method, expressed in an ideal form, where participants are given an exhaustive set of questions:

- Step 1** For each pair of criteria in the network we ask the participant/expert indicate their importance for the decision problem, rating one criterion's importance as 1 or more times the importance of the other. This gives a relative importance weight for each pair.
- Step 2** Next, they are asked to provide confidence they have in the assessment given in Step 1.
- Step 3** Next, participants respond to case scenarios that involve each combination of pairs of criteria, together with a decision, and are asked how satisfied they are with that decision.
- Step 4** Finally, we aggregate the information elicited from each participant in Steps 1-3, namely the relative importance weights, their confidence, and the satisfaction with scenario decisions. This aggregated information is used to specify the utility functions for the BDN, reflecting the aggregation of all participants ratings and decisions.

We now describe the information elicited in Steps 1-3, and the calculations using that information performed in Step 4.

4.2.1 Step 1: Eliciting pairwise preferences

For any MCDA problem, we call m the number of criteria to be considered while making a decision, n the number of stakeholders, decision makers or experts (denoted by E) making a single decision.

In the first step of the Baseline preference elicitation method, we ask pairwise comparison questions from the participants to weight each criterion with respect to every other criterion in the MCDM problem based on its relative importance. Specifically, for each pair of criteria, C_i and C_j , participants are asked whether criterion C_i is more important, or whether C_j is more important, or whether the two criteria are equally important.

Once the participant chooses one criterion, say C_i , as being more important than the other, they are then asked to indicate “how many times” more important by providing a real number in the range of $(1.0, Max]$, that is, a number greater than 1, up to a maximum, Max .² This upper bound is used to ensure that the preferences from different participants are on the same scale. The number provided is then used as a weight in the scoring calculation in Step 4 (see below). If the participant answers that the given pair of criteria are equally important, a weight of 1 is assigned to both criteria.

4.2.2 Step 2: Confidence Elicited in the Baseline Method

In step 2, a confidence W on the scale of 1 to 5 (where 1 represents the lowest confidence level and 5 represents the highest level of confidence) is provided by the participant when rating one criterion over the other based on its relative importance.

4.2.3 Step 3: Eliciting Satisfaction With the Decision in the Baseline Method

In Step 3 of the Baseline method, participants are asked about their satisfaction level about a decision made by a decision maker based on an example scenario given in the actual survey. More specifically, they are given a number of scenarios, where each scenario involves a specific value for each of the m criterion, $C_1 = v_1, \dots, C_m = v_m$, and told the decision, i.e. which *one* of the scenarios has been chosen. The participants are then asked how satisfied they are with the specific decision, and must answer using a Likert scale.

In a Likert scale method, we present a statement to the participant, in this case the decision made, and asks her/him to rate the extent to which she/he agrees to that statement.

In order to capture preferences as utility functions in our BDN, in our methods we must obtain a numerical value. In the Baseline method, the satisfaction is elicited by having the survey participants provide their satisfaction on a 5 valued Likert verbal scale, from very satisfied to very dissatisfied. To map these to the required numerical scale, we assign the highest value 5 to the highest satisfaction level and lowest value 1 to the lowest level of satisfaction of the participant. Table 4.1 shows the numerical Likert scale for the verbal satisfaction scale we used in our case study. The method can of course be used with any verbal scale, and participants could also be asked to specific satisfaction directly as a number.

²For the case study presented in Chapters 5 and 6, we used $Max=10$, but this is maximum “how many times more important” is of course certainly problem dependent and in practice would likely be influenced by cultural factors, personality and so on. We did not investigate how to set such a maximum, or how changing the maximum might influence the utilities and hence the decisions of the elicited BDN; we leave that as an interesting topic for future research, as discussed in Section 7.2.2.

Verbal Scale	Numerical Scale
Very Satisfied	5
Satisfied	4
Neither satisfied nor dissatisfied	3
Dissatisfied	2
Very dissatisfied	1

Table 4.1: Verbal Scale into Numerical Scale

4.2.4 Step 4: Score and Weight Calculation for Each Criterion in the Baseline Method

In step 4 of the Baseline method, we calculate the score S_C of each pair of criteria given by n experts in the MCDA network using the following algorithm.

ALGORITHM 4.1 (CALCULATE SCORE FOR EACH PAIR OF CRITERIA)

```

Call: calculateScore(C)
Input:  $C$ : Set of criteria
Output:  $S_C$ : Score for each pair of criteria
1 begin
2   foreach pair of different criteria  $g$  and  $h$  do
3     /* Set counter  $g$  to count from first criterion to the second last criterion  $m - 1$  */
4      $h = g + 1$  /* Set counter  $h$  to count from second criterion  $g + 1$  to the last criterion
5        $m$  for pairwise comparison of all criteria */
6      $S_{C_{g,h}} \leftarrow \sum_{e \in E=1}^n (S_{e \in E, C_{g,h}} \times W_{e \in E, C_{g,h}})$ 
7      $S_{C_{h,g}} \leftarrow \sum_{e \in E=1}^n \left( \frac{1}{S_{e \in E, C_{g,h}}} \times W_{e \in E, C_{g,h}} \right)$ 
8     if  $h < m$  then
9       |  $h = h + 1$ 
10    else
11      | stop
12    if  $g < m - 1$  then
13      |  $g = g + 1$ 
14    else
15      | stop
16  return  $S_{C_{g,h}}, S_{C_{h,g}}$ 

```

In Algorithm 4.1, the score $S_{C_{g,h}}$ is the total expert-weighted score for criterion g relative to criterion h , where $S_{e \in E, C_{g,h}}$ is the individual score of criterion g (relative to h) given by expert $e \in E$. $W_{e \in E, C_{g,h}}$ is the confidence level of each expert $e \in E$ in the score she/he has given to

criterion g relative to criterion h .

This comparative score method implies that there will be a minimum of m total scores, $S_{C_1}, S_{C_2}, \dots, S_{C_m}$ when the preference elicitation involves m criteria, and that for every expert's score $S_{e \in E, C_{g,h}}$, there is a corresponding score $S_{e \in E, C_{h,g}} = \frac{1}{S_{e \in E, C_{g,h}}}$. Note also that the same weight $W_{e \in E, C_{g,h}}$ is used to calculate both $S_{C_{g,h}}$ and $S_{C_{h,g}}$, as there is only one weight (confidence answer) given for each criteria pairwise comparison score elicited.

$$S_{C_{h,g}} = \sum_{e \in E=1}^n \left(\frac{1}{S_{e \in E, C_{g,h}}} \times W_{e \in E, C_{g,h}} \right) \quad (4.1)$$

Once we calculate all the scores for each pair of criteria, we aggregate those scores for each criterion, C_i using the following formula:

$$S_{C_i} = \sum_{z \in \{1, m\} \setminus i} S_{C_{i,z}} \quad (4.2)$$

where z in equation 4.2 is the counter to sum up all the scores of criterion C_i relative to all other $m - 1$ criteria in the network (already calculated by algorithm 4.1).

This gives us the weight of all criteria based on their relative importance in a multi-criteria decision problem. These weights need to be normalized before incorporating into the decision network:

$$NW_{C_i} = \frac{S_{C_i}}{\sum_{j=1}^m S_{C_j}} \quad (4.3)$$

So, NW_C denoted the normalized weight for each criterion C , i.e., the total score S_C of that criterion divided by the sum of all the scores of m number of criteria in the network.

The follow up questions elicit responses in the form of Likert verbal scale, where we present each combination of the states of a pair of criteria in the form of example decision questions and ask the participants to provide their satisfaction level to that decision statement (see step by step elicitation in Algorithm 5.3. Those verbal Likert scale elicited from all the participants are then converted into numerical scale as shown in Table 4.1 and geometric mean is calculated for each of the combination of states, for example, we have 2 criteria C_1 and C_2 and both having 3 states *High*, *Medium* and *Low*. So the geometric mean is calculated for (*High*, *High*), (*High*, *Medium*), etc. Once we calculate those values, we get them in the form of a resultant table (see detailed example Table B.11 in Appendix B) to sum up the cells to get a score S_{C_s} for each state s of criteria C . Lastly, we normalize those scores for each state of the parent criteria using the formula given below:

For each state s in criteria C we do

$$NW_{C_s} = \frac{S_{C_s}}{\sum_{i=1}^m S_{C_{s_i}}} \quad (4.4)$$

where m is the total number of criteria in the decision network.

The network in Figure 4.1 is the abstracted BDN fragment showing how the utility function is connected to the criterion nodes. In this figure, we can see that the top node is the decision node D which is the parent of chance nodes. For each criterion, there is a chance node $C_1, C_2, C_3, \dots, C_m$, and that chance node is a parent of utility node $U_1, U_2, U_3, \dots, U_m$. The total utility node T_U has parents $U_1, U_2, U_3, \dots, U_m$.

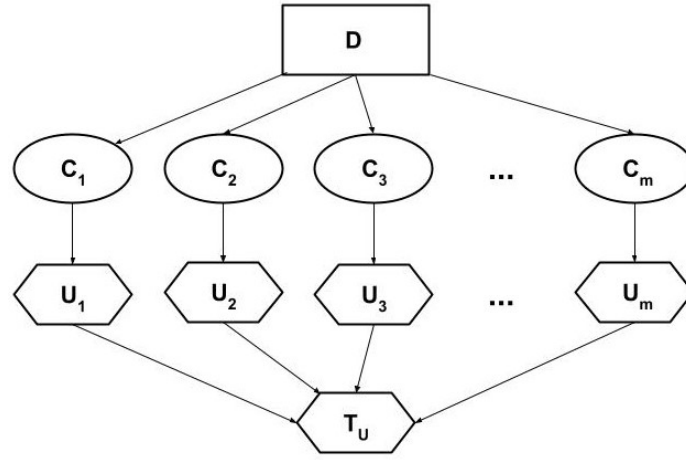


Figure 4.1: Abstract BDN Fragment

For each state of the parent criteria we have the following Table 4.2 which shows where the utility function go into the BDN.

Parent State	Utility
State s_1	$NW_{C_{s_1}}$
State s_2	$NW_{C_{s_2}}$
...	...
State s_n	$NW_{C_{s_n}}$

Table 4.2: Function table for U_i

To create the function for the total utility node T_U , we note that, for each utility node that has parent criterion C_i , the normalized weight is the final weight NW_{C_i} . This is calculated for each U_i (see Table 4.3) and finally combined in the function for the total utility node T_U as follows:

$$T_U = \sum_i (U_i \times NW_{C_i}) \quad (4.5)$$

The Baseline method uses the weight for each criterion with respect to every other criterion, how many times one criterion is better than other, the satisfaction level of the participant about

Parent Utility Node	Weight
U_1	NW_{C_1}
U_2	NW_{C_2}
...	...
U_m	NW_{C_m}

Table 4.3: Weights for each parent utility node in U_T

a given case scenario and the confidence they have in their given answer. We calculate values for each pair of criteria obtained from each participant using the Algorithm 4.1 and combine all the responses from all the participants to get the utility functions for each criterion and their relative overall weights. We parameterize our Baseline BDN using the information we get after calculating the utility functions and the weights for each criterion.

While the Baseline method is straightforward and easy to implement (see Appendix B), unfortunately, it provides no clear method of assessing or mitigating over-confidence, which is a well established problem in elicitation [87,88]. Therefore, we looked to other methods.

4.3 The 4 Point Elicitation Method

Given our decision to reject the Baseline method due largely to anticipated overconfidence, we turned to a recently developed, but empirically validated, elicitation method that appears to cope well with confidence issues [87], namely, the 4 Point Elicitation method [54], presented in Section 2.9.1.1. It distills the most useful elements from existing expert elicitation approaches, and combines them into a novel and practical protocol.

4.3.1 An Overview of the 4 Point Elicitation Method

In standard usage, the 4 Point Elicitation method produces for each estimate an interval or bracket for the estimated value:

1. Lowest plausible value
2. Highest plausible value
3. Best estimate
4. Confidence that the interval captures their value.

Typically, the 4pt method is used to elicit probabilities, or real-valued quantities.

Our use of the 4 Point Elicitation method is unique in a number of respects. First, while personal utilities are typically represented as real-valued numbers in decision analysis [89], the

4 Point Elicitation method has never previously been applied to eliciting personal utilities, to the best of our knowledge. In this application, our participants were required to answer a set of questions to provide their lowest plausible *preference value*, highest plausible *preference value*, their best *preferred value* and their degree of confidence that the interval they have provided contains their real preference value.

When eliciting expert judgments using the 4 Point Elicitation method, experts are asked (either by a facilitator, or in a survey) to provide their degrees of confidence that the intervals they have created, from lowest to highest, contain the true values, e.g., the actual probabilities of events occurring or the real valued quantity of interest. These confidence levels are constrained to lie between 50% and 100% [55] (if their confidence is less than 50% then the intervals haven't been estimated properly [53]). The true values or true probabilities are unknown actual values or probabilities, which, however, can be estimated by measurement or modelling. In our usage, we replaced "true value" with "*your value*", representing the participant's best preferred estimate. This value, as for the Baseline method, is expressed as the relative importance, in terms of "how many times more important".

We used this method to prepare an exhaustive question set to elicit preferences from participants, which was then used to parameterize the BDN. This involved designing new question formats based on the 4 Point Elicitation method. These questions were prepared to elicit the participants' preferences using the four point method mentioned above. We asked the participants' weights for criteria, in terms of relative importance, as for the Baseline method described above, with the weights being real-valued. The participants provided their lowest and highest values for criteria in terms of their importance for decision making. And unlike in the Baseline method, where the participants are asked simply for their confidence in their relative importance value, in the 4 Point Elicitation method, participants are asked for their confidence that the interval they have created contains their value.

4.3.1.1 Standardizing the Intervals

In the 4 Point Method, we elicit a sub-interval of $(1.0, Max]$ representing the participant's credible interval for the ratio of weights between each pair of criteria (value dimensions). Once we obtain this (sub)interval from the participants for each pair of criteria, we clean and preprocess the data in order to use them to calculate the utility functions. Elicited responses may have common mistakes such as assigning values in wrong places, wrong intervals, etc. As we aim to get the weights for criteria in the form of numbers great than 1.0 and less than or equal to Max , we standardize the lower intervals and upper intervals to 80% credible intervals, in order to get the uncertainties of the participants on a consistent scale. Algorithm 4.2 calculates

the standardized intervals for each participant using the linear extrapolation [90, 91] given in equations 4.6 and 4.7.

Lower Standard Interval:

$$L_S = R_B - (R_B - R_L) \times \frac{S}{T} \quad (4.6)$$

Upper Standard Interval:

$$U_S = R_B + (R_H - R_B) \times \frac{S}{T} \quad (4.7)$$

Where L_S represents the standardized lower interval value, U_S represents the standardized upper interval value, R_B is the best estimate from the participant, R_L is the lowest plausible value, R_H is the highest plausible value, S is the credible interval we use to standardize these intervals to 80% to view the uncertainties on a consistent scale and T is the confidence given by the participant for each interval they have created.

Let us suppose we have two criteria g and h to compare with each other based on their relative importance given a scenario, a participant says that g is more important than h and assigns a lowest plausible value $R_{L_{g,h}} = 2$, best estimate $R_{B_{g,h}} = 3$ and highest plausible value $R_{H_{g,h}} = 7$ to g being more important than h . The participant is 90% confident that the interval (from lowest to highest value) she/he has created holds her/his value. Using Equations 4.6 and 4.7, we calculate the standardized lower and upper interval values from the elicited data. If the participants give an equal importance E_S to a pair of criteria then the next question is to provide their confidence T in their answer. We calculate the standardized equal importance E_S for a pair of different criteria using the formula $E_S = \frac{S}{T}$.

$$L_S = R_{B_{g,h}} - (R_{B_{g,h}} - R_{L_{g,h}}) \times \frac{S}{T_{g,h}}$$

$$L_S = 3 - (3 - 2) \times \frac{80}{90}$$

$$L_S = 3 - (1) \times 0.88$$

$$L_S = 2.12$$

$$U_S = R_{B_{g,h}} + (R_{H_{g,h}} - R_{B_{g,h}}) \times \frac{S}{T_{g,h}}$$

$$U_S = 3 + (7 - 3) \times \frac{80}{90}$$

$$U_S = 3 + (4) \times 0.88$$

$$U_S = 6.52$$

So the standardized lower interval and upper interval using the above mentioned example are 2.12 and 6.52 respectively.

ALGORITHM 4.2 (STANDARDIZE THE INTERVALS)

```

Call : StandardizeIntervals( $g, h$ )
Input:  $R_{H_{g,h}}$ : Highest plausible value,  $R_{L_{g,h}}$ : Lowest plausible value,  $T_{g,h}$ : Confidence
Output:  $U_S$ : Standardized upper interval,  $L_S$ : Standardized lower interval
1 begin
2    $U_S \leftarrow R_{B_{g,h}} + (R_{H_{g,h}} - R_{B_{g,h}}) \times \frac{80}{T_{g,h}}$ 
3   /* To standardise these intervals we use 80% credible interval to view uncertainties
   on a consistent scale */
4    $L_S \leftarrow R_{B_{g,h}} - (R_{B_{g,h}} - R_{L_{g,h}}) \times \frac{80}{T_{g,h}}$ 
5    $E_S \leftarrow \frac{80}{T_{g,h}}$ 
6   return  $U_S, L_S$ 

```

4.3.1.2 Calculating Score Via 4 Point Elicitation Method

In this section, we calculate the weights for each criterion involved in a multi-criteria decision making problem using the 4 Point Elicitation method. In this method, we use the standardised lower interval, highest interval and best estimate of each criterion based on its relative importance to any other criterion, as well as the confidence of the participant about their created interval holding their value and the satisfaction level of the participant regarding a given case scenario. If two experts/participants give the same response, they are considered as just two responses that happen to be the same. We calculate the utility functions after aggregating the responses from all the participants for each criterion and populate the 4 Point Elicitation method based BDN. The weights in the 4 Point Elicitation based method are calculated using the preferences elicited from the participants in the form of intervals.

The elicited intervals from different participants are standardized using Algorithm 4.2, with those standardized lower intervals L_S , standardized upper intervals U_S and standardized equal importance E_S scores of each pair of different criteria g and h , we calculate the weights for each criteria using Algorithm 4.3. This algorithm calculates the scores for each criterion with respect to every other criterion in the network and aggregate the weights for each criterion to get an overall weight in order to parameterize the BDN accordingly. For a given set of m number of criteria in a decision problem, g and h represent a pair of different criteria in m . $T(L_{S_{g,h}})$ is the average of all the lower standard interval values and $T(U_{S_{g,h}})$ is the average

of all the upper interval values where g is more important than h . While $S_{C_{g,h}}$ represents the weight of criteria by taking the average of $T(L_{S_{g,h}})$ and $T(U_{S_{g,h}})$ where criterion g is more important than criterion h . Similarly, $T(L_{S_{h,g}})$, $T(U_{S_{h,g}})$ and $S_{C_{h,g}}$ hold the average values as mentioned above, but instead for criterion h being more important than criterion g .

ALGORITHM 4.3 (CALCULATE SCORE VIA 4 POINT ELICITATION METHOD)

```

Call : calculateScore(C)
Input:  $\mathbb{C}$ : Set of criteria,  $\mathbb{L}_S$ : Standardized lower interval,  $\mathbb{U}_S$ : Standardized upper
        interval,  $\mathbb{E}_S$ : Standardized equal importance
Output:  $S_C$ : Score for each pair of criteria
1 begin
2   foreach pair of different criteria  $g$  and  $h$  do
3      $h = g + 1$  in  $m$ 
4      $T(L_{S_{g,h}}) \leftarrow L_{S_{g,h}} \times E_{S_{g,h}}$ 
5     /* Take average of  $L_{S_{g,h}}$  and  $E_{S_{g,h}}$  where the criterion  $g$  is equally important to
       criterion  $h$  */
6      $T(U_{S_{g,h}}) \leftarrow U_{S_{g,h}} \times E_{S_{g,h}}$ 
7      $S_{C_{g,h}} \leftarrow \text{Average}(T(L_{S_{g,h}}), T(U_{S_{g,h}}))$ 
8      $T(L_{S_{h,g}}) \leftarrow L_{S_{h,g}} \times E_{S_{h,g}}$ 
9      $T(U_{S_{h,g}}) \leftarrow U_{S_{h,g}} \times E_{S_{h,g}}$ 
10     $S_{C_{h,g}} \leftarrow \text{Average}(T(L_{S_{h,g}}), T(U_{S_{h,g}}))$ 
11    if  $h < m$  then
12       $h = h + 1$ 
13    else
14      stop
15    if  $g < m - 1$  then
16       $g = g + 1$ 
17    else
18      stop
19  return  $S_{C_{g,h}}, S_{C_{h,g}}$ 

```

Once we get these values, we aggregate them to get an overall weight for each criterion to populate the 4 Point Elicitation method based BDN. The 4 Point Elicitation method can also be reduced to the Baseline method as the best estimate R_B can be used as the value elicited from a participant preferring one criterion over another. In order to use the elicited responses with multiple elicitation methods, we ask the participants to provide their values in a range of $(1.0, Max]$ to create their intervals.

4.4 AHP based BDNs

The Analytical Hierarchy Process (AHP) is a commonly used technique for MCDA problems. This technique is very simple and easy to use partly because it doesn't take criterial dependencies into account; details can be found in Section 2.6.1 of Chapter 2. In order to evaluate our proposed methods on equal terms, we describe a method for representing AHP in the form of a BDN (an AHP based BDN), capable of yielding the very same decisions as the AHP would in any other representation.

Previously, the AHP has been used in combination with BNs to improve the decision making process in MCDA problems [92,93] but we are using it in a unique way. According to the best of our knowledge, the AHP method has not been used previously for building BDNs with decisions and utilities; we perform a translation of the AHP representations to BDNs ourselves. Of course, the AHP elicitation technique could itself be used directly to elicit comparisons and calculate AHP weights, which would then be converted to utilities. Here, however, we described a method for deriving AHP matrices from data.

First of all, we measure the relative importance of each criterion to observed decisions using mutual information (MI). Let's suppose, if there are four criteria (*Cost*, *Safety*, *Customer Satisfaction* and *Environmental Impact*) in a vehicle purchase problem, each criterion has its MI showing its importance to the (observable) decision in a BDN. That is, we calculate $MI(\text{Criterion}, \text{Decision})$ for each criterion. We can then calculate the ratio of MI for every pair of criteria as required by AHP. Since $MI(\text{Criterion}, \text{Decision})$ is a direct measure of a criterion's importance to a decision, the *ratios* specify how much more important one criteria is relative to another, in precisely the way required by AHP. We can see that there will be a total of six pairwise comparisons of four criteria i.e., mutual information with the decision for each of *Cost vs Safety*, *Cost vs Customer Satisfaction*, *Cost vs Environmental Impact*, *Safety vs Customer Satisfaction*, *Safety vs Environmental Impact*, *Customer Satisfaction vs Environmental Impact*. We generate a symmetric matrix (details of comparison matrix calculation is explained in Section 2.6.1) that is 4 X 4 in size based on these mutual information ratios. The following matrix 4.4 is the resultant comparison matrix for the above mentioned criteria, where *MI* denotes *Mutual Information*, *C* is *Cost*, *S* is *Safety*, *CS* is *Customer Satisfaction* and *EI* is *Environmental Impact*.

$$A = \begin{bmatrix} & \mathbf{C} & \mathbf{S} & \mathbf{CS} & \mathbf{EI} \\ \mathbf{C} & 1 & \frac{M.I(C \text{ vs } D)}{M.I(S \text{ vs } D)} & \frac{M.I(C \text{ vs } D)}{M.I(CS \text{ vs } D)} & \frac{M.I(C \text{ vs } D)}{M.I(EI \text{ vs } D)} \\ \mathbf{S} & \frac{1}{\frac{M.I(C \text{ vs } D)}{M.I(S \text{ vs } D)}} & 1 & \frac{M.I(S \text{ vs } D)}{M.I(CS \text{ vs } D)} & \frac{M.I(S \text{ vs } D)}{M.I(EI \text{ vs } D)} \\ \mathbf{CS} & \frac{1}{\frac{M.I(C \text{ vs } D)}{M.I(CS \text{ vs } D)}} & \frac{1}{\frac{M.I(S \text{ vs } D)}{M.I(CS \text{ vs } D)}} & 1 & \frac{M.I(CS \text{ vs } D)}{M.I(EI \text{ vs } D)} \\ \mathbf{EI} & \frac{1}{\frac{M.I(C \text{ vs } D)}{M.I(EI \text{ vs } D)}} & \frac{1}{\frac{M.I(S \text{ vs } D)}{M.I(EI \text{ vs } D)}} & \frac{1}{\frac{M.I(CS \text{ vs } D)}{M.I(EI \text{ vs } D)}} & 1 \end{bmatrix}$$

In the more general form of the problem, we have an $m \times m$ matrix, where the entry in the i th row and the j th column represents the comparison between criteria C_i and C_j .

We can then provide the matrix 4.4 as input to a standard AHP software³, which performs the calculations presented in Section 2.6.1, including normalizing the Eigenvectors of the comparison matrix, where the normalized totals are the weights for each criterion. (In Chapter 5, for convenience, we use a geometric mean approximation of the AHP method used by the software [94, 95].) These resultant weights provide the values required to specify the utility functions in our BDN models. Appendix D provides a detailed example of this translation of the AHP method into a BDN.

In this thesis, we apply this method just to artificial data, since there we have sufficient data along with (simulated) observed choices to compute meaningful mutual information values. This is primarily for comparison with the causal discovery method, which is also data-dependent (described in the next section). However, there is no reason why this method cannot be used as a general method for developing AHP based BDNs where appropriate data is available.

4.5 Causal Discovery with Preference Data

The final method we present in this chapter involves using causal discovery, specifically the CaMML software⁴, to learn a BN from the preference data. As detailed in Section 2.2.2.1, utilities or preferences are modelled with “utility nodes” in BDN modelling, which, thus far,

³<https://bpmg.com/ahp/>

⁴<https://bayesian-intelligence.com/software/>

have only been developed by eliciting preference information from experts, with BN modellers capturing these in the utility function (usually tables) associated with the corresponding utility nodes. To our knowledge, the idea of treating the preference information as data for a causal discovery process is new. It is not, however, entirely unheard of in the sense that attributed preferences are commonly reasoned about and subject to Bayesian updating or inference. For example, models of enemy behaviour in warfare commonly include representations of their preferences or goals, which are updated based upon their observable actions. Treating utilities as the values of chance nodes for the purposes of causal discovery, as we propose here, is not different in principle. In order to use BN tools to perform expected value computations, these chance nodes can be subsequently converted into the usual utility nodes, of course.

4.5.1 Why Use Artificial Data?

In this thesis, we use an actual survey to elicit preferences / utilities from the participants / experts to be used by our preference elicitation methods. We create a survey based on an MCDA problem where we have n alternatives to choose from, and m criteria needed to be considered while making a decision to choose an alternative from n . We create a survey questionnaire based on such MCDA problem and ask the participants / respondents to rate a pair of criteria in terms of its relative importance to the decision. The survey presents some example scenario descriptions followed by an exhaustive set of pairwise questions to the participants, and ask them to rate a criterion in terms of its relative importance. The responses to such questions result in linear dependencies or pairwise dependencies between the utilities, which is not what we want. The elicited preferences do not cover the multiple dimensions of utility, which makes it impossible to learn higher levels of dependency.

Ideally, we would have been able to conduct a real case study, we would get real data from real experts that might exhibit such preference dependencies. However due to time limitations, and also because the real experts we had planned to use for variants of the land combat procurement case study were not available in 2020 and 2021 due to the COVID pandemic, we were not able to collect such data with preference dependencies. Therefore, for this thesis, we limit ourselves to a proof of concept study of interdependent utilities, utilizing artificial utility data which exhibit such dependencies.

In order to overcome the survey limitations, we took the approach of generating artificial data from models that are based on some practical scenarios. The scenarios aim to cover the multiple utility dimensions which exhibit dependencies that are useful to the decision itself. We create some preference-based scenarios and produce BDNs based on those scenarios. These scenario based preference models are then used to generate several cases where different par-

ticipants make different decisions.

4.5.2 Utility Dependencies in Expert Elicited Networks

Prior treatments of MCDA have assumed that utilities associated with different attributes are, amongst themselves, independent. But there is no necessity in that. For this research, we come up with scenarios where the dependencies between the utilities make a practical and real difference to the decision. Let suppose a scenario example 4.1 to illustrate the dependencies among the utilities.

DEFINITION 4.1 : SCENARIO 2

A private company who use a fleet of cars for executive perquisites. Here, the overriding goal is customer satisfaction, since it's part of the remuneration. However, amongst managers who are also highly concerned about the environment, there is reduced concern about cost and a greater willingness to purchase expensive EV options. The combination makes for dependencies between cost, environmental impact and customer satisfaction.

In this example scenario (which is a version of the case study we present in Chapter 5), we can see that different managers have different preferences when making a fleet purchase decision. The utilities are dependent; if they prefer environment friendly vehicles, they will have to compromise on cost. Such dependencies between the utilities are crucial in many decision making problems and need to be treated.

Causal discovery methods are specifically designed to be sensitive to interdependencies within the data and to explain them, and so they make a promising class of techniques for dealing with interdependent utilities. In order to try CaMML on this problem, we added invented utility dependencies to the expert elicited network based on scenarios like that described above (these are described in detail in Section 5.4). These modified networks describe possible worlds where those dependencies exist, and what we want is to develop methods that will rediscover them from data, model decision problems using them, and — hopefully — get resulting decisions more generally right than MCDA techniques that ignore the dependencies. For example, the AHP method will generate different decisions in at least some cases, because it has no capability for representing or reasoning about utility dependencies. As well as comparing different modelling methods using hacked networks, we compare results from the expert elicited network with and without dependencies, to assess the extent to which introduced dependencies affect the different methods.

4.5.3 The Method

Let us suppose there exists a BN that can represent the preferences of an individual or group, where:

- For each preference criterion, C_i , there is a chance node in the BN that represents it.
- There is a chance node, D , that represents an (observable but not intervenable) decision, that is directly affected by each criterion C_i .
- Any dependencies between the criteria in their effects on the decision are captured by the decision's conditional probability distribution (CPD), e.g. $C_i \rightarrow D$ for all C_i and $P(D|C_1, C_2, \dots)$.

Data that is generated from a process described by this BN (whether a real world process or some BN) would consist of rows, each containing a (non-unique) scenario that records a value for each of the chance nodes. These scenarios would appear at a frequency that accords with the probability distributions in the network such that the resulting dataset represents the joint distribution represented by the BN. With no evidence, there would be no dependencies amongst the criteria, only in the CPD of the decision; however, if we restrict the generated data to a particular choice, the data will manifest the dependencies between criteria nodes that are due to that choice.

CaMML includes functionality that allows us to provide priors on the structure that it learns. We use this functionality in our method to fix the structure relating to other 'non-criterion' nodes in the network, by giving a prior of 1 to all the arcs between those nodes, and from those nodes to the criterion nodes, and giving a prior of 0 to all potential arcs, other than between the criterion nodes. This means that when given the generated dataset, CaMML *only* considers the relationships between the criterion nodes, i.e. is only considering dependencies *between* the utilities.

CaMML can produce more than one BN, sorted by the MML score; our method uses the top BN, i.e. the model with the lowest MML score. This is converted back into a BDN simply by attaching utility nodes to each learned criteria node, assigning -1, 0, 1 to each utility node, and then the final "Total Utility" node is simply the sum of the utility nodes. See Figure 4.2, with a dependency illustrated (circled by red color). Because we provided CaMML with structural priors that were hard constraints, the results BN is identical to the original BDN used to generate the artificial data, other than the relationships between the criterion chance nodes.

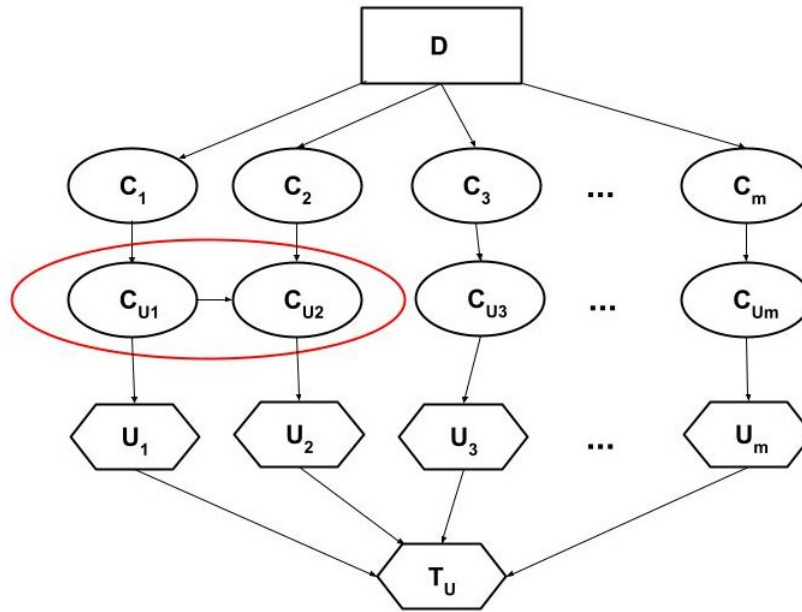


Figure 4.2: BDN Example via CaMML

4.6 Conclusion

In this chapter, we presented different kinds of preference elicitation methods to be used in this thesis. The first method was the Baseline method which incorporates a Likert scale response to assess participants' confidence. This allows participants to compare and rate each pair of criteria in a decision making problem and help them to make the best decision among multiple alternatives. The elicited preferences from the participants are then used to calculate the utility functions of a BDN. The second method was the 4 Point elicitation method, inspired by IDEA, which we have adapted to elicit preferences from participants, instead of probabilities or quantities, and which are then used to calculate the utility function in our BDNs. Our third preference elicitation method presented is an extended version of our 4 Point Elicitation method, which works on small subsets of data and helps to assist when we have fewer responses from the experts. We also described the translation of the traditional the AHP method into a BDN by calculating the weights from preference data using the AHP method. Lastly, we explained the purpose of using artificial data and applying causal discovery to that data. In the next chapters, we describe a case study conducted to explore the usefulness of the proposed method, including details of the case study, such as the survey used for data collection, the preprocessing of the survey data, the application of the methods, including causal discovery with scenario based preference data, and finally evaluating the proposed methods on a range of methods.

Modelling with Multi-Criteria Preference Data: A Fleet Purchase Case Study

We presented four different elicitation methods in Chapter 4, namely, the Baseline preference elicitation method, the 4 Point Elicitation method for preferences, CaMML and the AHP method. All four methods have distinctive elements. The Baseline method is the most straightforward elicitation method, but requires an exhaustive set of questions across all pairwise combinations of criteria, in terms of relative importance as well as their confidence. The 4 Point Elicitation method deals with biases but requires additional information from participants, namely upper and lower bounds on relative importance, as well as a confidence estimate. The AHP method is presented to use within a BN with decisions and utilities, by translating the method into BDNs themselves. Lastly, CaMML learns the causal models from the preference information by treating it as data, which allows dependency information amongst criteria and utilities to be discovered. In this chapter, we describe in detail how to elicit preferences using an actual survey, for a Fleet Purchase case study, as well as how to process the survey data into a form that can be used by the methods.

In Section 5.1 of this chapter, we introduce the Fleet Purchase case study and the associated BDN and explain why we chose this particular case study. We present and explain the survey methodology, data collection and preprocessing of data in Sections 5.3. We explain the utility parameterization of our BDNs with the 3 methods, using first the elicited data (Section 5.3.4), then using artificial data generated from scenario based preference models in 5.4. Lastly, in Section 5.5 we discuss the evaluation metrics we use to validate and evaluate the resultant BDNs, in the experimental results presented in the following chapter.

5.1 Case Study

Our case study is a version of the Fleet Vehicle Purchase problem that applies to any large organization that provides a fleet of vehicles for use by its employee. We chose to use Monash University as a specific example organization to (1) simplify access to genuine experts who helped us build a realistic BDN structure, and also (2) so that we could recruit Monash post-graduate students and staff members as pseudo-experts. The fleet purchase problem is of the same general nature as Land Combat Vehicle (LCV) Procurement Problem used in Chapter 3, involving multiple criteria as well as multiple Decision Makers (DMs) with different preferences and needs.

We first hand-crafted a small but realistic BN structure for the fleet purchase problem. Keeping in mind that the purpose of the case study was to conduct a survey to test the proposed preference elicitation methods, we needed to ensure that the problem was simple enough so that non-experts could understand it, and that the time required of each survey participant would be relatively short. Thus we limited the decision to a choice between three different car models, with the decision being to select the best option based on a variety of possible criteria, including upfront and ongoing costs, safety, environmental impact and customer satisfaction (in this case, the satisfaction of the employees who will be driving the vehicle). We constructed an initial BDN, then modified and finalized our BDN structure based on the information provided by the domain expert *Erick Arines* from the Procurement Department at Monash University.

5.2 Fleet Vehicle Purchase BN

The final BDN structure for the Fleet Vehicle Purchase problem used in our case study is shown in Figure 5.1, with Table 5.1 show nodes and their node types, states and parents. The network captures the high-level features of the purchasing problem and is organized into following tiers.

1. The top tier is simply the decision node: which vehicle brand to purchase.
2. The next tier consists of various “inputs” into the decision, that is the decision criteria: the different kinds of cost associated with the purchase, as well as safety, environmental impact and customer satisfaction. These are filled in (parameterized) based upon the actual data about costs, etc. per brand in the scenario.
3. The final tier reflects the utilities associated with these inputs, which is where the different weights associated with different costs would be reflected. For example, if an

organization values prestige over operating costs relative to another organization, these differences are reflected in the utilities.

4. Our simple scenario currently is for a 4 years purchase/maintenance problem.

In this fleet network, the root node is the decision node named as **Fleet Vehicle Purchase**. The different values (such as: Toyota Kluger, Ferrari Portofino, Fiat 500) in the decision node can be selected and the network automatically computes the expected utility named as **Overall Weight** of that particular choice. Chance nodes are set to particular values, or they can be updated based upon other values being selected. For example, in the current network, if the Decision Node is set to “Ferrari Portofino”, the purchase cost automatically updates to “High”.

5.2.1 Network Structure

The network of Fleet Vehicle Purchase problem consists of 7 criteria, that are each represented with 3 states. The Fleet problem network comprises of the biggest problem that is obviously the complexity e.g. $3^7 = 2187$ questions which are too many questions. So, we dealt with this in two parts. First was to use the standard BN modelling technique of “divorcing parents” [8, Sec. 10.3.6]; in particular, divorce off all the money-related criteria. Secondly, we say a) money is money (i.e. money is additive and independent) and b) we are ignoring time-discounting effects, so given both those things, we can just sum up the money-related criteria, rather than asking pairwise questions. These kinds of standard BN knowledge engineering techniques help us to reduce redundancies and the elicitation load, by substantially reducing the number of questions. So that resulted in $3^4 = 81$ questions (plus $4 \times 3 = 12$ for weighting the criteria themselves), which is still too much, but getting manageable. This combining of the money related criteria into one criterion can be seen in the “Overall Cost” node, with the four parent cost nodes (Purchase, Insurance, Maintenance and Operational) in Figure 5.1. In this figure we can see that we have one direct interdependency between *Safety* and *Customer Satisfaction*, and three arcs going up (shown via grey color) are indirect dependencies driven by the nature of the system. These nature node interdependencies are different to utility interdependencies (which are driven by psychology/intention/goals/values).

The detailed description of the nodes in the network, their types, states and parent nodes are outlined in Table 5.1.

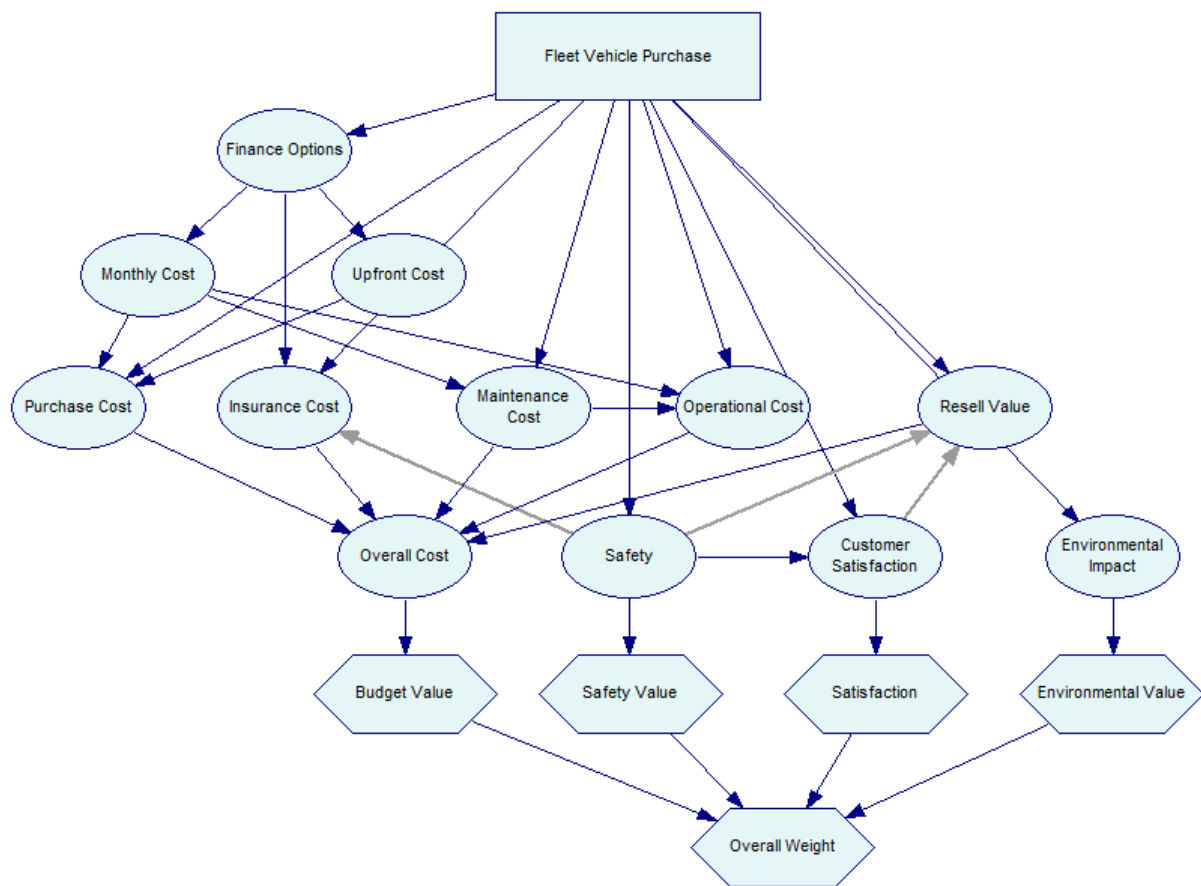


Figure 5.1: Fleet Vehicle Purchase BN after Combining Money-Related Criteria

Node	Type	States	Parents
Fleet Vehicle Purchase	Decision Node	Toyota Kluger, Ferrari Portofino, Fiat 500	None
Finance Options	Chance Node	Yes, No	Fleet Vehicle Purchase
Monthly Cost	Chance Node	Yes, No	Finance Options
Upfront Cost	Chance Node	Yes, No	Finance Options
Safety	Chance Node	High, Medium, Low	Fleet Vehicle Purchase
Environmental Impact	Chance Node	Good, Neutral, Worst	Fleet Vehicle Purchase
Customer Satisfaction	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Safety
Maintenance Cost	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Monthly Cost
Operational Cost	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Monthly Cost, Maintenance Cost
Insurance Cost	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Finance Options, Safety
Purchase Cost	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Monthly Cost, Upfront Cost
Resell Value	Chance Node	High, Medium, Low	Fleet Vehicle Purchase, Safety, Customer Satisfaction
Overall Cost	Chance Node	High, Medium, Low	Maintenance Cost, Operational Cost, Insurance Cost, Purchase Cost, Resell Value
Budget Value	Utility Node	Numeric	Overall Cost
Safety Value	Utility Node	Numeric	Safety
Satisfaction	Utility Node	Numeric	Customer Satisfaction
Environmental Value	Utility Node	Numeric	Environmental Impact
Overall Weight	Utility Node	Numeric	Budget Value, Safety Value, Satisfaction, Environmental Value

Table 5.1: Fleet Vehicle Network Structure

5.2.2 Network Parameters

The chance nodes in the Fleet network were parameterized based on the information available on internet. Some chance nodes were also parameterized with the help of an interpolator tool called the Beta Interpolator tool which is used for filling in large CPTs, when the parents and children are ordered or discretised-continuous nodes (it also works with binary nodes) [96]. To parameterize the utility nodes of the fleet vehicle network, we surveyed some ordinary people who do not necessarily have strong expertise but are familiar with similar problems and we surveyed enough of them to get a range of preferences, and sufficient coverage of all the questions. The elicited preferences were then used to calculate the utility functions using different elicitation methods discussed in Chapter 4.

5.3 Survey Methodology

In order to get a realistic set of preferences from experts/participants, we conducted an online survey to get actual data. Those elicited preferences from the participants are used to calculate the utility function using our proposed preference elicitation methods. The results are then incorporated into our expert elicited fleet BDN to help make a single best decision of which fleet to purchase for an organization. Before running the actual survey, we were required to go through the following steps.

5.3.1 Survey Questionnaire and Consent Form

First of all, we prepared pairwise sets of questions based on the fleet vehicle purchase model to ask from the participants. Those exhaustive set of questions involved every combination of all four criteria namely: *Overall Cost*, *Safety*, *Customer Satisfaction* and *Environmental Impact*, but those questions were prepared using pairwise comparison of criteria based on some example scenarios. For example, we defined an example scenario for the participants and asked them to rate pairwise criteria let's say *Overall Cost* and *Safety* followed by an exhaustive set of questions based on each combination of the states involved in *Overall Cost* and *Safety*. This was presented using a single form at a time and, once completed, directed the participants to attempt other forms having other pairs of criteria.

In addition to creating the questions for the survey questionnaire, another important task was creating the consent form and explanatory statement to provide to the participants to read and agree before they attempted the questions. The consent form outlined that the participants give their consent to use their provided information in this research project and the explanatory statement described the aim and purpose of the survey along with the confiden-

tiality of information and the possible benefits and risks to the participants (see more details in Appendix A).

5.3.2 Ethics Approval

Once all the questions were finalized and the consent form and explanatory statement were created for the survey, the survey was submitted for ethics approval from Monash University. Ethics approval is required when there needs to be interactions with humans to get information and collect any demographic data from them to be used in our research project.

5.3.3 Survey Setup

The survey consists of four different scenarios to be presented to participants that provide the problem description, allowing participants to understand the underlying issues before attempting the questions. All scenarios provide different problem descriptions based on the different needs of various fictional organizations, which help the participants to analyse a situation, consider the important criteria before making any purchase decisions and act as a head of an organization who reviews the purchase decision of the fleet manager as outlined in each scenario (see details in Section A.2 of Appendix A). We use Google Forms to create our survey forms; each case scenario asks participants to weight a pair of criteria followed by questions based on every possible combination of those criteria.

5.3.4 Data Collection

In order to elicit the preferences from the participants, we shared the links to the forms (containing our survey questionnaire) using different social media platforms and email invitations. In addition to that, we also made contact with different organizations, particularly the kind of organizations we had included in our example scenarios for the survey, such as NGOs, medical centres and fleet departments to participate in the survey. We left the survey open for about a month and sent reminder emails to the audience regularly to participate in our survey. After a month, we closed the survey and gathered the responses from the participants to clean up that data.

Our survey consists of two main parts, i.e., ratings of criteria relative to other criteria based on their relative importance in a scenario and a Likert scale response for assessing confidence for each combination of any two particular criteria. The participants are given an example scenario and then asked to provide answers to the questions in the survey based on the information given in that specific scenario. The participants are asked to rate one criterion against

another based on their relative importance; they can give any rating greater than 1 but less than or equal to 10 (bounds are given to put the answers onto the same scale).

In addition to eliciting weights for value dimensions directly, we ask participants to assess particular purchase decisions, which indirectly records their values by way of how highly, or how poorly, they rate particular purchases potentially reflecting those values. In order to do this, we present the decisions as those made by a fleet manager, while the participant is asked to assess those decisions in light of the scenario, acting as the head of the organization, using the Likert scale of 1 to 5 to rate her/his satisfaction level after the fleet manager's decision (see Table 5.2).

Very dissatisfied	Dissatisfied	Neither satisfied nor dissatisfied	Satisfied	Very Satisfied
1	2	3	4	5

Table 5.2: Likert Scale used in Survey

For example, Safety and Cost have three states, High, Medium and Low, so the questions cover each combination of these states and the participants are required to answer according to their satisfaction using the above Likert scale (see Appendix A for detailed questions asked in our survey).

The data we collect from the participants is translated to the same scale before using in any of our methods to calculate the utility functions.

The detailed experimental results for these methods can be found in Chapter 6.

5.3.5 Elicitation Question Formats

In this section, we present the step by step process of preference elicitation from the participants in the form of algorithms. These algorithms are used for the Baseline and 4 point elicitation methods and, coupled with demographic data, used to generate models weighted by participant expertise. Algorithm 5.1 is Step 1 of eliciting data from the participants, where the participants are presented with a couple of different criteria g and h to rank them based on their relative importance followed by their confidence T in the answer they have given. In a Multi-Criteria decision problem, there are m number of total criteria involved where g and h represent a pair of different criteria to be compared (as discussed in Section 4.3.1.2). Algorithm 5.1 starts with the comparison of g and h , where g starts from first criterion in the set of m criteria while h is always $g + 1$. The algorithm compares all the criteria with every other criteria in m based on their relative importance.

If the participant prefers criterion g over h , we call Algorithm 5.2 to elicit the highest plausible value $R_{H_{g,h}}$, lowest plausible value $R_{L_{g,h}}$ and best estimate $R_{B_{g,h}}$ of criterion g being more important than criterion h . $R_{H_{g,h}}$, $R_{L_{g,h}}$ and $R_{B_{g,h}}$ ranges from 1 to 10 assigned by the

participant as the highest plausible value, lowest plausible value and the best estimate of criterion g being more important than criterion h . The participants are also asked to provide their confidence $T_{g,h}$ (which ranges from 50 to 100) that the interval they have created from $R_{L_{g,h}}$ to $R_{H_{g,h}}$ captures their value.

If the participant prefers criterion h over g , then Algorithm 5.2 is called again, but with h and g swapped.

Algorithm 5.3 is Step 2 of the elicitation process of our survey, where the participants are given an example scenario of a large organization planning to buy a fleet vehicle for their organization. The participants are asked to imagine acting as the CEOs or head of the organizations and are presented with some purchase decision statements where they have to provide their satisfaction level based on the decisions being made by the fleet manager. An exhaustive set of questions are asked for each combination of different decisions and each pairwise set of criteria and the participants respond to them using the Likert scale. The elicited responses from the participants are stored in the form of a matrix to be used for calculating criteria weights. For example, for a criterion g in a set of m criteria, there could be q states; for example a criterion *Cost* with *High*, *Medium* and *Low* states comprises of $q = 3$, similarly for criterion h in m there could be p states. In Algorithm 5.3, we elicit participants' responses in the form of Likert scale for each combination of two criteria and store those values in the form of a matrix $L_{[i][j]}$ where i represents a row and j represents a column in L .

Once the preference information is collected using these algorithms, the Baseline method and the 4 Point Elicitation method are used to calculate the utility functions (see the details about these methods in Sections 4.2 and 4.3 of Chapter 4).

ALGORITHM 5.1 (ELICIT DATA FROM PARTICIPANTS (STEP 1))

Call : *ElicitRatingsOfCriteria(C)*

Input: C : Set of criteria

Output: $g_{Values}, h_{Value}, T_{g,h}$: Rating of Criteria and Confidence of participants

```

1 begin
2   foreach pair of different criteria  $g$  and  $h$  do
3      $h = g + 1$  in  $m$ 
4     if  $g > h$  then
5       /* If  $g$  is more important than  $h$  */
6        $g_{Values} \leftarrow \text{call}(\text{ElicitValues}(g, h))$ 
7     if  $h > g$  then
8       /* If  $h$  is more important than  $g$  */
9        $h_{Values} \leftarrow \text{call}(\text{ElicitValues}(h, g))$ 
10    else
11       $T_{g,h} \leftarrow$  Assign a score where  $50 \geq T_{g,h} \leq 100$  /* Your confidence  $T$  that
12        criteria  $g$  and  $h$  are equally important */
13    if  $h < m$  then
14       $h = h + 1$ 
15    else
16      stop
17  if  $g < m - 1$  then
18     $g = g + 1$ 
19  else
20    stop
21 return  $g_{Values}, h_{Value}, T_{g,h}$ 

```

ALGORITHM 5.2 (RATE CRITERION g OVER h)**Call :** ElicitValues(g, h)**Input:** g, h : Pair of different criteria**Output:** $R_{H_{g,h}}, R_{L_{g,h}}, T_{g,h}$: Participants' highest plausible value, lowest plausible value and confidence

```

1 begin
2    $R_{H_{g,h}} \leftarrow$  Assign a score where  $1 < R_{H_{g,h}} \leq 10$ 
3    $R_{L_{g,h}} \leftarrow$  Assign a score where  $1 < R_{L_{g,h}} \leq 10$ 
4    $R_{B_{g,h}} \leftarrow$  Assign a score where  $1 < R_{B_{g,h}} \leq 10$ 
5    $T_{g,h} \leftarrow$  Assign a score where  $50 \geq T_{g,h} \leq 100$ 
6   return  $R_{H_{g,h}}, R_{L_{g,h}}, T_{g,h}$ 

```

ALGORITHM 5.3 (ELICIT DATA FROM PARTICIPANTS (STEP 2))**Call :** ElicitResponses(g, h)**Input:** g, h : Pair of different criteria**Output:** $L_{[i][j]}$: Participants' responses for each combination of two criteria

```

1 begin
2   foreach  $o$  in  $q$  states of Criterion  $g$  do
3     Set  $i$  and  $j$  to 0
4     /* where  $i$  and  $j$  are indices to hold the values and return in the end */
5     foreach  $n$  in  $p$  states of Criterion  $h$  do
6        $Likert_{g_o, h_n} \leftarrow$  Assign a scale from the given options: Very dissatisfied,
7         Dissatisfied, Neither satisfied nor dissatisfied, Satisfied, Very Satisfied
8        $L_{[i][j]} \leftarrow L_{g_o, h_n}$ 
9       if  $n == p$  then
10        Stop
11      else
12         $j = j + 1$ 
13        Continue
14    if  $o == q$  then
15      Stop
16    Else
17       $i = i + 1$ 
18      Continue
19  return  $L_{[i][j]}$ 

```

5.3.6 Expert-Weighted Resampled Survey Data for the 4 Point Elicitation Method

We also generate a larger data set for the 4 Point Elicitation method by resampling the collected survey data after restricting it to subsets of participants based on expertise. The 4 Point Elicitation method was used because it provides intervals that can be used in the resampling, which are not available and would need to be arbitrarily estimated with the Baseline method. This helps us to evaluate cases where we may only have a small number of participants who are “experts/stakeholders”. In many real world problems, we will not be able to recruit enough experts to collect the data required by the exhaustive question set method, and hence this approach can help with evaluation of this and similar elicitation methods. We validate the method on our Fleet Problem case study as mentioned in Section 6.1.3 of Chapter 6. Rather than running a separate data collection survey, we use a subset of the original data, in effect “simulating” some of the participants who completed the survey being experts/stakeholders. We collect basic demographic information such as age and background in purchasing in our survey A, which we use to determine the subset (and we vary the experiment with different subsets of participants).

We categorize the participants into four different groups i.e., least experienced, basic, intermediate and most experienced participants as shown in Table 5.3. Participants with no or one vehicle purchase experience are assigned to *Least* group, those who have purchased more than one but less than five vehicles belong to *Basic* group, participants with a purchase experience of more than five but less than twenty vehicles goes to *Intermediate* group and *Most Experienced* group consists of participants with any number of purchased vehicles but having vehicle purchase experience for an organization.

Group	Vehicle Purchase Experience
<i>Least</i>	Purchased 0 or 1 vehicle
<i>Basic</i>	Purchased more than 1 but less than 5 vehicles
<i>Intermediate</i>	Purchased more than 5 but less than 20 vehicles
<i>Most Experienced</i>	Organizational purchase experience of ≥ 1 vehicle/s

Table 5.3: Categories Based on Purchase Experience

Different subsets of participants provide different decisions based on their level of background experience, personal purchases and other variants. Once we group the participants based on their vehicle purchase background, we use the the 4 Point Elicitation method to calculate the utility functions and parameterize our hand crafted BDN. In this method, we use two different subsets of our original dataset; one is based on *Basic*, *Intermediate* and *Most Experienced* group of participants, and other is based on the data elicited from *Intermediate* and *Most Experienced* participants. For both of these subsets of data, we calculate the standardized

upper and lower intervals, combine them and aggregate the values to get overall weights of criteria put into our BDNs.

5.4 Scenario Based Preference Models

As our elicited data from human participants in surveys was not rich enough to capture dependencies among the utilities, we created data sets that did contain dependencies. We did this by inventing models based on artificial scenarios where there are such dependencies. The focus of this thesis is to develop promising methods for dealing with MCDA problems, rather than to demonstrate our techniques on real-world problems. That is to say, artificial scenarios and models used to represent them suffice for a proof-of-concept argument that the methods show promise.

The basic idea here is to create preference models that exhibit dependent utilities, where the dependencies are such that they make an actual difference in some decision problem, so that techniques (like CaMML) which can discover them might do better in some actual problems. We began with the original expert-elicited model, since the choice of model is separate from any elicitation method or data analysis method, and not involving those methods makes later discussion a bit clearer.

We created three scenarios where there are clear dependencies among two or more criteria. We then created preference models for each scenario, which we call “true models”, since they provide a ground truth for comparing to learned models.

For each set of one or more dependencies below, we describe a preference scenario that makes the dependencies sensible, i.e., explains why a fleet manager, or a group of similar fleet managers, might exhibit those dependencies within their preferences.

DEFINITION 5.1 : SCENARIO 1 (COST, SAFETY AND ENVIRONMENTAL IMPACT)

An underfunded medical NGO is purchasing vehicles to deliver medical care in remote areas. The NGO is concerned about the reduction of carbon dioxide emissions and wants to switch to Electric Vehicles (EVs). There are two kinds of fleet managers: those who emphasize safety and de-emphasize vehicle cost; and those who emphasize maximizing medical care delivery and so cost minimization, and tend to de-emphasize safety and environmental impact. The combination makes for a dependency between cost and safety, and cost and environmental impact.

DEFINITION 5.2 : SCENARIO 2 (COST AND ENVIRONMENTAL IMPACT)

The Department of Environment wants to reduce the carbon dioxide emissions by purchasing Electric Vehicles (EVs). The outright purchase cost and the costs associated with batteries (lifespan and the servicing costs) are common financial drawbacks. In addition to these costs, fleet managers also feel uncertain about the resale value of EVs. If, for such organizations, environmental impact of the fleet is the most important factor then it would most likely reduce the importance of cost. In other words, if they want to switch to EVs, they will have to compromise on cost. The combination makes for a dependency between cost and environmental impact.

DEFINITION 5.3 : SCENARIO 3 (COST AND SAFETY)

A federal government organization considers safety as the most important criterion to achieve when making a fleet purchase decision. In order to achieve maximum safety, they emphasize a larger budget for safer, more reliable vehicles. This makes for a dependency between cost and safety.

Each true model based on the above preference scenarios is represented using all the four utility nodes (*Budget Value*, *Safety Value*, *Satisfaction* and *Environmental Value*), which are now converted into chance nodes, and made to affect a final *Revealed Decision* chance node (named in the spirit of revealed preference theory [10]). The Fleet Vehicle Purchase decision node is no longer appropriate (because we want to understand what drives the decision, rather than understand the impact of the decision), and hence has been replaced instead by a ‘Car’ chance node, with uniform prior probabilities. Dependencies are not explicitly represented between the criteria; instead the dependent effects of the criteria on *Revealed Decision* for a specific scenario are captured in that node’s CPT (Conditional Probability Table). Figure 5.2 shows the common structure for all three models (with the four ‘utility’ nodes and the *Revealed Decision* node highlighted); Table 5.4 shows an extract of the CPT for Scenario 5.1, demonstrating how the dependent effects have been represented. Note that the first layer of nodes in the figure depicts the chance nodes — the dependency between *Safety* and *Customer Satisfaction* here is part of the system itself, *not* a dependency between utilities. Utility dependencies are captured only in *Revealed Decision*.

As can be seen in the extract, individuals to whom this preference model applied would choose Toyota most of the time. In particular, they would choose Toyota all of the time when *Budget Value* is High as indicated in the first row of the table (regardless of other values, as represented by * in the table). However, when they consider *Safety Value* more important than *Budget Value*, things begin to change. When *Budget Value* is Low, *Safety Value* is High, and

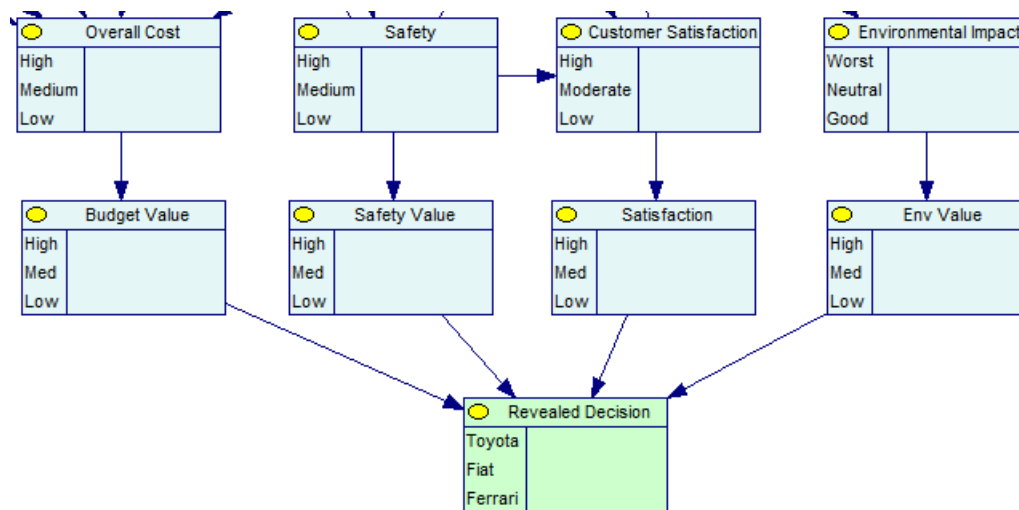


Figure 5.2: True Model based on Preference Scenario 5.1. The top layer shows chance nodes (the dependency here is *not* a utility dependency), the second layer shows utility nodes converted to chance nodes, and the third layer shows just *Revealed Decision*

Satisfaction and *Env Value* can be anything, the individual would choose Ferrari (two thirds of the way down). If, instead, *Budget Value* is more important than *Safety Value* (again, where *Satisfaction* and *Env Value* can be anything), the individual would choose Fiat as shown in the last row of the table.

Utility Nodes				Revealed Decision		
Budget Value	Safety Value	Satisfaction	Env Value	Toyota	Fiat	Ferrari
High	*	*	*	1	0	0
Med	High	High	High	0.99	0	0.01
Med	High	High	Med	0.8	0	0.2
Med	High	High	Low	0.0485	0	0.951
Med	High	Med	High	0.8	0	0.2
Med	High	Med	Med	0.8	0	0.2
Med	High	Med	Low	0.8	0	0.2
Med	High	Low	High	0.3	0	0.7
Med	High	Low	Med	0.3	0	0.7
Med	High	Low	Low	0.3	0	0.7
Med	Med	High	High	0.989	0.00099	0.0099
Med	Med	High	Med	0.989	0.00099	0.0099
Med	Med	High	Low	0.989	0.00099	0.0099
Med	Med	Med	High	0.333	0.333	0.333
Med	Med	Med	Med	0.333	0.333	0.333
Med	Med	Med	Low	0.333	0.333	0.333
Med	Med	Low	High	0.0476	0.4761	0.4761
Med	Med	Low	Med	0.0476	0.476	0.476
Med	Med	Low	Low	0.0476	0.476	0.4761
Med	Low	*	*	0	1	0
Low	High	*	*	0	0	1
Low	Med	High	High	0.0588	0.000117	0.941
Low	Med	High	Med	0.0588	0.000117	0.941
Low	Med	High	Low	0.0588	0.000117	0.941
Low	Med	Med	High	0.0104	0.15625	0.833
Low	Med	Med	Med	0	0.1666	0.833
Low	Med	Med	Low	0	0.1666	0.833
Low	Med	Low	High	0.1666	0.666	0.1666
Low	Med	Low	Med	0	0.8	0.2
Low	Med	Low	Low	0	0.8	0.2
Low	Low	*	*	0	1	0

Table 5.4: An Extract of CPT Based on Scenario 5.1

5.4.1 Generating Data from the Preference Models for each Method

Generated data is used by the causal discovery algorithm to learn the causal structure, and we also apply similar methods to generate appropriate data for the other elicitation methods, allowing us to calculate their utility functions and ultimately enabling all methods to be compared on data that derives from the same true preference model.

In general, to generate data from the true preference models, we set the *Revealed Decision* node to one of the car choices in order to induce dependencies amongst the criteria (due to *Revealed Decision* forming v-structures with all the criteria nodes). With this evidence set, we then generate data (i.e., simulate cases) for the 4 criteria nodes along with the *Revealed Decision* node — creating an artificial data set for each scenario, where each case specifies the value of

the criteria along with the (simulated and randomised) choice made when the criteria have those values. *Revealed Decision* is stripped from the data prior to applying each of the methods described in Chapter 4, producing one model per method.

Data is generated in slightly different ways for each method, so as to produce data that is appropriate for use by each method. Of course, the preference model used in each case is the same. These methods are described below.

5.4.1.1 Generating Data for Causal Discovery

For causal discovery, we use the CaMML causal discovery software described in Section 2.10.3. The version of CaMML that we use (and is applicable here) can only learn BN structures for discrete, categorical data; fortunately, our preference model is already in a suitable form since it is generated after transforming all utility and decision nodes in a preference model into discrete chance nodes as described in the previous section. Data is then generated directly from the true preference models, since they are already in the right form. As described in the previous section (and as applies to all methods), this data is generated after setting evidence on *Revealed Decision*.

5.4.1.2 Generating Data for the Baseline and 4 Point Methods

For the Baseline method and the 4 Point Elicitation method, we begin by generating survey-like data that allows us to apply these methods unaltered, calculating their utility functions and populating the hand crafted networks with those functions. There are two stages to generating this data:

Pairwise comparison questions First of all, we use the mutual information between a criterion and *Revealed Decision* in the true preference model, which captures the importance of each criterion for the decision node. The pairwise *ratios* of these mutual information values is used to generate normalized random responses for each pair of criteria which act as survey responses from the participants about how many times one criterion is more important than another. (Here, this translates to how many times higher the mutual information of one criterion with *Revealed Decision* is over another.)

Satisfaction questions We additionally generate data from the true preference model in such a way that reflects the survey-like responses for the satisfaction level of the participants about one criterion being more important than another. This is done by mapping every pair of criteria nodes states (e.g. *HH* (High, High), *HM* (High, Medium), *HL* (High, Low) etc.), to a

distribution over satisfaction levels, and randomly generating a satisfaction response for that pair of criteria using this distribution. In particular, HH is very likely (0.8 probability) to produce a maximum satisfaction response and LL is very likely (0.8 probability) to produce a minimum satisfaction response.

Once we get all the survey-like responses from the true model, we use our proposed Baseline and the 4 Point Elicitation methods (as discussed in Section 4.2 and 4.3.1 of Chapter 4) to calculate the utility functions and apply them to their respective BDNs.

5.4.1.3 Generating Data for the AHP based BDN Method

AHP (Analytical Hierarchy Process) is based on the pairwise comparison of criteria in a hierarchical way, which uses a judgment scale to rate criteria based on their relative importance to the decision as discussed in detail in Section 2.6.1 of Chapter 2. Here we directly use the mutual information of criteria in the true preference model to calculate the AHP weights. Firstly, we generate a comparison matrix for each pair of criteria and then combine all the resultant matrices to get a complete and final 4 X 4 comparison matrix. Using the final comparison matrix, we calculate the normalized EigenVectors and get the normalized total weights for each criterion. These weights are used to populate the utilities in our BDNs which are called the AHP based BDNs. The detailed explanation of the translation of the AHP method into BDNs can be found in Section 4.4.

5.5 Evaluation

Evaluation of the proposed model is an integral part of the development process of the model. The evaluation process helps us to find the best model to represent our data and provides guidance on which approaches may work best and when in the future as well. This section is an overview of the evaluation techniques we use to do an evaluative comparison of all the methods.

We employ multiple evaluative measures in order to cover both the most popular measures (predictive accuracy and F score), despite their serious flaws in assessing probabilistic model performance, as well as measures which do a much better job of assessing model performance, as will become clear from their presentation.

5.5.1 Predictive Accuracy

Predictive accuracy is the most widely used technique to assess predictions across the disciplines, it is the measure of difference between the actual values/observed values and the

predicted values. It is calculate using the formula given below:

$$\text{Predictive Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

In our case, this is simply the frequency with which the recommended purchase matches the actual purchase.

For our fleet vehicle purchase case study, we generate data from the true models based on scenarios for three different vehicles i.e., *Car1*, *Car2*, and *Car3*. We have four different methods; let's say (Method 1, Method 2, Method 3 and Method 4) to compare and evaluate against the true model for each scenario based model. We generate the data for several cases and evaluate each method based on the choice it makes. Let's assume an example case as shown in Table 5.5, *Method 1* chooses *Car1*, *Method 2* chooses *Car3*, *Method 3* also chooses *Car3*, *Method 4* chooses *Car2*, and the *True Model* chooses *Car1*.

Case No.	Method 1	Method 2	Method 3	Method 4	True Model Choice
1	Car1	Car3	Car3	Car2	Car1
Score	1	0	0	0	-

Table 5.5: Scoring Method for Predictions

We can give 1 point to *Method 1* for its correct prediction but 0 points to the other methods as they have made wrong predictions. We repeat this for all the cases and accumulate the points at the end to get an average score for each method, as shown via the example in Table 5.6.

Method	Predictive Accuracy	Confidence Interval
Method 1	0.94	[0.92, 0.96]
Method 2	0.67	[0.62, 0.71]
Method 3	0.46	[0.42, 0.51]
Method 4	0.45	[0.41, 0.50]

Table 5.6: Predictive Accuracy Example

In this table, the last column shows the Confidence Intervals (CI) around the mean accuracy of each method. A CI is an interval (with a lower and upper bound) that covers the statistic (in this case, the mean value) with a certain expected frequency over repeated samples. It tells us about the reliability of a statistical estimate based on the sample size; wider intervals mean that the model at the given sample size is poor and the data is very noisy. CIs are often calculated using a 95% or 99% confidence level (which means that we are 95% or 99% certain that the method we used would produce a CI that holds the true value); we use a 95% confidence level to construct the CIs in our methods. CIs can be calculated using different methods, such as z-scores, t-scores and bootstrap, depending on the quality of the data and the ease of calculating

the statistic and its associated sampling distribution. If the population size is smaller, variance is unknown, and the sampling distribution approximately normal, it is better to use t-score; however, if the population size is greater than 30 and the standard deviation is known then z-score is recommended to use. In cases where the sampling distribution is unknown or difficult to use for exact calculation, the bootstrap method (which estimates the CI by computing the statistic on repeated re-samples of the available data) can also be used. Here, since z-score gives us an estimate of how far the values lie from the mean, we use the z-score as our population is large and the variance is known.

The example in Table 5.6 shows that the *Method 1* learned causal model reports significantly higher predictive accuracy of 0.94 with a CI of [0.92, 0.96]. We measure predictive accuracy of all of our methods and scenario based preference models to do an evaluative comparison against the true models.

5.5.2 F Score

F Score is a measure of the accuracy of a model on a dataset; it is also called F Measure and (when balanced) F1 Score. F Score considers both precision and recall of a model to evaluate many kinds of machine learning models. *Precision* is the number of true positives divided by the number of both true and false positives produced by a model.

$$Precision = \frac{True\ Positives}{False\ Positives + True\ Positives}$$

In words, *Precision* is the estimated probability the model is asserting the truth when identifying a positive case. In our example, restricting our attention to cases where the *True Model's* choice is *Car1*, the precision is the number of times our method's chosen car is also *Car1* divided by the total number of cases.

Recall is equal to the number of true positive predictions divided by the number of false negatives and true positive predictions made by a method. In other words, it's an estimate of how likely it is the model will find a true positive that's there to be found.

$$Recall = \frac{True\ Positives}{False\ Negatives + True\ Positives}$$

False negatives count the number of times our method's chosen car is not *Car1* where the *True Model's* chosen car is *Car1*. We calculate precision and recall for each method and then compute the F1 score of all of our methods using the following equation 5.1 to evaluate and compare the results.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (5.1)$$

This is equivalent to the harmonic mean of precision and recall (with precision and recall weighted equally).

Method	F1 Score			Mean	Confidence Interval
	Car1	Car2	Car3		
Method 1	0.92	0.90	0.90	0.90	[0.89, 0.91]
Method 2	0.47	0.40	0.30	0.39	[0.29, 0.48]
Method 3	0.47	0.45	0.39	0.43	[0.38, 0.48]
Method 4	0.83	0.80	0.78	0.80	[0.77, 0.83]

Table 5.7: F1 Score based on Scenario 5.3

Table 5.7 represents the F1 score for each method based on scenario 5.3 where we can see statistically significantly highest scores for each car choice made by *Method 1* highlighted by green color.

5.5.3 Kullback Leibler Divergence

The Kullback Leibler Divergence (KLD) score is the measure of the divergence of an arbitrary distribution from the true distribution over the same variables. It can be thought of as the excess number of bits (given $\log_2$) expected for an optimal code using the learned model to encode a random sample from the true model, which ranges from 0 (when the learned model is the true model) to infinity. Any two probability distributions can be compared using KLD, including probabilistic models or subsets of probabilistic models, such as restrictions to target variables of interest. While it is not a true distance metric (violating both the triangle inequality and symmetry), it is in many ways an ideal measure of how well or poorly a learning or modelling method has done in capturing the probability distribution implied by a known, true model.

We use KLD to measure the dissimilarity between the probability distribution produced by our methods and the probability distribution of the true model as explained in Section 2.11.5 of Chapter 2. This metric allows us to compare and evaluate the difference between the probability distributions of our models with respect to the true model based on three different car choices.

Figures 5.3, 5.4, 5.5 and 5.6 show the bar graphs based on the distributions of all the models in comparison to the True model using Scenario 5.3. (These results are shown for illustration only, the real results are outlined in Section 6.2.3 of Chapter 6. The blue bar shows the distribution based on the three car choices (*Car1*, *Car2* and *Car3*) by our methods (Method 1, Method

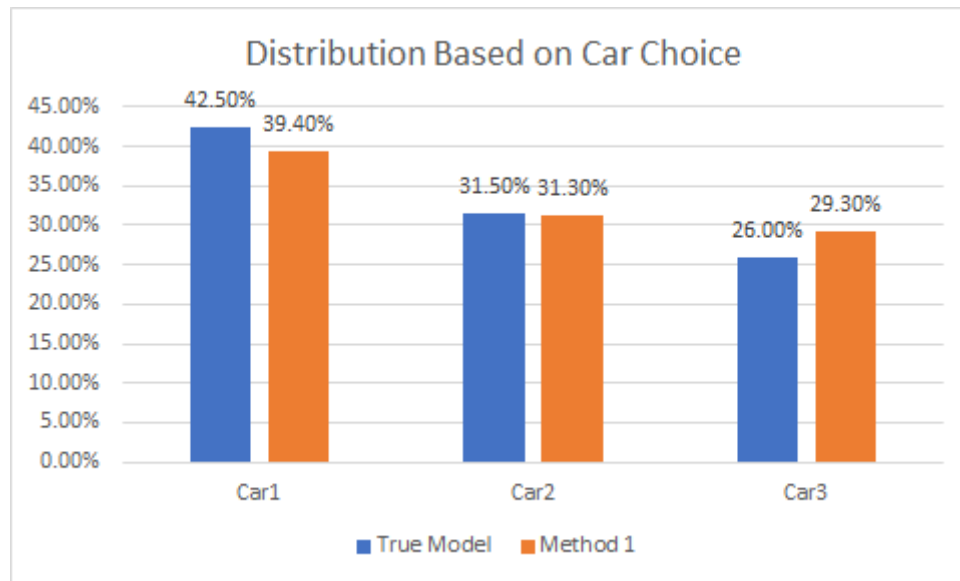


Figure 5.3: Distribution of Method 1 vs True Model

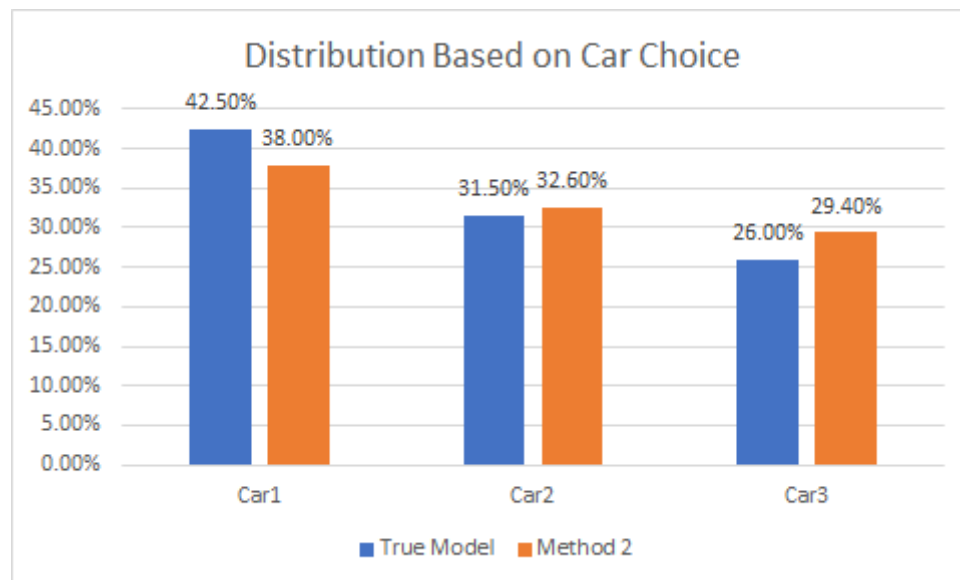


Figure 5.4: Distribution of Method 2 vs True Model

2, Method 3 and Method 4). The orange bar represents the car choice distribution of the True model based on Scenario 5.3.

Using KLD metric, we calculate the distance of our methods' distribution from the true model's distribution as shown in Table 5.8. P represents the distribution based on three different car choices by our methods and Q represents the distribution of the True model using Scenario 5.3.

The table highlights that any method (*Method 1* in this example) with the lowest KL divergence value is the best method. The distribution of *Method 1* in this example has the lowest distance of 0.458 bits from the True model's distribution.

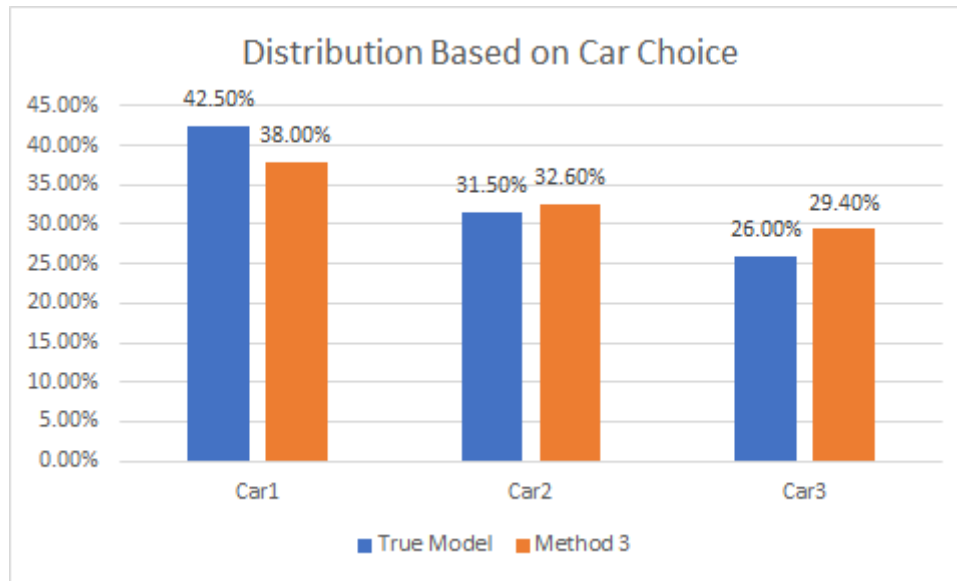


Figure 5.5: Distribution of Method 3 vs True Model

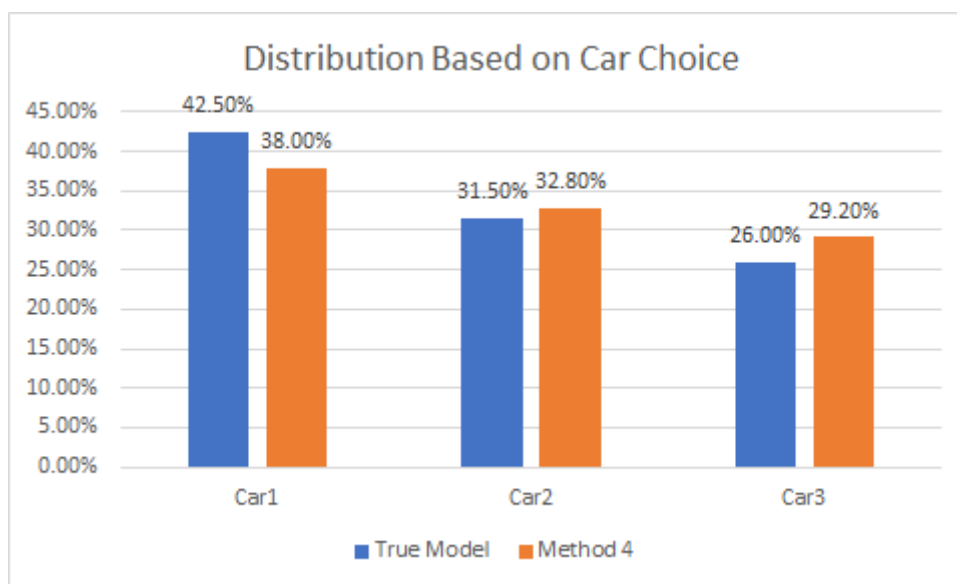


Figure 5.6: Distribution of Method 4 vs True Model

Method	Kullback Leibler Divergence (in bits)
Method 1	KLD($P \parallel Q$): 0.458 bits KLD($Q \parallel P$): 0.451 bits
Method 2	KLD($P \parallel Q$): 0.691 bits KLD($Q \parallel P$): 0.692 bits
Method 3	KLD($P \parallel Q$): 0.691 bits KLD($Q \parallel P$): 0.692 bits
Method 4	KLD($P \parallel Q$): 0.668 bits KLD($Q \parallel P$): 0.670 bits

Table 5.8: Kullback Leibler Divergence

5.5.4 Logarithmic Score

Logarithmic Score (log score) was proposed by I.J. Good in 1952 and since then it has been widely used in various disciplines to evaluate predictive models. The log score rule is a proper scoring rule, which means that a forecaster can maximize their expected reward by forecasting the true distribution. A scoring rule is *strictly* proper if the forecaster can *only* maximize their expected score by reporting the true probabilities. It encourages the forecaster to make careful assessments and give honest forecasts, as reporting any other probabilities yield a lower expected reward. The log score rule is also strictly proper. In general, it assigns a harsh penalty to incorrect low probability cases and is extremely sensitive to extreme cases, and is calculated as the log of the probability estimate for the actual outcome [97].

We use log score to measure the expected reward by adding the $\hat{p}_i$ across all the predictions. It is always negative, with the larger magnitudes representing worse results. We generate a variety of cases from the scenario based preference models where our models make different decisions. Good's IR of a binary classification with probability $\hat{p}_i$ is defined by the following formulas.

$$I^+ = 1 + \log_2 \hat{p}_i \quad (\text{for correct classifications})$$

$$I^- = 1 + \log_2(1 - \hat{p}_i) \quad (\text{for incorrect classifications})$$

IR has a range of $[-\infty, 1]$. Let's have a look at the example log score values for our models outlined in Table 5.9.

Method	Mean Log Score	Confidence Interval
Method 1	-1.57	[-1.574, -1.573]
Method 2	-5.04	[-5.05, -5.04]
Method 3	-0.99	[-1.03, -0.96]
Method 4	-0.64	[-0.66, -0.62]

Table 5.9: Log Score Example

The method with the lowest log score value is the best method reporting the minimum expected surprise. In this example 5.9, we can see that *Method 4* has the lowest log score among all, indicating that the maximum expected reward has been assigned to the method closest to the true distribution.

5.5.5 Bayesian Information Reward (BIR)

Bayesian Information Reward (BIR) is the generalized form of the original log score — Information Reward (IR) — that was proposed by Good in 1952. The original IR positively rewards binary classifications which are informed relative to an uninformed uniform prior, while BIR generalizes the binary case to multinomial predictions, taking non-uniform priors into account. BIR also appropriately punishes miscalibration; that is, it will penalize any *correct* prediction with a confidence different from its actual probability over the long term, meaning it is a strictly proper scoring function [66].

In this thesis, we use the BIR metric to measure the expected reward of each method using the formula given below:

$$IR_B = \sum \frac{I_i}{n}$$

where n is the possible target values, $I_i = I_i^+$ for true values

$$I_i^+ = \log \frac{\hat{p}_i}{p_i}$$

and $I_i = I_i^-$ otherwise

$$I_i^- = \log \frac{1 - \hat{p}_i}{1 - p_i}$$

where $\hat{p}_i$ is the posterior probability and p_i is the prior probability of the variable taking the value.

Table 5.10 shows an example of BIR score for our methods, the highest value is highlighted by green color showing the maximum expected reward using BIR method.

Method	Mean BIR	Confidence Interval
Method 1	0.008	[0.007, 0.008]
Method 2	-2.18	[-2.181, -2.18]
Method 3	0.37	[0.35, 0.39]
Method 4	0.595	[0.575, 0.615]

Table 5.10: Bayesian Information Reward Example

These table highlights the method with the highest BIR score, as it reports the closest distribution to the true distribution and maximizes the expected reward.

5.5.6 Expected Value

Expected Value (EV) is the weighted average of the probabilities of all possible outcomes. In statistics, EV is calculated by multiplying the value of each possible outcome by the likelihood of each outcome, and then summing. We accumulate the actual expected utilities for all the

decision choices made by a method using multiple cases. In order to measure the EV for each method, we generate several cases from our scenario based preference models and run all our elicitation methods over the generated cases. Once we get a method's decision for each case, we plug it into the true BDN model and extract the actual EV. These actual EVs are then aggregated via the geometric mean to compare them with those of other methods.

Table 5.11 shows an example of the mean EV for each method along with their confidence intervals. The highest expected utility value is highlighted by green color i.e., *Method 1* in this example. The real results are based on our three Scenarios 5.1, 5.2 and 5.3, which are shown in Section 6.2.6 of Chapter 6.

Method	Mean EV	Confidence Interval
Method 1	1.966	[1.91, 2.01]
Method 2	1.957	[1.90, 2.01]
Method 3	1.953	[1.90, 2.0]
Method 4	1.927	[1.87, 1.97]

Table 5.11: Expected Value Example

5.6 Conclusion

In this chapter, we explained the fleet vehicle purchase case study along with the Fleet purchase BN and the network parameters. Then we discussed the survey methodology that we applied to this case study including the way in which the survey was setup. Using Algorithms 5.1 and 5.3 we presented the step by step approach for how we elicited responses from the participants. As the survey was limited to pairwise comparisons between the criteria and had a relatively small number of participants, we looked to alternatives for the large data-dependent methods. We presented some scenarios to generate preference models to cover multiple criteria and used those scenario based preference models to generate artificial data so that we could apply CaMML to learn the causal structure. We also used the artificial data to calculate utility functions in the Baseline method and the 4 Point Elicitation method. In addition to that, we used the AHP method to calculate the utilities using the artificial data and translated those results into a BDN, called the AHP based BDN. Lastly, we described different evaluation and validation techniques to compare our models against the scenario based preference models.

In the upcoming chapter, we cover the detailed experimental results of all the methods using survey data and using virtual data that is generated from scenario based preference models.

Experimental Results

We presented preference elicitation methods in Chapter 4 and explained how we perform modelling with multi-criteria preference data in chapter 5, where we also described methods for evaluating that modelling. This chapter presents the empirical evaluation of our methods using the case study of Fleet Vehicle Purchasing for an arbitrary large organization. As we will see, the elicitation methods based on survey data tend to produce results that are quite similar, but there is more variation in performance for methods tested on the generated data.

In Section 6.1, we explain the experimental results for the Baseline method, the 4 Point Elicitation method, as well as the 4 Point Elicitation method using a larger expert-weighted resample of the survey data. A head-to-head comparison of the Baseline method and the 4 Point Elicitation method is performed in Section 6.1.4 using the full set and subsets of survey data. In Section 6.2, we turn to the evaluations based on generated data, presenting the virtual experimental results of all the methods including causal discovery and the AHP method using data generated from scenario-based preference models (a detailed description of scenario based preference models is in Section 5.4 of Chapter 5).

6.1 Survey Experiment for Baseline and 4 Point Elicitation Method

We ran an actual survey to elicit preferences from the experts (details are provided in Chapter 5) and the collected data was preprocessed to be used in our elicitation methods i.e., Baseline method and the 4 Point Elicitation method. This section presents and explains the experimental results for these methods on the survey data in detail.

6.1.1 The Baseline Method

The Baseline method was our original proposed preference elicitation method which helped us to elicit weights from multiple participants for each criterion in terms of its relative importance to other criteria. It also elicited the participants' confidence about the weights they had given to a specific criterion; once we received the responses from the participants, we combined them

and calculated the utility function as explained in Section 4.2 of Chapter 4. Using our actual survey for our Fleet Vehicle Purchase case study, we gathered responses from the participants and calculated the utility function using this method and got the results shown in Table 6.1 below.

Utility Node	States of Parent Chance Node	Value
Budget Value	High	0.065
	Medium	0.093
	Low	0.1
Safety Value	High	0.096
	Medium	0.083
	Low	0.057
Satisfaction Value	High	0.12
	Medium	0.082
	Low	0.053
Environmental Value	Poor	0.058
	Neutral	0.086
	Good	0.107

Table 6.1: Values for each state of every parent chance node after calculating utility functions via the Baseline Method (Highest values are highlighted in green and lowest values are highlighted in red).

In this table, the first column shows the utility nodes involved in our expert-elicited BDN, the second column represents the states of each utility node's parent node (i.e., *Overall Cost* is the parent node of *Budget Value*, *Safety* node is the parent of *Safety Value*, *Customer Satisfaction* is the parent node of *Satisfaction Value* and *Environmental Impact* is the parent of *Environmental Value*) and the last column shows the values calculated by the Baseline method for each utility node. As we can see, the results of applying the Baseline method to survey data are in the order of what we would expect. *Budget Value* has a high score when the overall cost is low, *Safety Value* has a maximum score for high safety, *Satisfaction* is high when customer satisfaction is high and *Environmental Value* has a high score for vehicle's good environmental impact. These values are highlighted in green for calculations using the Baseline method from the expert elicited data. On the other hand, the values in red cells are the minimum scores representing high overall cost, low safety, low customer satisfaction and poor environmental impact of the vehicle. These calculated utilities become the direct inputs to the individual criterion utility nodes in our expert-elicited BDN. We also calculate the overall weights for each criterion based on its importance using the Baseline method shown in Table 6.2. The results in this table show that *Safety Value* has the highest score of 0.615 among all criteria, which means that given the problem scenarios, participants value safety more than other criteria. The second highest score is 0.17 for *Environmental Value* followed by *Satisfaction* of the customer with a weight of 0.125. However, *Budget Value* of the vehicle is least important to participants, with a lowest weight of

0.088. It is interesting to see that *Safety Value* is weighed almost 7× more important than *Budget Value*, but 5× and 3.6× more important than *Satisfaction* and *Environmental Value* respectively. It is to be noted that the absolute values of the utilities of each criterion are not meaningful, it is just the relative values of the orders which is important to reflect the *Overall Weight*.

These weights in Table 6.2 were used as the inputs into the *Overall Weight* node of the expert-elicited network to specify the weight of each criterion.

Utility Node	Overall Weight
Budget Value	0.088
Safety Value	0.615
Satisfaction	0.125
Environmental Value	0.17

Table 6.2: Overall Weight of Each Utility Node via the Baseline Method (Maximum score is highlighted by green color)

We produced all these required values using our Baseline method and populated our expert-elicited BDN for the Fleet Vehicle Purchase network to get the network shown in Figure 6.1.

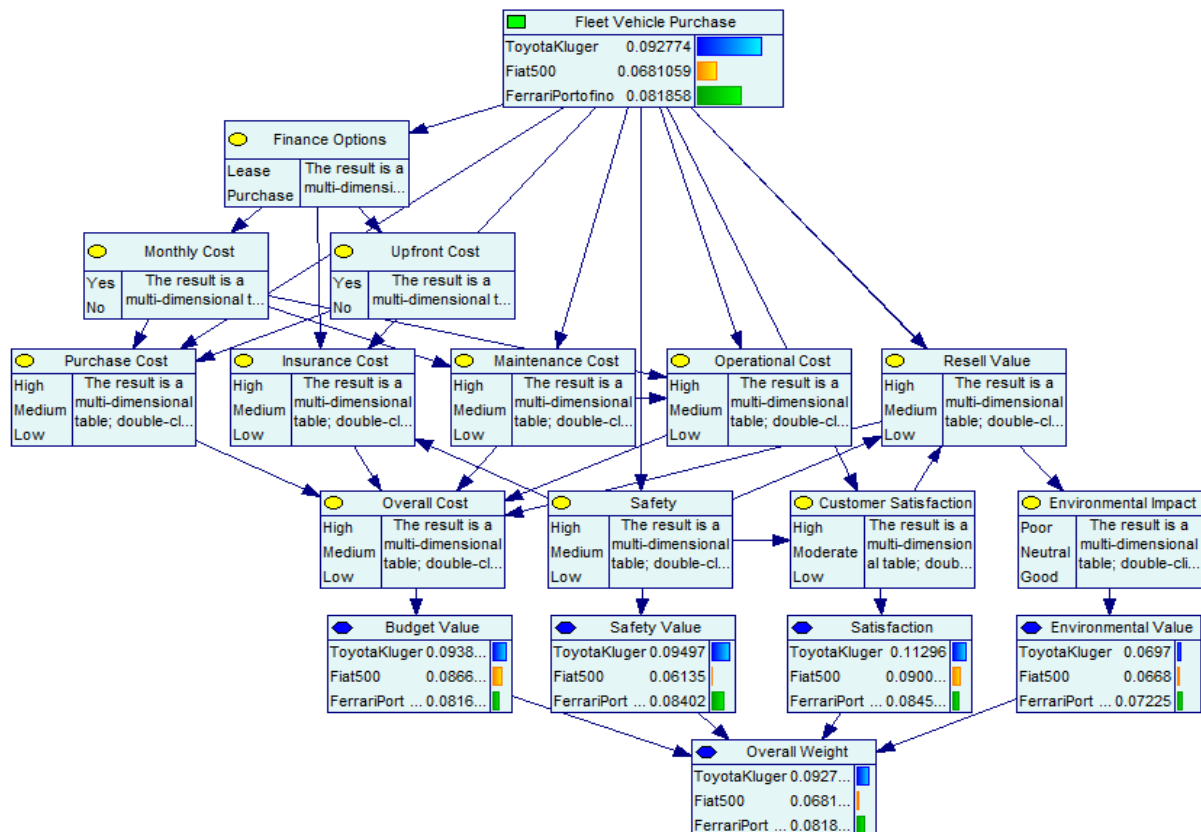


Figure 6.1: Parameterizing Expert-Elicited BN Using the Baseline Method

Looking at the network in Figure 6.1, we can see the Expected Utility (EU) for each of three car types i.e., *Toyota Kluger*, *Ferrari Portofino* and *Fiat 500*. These EUs are shown in Table 6.2

where *Toyota Kluger* has the highest EU of 0.092 followed by *Ferrari Portofino* while *Fiat 500* has the lowest EU of 0.068. Of course, the absolute numbers do not matter since this is an interval scale; only the order and relative weight matter.

Options	Expected Utility
Toyota Kluger	0.092
Ferrari Portofino	0.081
Fiat 500	0.068

Table 6.3: Expected Utilities Using the Baseline Method

So according to the Baseline method as applied to the survey data, *Toyota Kluger* seems to be the best car with the highest EU among all three cars to choose for our Fleet Vehicle Purchase problem.

6.1.1.1 Sensitivity Analysis of the Baseline Method

The network in Figure 6.2 represents the sensitivity analysis of expert elicited BDN using the Baseline method. It is the sensitivity of the network to changes in the parameters (in GeNIe, the default is a change of 10%, which is what we use here). That is, it examines how the final utility node — in this case, *Overall Weight* — changes when we change CPT parameters in any of the nodes, including in “Overall Weight” itself.

The redness of the nodes in this figure indicate the degree to which changes in the parameters of these nodes produce changes in Overall Weight; in this case *Budget Value*, *Satisfaction Value*, *Safety Value* and *Environmental Value* are the most sensitive nodes. The analysis also highlights that changes to Overall Weight’s definition (i.e. alteration of the weights) will have less impact than changes to any of the independent criteria.

Figure 6.3 below shows the visualisation of the strength of influence of each arc. It’s computed using various distribution distance metrics (here, just the Euclidean average distribution difference), with the thickness of the arrows showing the strength of the influence between the nodes. The decision node has a strong influence on *Finance Options* and *Safety*, and a strong influence can be noted between the criteria nodes and their associated utility nodes.

6.1.1.2 What-if Analysis of the Baseline Method

As we know that causal BNs allow arbitrary hypothesis and counterfactual analyses, we ask different types of what-if questions to see how the network react to those questions.

For example, we can see the networks in Figures 6.4 and 6.5 where we ask two different questions; firstly, what-if we set the criteria nodes to the ideal case (where the *Overall Cost* is

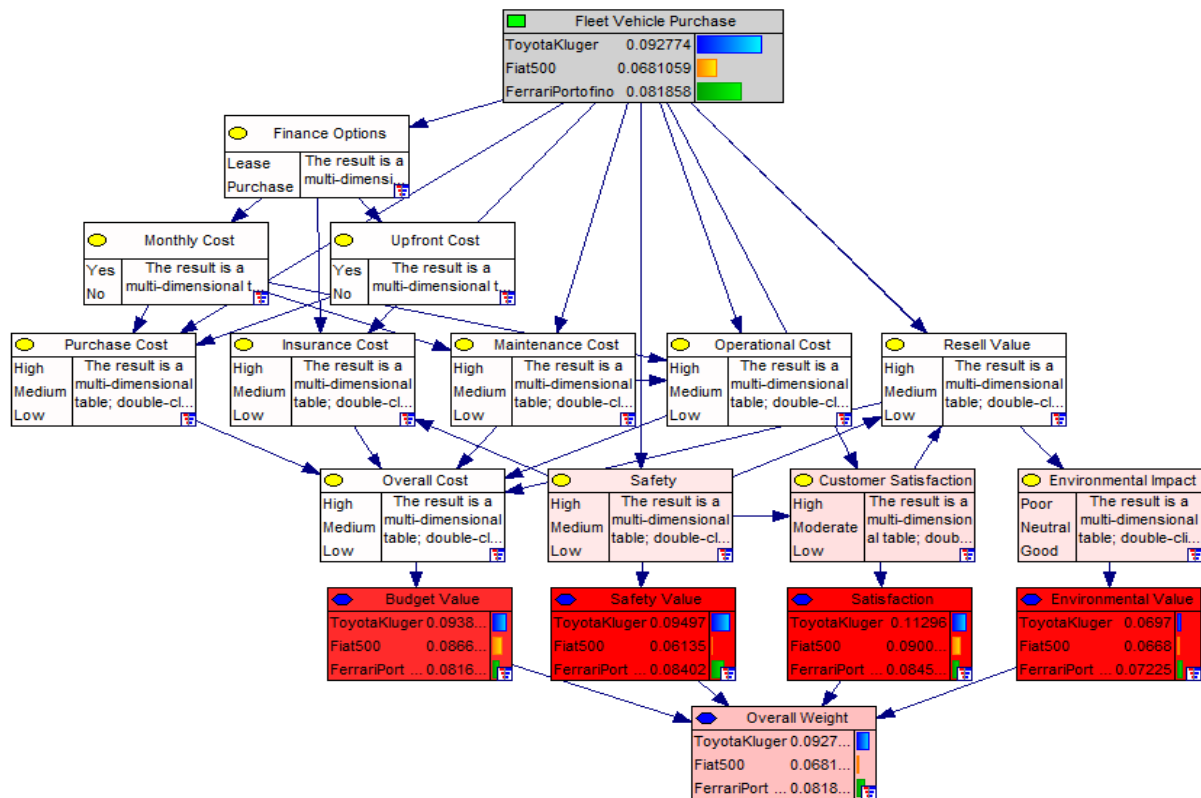


Figure 6.2: Sensitivity Analysis Using The Baseline Method

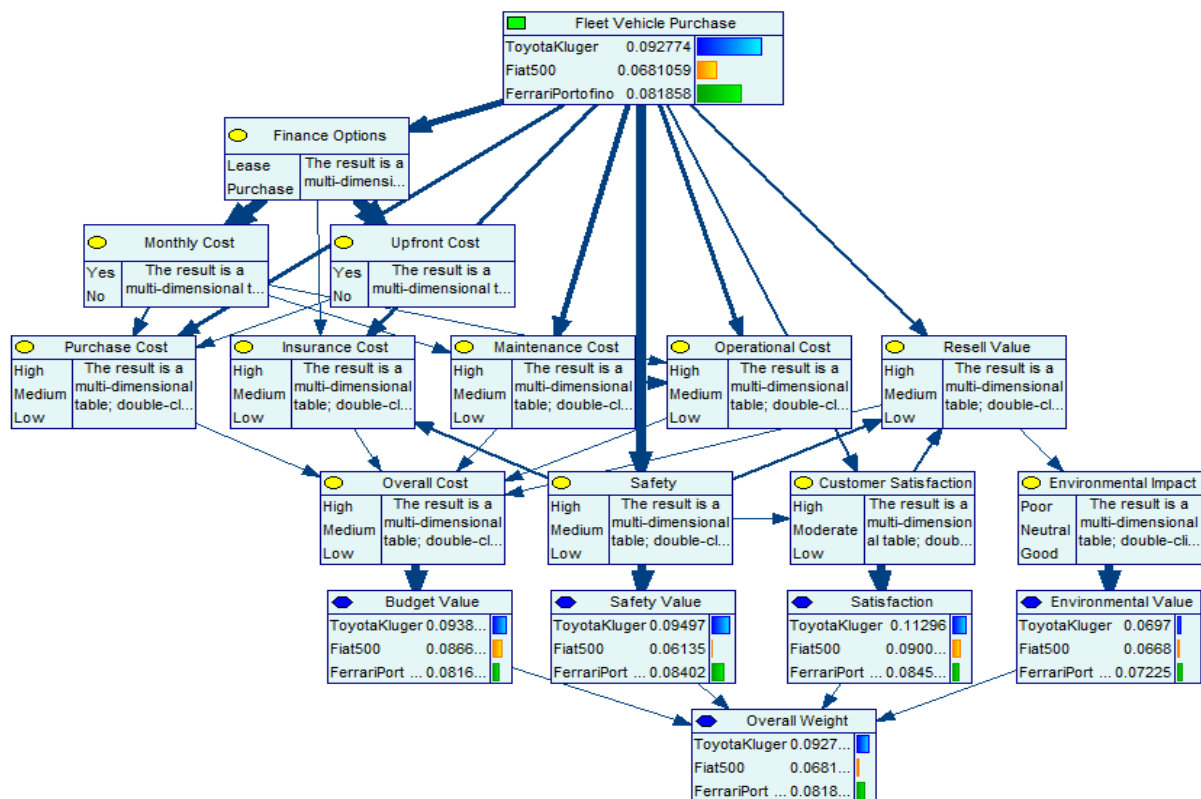


Figure 6.3: Visualization of the Strength of Influence of Each Arc Using the Baseline Method

set to *Low*, *Safety* to *High* and so on) and secondly, what-if we set the states to the worst case (where the *Overall Cost* is set to *High*, *Safety* to *Low* and so on).

The best case shows a higher overall expected utility of 0.1 while the worst case reports an expected utility of 0.05. This kind of hypothetical analysis in the BDNs allows the user to look at the correlation between the criteria and the decision node.

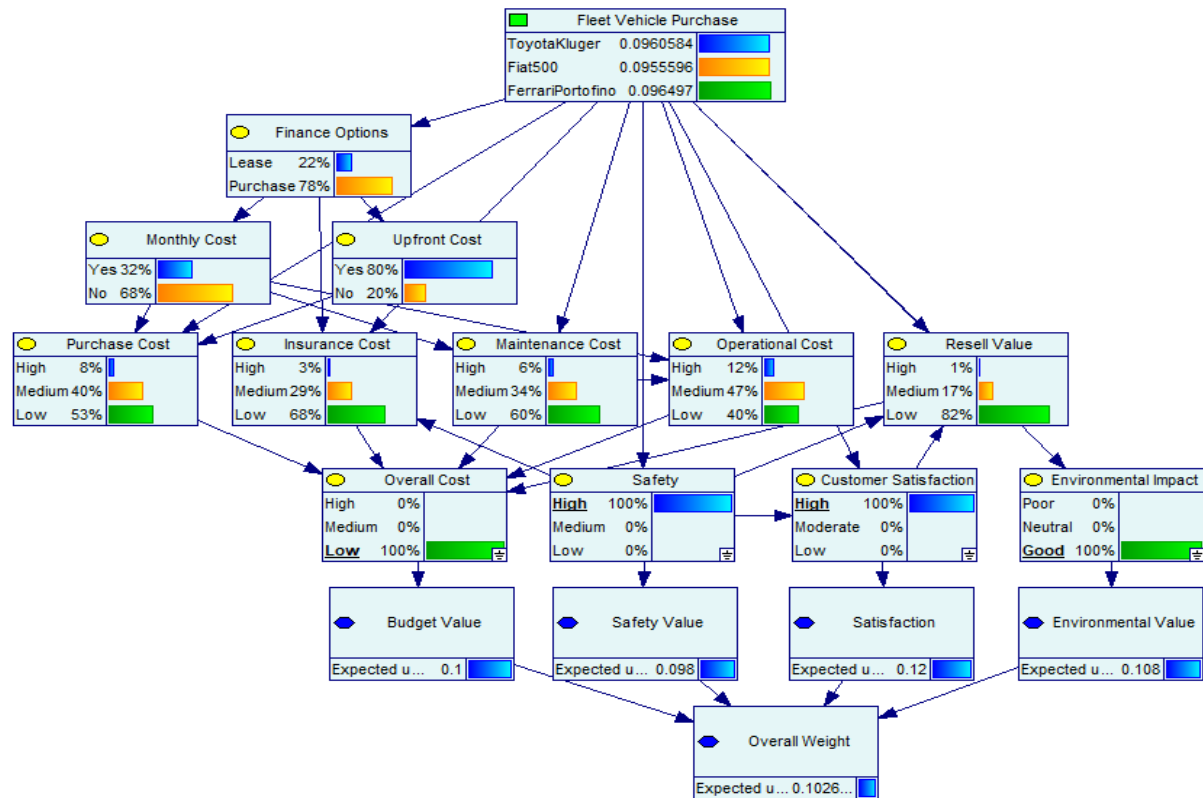


Figure 6.4: What-if Analysis Using the Baseline Method (Best Case Scenario)

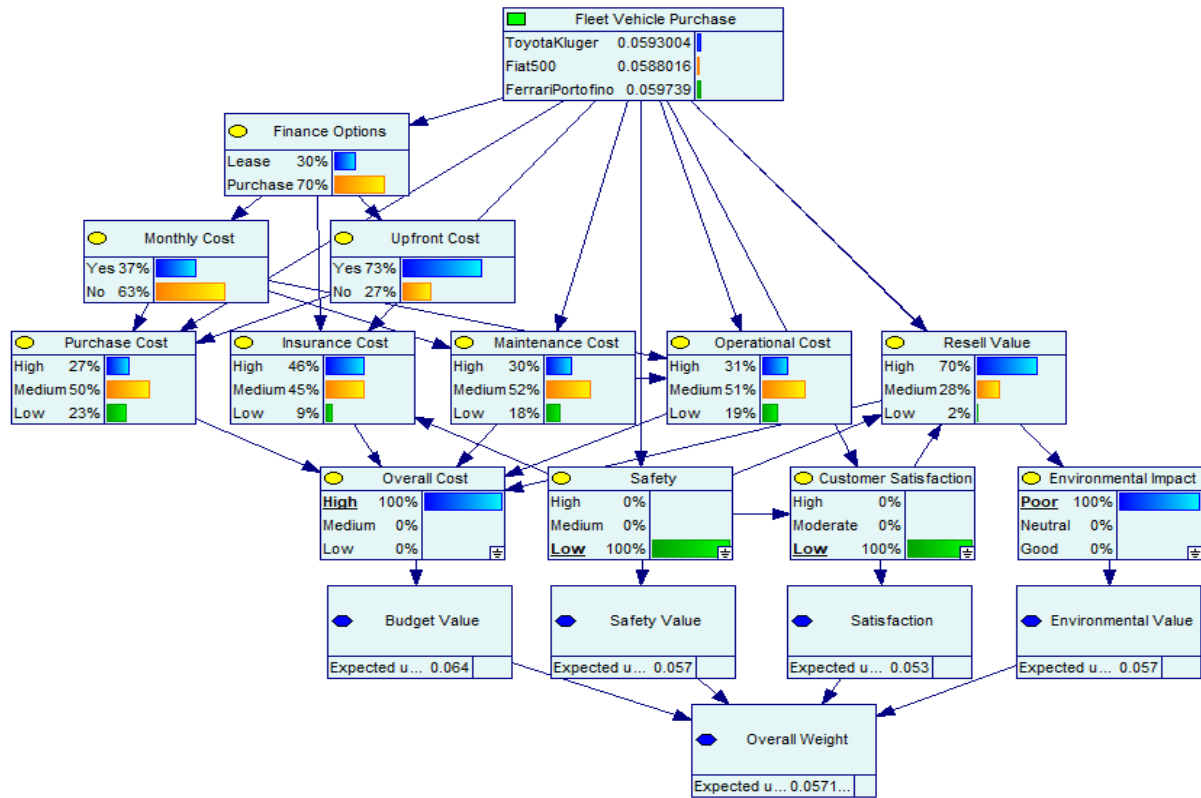


Figure 6.5: What-if Analysis Using the Baseline Method (Worst Case Scenario)

6.1.2 The 4 Point Elicitation Method

The 4 Point Elicitation method is based on the traditional 4 point elicitation method of IDEA protocol which is widely used for expert elicitation purposes [9]. Using the 4 Point Elicitation method, we elicited the lowest plausible value, highest plausible value and best estimate from the participant to weight a criterion based on its relative importance, which created an interval from lowest to highest value, followed by their confidence that their created interval holds their true value, as discussed in Section 4.3.1. Once we elicited the required data from the participant, we standardized the intervals using the process explained in Section 4.3.1.1 of Chapter 4. The values of the criteria involved in our expert-elicited BN were calculated using the 4 Point Elicitation method as shown in Table 6.4.

In this table, just like the Baseline method we have the highest score for *Budget Value* when the parent node *Overall Cost* is low, a highest score for *Safety Value* when the *Safety* ranking of the vehicle is high, highest score for *Satisfaction* when *Customer Satisfaction* rating of the vehicle is high and maximum score of *Environmental Value* when the *Environmental Impact* rating of the fleet vehicle is good. The highest values for all utility nodes are highlighted by green color and lowest values are highlighted by red cells that are representing high overall cost, low safety, low customer satisfaction and poor environmental impact. The calculated utility

Utility Node	States of Parent Chance Node	Value
Budget Value	High	2.23
	Medium	3.08
	Low	3.19
Safety Value	High	3.11
	Medium	2.73
	Low	1.99
Satisfaction	High	3.61
	Medium	2.84
	Low	2
Environmental Value	Poor	1.95
	Neutral	2.72
	Good	3.52

Table 6.4: Values for each State of Parent Chance Node after Calculating Utility Functions via the 4 Point Elicitation Method (Highest values are highlighted by green color and lowest values are highlighted by red color)

functions were the inputs to the states of parent chance nodes in our fleet vehicle BDN. In addition to that, we calculated the weight of each criterion using the 4 Point Elicitation method to parameterize the BDN as outlined in Table 6.5.

Utility Node	Overall Weight
Budget Value	0.38
Safety Value	1.10
Satisfaction	0.43
Environmental Value	0.76

Table 6.5: Overall Weight of Each Utility Node via the 4 Point Elicitation Method (Maximum score is highlighted by green color)

This table shows the weights of each criterion using the the 4 Point Elicitation method, where *Safety Value* has the highest weight among all as given by the participants according to the given problem scenarios. On the other hand, *Budget Value* has the lowest weight of 0.38, alternatively, they do not care about the overall cost of the vehicle to be purchased by the fleet manager.

After calculating all the values using the 4 Point Elicitation method, we parameterized our expert-elicited BDN and got the network shown in Figure 6.6.

According to the above mentioned network details calculated by the 4 Point Elicitation method, we got the EUs for each of three cars as shown in Table 6.4.

The results in this table show that *Toyota Kluger* has the high EU of 7.76 followed by *Ferrari Portofino* with an EU of 7.09 while the *Fiat 500* has the lowest EU of 6.38 as calculated by the 4 Point Elicitation method. We can say that according to the 4 Point Elicitation method, *Toyota Kluger* seems to be the best option to be chosen for Fleet Vehicle Purchase based on the

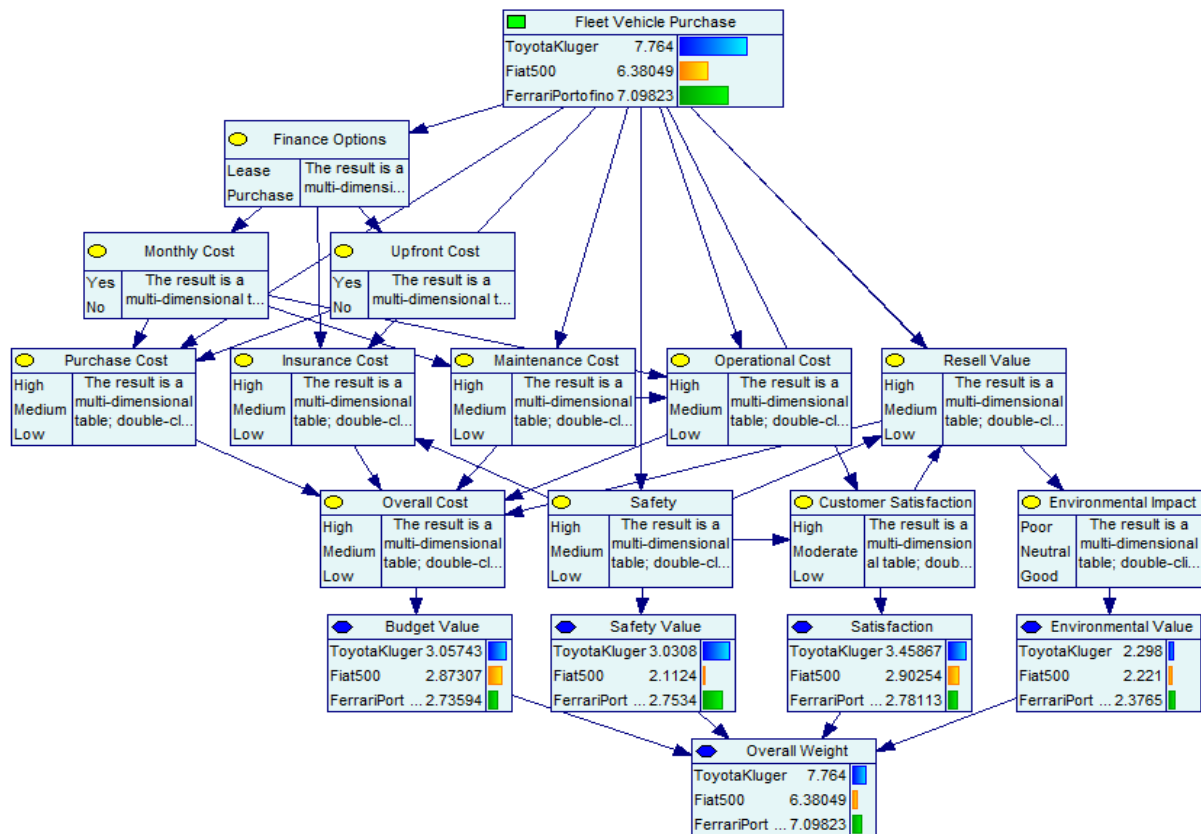


Figure 6.6: Parameterizing Expert-Elicited BN Using the 4 Point Elicitation Method

Options	Expected Utility
Toyota Kluger	7.76
Ferrari Portofino	7.09
Fiat 500	6.38

Table 6.6: Expected Utilities Using the 4 Point Elicitation Method

problem scenarios presented to the participants in the survey. We can see that the range of the values generated by Baseline and the 4 Point Elicitation methods are not the same because we normalized the values for the Baseline method but not for the 4 point method, but the range is not important, it is the order and the ratio¹ which matters.

The network in Figure 6.7 shows the 4 Point Elicitation method based network with continuous nodes, showing the uncertainty in responses. We calculated the mean and standard deviation for each state of the criteria from the elicited data to know how uncertain peoples' responses are. As we can see the distribution in this network, we have a larger variation in the responses we had got from participants.

¹That is, the same place in the order and same ratio to other estimates.

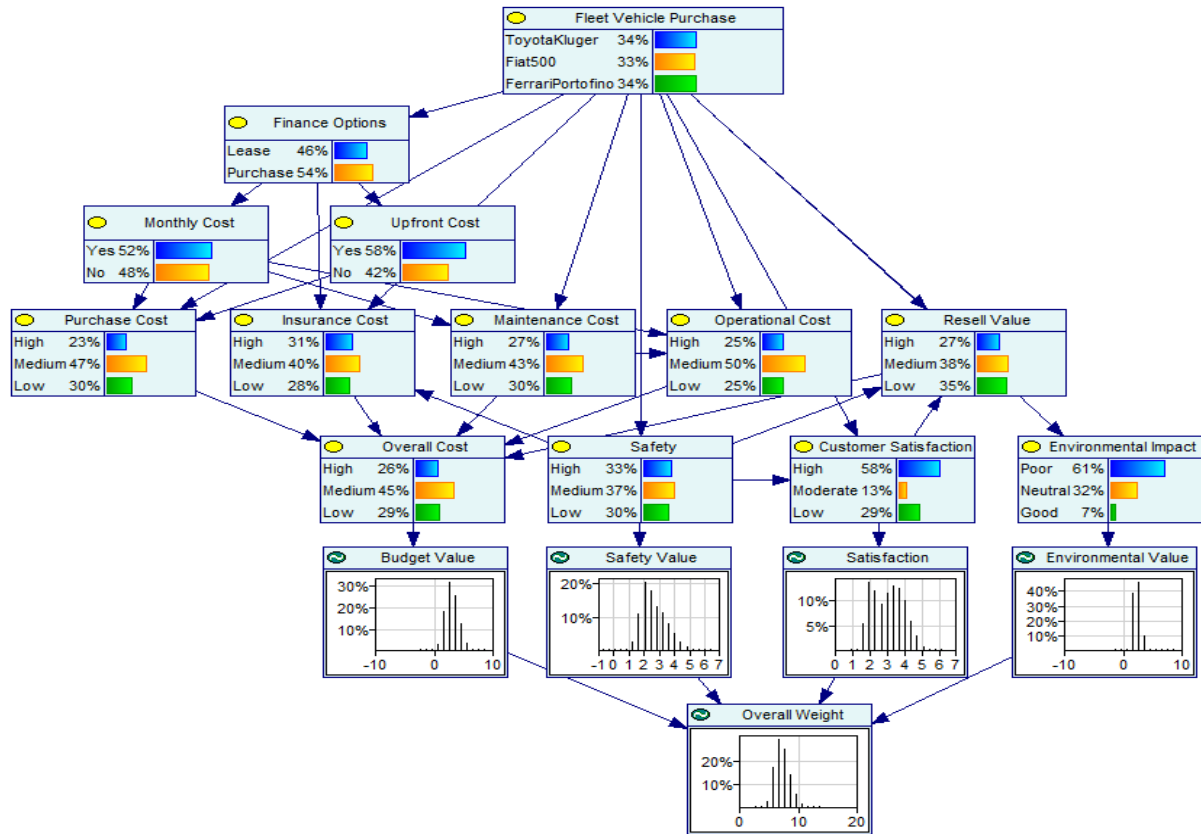


Figure 6.7: The 4 Point Elicitation Method Based Network with Continuous Nodes

6.1.2.1 Sensitivity Analysis of the 4 Point Elicitation Method Based Network

The network in Figure 6.8 represents the sensitivity analysis of expert elicited BDN using the 4 Point Elicitation method. The red nodes indicate that they are very sensitive to the decision, in this case *Overall Weight* of the criteria is the most sensitive node to the decision node, the second level sensitivity is shown by *Safety*, *Customer Satisfaction* and *Environmental Impact* nodes in this network.

Network in Figure 6.9 shows that causal visualization based on the 4 Point Elicitation method based network, where the thickness of the arrows show the strength of the influence between the nodes. The decision node has a strong influence on *Finance Options* and *Safety*, there is also an influence between the criteria nodes and their associated utility nodes.

6.1.2.2 What-if Analysis of the 4 Point Elicitation Method Based Network

Using the 4 Point Elicitation method based network, we are also looking at different types of what-if questions to see how this network react to those questions. For example, we can see the networks in Figures 6.10 and 6.11 where we look at two extreme cases; firstly, what-if we set the criteria nodes to the ideal case (where the *Overall Cost* is set to *Low*, *Safety* to *High* and so on) and secondly, what-if we set the states to the worst case (where the *Overall Cost* is set to

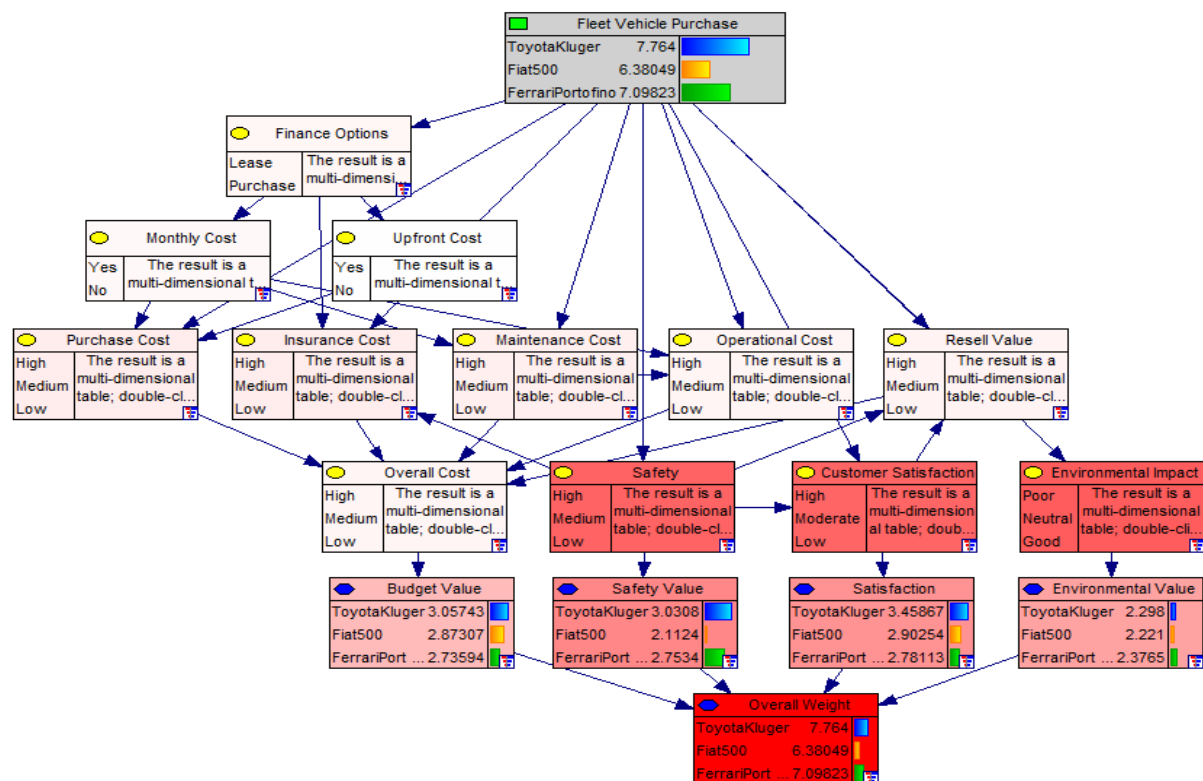


Figure 6.8: Sensitivity Analysis of the 4 Point Elicitation Method Based Network

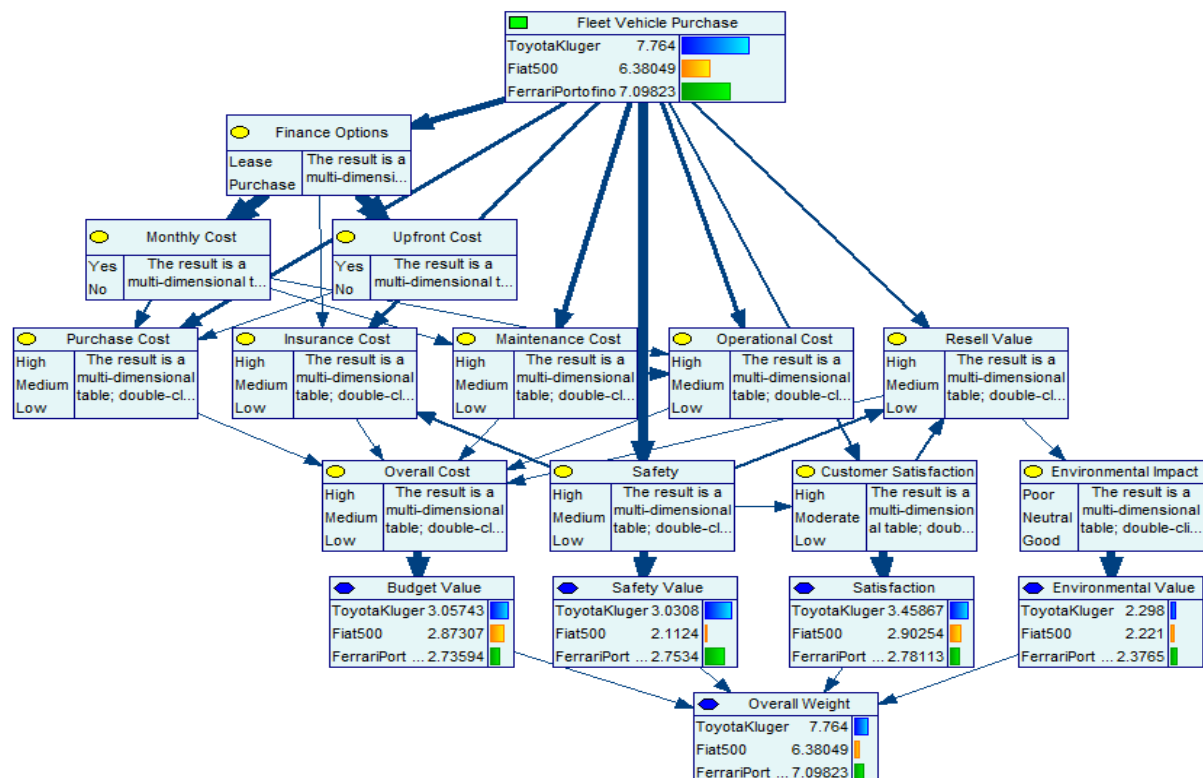


Figure 6.9: Visualization of the Strength of Influence of Each Arc Using the 4 Point Elicitation Method

High, Safety to Low and so on).

The best case shows a higher overall expected utility of 8.89 while the worst case reports an expected utility of 5.39. By setting the states of interest, we discover that intervening the criteria nodes lead to a higher expected utility.

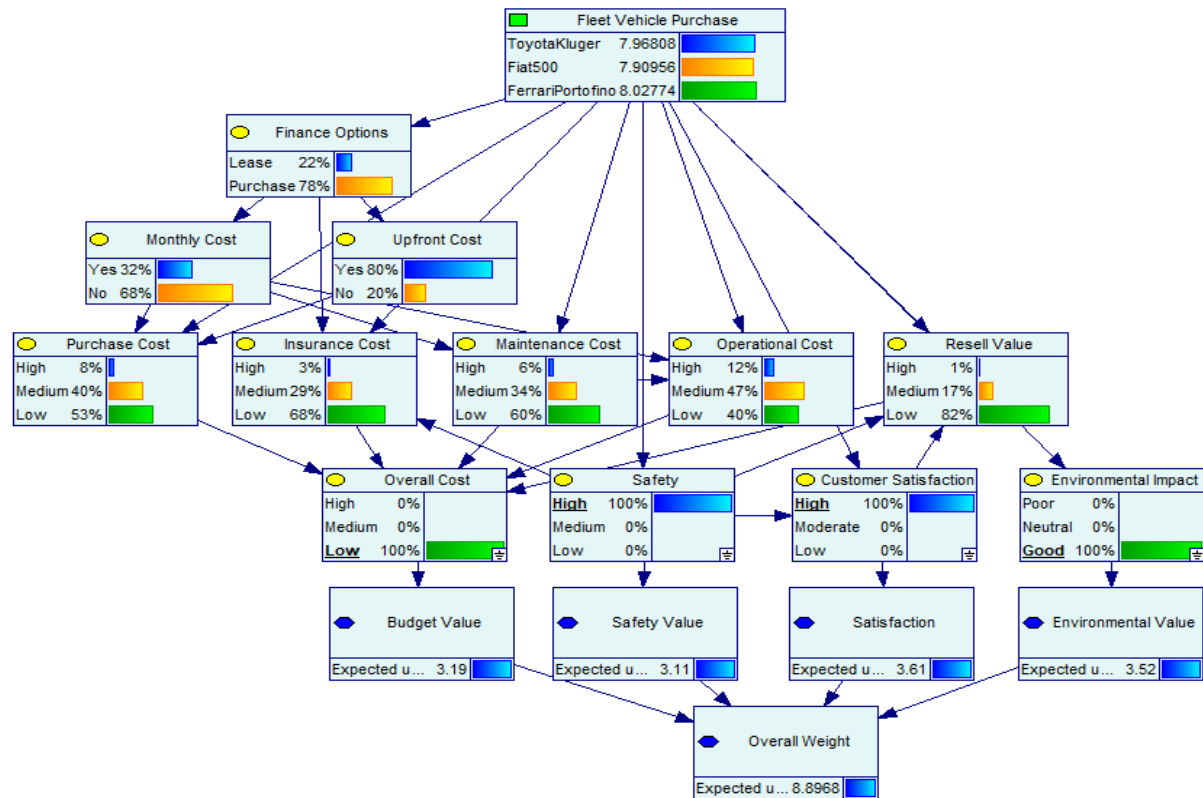


Figure 6.10: What-if Analysis Using the 4 Point Elicitation Method (Best Case Scenario)

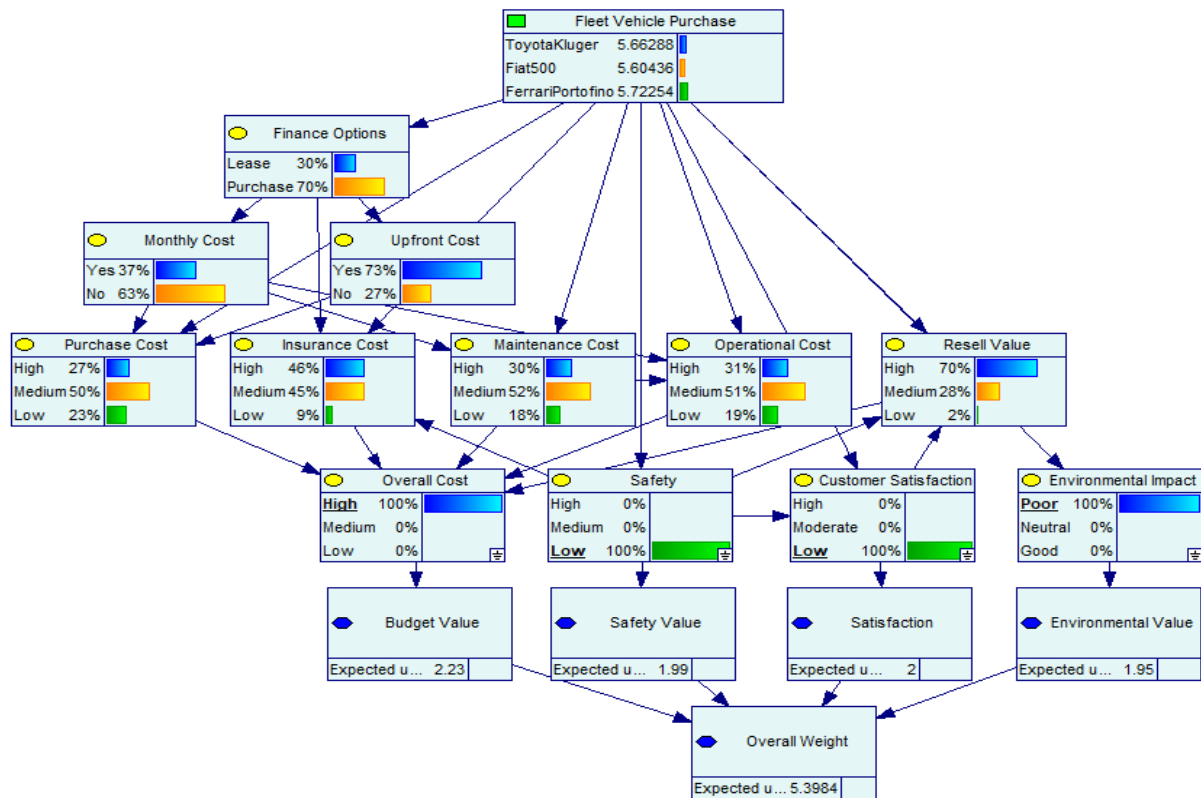


Figure 6.11: What-if Analysis Using the 4 Point Elicitation Method (Worst Case Scenario)

6.1.3 Using Expert-Weighted Resampling Data

Here we generated datasets based on a larger resampling of collected survey data, that helped us with evaluation when having only a small number of participants (see details in Section 5.3.6 of Chapter 5). In addition to applying the method with less data, we also give more weight to the data from some participants compared to others, showing how the method can be used when some people are more expert than others, or their preferences are more important.

More specifically, using the demographic information we collected in our survey, such as fleet vehicle purchase experience and the number of vehicles purchased; we divided the participants into four groups; *Least* experienced group, *Basic*, *Intermediate* and *Most Experienced* groups. We used the *Most Experienced* group of participants' responses as seeds for stochastic sampling to develop a larger "expert" sample via adding Gaussian noise. Then we used the 4 Point Elicitation method to calculate the utility functions and the weights of criteria. Table 6.7 shows the values calculated for three group of subsets i.e., *Basic*, *Intermediate* and *Most Experienced* group, while Table 6.8 shows the values calculated for *Intermediate* and *Most Experienced* group using the 4 Point Elicitation method.

Networks in Figures 6.12 and 6.14 show the respective expert-elicited networks using two types of subsets as mentioned above along with the networks with continuous nodes in Figures

Utility Node	Overall Weight
Budget Value	0.43
Safety Value	0.83
Satisfaction	0.27
Environmental Value	0.47

Table 6.7: Weights of Criteria using Basic, Intermediate and Most Experienced Participant's Data (Maximum score is highlighted by green color)

Utility Node	Overall Weight
Budget Value	0
Safety Value	0.92
Satisfaction	0
Environmental Value	0.31

Table 6.8: Weights of Criteria using Intermediate and Most Experienced Participant's Data (Maximum score is highlighted by green color)

6.13 and 6.15 respectively.

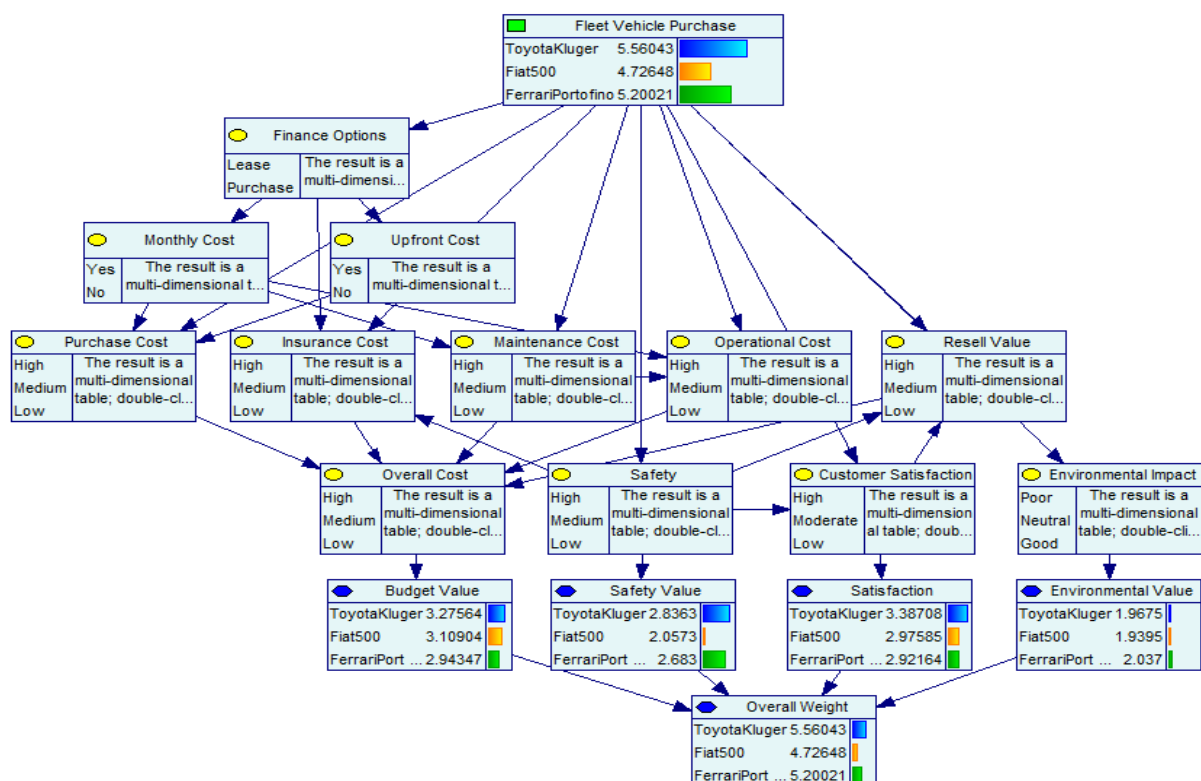


Figure 6.12: Parameterizing Expert-Elicited BDN Using Basic, Intermediate and Most Experienced Participants' Data via The 4 Point Elicitation Extension Method (Network with Discrete Nodes)

Table 6.9 shows the expected utilities obtained by using three groups i.e., *Basic*, *Intermediate* and *Most Experienced* participants in vehicle purchase procurement. This highest expected utility is recorded as 5.56 for *Toyota Kluger* option while the lowest expected utility is 4.73 for *Fiat 500*. In this case, the data from three groups suggest that the best choice is *Toyota Kluger*

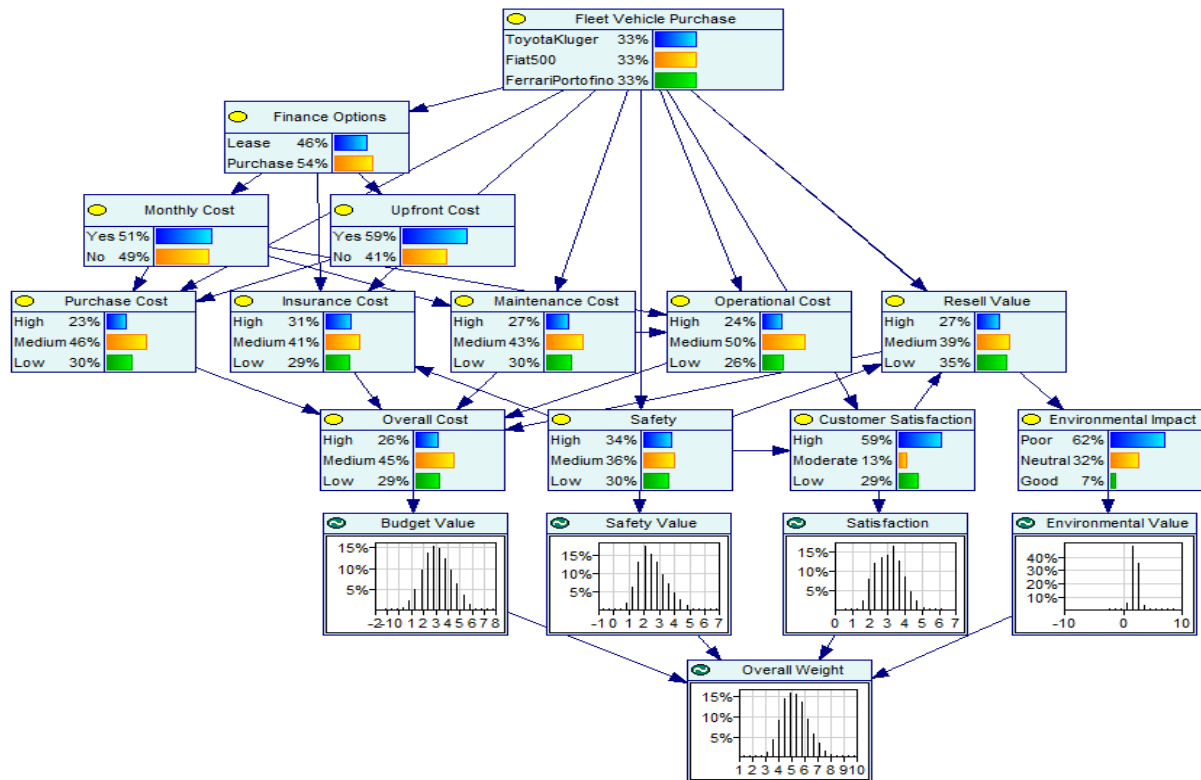


Figure 6.13: Parameterizing Expert-Elicited BDN Using Basic, Intermediate and Most Experienced Participants' Data via The 4 Point Elicitation Extension Method (Network with Continuous Nodes)

Options	Expected Utility
Toyota Kluger	5.56
Ferrari Portofino	5.2
Fiat 500	4.73

Table 6.9: Expected Utilities Using the Basic, Most and Intermediate Participants Data

with the highest EU among all three options.

However, Table 6.10 shows the results obtained from using the data from *Intermediate* and *Most Experienced* group. We can see that values are very different from the values calculated using the other subset of data. Here the *Ferrari Portofino* has the highest expected utility of 2.83 followed by *Toyota Kluger* with 2.7, however, *Fiat 500* has still the lowest expected utility of 2.14. This subset of data decides *Ferrari Portofino* to be the best choice with the highest EU and the *Fiat 500* to be the least preferred option with the lowest EU value. These results show that the results may vary when the different subsets are used based on the expertise level of the participants.

6.1.3.1 Sensitivity Analysis on the BDN created using Expert-Weighted Resampling Data

The networks in Figure 6.16 and 6.17 represent the sensitivity analysis of two networks based on different groups of expert elicited data. The red nodes indicate that they are very sensitive to

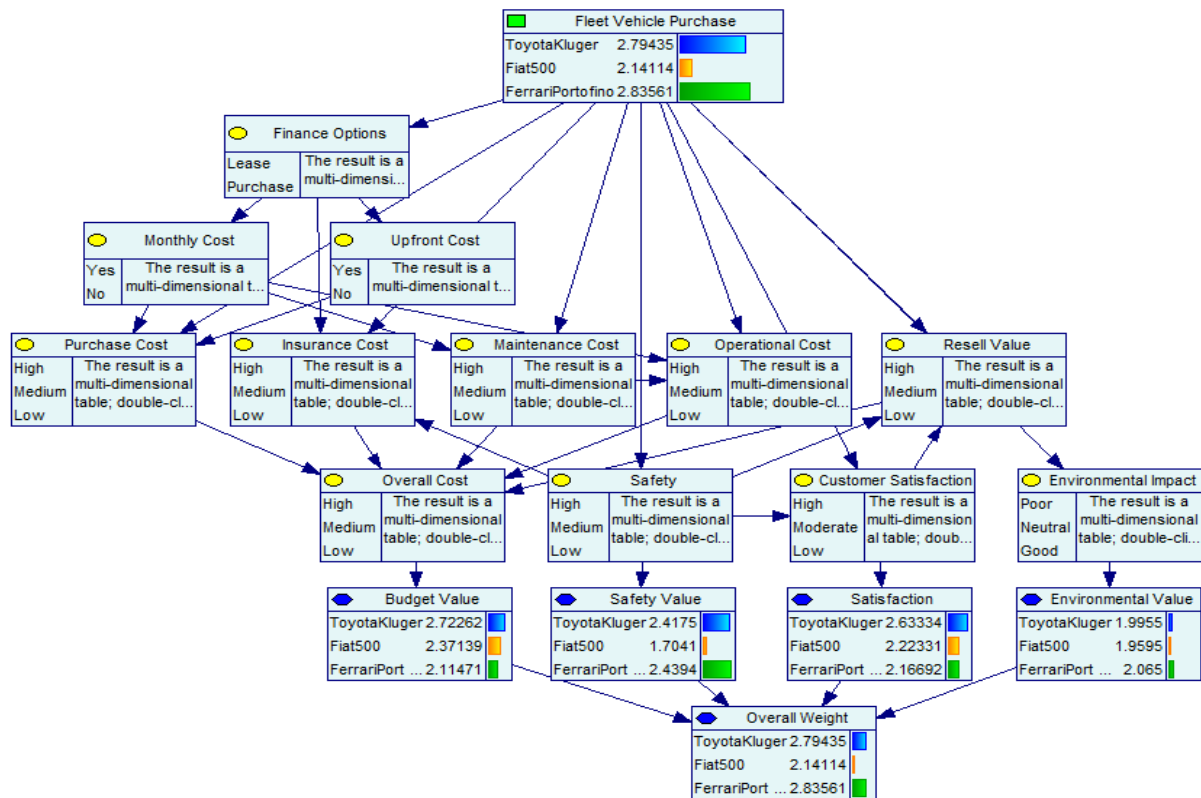


Figure 6.14: BDN using Intermediate & Most Experienced Group (Discrete Nodes)

Options	Expected Utility
Toyota Kluger	2.79
Ferrari Portofino	2.83
Fiat 500	2.14

Table 6.10: Expected Utilities Using the Most and Intermediate Participants Data

the decision, in this case *Overall Weight* of the criteria is the most sensitive node to the decision node, the second level sensitivity is shown by *Safety Customer Satisfaction* and *Environmental Impact* nodes in this network.

6.1.3.2 What-if Analysis Using Expert-Weighted Resampling Data

Using both networks, we ask what-if questions to see how this network react to those questions as mentioned in previous sections. For example, the networks in Figures 6.18 and 6.20 represent the ideal case scenario, where we set the states of of our interest such as *Overall Cost* is set to *Low*, *Safety* to *High* and so on. On the other hand, the networks in Figures 6.19 and 6.21 show the worst case, where we *Overall Cost* is set to *High*, *Safety* to *Low* and so on.

When we set the states in this way we can clearly see the highest expected utilities of 6.18 and 3.18 for two networks based on two different groups in Figures 6.18 and 6.20 respectively. While the worst cases in both the networks report expected utilities of 4.0 and 1.9.

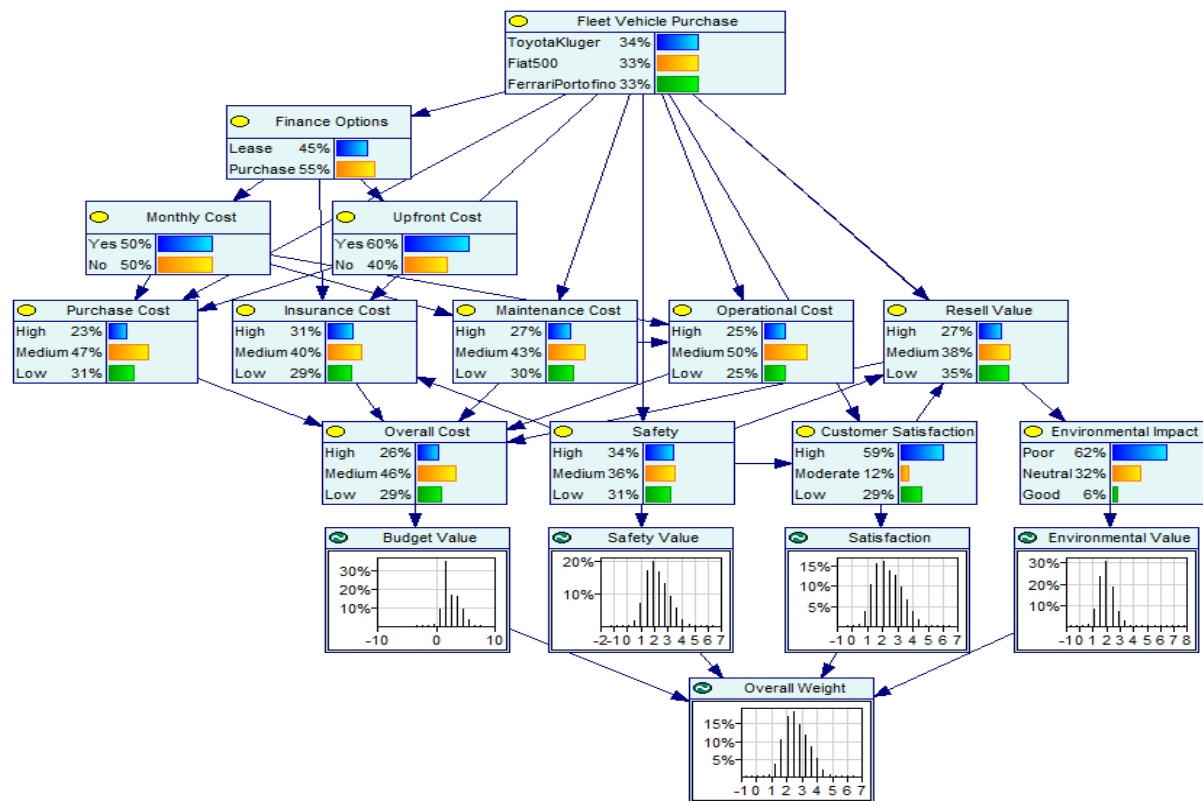


Figure 6.15: BDN using Intermediate & Most Experienced Group (Continuous Nodes)

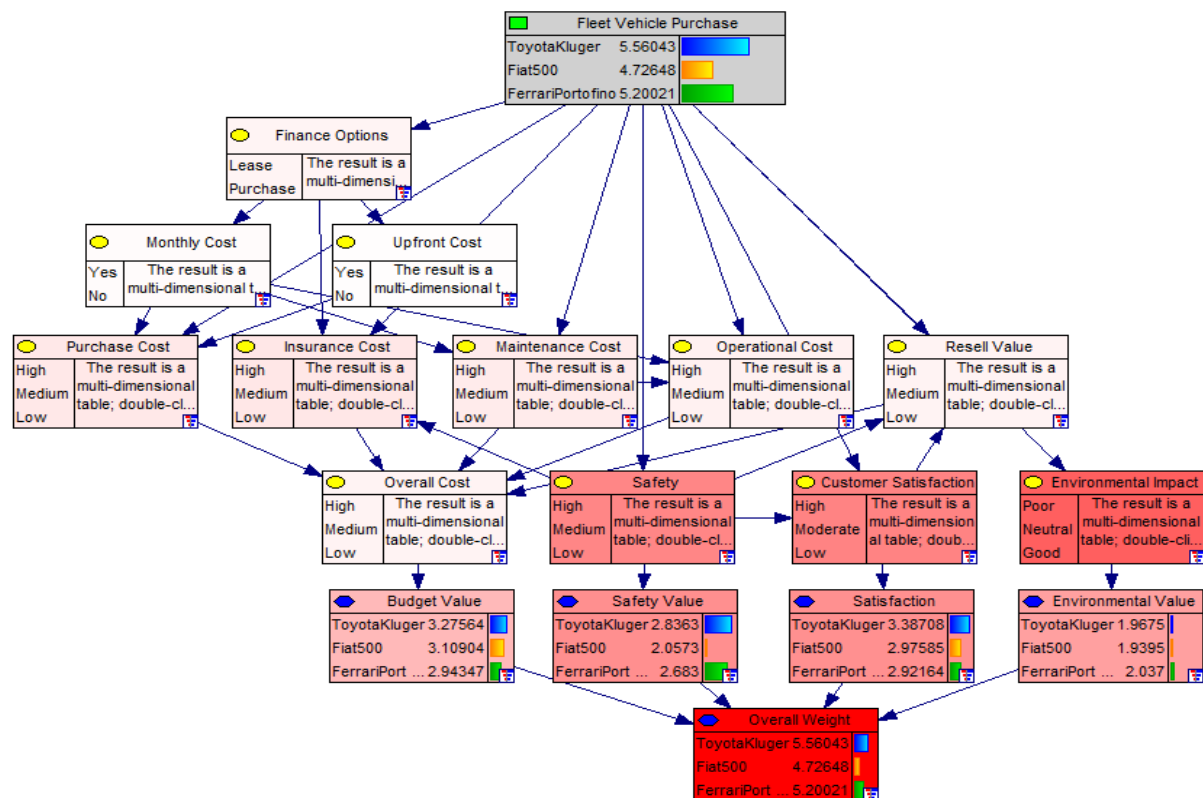


Figure 6.16: Sensitivity Analysis of BDN Based on Basic, Intermediate & Most Experienced Groups

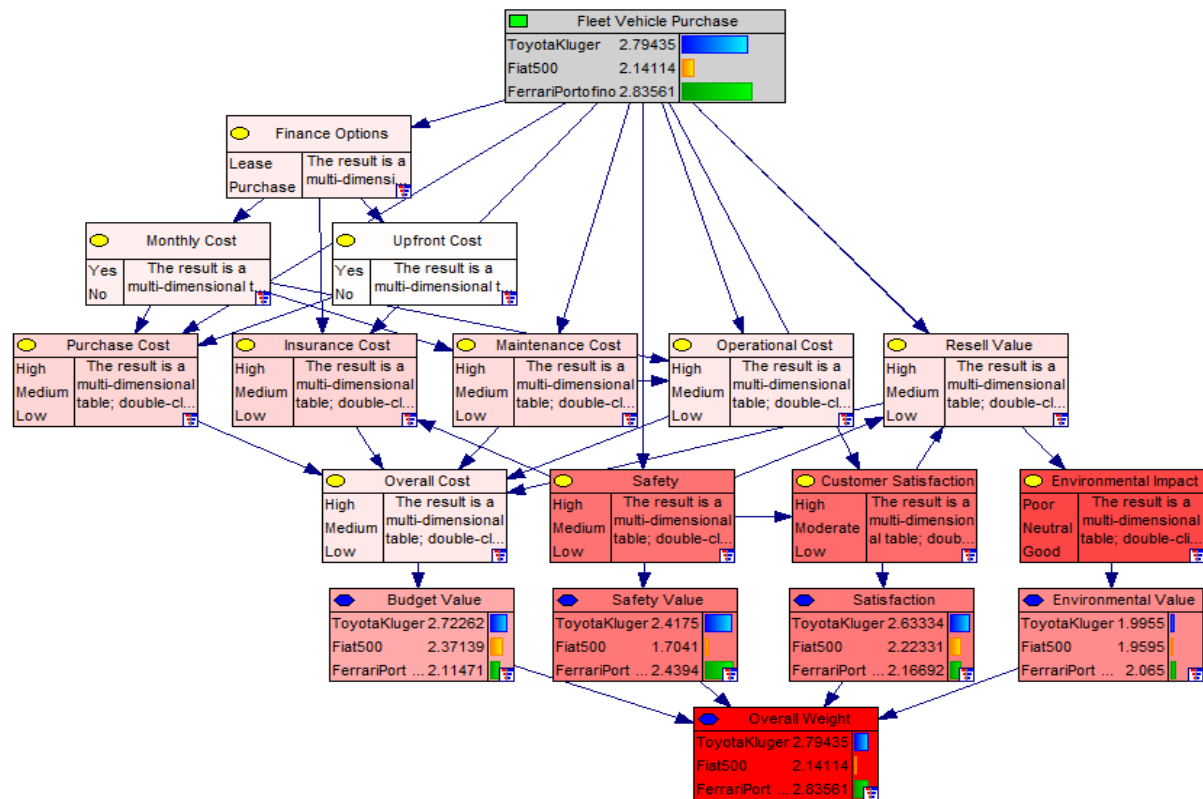


Figure 6.17: Sensitivity Analysis of BDN Based on Intermediate & Most Experienced Groups

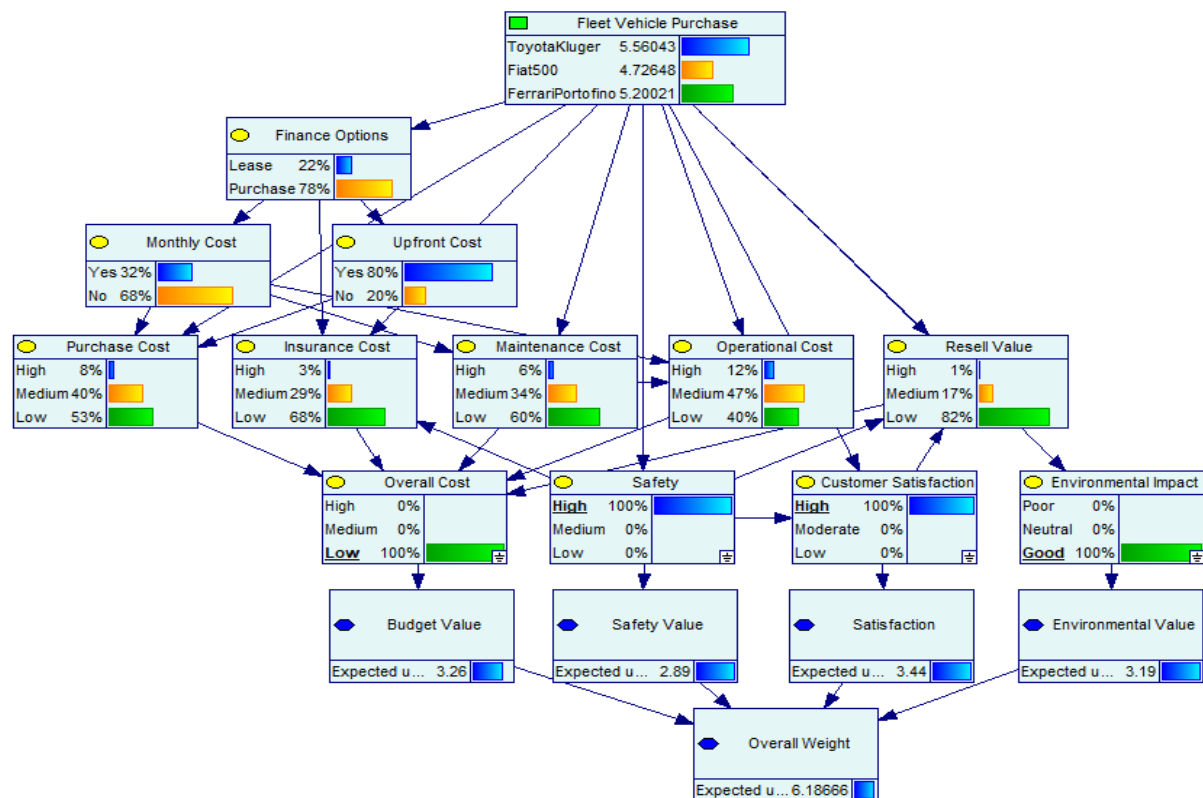


Figure 6.18: What-if Analysis Using Basic, Intermediate & Most Experienced Groups (Best Case)

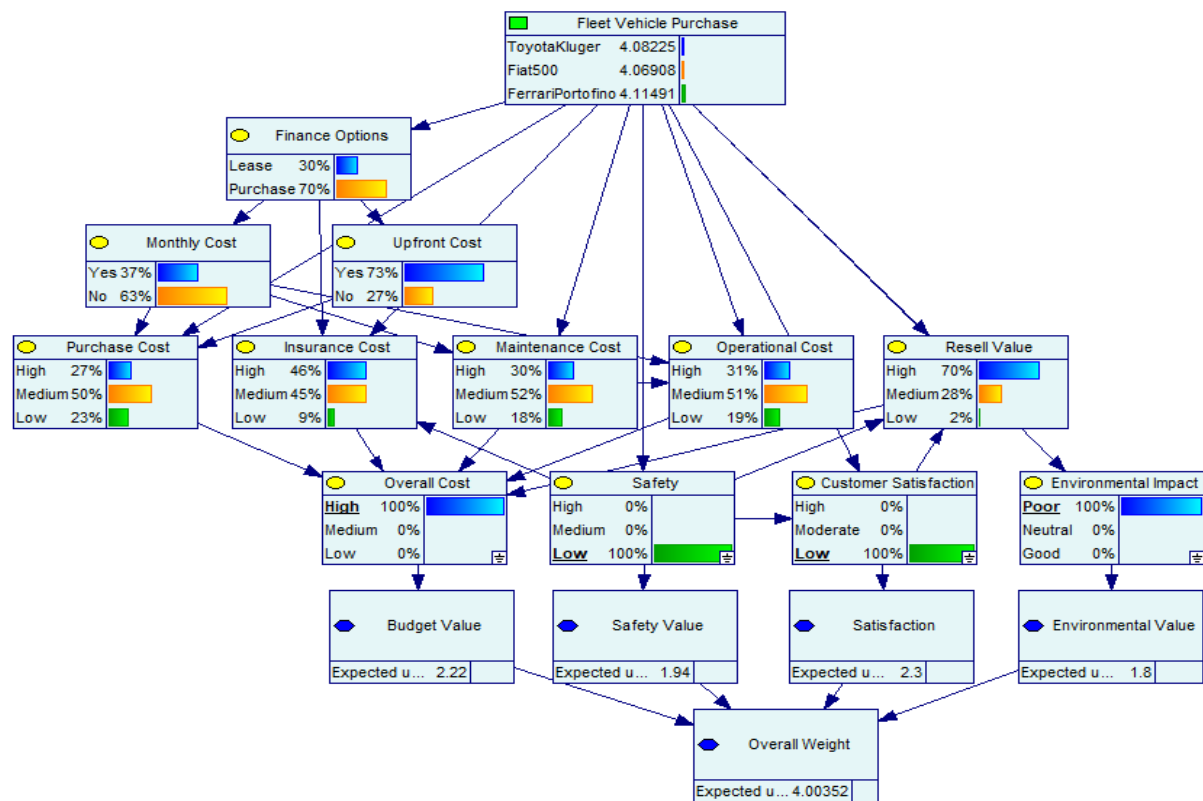


Figure 6.19: What-if Analysis Using Basic, Intermediate & Most Experienced Groups (Worst Case)

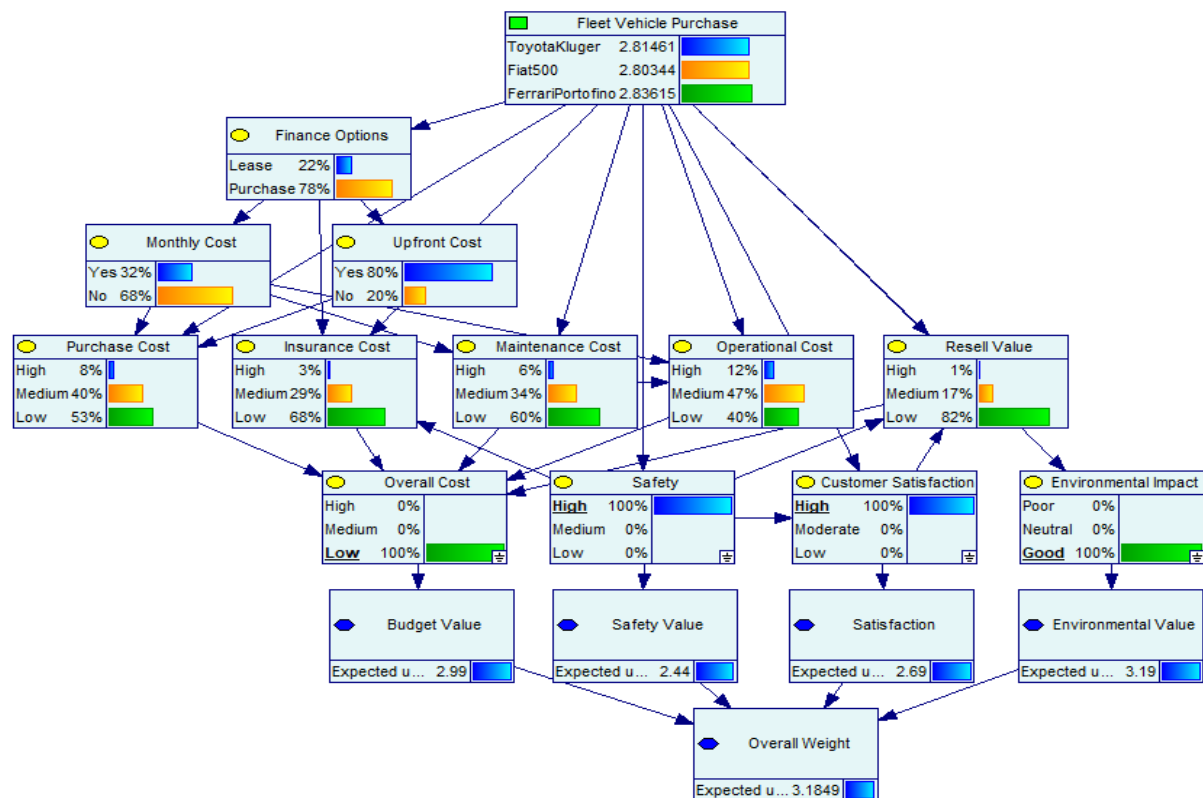


Figure 6.20: What-if Analysis Using Intermediate and Most Experienced Experts (Best Case)

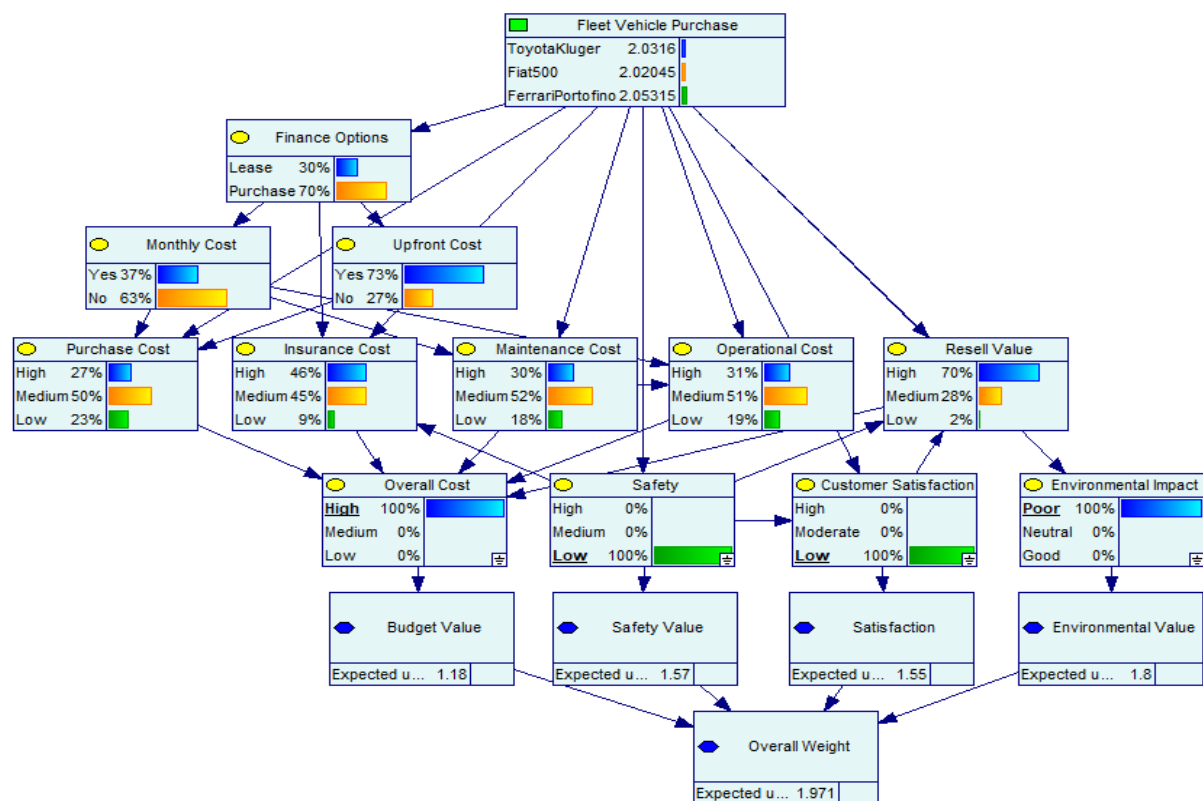


Figure 6.21: What-if Analysis Using Intermediate and Most Experienced Experts (Worst Case)

6.1.4 Comparative Experimental Results

In this section we study the comparative analysis of the experimental results for the Baseline and the 4 Point Elicitation methods using full and subsets of the data set.

In this section we look at the comparative experimental results of both the Baseline method and the 4 Point Elicitation method using full data set and the expert-weighted BDNs. The bar graphs in Figures 6.22a and 6.22b represent the utility values for all of the states of parent chance node of each criterion in the network using full dataset and the Figures 6.23a and 6.23b represent the values using subsets of the data. Both the methods give us almost similar comparative results, where the green bars indicate highest values for low overall cost, high safety ranking, high customer satisfaction and good environmental impact of the fleet vehicle to be purchased. On the other hand, the red bars record the lowest values that are recorded for high overall cost, low safety ranking, low customer satisfaction ranking and poor environmental impact in the decision network. Blue bars in these graphs show the medium ranked options for each criterion using both elicitation methods.

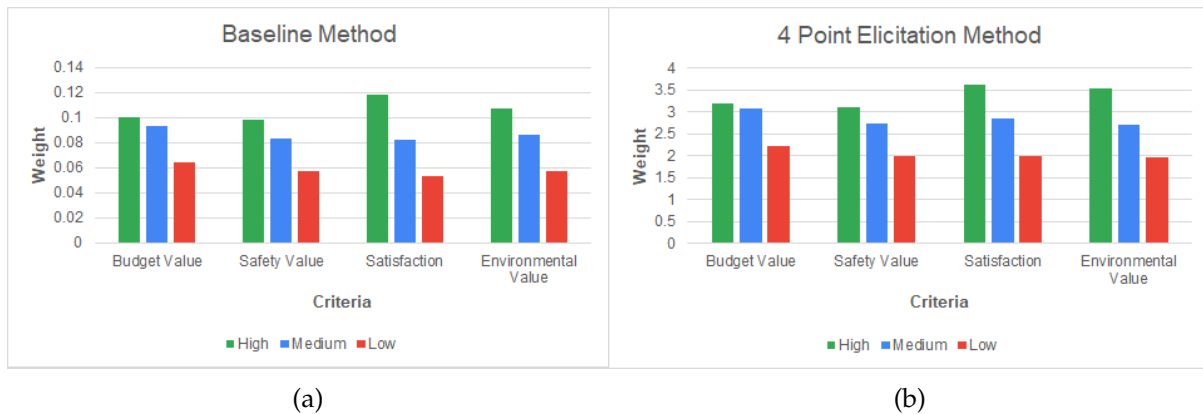


Figure 6.22: Comparison of Values for each Criterion Using Full Dataset

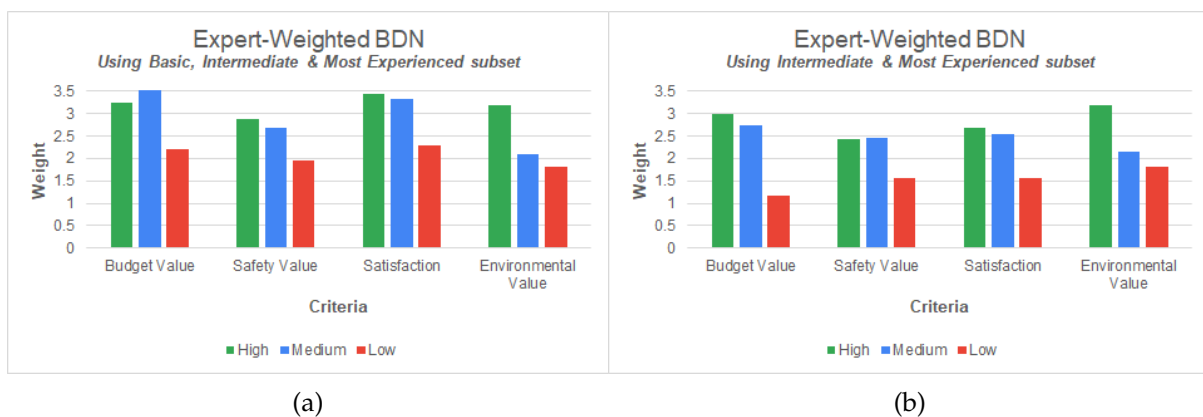


Figure 6.23: Comparison of Values for each Criterion Using Expert-Weighted Resampling Data

In Figure 6.22 we can see that both the Baseline and the 4 Point Elicitation methods produce

highest weights for safety criteria followed by environmental impact and customer satisfaction respectively. Overall cost of the vehicle which includes maintenance cost, purchase cost, operational cost, insurance cost and resell value; has the lowest weight among all criteria. However, the results of the expert-weighted BDNs show slightly different values shown in Figure 6.23b of 6.23, here we can see that the experts prefer the environment friendly vehicles followed by those with high budget value.

Figures 6.24 and 6.25 show the vehicle purchase choice made by each of the models based the highest expected utility.

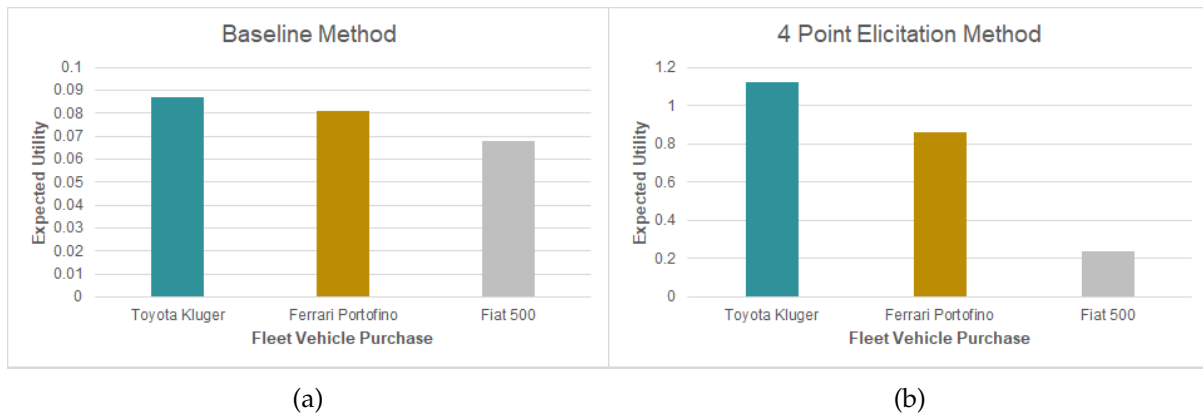


Figure 6.24: Comparison of Expected Utility Using Full Dataset

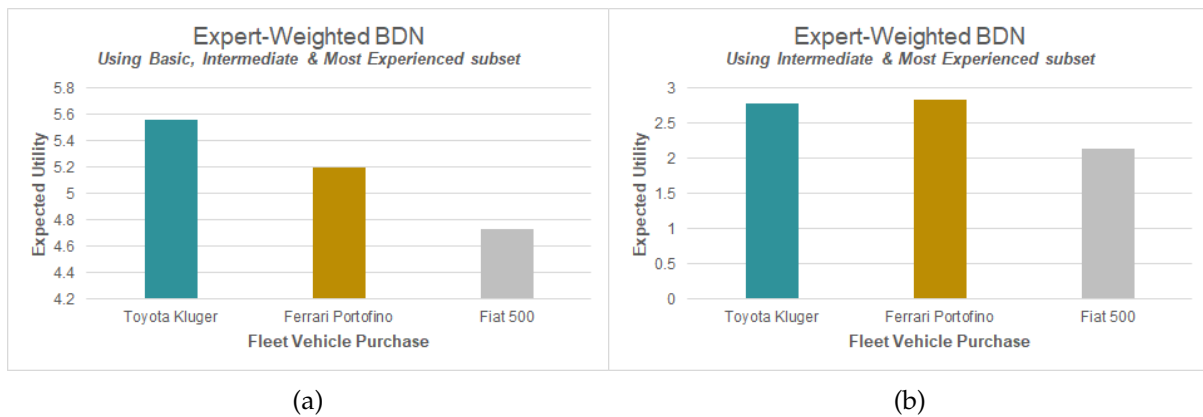


Figure 6.25: Comparison of Expected Utility using Two Subsets of Data

A comparative analysis of the expected utility for Baseline method, the 4 Point Elicitation method and the Expert-Weighted BDN is represented in Figure 6.26. In this figure, the results were obtained from using the subset of data elicited from *Intermediate* and *Most Experienced* participants. Here we can see that the decision choice is different from all other models. The results show that the expert-weighted BDN chooses *Ferrari Portofino* while the Baseline and the 4 Point Elicitation methods choose *Toyota*.

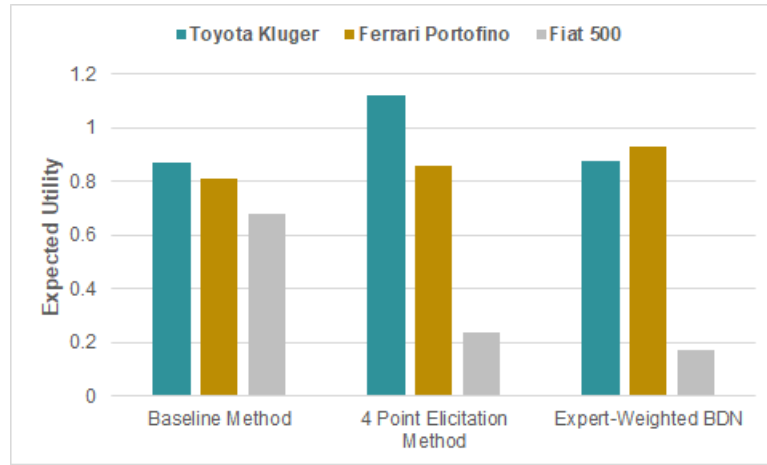


Figure 6.26: Comparing Weights of the Criteria Using The Baseline, The 4 Point and Expert-Weighted BDNs (Normalized)

6.2 Virtual Experiment Using Scenario-Based Preference Models

We discussed the use of scenario based preference models in Section 5.4 of Chapter 5 to generate artificial data which can exhibit dependencies among the criteria that make a real difference to the decision. Using the generated data from three different scenario based preference models, we use the Baseline method, the 4 Point Elicitation method and the AHP based method to produce BDNs, in addition to that, we use CaMML to apply on the preference data to learn the causal structures. This produces 3 scenarios by 4 methods, equal to 12 different models; these are given in Appendix E. Here we present a comparative evaluation of these models using the techniques described in Section 5.5 of Chapter 5.

6.2.1 Predictive Accuracy Results

We calculate predictive accuracy for each of the four models, i.e., CaMML, the Baseline model, the 4 Point Elicitation model, and the AHP based model using the data generated from our scenario-based preference models. Tables 6.11, 6.12 and 6.13 represent the predictive accuracy of our models along with the confidence intervals for Scenarios 5.1, 5.2 and 5.3 respectively.

Method	Predictive Accuracy	Confidence Interval
Baseline Method	0.49	[0.459, 0.521]
4 Point Elicitation Method	0.502	[0.47, 0.53]
AHP Based Model	0.84	[0.81, 0.86]
CaMML	0.913	[0.89, 0.93]

Table 6.11: Predictive Accuracy based on Scenario 5.1

Table 6.11 shows the predictive accuracy scores of each method based on Scenario 5.1 (with dependencies in the true model between Cost & Safety and Cost & Environmental Impact),

reporting the highest predictive accuracy for the CaMML's BDN of 0.913 that is higher than the other methods with a confidence interval of [0.89, 0.93]. On the other hand, the Baseline method is the worst performing method with the lowest predictive accuracy of 0.49 followed by the 4 Point Elicitation method having an accuracy of 0.502, although their confidence intervals overlap.

Method	Predictive Accuracy	Confidence Interval
Baseline Method	0.482	[0.45, 0.51]
4 Point Elicitation Method	0.492	[0.46, 0.52]
AHP Based Model	0.666	[0.63, 0.69]
CaMML	0.93	[0.91, 0.94]

Table 6.12: Predictive Accuracy based on Scenario 5.2

Table 6.12 results are based on the data generated from the preference model using Scenario 5.2 (with dependencies in true model between Cost & Environmental Impact). These results show that CaMML has the highest predictive accuracy of 0.93 while other models' predictive accuracies are much lower than CaMML's accuracy.

Method	Predictive Accuracy	Confidence Interval
Baseline Method	0.406	[0.37, 0.43]
4 Point Elicitation Method	0.406	[0.37, 0.43]
AHP Based Model	0.813	[0.79, 0.83]
CaMML	0.913	[0.89, 0.93]

Table 6.13: Predictive Accuracy based on Scenario 5.3

Table 6.13 results are based on the data from Scenario 5.3's preference model (with dependencies between Cost & Safety) which show that again CaMML beats all other methods in term of predictive accuracy. The highest accuracy is 0.913 as reported by CaMML, followed by 0.813 for the AHP based BDN while the Baseline and the 4 Point Elicitation based methods are the worst performing methods with a predictive accuracy of 0.406.

As can be seen in Section E.4 of Appendix E, the three different scenarios produced three different causal structures via CaMML comprising three different dependencies — one BDN capturing a dependency from *Budget Value* to *Safety Value* for Scenario 5.1 (missing the dependency between *Environmental Value* and *Budget Value*), second BDN capturing a dependency from *Environmental Value* to *Budget Value* for Scenario 5.2 and in last BDN *Safety Value* to *Budget Value* for Scenario 5.3. Looking at only the predictive accuracy results, we seem to have scenarios where these discovered dependencies help CaMML perform best. The AHP based method seems to be second best performing method while the other methods are not doing so well.

6.2.2 F Score Results

F Score is calculated for our methods to measure the accuracy on datasets that are generated using our scenarios as explained in Chapter 5. Following Tables 6.14, 6.15 and 6.16 are the experimental results of the F Score values of our methods based on Scenario 5.1, 5.2 and 5.3 respectively.

Method	F1 Score			Mean	Confidence Interval
	Car1	Car2	Car3		
Baseline Method	0.53	0.50	0.42	0.48	[0.37, 0.59]
4 Point Elicitation Method	0.54	0.51	0.43	0.49	[0.39, 0.60]
AHP Based Model	0.84	0.84	0.83	0.84	[0.83, 0.85]
CaMML	0.91	0.91	0.91	0.91	[0.91, 0.915]

Table 6.14: F1 Score based on Scenario 5.1

The results in Table 6.14 show the F Score measures of all the methods based on the distribution over the decision choices i.e., *Car1*, *Car2* and *Car3*. CaMML has the highest F score measure of 0.91, 0.91, 0.91 for each car type with a mean value of 0.91. AHP based model's F scores are not significantly different, but are slightly lower with a mean of 0.84. The Baseline method is the worst performing method with a mean F score measure of 0.48 followed by the 4 Point Elicitation method having 0.49 value for mean F score. Based on the F score results, CaMML seems to outperform other methods.

Method	F1 Score			Mean	Confidence Interval
	Car1	Car2	Car3		
Baseline Method	0.52	0.47	0.43	0.478	[0.43, 0.52]
4 Point Elicitation Method	0.52	0.48	0.46	0.489	[0.45, 0.52]
AHP Based Model	0.69	0.66	0.63	0.66	[0.63, 0.69]
CaMML	0.93	0.93	0.91	0.93	[0.91, 0.94]

Table 6.15: F1 Score based on Scenario 5.2

Looking at Table 6.15 F scores, we can see that CaMML beats other methods with an accuracy of 0.93, 0.93, 0.91 for each car type with a mean F score of 0.93 based on Scenario 5.2. The second best performer method is reported by the AHP based model with F score measures of 0.69, 0.66, 0.63 for each car choice and a mean score of 0.66. On the other hand, the Baseline method is the worst performing method with a mean score of 0.478 followed by 0.489 score for the 4 Point Elicitation based method.

The last Table 6.16 is based on the results obtained from Scenario 5.3 based preference model which demonstrates that again CaMML has the highest F scores of 0.92, 0.90, 0.90 for each car type and a mean value of 0.91 followed by the AHP based model having a mean F score of 0.809. However, the Baseline and the 4 Point Elicitation based methods are the worst performing methods based on the F score measure results having a mean value of 0.39 for both

Method	F1 Score			Mean	Confidence Interval
	Car1	Car2	Car3		
Baseline Method	0.47	0.40	0.30	0.39	[0.29, 0.49]
4 Point Elicitation Method	0.47	0.40	0.30	0.39	[0.29, 0.49]
AHP Based Model	0.83	0.80	0.78	0.809	[0.77, 0.84]
CaMML	0.92	0.90	0.90	0.91	[0.89, 0.92]

Table 6.16: F1 Score based on Scenario 5.3

of them.

Looking the F score measure, the overall results are somehow similar to the predictive accuracy results where CaMML is the best performing method using Scenario 5.1 and 5.2 and 5.3.

6.2.3 Kullback Leibler Divergence Results

We also calculate the KL Divergence for all elicitation methods to measure the dissimilarity of their distribution from true model's distribution. As predictive accuracy and F score measure are unable to measure calibration, we use KLD as it takes calibration into account. Calibration is very important when the aim is to support decision making [98]. The distribution is based on three decision options (*Car1*, *Car2* and *Car3*) as shown in Tables 6.17, 6.18 and 6.19.

Method	Kullback Leibler Divergence (in bits)
Baseline Method	KLD($P \parallel Q$): 0.159 bits KLD($Q \parallel P$): 0.161 bits
4 Point Elicitation Method	KLD($P \parallel Q$): 0.141 bits KLD($Q \parallel P$): 0.143 bits
AHP Method	KLD($P \parallel Q$): 0.166 bits KLD($Q \parallel P$): 0.164 bits
CaMML	KLD($P \parallel Q$): 0.577 bits KLD($Q \parallel P$): 0.582 bits

Table 6.17: Kullback Leibler Divergence based on Scenario 5.1

Table 6.17 shows the KLD results using Scenario 5.1 based preference model for all methods. The 4 Point Elicitation method's distribution seems to have minimum distance of 0.141 bits from the true model's distribution based on the target node, followed by the Baseline method having a distance of 0.159 bits. However, CaMML learned model's distribution has the maximum difference of 0.577 bits from the true model's distribution, making it the worst performing method followed by the AHP based method with a dissimilarity of 0.166 bits.

The KLD results in Table 6.18 illustrate that the Baseline method outperforms other methods with a minimum divergence of 0.039 bits based on Scenario 5.2. The second best performing method is the 4 Point Elicitation method with a difference of 0.104 bits from the true distribution. On the other hand, the worst performing method is again CaMML with a differ-

Method	Kullback Leibler Divergence (in bits)
Baseline Method	KLD($P \parallel Q$): 0.039 bits KLD($Q \parallel P$): 0.039 bits
4 Point Elicitation Method	KLD($P \parallel Q$): 0.104 bits KLD($Q \parallel P$): 0.104 bits
AHP Method	KLD($P \parallel Q$): 0.423 bits KLD($Q \parallel P$): 0.425 bits
CaMML	KLD($P \parallel Q$): 0.526 bits KLD($Q \parallel P$): 0.531 bits

Table 6.18: Kullback Leibler Divergence based on Scenario 5.2

ence of 0.526 bits from the true model's distribution followed by the AHP based method with a KLD score of 0.423

Method	Kullback Leibler Divergence (in bits)
Baseline Method	KLD($P \parallel Q$): 0.691 bits KLD($Q \parallel P$): 0.692 bits
4 Point Elicitation Method	KLD($P \parallel Q$): 0.691 bits KLD($Q \parallel P$): 0.692 bits
AHP Method	KLD($P \parallel Q$): 0.668 bits KLD($Q \parallel P$): 0.670 bits
CaMML	KLD($P \parallel Q$): 0.458 bits KLD($Q \parallel P$): 0.451 bits

Table 6.19: Kullback Leibler Divergence based on Scenario 5.3

The last and third Table 6.19 shows the results for all four methods based on Scenario 5.3. CaMML seems to have the minimum distance among all with a divergence of 0.458 bits from true model's distribution. The second best method in this table is the AHP based method with a distance of 0.668 bits while the Baseline and the 4 Point Elicitation methods are the worst methods this time with a same KLD score of 0.691 bits.

This metric shows three different methods having best performance based on the data generated from the preference models based on different scenarios. We can see that the 4 Point Elicitation method is the best performing method based on Scenario 5.1, the Baseline method is the best performing method based on Scenario 5.2 and CaMML is the best performing method based on Scenario 5.3. The KLD results show that the 4 Point Elicitation method and the Baseline method are the winner methods based on having a minimum distance from the true distribution using Scenario 5.1 and 5.2, however, these results are different for predictive accuracy and F score measure where the best performing method is CaMML for same scenarios.

6.2.4 LogScore Results

We calculate the log score measure for all the methods and see the expected reward across all predictions. Tables 6.20, 6.21 and 6.22 represent the mean log score values for each method based on Scenario 5.1, 5.2 and 5.3.

Method	Mean Log Score	Confidence Interval
Baseline Method	-1.55	[-1.56, -1.55]
4 Point Elicitation Method	-1.76	[-1.76, -1.75]
AHP Based Model	-0.452	[-0.48, -0.42]
CaMML	-0.638	[-0.66, -0.61]

Table 6.20: Log Score based on Scenario 5.1

Table 6.20 are the log score measures for all the methods based on Scenario 5.1 which shows that the AHP based method is the best performing method having a maximum expected reward of -0.452 as its mean log score. The second best method is CaMML with a log score of -0.638 while the worst performing method is the 4 Point Elicitation method with a lowest log score value of -1.76 followed by the Baseline method with that of -1.55.

Method	Mean Log Score	Confidence Interval
Baseline Method	-1.57	[-1.573, -1.572]
4 Point Elicitation Method	-5.04	[-5.046, -5.045]
AHP Based Model	-0.979	[-1.014, -0.944]
CaMML	-0.642	[-0.66, -0.619]

Table 6.21: Log Score based on Scenario 5.2

Scenario 5.2 based log score results in Table 6.21 show that CaMML is the best performing method with a highest log score of -0.642 followed by the AHP based method with a second highest log score measure of -0.979. On the other hand, the 4 Point Elicitation method is the worst method having a significantly low log score of -5.04 followed by the Baseline method having a score of -1.57.

Method	Mean Log Score	Confidence Interval
Baseline Method	-1.535	[-1.538, -1.532]
4 Point Elicitation Method	-3.664	[-3.668, -3.661]
AHP Based Model	-0.424	[-0.45, -0.39]
CaMML	-0.636	[-0.66, 0.61]

Table 6.22: Log Score based on Scenario 5.3

The results in Table 6.22 show that the AHP based method is again the best performing method having a log score of -0.424. CaMML is the second best method based on the log score results based on Scenario 5.3 having a mean score of -0.636 while the 4 Point Elicitation method is the worst method having a minimum expected reward of -3.664 followed by the

Baseline method having -1.535 score.

Based on the log score measures, we cannot decide which method is the winner based on their performance using three different scenarios. The results vary between the AHP based method and the CaMML method whereas the other two methods seem to be performing poorly.

6.2.5 BIR Results

Bayesian Information Reward (BIR) are calculated for our methods to measure the expected reward for each method based on the predictions. Tables 6.23, 6.24 and 6.25 outline the BIR values for all the methods based on Scenarios 5.1, 5.2 and 5.3.

Method	Mean BIR	Confidence Interval
Baseline Method	0.02	[0.018, 0.02]
4 Point Elicitation Method	-0.11	[-0.11, -0.10]
AHP Based Model	0.714	[0.69, 0.73]
CaMML	0.59	[0.58, 0.61]

Table 6.23: Bayesian Information Reward Results

The BIR scores in Table 6.23 show that the AHP based method is the best method based on making correct predictions with a maximum BIR score of 0.714 followed by CaMML having a score of 0.59. The 4 Point Elicitation method is the worst method with a lowest BIR score of -0.11 followed by a score of 0.02 made by the Baseline method using Scenario 5.1 based preference model's data.

Method	Mean BIR	Confidence Interval
Baseline Method	0.008	[0.0079, 0.008]
4 Point Elicitation Method	-2.18	[-2.181, -2.180]
AHP Based Model	0.38	[0.36, 0.40]
CaMML	0.595	[0.57, 0.61]

Table 6.24: Bayesian Information Reward based on Scenario 5.2

In Table 6.24, CaMML has the maximum BIR score of 0.595 which makes it the best performing method among all the methods based on Scenario 5.2. The second best performing method is the AHP based method with a score of 0.38 followed by the Baseline method having a score of 0.008 while the 4 Point method has the lowest BIR score of -2.18.

Table 6.25 reports the BIR scores for our methods based on Scenario 5.3 showing a maximum reward for the AHP based model with a BIR score of 0.732 followed by CaMML having a score of 0.60. On the other hand, the 4 Point Elicitation method is again the worst performing method among all with a lowest score of -1.31 followed by the Baseline method with a BIR score of 0.032.

Method	Mean BIR	Confidence Interval
Baseline Method	0.032	[0.030, 0.034]
4 Point Elicitation Method	-1.31	[-1.312, -1.308]
AHP Based Model	0.732	[0.71, 0.75]
CaMML	0.60	[0.57, 0.61]

Table 6.25: Bayesian Information Reward based on Scenario 5.3

BIR results show that the AHP based method is the best performing method twice based on the BIR score we got from Scenario 5.1 and 5.3, however, CaMML is the best performing method having a maximum BIR score among all based on Scenario 5.2 results.

6.2.6 Expected Value Results

We also evaluate our models based on the Expected Value (EV) using different models generated from the data based on the scenario models. The mean of EVs across multiple cases for each method using three different scenarios are shown in Tables 6.26, 6.27 and 6.28.

Method	Mean Expected Value	Confidence Interval
Baseline Method	1.962	[1.92, 2]
4 Point Elicitation Method	1.963	[1.92, 2]
AHP Method	1.946	[1.90, 1.98]
CaMML	1.941	[1.90, 1.97]

Table 6.26: Mean Expected Value based on Scenario 5.1

The results in Table 6.26 show that the 4 Point Elicitation is the best performing method having the highest mean EV of 1.963 based on Scenario 5.1. The Baseline method is the second best performing method with a value of 1.962 which is slightly less than the 4 Point Elicitation method. On the other hand, CaMML is having the lowest value of 1.941 followed by the AHP based method having a mean EV of 1.946.

Method	Mean Expected Value	Confidence Interval
Baseline Method	1.956	[1.92, 1.99]
4 Point Elicitation Method	1.951	[1.91, 1.98]
AHP Method	1.93	[1.89, 1.96]
CaMML	1.951	[1.91, 1.98]

Table 6.27: Mean Expected Value based on Scenario 5.2

Table 6.27 are the mean EVs of all the methods based on Scenario 5.2 showing the highest mean EV for the Baseline method with a score of 1.956 followed by the 4 Point Elicitation method having the second highest value of 1.951. CaMML method is having the third highest value of 1.951 followed by the AHP based method reporting the lowest mean EV of 1.93.

Lastly, the results in Table 6.28 illustrate that CaMML is the best performing method in

Method	Mean Expected Value	Confidence Interval
Baseline Method	1.955	[1.91, 1.99]
4 Point Elicitation Method	1.955	[1.91, 1.99]
AHP Method	1.95	[1.91, 1.99]
CaMML	1.97	[1.93, 2.0]

Table 6.28: Mean Expected Value based on Scenario 5.3

terms of EV results based on Scenario 5.3 with a highest mean EV of 1.97 among all the methods. The second highest values are reported for the Baseline and the 4 Point Elicitation methods which is 1.955 for both of them while the AHP the worst performing method with a lowest mean EV of 1.95.

The results obtained from EV scores are similar to those of KLD, showing that the 4 Point Elicitation method, the Baseline method and CaMML are the best performing methods having the maximum values based on Scenario 5.1, 5.2 and 5.3 respectively. However, the EVs for all the methods are very similar, and the confidence intervals are so wide, that there are no real statistically significant differences here.

6.2.7 Comparative Analysis of Evaluation Metrics

Here we make a comparative discussion of the results for all methods (including the Baseline method, the 4 Point Elicitation method, the AHP and CaMML) using predictive accuracy, F Score, KL divergence, Log Score, Bayesian information reward and expected value. The results are outlined in Tables 6.29, 6.30 and 6.31 using three different scenarios explained in previous chapters.

Scenario 1	Evaluation Metrics					
Method	PA	F Score	KLD	Log Score	BIR	EV
Baseline	0.49 [0.459, 0.521]	0.48 [0.37, 0.59]	0.159	-1.55 [-1.56, -1.55]	0.02 [0.018, 0.02]	1.962 [0.92, 2]
4 Point	0.50 [0.47, 0.53]	0.49 [0.39, 0.60]	0.141	-1.76 [-1.76, -1.75]	-0.11 [-0.11, -0.10]	1.963 [1.92, 2]
AHP	0.84 [0.81, 0.86]	0.84 [0.83, 0.85]	0.166	-0.45 [-0.48, -0.42]	0.71 [0.69, 0.73]	1.946 [1.90, 1.98]
CaMML	0.913 [0.89, 0.93]	0.91 [0.91, 0.915]	0.577	-0.64 [-0.66, -0.61]	0.59 [0.58, 0.61]	1.941 [1.92, 1.97]

Table 6.29: Resultant Scores of Each Method Based on Scenario 5.1 (Green cells show the best scores)

Scenario 2	Evaluation Metrics					
Method	PA	F Score	KLD	Log Score	BIR	EV
Baseline	0.48 [0.45, 0.51]	0.47 [0.43, 0.52]	0.039	-1.57 [-1.573, -1.572]	0.008 [0.0079, 0.008]	1.956 [1.92, 1.99]
4 Point Elicitation	0.49 [0.46, 0.52]	0.48 [0.45, 0.52]	0.104	-5.04 [-5.046, -5.045]	-2.18 [-2.181, -2.180]	1.951 [1.91, 1.98]
AHP	0.66 [0.63, 0.69]	0.66 [0.63, 0.69]	0.423	-0.979 [-1.014, -0.944]	0.38 [0.36, 0.40]	1.93 [1.89, 1.96]
CaMML	0.93 [0.91, 0.94]	0.93 [0.91, 0.94]	0.526	-0.642 [-0.66, -0.619]	0.595 [0.57, 0.61]	1.951 [1.91, 1.98]

Table 6.30: Resultant Scores of Each Method Based on Scenario 5.2

It is worth noting that the different measures applied do not speak with a uniform voice. We would argue that KLD and Expected Value are, in principle, the best measures to apply, when they are available. KLD is an ideal measure for the relation of a learned or expert-

Scenario 3	Evaluation Metrics					
Method	PA	F Score	KLD	Log Score	BIR	EV
Baseline	0.406 [0.37, 0.43]	0.39 [0.29, 0.49]	0.691	-1.53 [-1.538, -1.532]	0.032 [0.030, 0.034]	1.955 [1.91, 1.99]
4 Point Elicitation	0.406 [0.37, 0.43]	0.39 [0.29, 0.49]	0.691	-3.66 [-3.67, -3.66]	-1.31 [-1.312, -1.308]	1.955 [1.91, 1.99]
AHP	0.81 [0.79, 0.83]	0.80 [0.77, 0.84]	0.668	-0.424 [-0.45, -0.39]	0.732 [0.71, 0.75]	1.954 [1.91, 1.99]
CaMML	0.913 [0.89, 0.93]	0.91 [0.89, 0.92]	0.458	-0.636 [-0.66, -0.61]	0.60 [0.57, 0.61]	1.97 [1.93, 2.0]

Table 6.31: Resultant Scores of Each Method Based on Scenario 5.3

built probabilistic model relative to a known truth. EV is the best assessment of the quality of decision making given a known utility function across available cases. It is not surprising that these two measures agree in all three scenarios. Aside from that, BIR (or Log Score, which, due to uniform priors, is equivalent to BIR in ranking in these cases) should be preferred when a ground truth for a set of cases is known. BIR in the long run will approximate KLD's rankings. PA and F scores are in a sense approximations to the other measures, which are, however, insensitive to calibration.

The most prominent feature of these results is that there is no uniform “winner”. Across scenarios, different modelling techniques perform better, as is the case for most problems in Machine Learning. A future investigation that explores varieties of problems where different methods may dominate performance would likely be worthwhile. In any case, our thesis here, that MCDA modelling that accommodates interdependent utilities is valuable is at least weakly vindicated in that CaMML performs the best of the trialled approaches in at least some of the cases, as assessed by the metrics responsive to a combination of accuracy and calibration (i.e., KLD, Log Score, BIR and EV).

Table 6.32 shows the number of times each method was the best performing method using results obtained from six evaluation metrics. These results are calculated using the data generated from three different scenario based preference models 5.1, 5.2 and 5.3. In this table, we can see that CaMML outperforms or ties the other methods in each scenario, and outperforms the other methods overall.

Method	Winner Based on Performance			
	Scenario 5.1	Scenario 5.2	Scenario 5.3	Total
Baseline Method	0	2	0	2
4 Point Method	2	0	0	2
AHP	2	0	2	4
CaMML	2	4	4	10

Table 6.32: Number of Times Each Method Was Best Performer Based on Scenario 5.1, 5.2 & 5.3

6.3 Conclusion

In this chapter, we discussed detailed experimental results for all elicitation methods presented in Chapter 4. Firstly, we used the full dataset elicited via online survey to calculate the utility values using the Baseline method and the 4 Point Elicitation method, in addition to that, we also generated datasets based on the larger resampling of the collected survey data and used that to calculate the utility functions via the 4 Point Elicitation method to perform a comparative analysis of the results. Then we looked at the sensitivity of the networks to changes in the parameters, and performed what-if analysis to see the correlation between the criteria and the decision node. The experimental results based on the survey data showed that both the Baseline and the 4 Point methods made almost same purchase decisions, however, the expert-weighted BDN results were quite interesting because the experts weighed the criteria differently based on their experience which resulted in a different decision option.

Secondly, we also explored the virtual experimental results using the data generated from three different preference scenarios. We used the Baseline method, the 4 Point Elicitation method, the AHP based method and CaMML to use that scenario based preference data and produce BDNs (BDNs are shown in Appendix E. Lastly, we evaluated all the methods using six different evaluation metrics which showed different answers on the superiority of different methods but the overall results showed that CaMML was the best performing method in the large majority of cases.

Conclusion

7.1 Research Contribution

Multi-Criteria Decision Making (MCDM) is one of the most important challenges when confronting complex decisions, whatever the domain. These MCDM problems often involve a large number of conflicting criteria, multiple and diverse Decision Makers (DMs), decision making uncertainty and interdependencies among the utilities. Traditional MCDM approaches are limited to handling a small number of criteria, assume those criteria are independent, and do not consider dependencies among the utilities from different experts. In addition to that, the existing MCDA techniques also do not handle uncertainty very well. Bayesian Networks (BNs) are one of the most suitable representations for making decisions under uncertainty. They allow us to learn and formally represent the dependencies among criteria using BN models and then incorporate the results in larger scale Bayesian Decision Networks (BDNs).

In Chapter 3, we provided an end-to-end process for developing BDNs suitable for decision support on top of existing simulation studies, explaining this method using an example based upon a Land Combat Vehicle Procurement (LCV) problem for the DST Group. We used two commonly used and widely available causal discovery algorithms, PC and K2, to learn the BN models from the simulation data. We performed sensitivity analyses to examine the associations between nodes and the strength of influence from one node to another. BNs also support arbitrary hypothetical and counterfactual analyses, so we asked different kinds of what-if questions to see how the networks would react in different situations. We extended the BNs to utilities by introducing utility nodes associated with each node of interest. Those utilities were initially elicited from a single expert to show how this information can be incorporated into our BDN in this approach. Ultimately, this same process can be applied to a variety of simulations, dealing effectively with the large amounts of data that simulations can produce, and hence we believe this process has broad value and applicability.

Our main goal in this thesis was to develop new methods for creating multi-criteria decision models with the help of multiple experts. We began by proposing a new method called

the Baseline method in Chapter 4 to address this issue. This method compares each criterion in the network to every other criterion using pairwise comparisons. It allows preference elicitation from multiple experts to weigh a pair of criteria followed by their confidence in the answer they have given. Once the preferences from the participants are elicited, this method combines all the responses by multiplying each criterion score and the confidence of the expert, and then summing up all the resultant values into a final score (see algorithm 4.1). The Baseline method may work in simple cases, but has a number of limitations. In particular, its requirement for an exhaustive set of pairwise questions over the criteria and the potential for uncontrolled participant biases, such as overconfidence, anchoring and mitigating confidence in group thinking make the Baseline approach dubious.

To address overconfidence and other bias issues, we turned to the 4 Point Elicitation method of the IDEA protocol (Investigate, Discuss, Estimate, Aggregate judgments), which is well validated and one of the most comprehensive structured approaches used to deal with biases during elicitation. To the best of our knowledge, this method has never been applied to eliciting personal utilities which are typically represented as real-valued numbers in decision analysis. Here we asked participants to rank a pair of criteria based on their relative importance and provide their highest plausible value, lowest plausible value, best estimate and their confidence that their created intervals hold their real preference values. Once the responses were gathered, we standardized the intervals (using algorithm 4.2) and calculated the overall weights of the criteria (using the Algorithm 4.3). However, this method still requires an exhaustive set of questions to be asked of the participants.

As we had an exhaustive set of survey questions, we split up the questions into smaller sets representing degrees of “expertise”. To distinguish groups we used their experience, such as the number of vehicles purchased, and demographic information (e.g., age) and collected their responses to the subsets of questions. The elicited responses were then used to calculate the utility functions using the 4 Point Elicitation method to populate the utility nodes in our BDN. This method was followed to generate more data that was used for evaluation purposes (see Section 5.3.6).

We also used the AHP method to calculate utility values and then translated those values into a BDN for evaluation purposes. More specifically, we have used the AHP method to aggregate the elicited responses from the participants using the Eigenvalues, and then translated those values into a BDN which produces the same decisions as the AHP. This gives us the so-called AHP based BDN. Our approach is novel in that while previously the AHP has been used in combination with BNs, it has not been used with BDNs with utilities and decision nodes. Our method can be considered a compact representation of the AHP method translated into a

BDN, allowing us to compare directly against our alternative methods yielding BDN models.

Lastly, we demonstrated an approach using a causal discovery algorithm, CaMML (Causal discovery via MML), on preference data to learn dependencies among the utilities. CaMML learns a BN structure from data using a search and score approach, with the score combining the model's complexity and fit to data using the MML principle. Causal discovery has not previously been applied to preference data, but here we show its utility for this purpose.

We used artificial data to learn causal structures, treating the utilities as chance nodes, allowing CaMML to learn dependencies among utilities. We also used the artificial data generated from the scenario-based preference models to calculate the utility functions in our hand crafted BDNs using the Baseline and the 4 Point Elicitation method. Moreover, we used the AHP method to calculate the utilities and translate the results obtained from the AHP into a BDN as discussed in Section 4.4. All of these allowed a comparison of all the methods for the scenarios we created. The 4 Point method elicited the responses in terms of interval and a best estimate for each pair of different criteria, while the Baseline method used a subset of the elicited responses (the best estimate only). Both the methods used different techniques to calculate the utilities resulting in different range of values, however the relative order of the values impacted the final decision not the absolute values. We evaluated the performance of our models using six different metrics which gave different answers on the superiority of different methods. Using the data generated from different scenarios, we saw that CaMML outperformed the other elicitation methods in the majority of cases.

7.2 Future Work

There are a number of directions for extending the research presented in this thesis. We propose the following areas as the most promising.

7.2.1 Implementing a Full IDEA Method

One important future extension would be to extend the 4 Point Elicitation method into a full IDEA protocol. The IDEA protocol provides some intrinsic protection and correction for overconfidence in that in the *Discuss* step, experts inform and correct each other as a part of the process. In a full implementation of our ideas, we would have included this step in our elicitation process. Unfortunately, we were unable to use a face-to-face elicitation process (in part, due to COVID-19 restrictions). Our elicitation was done entirely by online web surveys completed by participants in a single sitting. Whereas online surveys can be (and some have been, e.g., [53]) designed so as to incorporate discussion phases during elicitation, we did not have

the time to implement such an approach. Instead, in this thesis we relied on a truncated version of IDEA that included a single elicitation step from each expert and also included the interval/bracketing elicitation. With the full IDEA method, experts can inform and correct each other, and the estimated intervals would help to mitigate overconfidence and other biases. The full IDEA allows comparison between models created before and after discussion, presenting back the intervals to participants so that they can change them and other improvements based on IDEA (which could be done now) is incorporating uncertainty into the BDN models using the average upper and lower intervals.

In effect, participants are obliged to estimate their confidence twice, since confidence can be measured by the spread of their intervals and again directly by the fourth question of each estimate. These two confidence estimates can be compared with each other to provide a sanity check on any individual's estimates. We would take the survey method and develop it into an interactive workshop (either in person or online), finishing in aggregated utility estimates.

7.2.2 Comparing Criterial Dimensions

Another area for future research is the assessment of the relative importance of criteria, which is of course crucial in preference elicitation. In our method, we asked participants about pairwise relative importance using the phrase “how many times” more important. We allowed them to answer within a range of real numbers $[1, 10]$, constraining the answer to a maximum in order to facilitate the aggregation of responses. However, there is scope for further research to determine what the granularity of “how many times” answers should be – can people really differentiate down to one or more decimal places? – and what the value of the maximum should be. For the case study presented in Chapters 5 and 6, we used $Max=10$, but the validity of this is of course problem dependent, as well as dependent upon the participants, their culture, etc. We did not investigate how to set such a maximum, nor how changing the maximum might influence the utilities and hence the decisions of the elicited BDN; we leave that as an interesting topic for future research.

7.2.3 Design of Question Sets

A third potential direction of future research involves redesigning the question set to lift the current restriction to questions that cover pairwise dependencies among the criteria. Redesigning the question sets so that they cover multiple criteria (not just pairwise) would generate a richer representation of preferences, and would support methods like CaMML to learn the causal structures that exhibit the dependencies among multiple criteria. By using such question sets, we would no longer require artificial scenarios to create preference models that are

required for generating artificial data.

7.2.4 Developing a Variant to Work on Small Samples

The final potential area for future research would be extending the online survey into an elicitation workshop format, either online or in person. In such workshops, as for a survey method, it is not feasible to ask each participant/expert a large number of questions. In the survey method, we were able to partition the questions ahead of time to ensure full coverage of the question set and keep running the survey until we had sufficient responses to all questions, however in a workshop environment, it might not be possible to obtain exhaustive coverage of the questions. Thus, it will be important to extend the method to handle a non-exhaustive coverage of the pairwise relative importance questions.

Where we have simulations available, these can be used to parameterize the CPTs. Using that parameterized BN, we could then generate scenarios that experts then rate. So, we can reduce the elicitation burden by focusing the elicitation on questions around the scenarios that the simulations say occur frequently.

But if we do not have simulations then we could use an approach similar to those of the CPT interpolator, where extremes are elicited, as well as a range of values (within the CPT in that case), and other values are interpolated from these. This would involve developing a variant of the small sample method, that works with the workshop elicitation and uses the subsets of the redesigned question sets to cover all the criteria.

Glossary

TEXTBOX 1 (GLOSSARY TABLE (ALPHABETIC))

AHP	<u>A</u> nal ^y tical <u>H</u> ierarch ^y <u>P</u> rocess
ANP	<u>A</u> nal ^y tical <u>N</u> etwork <u>P</u> rocess
BDeu	<u>B</u> ayesian <u>D</u> irichlet <u>e</u> quivalent <u>u</u> niform
BN	<u>B</u> ayesian <u>N</u> etwork
BIR	<u>B</u> ayesian <u>I</u> nformation <u>R</u> eward
BDN	<u>B</u> ayesian <u>D</u> ecision <u>N</u> etwork
CaMML	<u>C</u> ausal discovery via <u>M</u> inimum <u>M</u> essage <u>L</u> ength
CPT	<u>C</u> onditional <u>P</u> robability <u>T</u> able
CT	<u>C</u> ombat <u>T</u> eam
DAG	<u>D</u> irected <u>A</u> cyclic <u>G</u> raph
DM	<u>D</u> ecision <u>M</u> aker
DST	<u>D</u> efence group of <u>S</u> cience and <u>T</u> echnology
e.g.	<u>e</u> xempli <u>g</u> ratia (for example)
ELECTRE	<u>E</u> Limination <u>E</u> t <u>C</u> hoix <u>T</u> raduisant la <u>R</u> Ealité
EU	<u>E</u> xpected <u>U</u> tility
FN	<u>F</u> alse <u>N</u> egative
FP	<u>F</u> alse <u>P</u> ositive
IDEA	<u>I</u> nvestigate <u>D</u> iscuss <u>E</u> stimate <u>A</u> ggregate
i.e.	<u>i</u> d <u>e</u> st (in other words)
IR	<u>I</u> nformation <u>R</u> eward
KLD	<u>K</u> ullback <u>L</u> eibler <u>D</u> ivergence
LCV	<u>L</u> and <u>C</u> ombat <u>V</u> ehicle
MCDA	<u>M</u> uti- <u>C</u> riteria <u>D</u> ecision <u>A</u> nal ^y sis
MAUT	<u>M</u> ulti- <u>A</u> tttribute <u>U</u> tility <u>T</u> heory
PC	<u>P</u> rototypical <u>C</u> onstraint-based algorithm
PGM	<u>P</u> robabilistic <u>G</u> raphical <u>M</u> odel
SMAA	<u>S</u> tochastic <u>M</u> ulti- <u>A</u> ccessibility <u>A</u> nal ^y sis

SMEs	<u>S</u> ubject <u>M</u> atter <u>E</u> xperts
TP	<u>T</u> ru <u>e</u> <u>P</u> ositiv <u>e</u>
w.r.t.	<u>w</u> ith <u>r</u> espect <u>t</u> o

Symbols and Notation

TEXTBOX 2 (NOTATION TABLE)

B	Best estimate	57
C	Criteria	51
CI	Consistency index	22
CR	Consistency ratio	22
D_{KL}	Kullback leibler divergence	33
E	Experts group for elicitation	53
EU	Expected utility	12
EV	Expected Value	35
ϵ	Belongs to	17
$f(\gamma)$	Density function	19
G	DAG representing an ordinary BN	11
I_i^+	Positive reward for true prediction	34
I_i^-	Negative reward for incorrect prediction	34
IR_B	Bayesian information reward	34
$\int$	Integral	17
L_S	Lower standard interval	57
NW	Normalized weight	54
O	Outcome	12
P	Probability	10
p_i	Prior probability	34
$\hat{p}_i$	Posterior probability	34
R^n	Rational numbers	17
S	Score	53
Σ	Summation	12
U	Utility	16
U_S	Upper standard interval	57
W	Weight	17
xPy	x is strictly preferred to y	26

xIy	x is indifferent to y	26
xRy	x is incomparable to y	26
z	Z score or standard score	86

Survey Details

An online survey was conducted to elicit the preferences from the participants using Google Forms. Participants from different countries and backgrounds, having different experiences, were welcomed to participate in the survey using various social media forums, personal emails, and elicitation platforms. The questions of the survey were divided into several Google Forms — each form containing five base questions followed by nine pairwise comparison questions (total 14 questions to answer). At the end of each form, the participants were provided a link to the next form having a different set of questions, if they were willing to attempt more questions. The survey had to start with eliciting demographic information, such as the number of total vehicles purchased by the participant, experience of fleet purchase etc.; this was then used for extracting the subsets of the data. The survey was open for about a month and reminders were sent to the participants on a regular basis to respond to the questions if they hadn't already. There were a total of 147 responses we collected from 106 participants during the elicitation process. The demographic information of the participants is shown in the following Figure A.1 base on their gender, qualification, total number of vehicles purchased and fleet vehicle purchase experience.

We divided those responses into various groups based on their demographic information as discussed in Section 5.3.6. The group of *Basic*, *Intermediate*, and *Most Experienced* participants had 63 responses and the group of *Intermediate* and *Most Experienced* had 20 responses. However, the group of only *Most Experienced* participants had only 2 responses which were not enough to use for our methods so we used only the other two groups (after folding *Most Experienced* into *Intermediate*) as subsets of our data.

Before running the survey, we provided the participants following explanatory statement along with their consent to participate in our survey. The explanatory statement outlines all the information regarding the research project, data collection, privacy, usage of data and confidentiality.

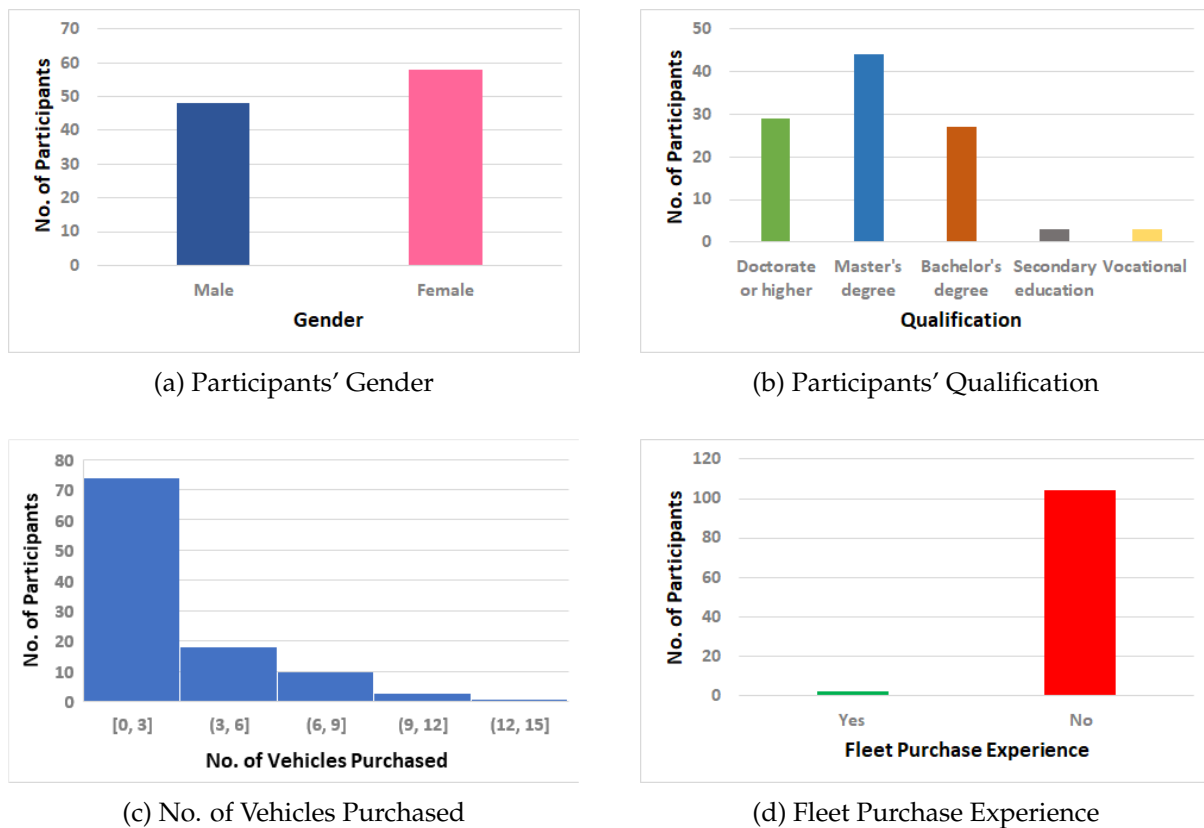


Figure A.1: Demographic Information of the Participants Involved in Survey

A.1 Explanatory Statement and Consent Form

You are being invited to take part in a research study based on the Fleet Vehicle Purchasing problem. This research study will result in a multi-criteria decision-making model which will assist the decision maker to select the best Fleet vehicle.

Project ID: 23526

Project Title: Multi-attribute Utility Elicitation - Fleet Purchasing Case Study

You are invited to take part in this study. Please read this Explanatory Statement in full before deciding whether or not to participate in this research. If you would like further information regarding any aspect of this project, you are encouraged to contact the researchers via the phone numbers or email addresses listed above.

A.1.1 What does the research involve?

This survey aims to elicit your preferences regarding a Fleet Vehicle Purchasing problem. There are multiple criteria involved in this case study such as safety, cost, environmental impact and customer satisfaction. You are asked to rank or score options based on your preferences for the described scenario. This survey questionnaire has around 10 questions and may take up to 10-12 minutes.

A.1.2 Why are you chosen for this research?

You are chosen for this survey either as a representative of a knowledgeable group or as a member of the general public.

A.1.3 Source of funding

This research is being funded by Monash University and Defence Group of Science and Technology as a collaborative project between Monash University and DST Group.

A.1.4 Consenting to participate in the project and withdrawing from the research

Your participation in this survey is the submission of an anonymous questionnaire and is voluntary. If you do not wish to take part, you do not have to. If you decide to take part and later change your mind, you are free to withdraw from the project at any stage, but it will not be possible to withdraw data once you have submitted your responses.

A.1.5 Possible benefits and risks to participants

You are invited to disclose only what feels comfortable for you and you can decline to answer any question if you so wish. You are asked to provide demographic information (any kind of purchase experience) without revealing any specific or personal information related to the institution/organization, but only general information. In addition to that, you need to provide preferences given a set of attributes/criteria while making an optimal car fleet vehicle selection. This research doesn't aim to collect data that might result in any psychological, physical, mental or financial impact. We will de-identify any data before analysing or using it in our research project. The contact details of the chief investigator and researcher are provided in case support or further information is necessary. You may leave contact information with the investigators to be notified of any of our publications, including theses, utilizing the collected data.

A.1.6 Payment

Your email address is collected for you to enter into a lucky draw to win a \$50 cash prize. Once the survey is complete, a lucky draw will be conducted and the winner will be contacted via email address to get the prize.

A.1.7 Confidentiality

Your participation and any information you provide will be kept confidential. To ensure confidentiality, all the data will be securely stored both during and after the project is completed in a de-identified form.

A.1.8 Storage of data

The de-identified data will be stored in the researchers' password protected PC and also in restricted access Monash google drive servers (for access by others on the research team, during the data analysis stage). Once the research project has been completed, the researchers will upload the data to the Open Science Foundation (osf.io) archives for open access by future researchers, not including any recordings. All the consent forms, and any recordings and electronic data collected from human subjects for the study will be transferred onto a labelled USB which will be handed to the Graduate Research Officer for secure storage. The USB will be securely stored for a duration recommended by the university policy. All consent forms will be received in digital form (no hard copy). Any printed versions of forms or data will be shredded. The researcher will remove all the consent forms, the recordings and identified data collected from participants from her computer, and from Monash Google drive, after transferring to USB drive for secure storage by Graduate Research Officer. The de-identified electronic data related to the study will be loaded to Open Science Foundation archives.

A.1.9 Results

This research project aims to uphold confidentiality and anonymity, to support and maintain the well-being of its participants, as well as to treat their views, and their person, with due respect. As it states, all the research outcomes will be presented in de-identified and summated format and participants will be informed before giving their consent about the matter in which the research will be published in peer reviewed journals or conferences.

A.1.10 Complaints

Should you have any concerns or complaints about the conduct of the project, you are welcome to contact the Executive Officer, Monash University Human Research Ethics Committee (MUHREC):

Executive Officer

Monash University Human Research Ethics Committee (MUHREC) Room 111, Chancellery Building D, 26 Sports Walk, Clayton Campus

Research Office
Monash University VIC 3800
Tel: +61 3 9905 2052
Email: muhrec@monash.edu
Fax: +61 3 9905 3831

A.1.11 Informed consent

You have read and understood the Explanatory Statement and hereby agree to participate in the Monash University research project specified above.

You consent to the following:

- The data being retained according to Monash University regulations for a period of time as anonymised data.
- The collection of information about the subjects such as educational demographic information.
- The data that you provide during this research may be used by Monash University in future research projects.

A.2 Scenarios

In our actual survey, we use four different problem scenarios in order to extract a wide range of information through different angles. These scenarios help the participants to better understand the problem and questions and act as fleet manager when deciding different purchase options. Detailed descriptions of the scenarios used in survey are given below.

A.2.1 Problem Description (Scenario 1):

You are the Chief Operating Officer at Monash University, reviewing the fleet manager's purchasing decision regarding an expansion of the car fleet for the next four years:

1. Ferrari Portofino - Ferrari Portofino is a grand touring sports car produced by Ferrari. It is a very expensive and luxurious car but has an average level of safety due to the lack of an auto emergency braking system, lane departure warning etc. (For car specs and price visit: [Ferrari Portofino](#))
2. Toyota Kluger - Toyota Kluger is a mid-size crossover SUV produced by Toyota. It is a spacious and comfortable car with a very high safety rating. It is not very expensive yet

offers auto braking and other advanced driver assistance features as standard. (For car specs and price visit: [Toyota Kluger](#))

3. Fiat 500 - Fiat 500 is a rear-engine, four seat, small city car that was manufactured by Fiat Automobiles. It is cheaper than Ferrari and Toyota but has high maintenance costs and some safety issues. (For car specs and price visit: [Fiat 500](#))

Your customers are the staff/people at Monash University who will use a fleet vehicle for different purposes. The university has a budget of half a million dollars for fleet vehicles over this 4 year period, and your fleet manager has to consider the following factors before deciding which vehicle to purchase:

- Overall Cost
- Safety
- Customer Satisfaction
- Environmental Impact

Ratings for the above mentioned criteria ranges from 1 ~ 5 where 1 is the lowest score and 5 is the highest.

A.2.2 Problem Description (Scenario 2):

Let's suppose you are the president of an NGO charity operating health services in Vietnam, reviewing the fleet manager's purchasing decision regarding an expansion of the car fleet for the next four years:

1. Ferrari Portofino - Ferrari Portofino is a grand touring sports car produced by Ferrari. It is a very expensive and luxurious car but has an average level of safety due to the lack of an auto emergency braking system, lane departure warning etc. (For car specs and price visit: [Ferrari Portofino](#))
2. Toyota Kluger - Toyota Kluger is a mid-size crossover SUV produced by Toyota. It is a spacious and comfortable car with a very high safety rating. It is not very expensive yet offers auto braking and other advanced driver assistance features as standard. (For car specs and price visit: [Toyota Kluger](#))
3. Fiat 500 - Fiat 500 is a rear-engine, four seat, small city car that was manufactured by Fiat Automobiles. It is cheaper than Ferrari and Toyota but has high maintenance costs and some safety issues. (For car specs and price visit: [Fiat 500](#))

Your customers are the staff/people at health centres who will use a fleet vehicle for delivering health services. The NGO is committed to providing a safe workplace for all employees and that extends to the company-leased vehicles. This commitment demonstrates the level of care the organisation has for the safety of its employees on the road and the safety of other road users. Your NGO needs to create a safe transport system, which involves safe driving, safe vehicles and safe road use by all people using the road. The NGO has a budget of half a million dollars for the fleet over this 4 year period, and your fleet manager has to consider the following factors before deciding which vehicle to purchase:

- Overall Cost
- Safety
- Customer Satisfaction
- Environmental Impact

Ratings for the above mentioned criteria ranges from 1 ~ 5 where 1 is the lowest score and 5 is the highest.

A.2.3 Problem Description (Scenario 3):

Assume that you are the CEO of a Local Government Authority (LGA), reviewing the fleet manager's purchasing decision regarding an expansion of the car fleet for the next four years:

1. Ferrari Portofino - Ferrari Portofino is a grand touring sports car produced by Ferrari. It is a very expensive and luxurious car but has an average level of safety due to the lack of an auto emergency braking system, lane departure warning etc. (For car specs and price visit: [Ferrari Portofino](#))
2. Toyota Kluger - Toyota Kluger is a mid-size crossover SUV produced by Toyota. It is a spacious and comfortable car with a very high safety rating. It is not very expensive yet offers auto braking and other advanced driver assistance features as standard. (For car specs and price visit: [Toyota Kluger](#))
3. Fiat 500 - Fiat 500 is a rear-engine, four seat, small city car that was manufactured by Fiat Automobiles. It is cheaper than Ferrari and Toyota but has high maintenance costs and some safety issues. (For car specs and price visit: [Fiat 500](#))

Your customers are the staff in local government who will use a fleet vehicle for varied purposes. The LGA has recently pledged to go to zero CO2 emissions in the near future, with

an emphasis on low-emissions buildings. The organization believes that it can reduce their CO2 emissions by as much as 50% over this four-year period. The organization has a budget of half a million dollars for the fleet over this 4 year period, and your fleet manager has to consider the following factors before deciding which vehicle to purchase:

- Overall Cost
- Safety
- Customer Satisfaction
- Environmental Impact

Ratings for the above mentioned criteria ranges from 1 ~ 5 where 1 is the lowest score and 5 is the highest.

A.2.4 Problem Description (Scenario 4):

Let's suppose you are the CEO of the Australian federal government, reviewing the fleet manager's purchasing decision regarding an expansion of the government's car fleet for the next four years:

1. Ferrari Portofino - Ferrari Portofino is a grand touring sports car produced by Ferrari. It is a very expensive and luxurious car but has an average level of safety due to the lack of an auto emergency braking system, lane departure warning etc. (For car specs and price visit: [Ferrari Portofino](#))
2. Toyota Kluger - Toyota Kluger is a mid-size crossover SUV produced by Toyota. It is a spacious and comfortable car with a very high safety rating. It is not very expensive yet offers auto braking and other advanced driver assistance features as standard. (For car specs and price visit: [Toyota Kluger](#))
3. Fiat 500 - Fiat 500 is a rear-engine, four seat, small city car that was manufactured by Fiat Automobiles. It is cheaper than Ferrari and Toyota but has high maintenance costs and some safety issues. (For car specs and price visit: [Fiat 500](#))

Your customers are the staff of the Australian federal government who will use a fleet vehicle for varied purposes. The political leadership believes electric vehicles are a waste of money and generally that environmental impact is a low priority. When selecting the fleet, your fleet manager must ensure that the vehicle selected is capable of performing a wide range of routine tasks, for both local and interstate travel. (Specialist purchases, including buses

and vans, will not be a part of this purchase decision.) The agency has a budget of half a million dollars for the fleet over this 4 year period, and your fleet manager has to consider the following factors before deciding which vehicle to purchase:

- Overall Cost
- Safety
- Customer Satisfaction
- Environmental Impact

Ratings for the above mentioned criteria ranges from 1 ~ 5 where 1 is the lowest score and 5 is the highest.

Please answer the following questions on the assumption you are actually in this scenario.

A.3 Example Question

This example is meant to show you quickly what's expected for the actual survey below.

Let's suppose you have to decide which fruit to select for lunch while hiking on a hot day, based on the cost and durability of the fruit. Your options are: Bananas, Apples and Persimmon. (Bananas are cheap but not durable, Apples are expensive but durable, Persimmon is expensive and durable).

- Is durability more important than the cost of the fruit?
- Is cost more important than the durability of the fruit?
- Are both equally important?

A.3.1 Example Problem Description

Let's say Mark is a hiker planning to hike in hot weather; normally he likes Bananas more than any other fruits because they are cheap, but he expects they might die on the trip, so instead he will prefer some durable fruit such as Apples. So he goes with an option (i.e., Durability is more important than the cost of the fruit).

Let's say: **Durability is more important than Cost** You have answered that durability is more important than the cost of the fruit. Next we want to know how much more important — that is, how many times more important is durability than cost. For these kinds of judgments, often it's often difficult to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then

your best guess for “how many times”.

Considering the plausible values for durability being more important than cost, realistically what do you think would be: (Please enter your answers as any real number greater than 1 but less than or equal to 10. If you answer “5”, it means that you think durability of fruit is 5 times more important than cost.)

1. Your lowest plausible value?

Answer: 4 (Durability is plausibly at least 4 times more important than the cost of the fruit)

2. Your highest plausible value?

Answer: 8 (Durability is plausibly at most 8 times more important than the cost of the fruit)

3. Your best estimate?

Answer: 6 (Your best estimate is that Durability is 6 times more important than the cost of the fruit)

4. How confident are you that the interval you created, from lowest to highest, captures your value for durability being more important than the cost of the fruit? (Enter a value between 50 and 100%)

Answer: 80%

A.4 Demographic Information and Ranking Questions

1. How many motorised vehicles have you purchased?
2. Have you ever purchased vehicles for an organisation? If so, what kind of organisation (company, government authority, NGO, etc.)? Also, what was the size of the purchase? (Optional)
3. What is your highest level of education achieved?
4. Gender?

5. **Cost vs Safety:**

Do you think Cost is more important than Safety when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.1](#))
- If No (Go to section [A.4.2](#))

- If Equally Important (Go to section [A.4.3](#))

Do you think Cost is more important than Customer Satisfaction when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.5](#))
- If No (Go to section [A.4.6](#))
- If Equally Important (Go to section [A.4.7](#))

Do you think Cost is more important than Environmental Impact when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.9](#))
- If No (Go to section [A.4.10](#))
- If Equally Important (Go to section [A.4.11](#))

Do you think Safety is more important than Customer Satisfaction when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.13](#))
- If No (Go to section [A.4.14](#))
- If Equally Important (Go to section [A.4.15](#))

Do you think Safety is more important than Environmental Impact when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.17](#))
- If No (Go to section [A.4.18](#))
- If Equally Important (Go to section [A.4.19](#))

Do you think Customer Satisfaction is more important than Environmental Impact when it comes to purchasing a vehicle for fleet purposes?

- If Yes (Go to section [A.4.21](#))
- If No (Go to section [A.4.22](#))
- If Equally Important (Go to section [A.4.23](#))

A.4.1 Cost is more important than Safety

You have answered that Cost is more important than Safety. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Cost being more important than Safety, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Cost is 5 times more important than Safety)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Cost being more important than Safety. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.4](#).

A.4.2 Safety is more important than Cost

You have answered that Safety is more important than Cost. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Safety being more important than Cost, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Safety is 5 times more important than Cost)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?

4. How confident are you that the interval you created, from lowest to highest, captures your value for Safety being more important than Cost. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.4](#).

A.4.3 Cost and Safety are equally important

You have answered that Cost and Safety are equally important. Next we would like to know your confidence in your answer.

How confident are you that Cost and Safety are equally important. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.4](#).

A.4.4 Cost vs Safety Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose purchase price is \$400k and the safety rating is 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose purchase price is \$400k but the safety rating is 3. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied

-
3. Your fleet manager decided to purchase a Ferrari whose purchase price is below \$400k but the safety rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose purchase price is \$60k and the safety rating is 5. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose purchase price is \$80k but the safety rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose purchase price is \$61k but the safety rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose purchase price is \$20k but the safety rating is 5. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

8. Your fleet manager decided to purchase a Ferrari whose purchase price is \$19k but the safety rating is 3. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

9. Your fleet manager decided to purchase a Toyota whose purchase price is below \$18k but the safety rating is 1. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

A.4.5 Cost is more important than Customer Satisfaction

You have answered that Cost is more important than Customer Satisfaction. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Cost being more important than Customer Satisfaction, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Cost is 5 times more important than Customer Satisfaction)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Cost being more important than Customer Satisfaction. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.8](#).

A.4.6 Customer Satisfaction is more important than Cost

You have answered that Customer Satisfaction is more important than Cost. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Customer Satisfaction being more important than Cost, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Customer Satisfaction is 5 times more important than Cost)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Customer Satisfaction being more important than Cost. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.8](#).

A.4.7 Cost and Customer Satisfaction are equally important

You have answered that Cost and Customer Satisfaction are equally important. Next we would like to know your confidence in your answer.

How confident are you that Cost and Customer Satisfaction are equally important. (Please

enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.8](#).

A.4.8 Cost vs Customer Satisfaction Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose purchase price is \$400k and the Customer Satisfaction rating is 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose purchase price is \$400k but the Customer Satisfaction rating is 3. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
3. Your fleet manager decided to purchase a Ferrari whose purchase price is below \$400k but the Customer Satisfaction rating is 1. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose purchase price is \$60k and the Customer Satisfaction rating is 5. How satisfied are you with this decision?

-
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose purchase price is \$80k but the Customer Satisfaction rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose purchase price is \$61k but the Customer Satisfaction rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose purchase price is \$20k but the Customer Satisfaction rating is 5. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
8. Your fleet manager decided to purchase a Ferrari whose purchase price is \$19k but the Customer Satisfaction rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied

- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

9. Your fleet manager decided to purchase a Toyota whose purchase price is below \$18k but the Customer Satisfaction rating is 1. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

A.4.9 Cost is more important than Environmental Impact

You have answered that Cost is more important than Environmental Impact. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Cost being more important than Environmental Impact, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Cost is 5 times more important than Environmental Impact)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Cost being more important than Environmental Impact. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.12](#).

A.4.10 Environmental Impact is more important than Cost

You have answered that Environmental Impact is more important than Cost. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Environmental Impact being more important than Cost, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Environmental Impact is 5 times more important than Cost)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Environmental Impact being more important than Cost. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.12](#).

A.4.11 Cost and Environmental Impact are equally important

You have answered that Cost and Environmental Impact are equally important. Next we would like to know your confidence in your answer.

How confident are you that Cost and Environmental Impact are equally important. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.4](#).

A.4.12 Cost vs Environmental Impact Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose purchase price is \$400k and the Environmental Impact rating is 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?

-
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose purchase price is \$400k but the Environmental Impact rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
3. Your fleet manager decided to purchase a Ferrari whose purchase price is below \$400k but the Environmental Impact rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose purchase price is \$60k and the Environmental Impact rating is 5. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose purchase price is \$80k but the Environmental Impact rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied

-
- ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose purchase price is \$61k but the Environmental Impact rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose purchase price is \$20k but the Environmental Impact rating is 5. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
8. Your fleet manager decided to purchase a Ferrari whose purchase price is \$19k but the Environmental Impact rating is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
9. Your fleet manager decided to purchase a Toyota whose purchase price is below \$18k but the Environmental Impact rating is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied

- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

A.4.13 Safety is more important than Customer Satisfaction

You have answered that Safety is more important than Customer Satisfaction. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Safety being more important than Customer Satisfaction, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Safety is 5 times more important than Customer Satisfaction)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Safety being more important than Customer Satisfaction. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.16](#).

A.4.14 Customer Satisfaction is more important than Safety

You have answered that Customer Satisfaction is more important than Safety. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Customer Satisfaction being more important than Safety, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If

you answer “5”, it means that you think Customer Satisfaction is 5 times more important than Safety)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Customer Satisfaction being more important than Safety. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.16](#).

A.4.15 Safety and Customer Satisfaction are equally important

You have answered that Safety and Customer Satisfaction are equally important. Next we would like to know your confidence in your answer.

How confident are you that Safety and Customer Satisfaction are equally important. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.16](#).

A.4.16 Safety vs Customer Satisfaction Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose safety rating is 5 and customer satisfaction is also 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose safety rating is 3 and customer satisfaction is 5. How satisfied are you with this decision?

-
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
3. Your fleet manager decided to purchase a Ferrari whose safety rating is 1 and customer satisfaction is 5. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose safety rating is 5 and customer satisfaction is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose safety rating is 3 and customer satisfaction is also 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose safety rating is 1 and customer satisfaction is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied

-
- ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose safety rating is 5 and customer satisfaction is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
8. Your fleet manager decided to purchase a Ferrari whose safety rating is 3 and customer satisfaction is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
9. Your fleet manager decided to purchase a Toyota whose safety rating is 1 and customer satisfaction is also 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied

A.4.17 Safety is more important than Environmental Impact

You have answered that Safety is more important than Environmental Impact. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three

questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Safety being more important than Environmental Impact, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Safety is 5 times more important than Environmental Impact)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Safety being more important than Environmental Impact. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.20](#).

A.4.18 Environmental Impact is more important than Safety

You have answered that Environmental Impact is more important than Safety. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Environmental Impact being more important than Safety, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Environmental Impact is 5 times more important than Safety)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Environmental Impact being more important than Safety. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.20](#).

A.4.19 Safety and Environmental Impact are equally important

You have answered that Safety and Environmental Impact are equally important. Next we would like to know your confidence in your answer.

How confident are you that Safety and Environmental Impact are equally important. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.20](#).

A.4.20 Safety vs Environmental Impact Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose safety rating is 5 and Environmental Impact is also 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose safety rating is 3 and Environmental Impact is 5. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
3. Your fleet manager decided to purchase a Ferrari whose safety rating is 1 and Environmental Impact is 5. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied

-
- ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose safety rating is 5 and Environmental Impact is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose safety rating is 3 and Environmental Impact is also 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose safety rating is 1 and Environmental Impact is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose safety rating is 5 and Environmental Impact is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied

- ☐ Satisfied
- ☐ Very Satisfied

8. Your fleet manager decided to purchase a Ferrari whose safety rating is 3 and Environmental Impact is 1. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

9. Your fleet manager decided to purchase a Toyota whose safety rating is 1 and Environmental Impact is also 1. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

A.4.21 Customer Satisfaction is more important than Environmental Impact

You have answered that Customer Satisfaction is more important than Environmental Impact. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Customer Satisfaction being more important than Environmental Impact, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Customer Satisfaction is 5 times more important than Environmental Impact)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?

4. How confident are you that the interval you created, from lowest to highest, captures your value for Customer Satisfaction being more important than Environmental Impact. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.24](#).

A.4.22 Environmental Impact is more important than Customer Satisfaction

You have answered that Environmental Impact is more important than Customer Satisfaction. Next we want to know how much more important, in terms of “how many times” more important. For these kinds of judgments, often it’s hard to be exact or certain, so please answer the following three questions, about the lowest plausible “how many times”, the highest plausible “how many times”, and then your best guess for “how many times”.

Considering the plausible values for Environmental Impact being more important than Customer Satisfaction, realistically what do you think would be:

*Please enter your answers as any real number greater than 1 but less than or equal to 10. (If you answer “5”, it means that you think Environmental Impact is 5 times more important than Customer Satisfaction)

1. Your lowest plausible value?
2. Your highest plausible value?
3. Your best estimate?
4. How confident are you that the interval you created, from lowest to highest, captures your value for Environmental Impact being more important than Customer Satisfaction. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.24](#).

A.4.23 Customer Satisfaction and Environmental Impact are equally important

You have answered that Customer Satisfaction and Environmental Impact are equally important. Next we would like to know your confidence in your answer.

How confident are you that Customer Satisfaction and Environmental Impact are equally important. (Please enter a value between 50 to 100%)

Attempt the followup questions in section [A.4.24](#).

A.4.24 Customer Satisfaction vs Environmental Impact Questions:

(Please answer the rest of the questions carefully based on your preference of one criteria over the other.)

1. Your fleet manager decided to purchase a Ferrari whose Customer Satisfaction rating is 5 and Environmental Impact is also 5 (on a scale of 1 to 5, where 5 is best). How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
2. Your fleet manager decided to purchase a Fiat whose Customer Satisfaction rating is 3 and Environmental Impact is 5. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
3. Your fleet manager decided to purchase a Ferrari whose Customer Satisfaction rating is 1 and Environmental Impact is 5. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
4. Your fleet manager decided to purchase a Toyota whose Customer Satisfaction rating is 5 and Environmental Impact is 3. How satisfied are you with this decision?
 - ☐ Very dissatisfied
 - ☐ Dissatisfied

-
- ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
5. Your fleet manager decided to purchase a Ferrari whose Customer Satisfaction rating is 3 and Environmental Impact is also 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
6. Your fleet manager decided to purchase a Toyota whose Customer Satisfaction rating is 1 and Environmental Impact is 3. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
7. Your fleet manager decided to purchase a Fiat whose Customer Satisfaction rating is 5 and Environmental Impact is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied
 - ☐ Satisfied
 - ☐ Very Satisfied
8. Your fleet manager decided to purchase a Ferrari whose Customer Satisfaction rating is 3 and Environmental Impact is 1. How satisfied are you with this decision?
- ☐ Very dissatisfied
 - ☐ Dissatisfied
 - ☐ Neither satisfied nor dissatisfied

- ☐ Satisfied
- ☐ Very Satisfied

9. Your fleet manager decided to purchase a Toyota whose Customer Satisfaction rating is 1 and Environmental Impact is also 1. How satisfied are you with this decision?

- ☐ Very dissatisfied
- ☐ Dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Satisfied
- ☐ Very Satisfied

Baseline Method Implementation

This chapter explains how this baseline method works by looking at a very simple example of two criteria problem which is a sub network of our main network we hand craft for our case study. The case study is based on Fleet Vehicle Purchase problem for any large organization and the detailed description along with the network structure is explained in chapter 6. The sub network of fleet vehicle purchase problem shown in figure B.1, where we have a decision node named Fleet Vehicle Purchase attached to two different criteria called Cost and Safety. Both criteria are further up associated with their respective utility nodes and then we have a total score utility showing an additive summary of both the utility nodes. There is a link between Cost and Safety showing their relationship as they are dependent upon each other. We choose a single best car among 3 different car models such as Toyota, Fiat and Ferrari, assume that Ferrari is a very expensive model with an average safety, Toyota's cost is average but has a good safety ranking while Fiat is very cheap and has a very low safety ranking. We have 2 different criteria (cost and safety) to consider before making a decision which car to purchase. Different decision makers may have different preferences based on their own needs and choices, and we have to get their preferences and combine them to make a single best decision.

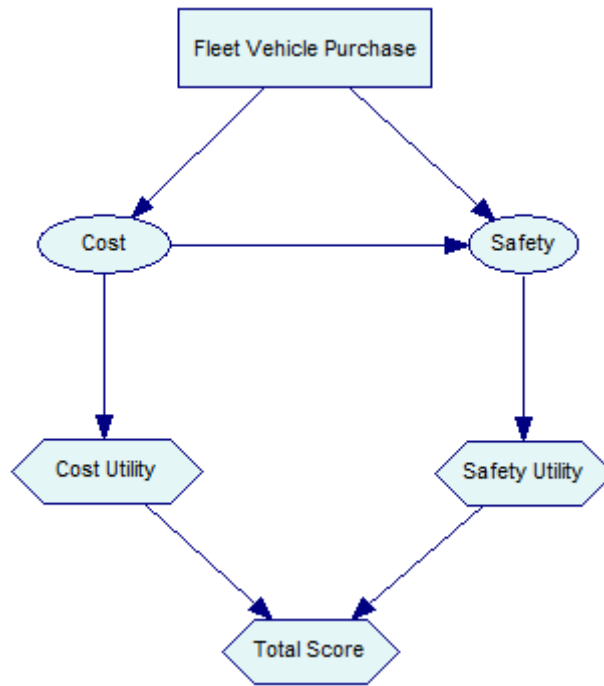


Figure B.1: Car Purchase Example for 2 Criteria

B.1 Elicitation Via Baseline Method

First of all we calculate the value for each criteria to feed into the utility node of our example BDN. Following are the notations we use in our formulas to calculate the utility score of each utility node.

C represents criteria.

m is the total number of criteria in the network.

E denotes the experts participating in the elicitation process.

n is the total number of experts involved in the elicitation process.

W is the weight based on the Likert scale confidence level in the comparative score assigned by the expert.

B.1.0.1 Likert Scale for Confidence

We use likert scale (1-5) for measuring the confidence level of the experts in providing their answers when ranking one criteria over other. Once they provide their confidence we map the values shown in table B.1.

Verbal Scale	Numerical Value
Not confident	1
Less confident	2
Average confidence	3
More confident	4
Fully confident	5

Table B.1: Likert Scale for Confidence

B.1.0.2 Eliciting Preferences for 2 Criteria Example

Let suppose we have 5 experts E_1, E_2, E_3, E_4 and E_5 , and a set of questions to ask from the experts to rank both the criteria in terms of its relative importance based on their own preferences or a given scenario.

Ranking Criteria

We compare and each pair of criteria in a given network based on their relative importance. We ask the following questions from participants to rank in terms of “how many times” one criteria is more important than other, before asking their confidence in their value.

1. Cost and Safety are two important factors to consider when buying a Fleet vehicle for an organization. Which one do you think is more important when it comes to purchasing a vehicle for fleet purpose?
 - (a) Cost is more important than safety. [Go to question 2]
 - (b) Safety is more important than cost. [Go to question 3]
 - (c) Cost and Safety are equally important. [Go to question 4]
2. How many times cost is more important than safety? Enter your answers as any real number greater than 1 but less than or equal to 10 (If you answer “5”, it means that you think cost is 5 times more important than safety).
3. How many times safety is more important than cost? Enter your answers as any real number greater than 1 but less than or equal to 10 (If you answer “5”, it means that you think safety is 5 times more important than cost).
4. How confident are you in the answer you have given to the previous question?
 - (a) Not confident
 - (b) Less confident
 - (c) Average confidence

(d) More confident

(e) Fully confident

For the sake of this 2 criteria example, we use likert scale to measure confidence. However, in elicitation process we ask participants their confidence in the range of 50–100.

After asking the above questions, let suppose we get the following responses from five experts:

- E_1 says cost is 3 times more important than safety and she/he is More Confident about her/his answer.
- E_2 say that cost is 4 times more important than safety and she/he is Less Confident of her/his answer.
- E_3 says safety and cost are equally important and she/he is Average Confident of her answer.
- E_4 says safety is 2 times more important than cost and she/he is Average Confident of her/his answer.
- E_5 say that safety is 10 times more important than cost and she/he is Fully Confident about her/his answer.

Follow up Questions

Suppose we get the following responses from five experts E_1, E_2, E_3, E_4 and E_5 to questions based on every combination of criteria. Below tables are representing the scales given to each combination of both criteria.

Cost	Safety		
	High	Medium	Low
High	3	2	1
Medium	4	4	2
Low	5	4	3

Table B.2: Expert E_1

Cost	Safety		
	High	Medium	Low
High	4	3	1
Medium	5	4	2
Low	5	5	2

Table B.3: Expert E_2

B.1.0.3 Calculate Score for Each Criteria

For each criteria in C , we find a total score S_C given by each expert in E by using the following formula.

$$S_C = \sum_{E=1}^n (S_{E,C} \times W_E) \quad (\text{B.1})$$

Cost	Safety		
	High	Medium	Low
High	3	2	1
Medium	3	4	1
Low	4	4	2

Table B.4: Expert E_3

Cost	Safety		
	High	Medium	Low
High	4	2	2
Medium	3	5	2
Low	5	4	1

Table B.5: Expert E_4

Cost	Safety		
	High	Medium	Low
High	2	3	1
Medium	4	3	1
Low	4	4	2

Table B.6: Expert E_5

When there are more than 2 criteria, we perform a pairwise comparison of all criteria in the network by setting loop.

Set a counter for first criteria $g = 1$ to second last criteria $m - 1$ and

For second criteria $h = g + 1$ to last criteria m in the network

$$S_{C_{h,g}} = \sum_{e \in E=1}^n \left(\frac{1}{S_{e \in E, C_g}} \times W_{e \in E} \right) \quad (\text{B.2})$$

$S_{C_{g,h}}$ is total score of criteria g relative to criteria h , where S_{E, C_g} is the score of criteria g (relative to criteria h) given by each expert in E . W_E is the confidence level of each expert E in the score she/he has given to criteria g relative to criteria h .

$$S_{C_{h,g}} = \sum_{e \in E=1}^n \left(\frac{1}{S_{e \in E, C_g}} \times W_{e \in E} \right) \quad (\text{B.3})$$

For 2 criteria problem, C has a set of $C = \{C_1, C_2\}$ representing cost and safety respectively. We calculate the score S_C for both criteria C_1 and C_2 .

Where S_C is the total score of a criteria C , E is expert index variable, n is the number of experts surveyed, $S_{E,C}$ is the score (an integer number of times given by an expert that criteria C is better than alternative criteria C_2) and W_E is a weight based on the Likert scale confidence level in the comparative score assigned by that expert.

This comparative score method implies that there will be a minimum of two total scores, S_{C_1} and S_{C_2} when the preference elicitation involves the minimum 2 criteria, and that for every

expert's score S_{E,C_1} , there is a corresponding score $S_{E,C_2} = \frac{1}{S_{E,C_1}}$.

Note also that the same weight W_E is used to calculate both S_{C_1} and S_{C_2} , as there is only one weight (confidence answer) given for each criteria pairwise comparison score elicited.

If we map the 5 value Likert scale answer for confidence to the integer weights 1 (least) to 5 (most), then we obtain table B.7:

Expert	Expert E's Score Cost w.r.t Safety $S_{E,C_1=Cost,C_2=Safety}$	Expert E's Score Safety w.r.t Cost $S_{E,C_1=Safety,C_2=Cost}$	Confidence weighting $W_{E(C_1,C_2)} = W_{E(C_2,C_1)}$
E_1	3	1/3	4
E_2	4	1/4	2
E_3	1	1	3
E_4	2	1/2	3
E_5	10	1/10	5

Table B.7: Expert elicited preferences

After applying the above formula (Equation 1) to the sample answers for the 5 experts E_1, E_2, E_3, E_4 and E_5 , we get the results shown in the table B.8.

Cost relative to Safety	Safety relative to Cost
$3 \times 4 = 12$	$0.33 \times 4 = 1.32$
$4 \times 2 = 8$	$0.25 \times 2 = 0.5$
$1 \times 3 = 3$	$1 \times 3 = 3$
$0.5 \times 3 = 1.5$	$2 \times 3 = 6$
$0.1 \times 5 = 0.5$	$10 \times 5 = 50$
$S_{Cost} = 25$	$S_{Safety} = 60.82$

Table B.8: Results of expert elicitation

Having more than 2 criteria in a BN gives us multiple scores S_C for each criteria as a result of pairwise comparisons. We combine all those scores S_C for each criteria using the formula below:

$$S_C = \sum_{cr=1}^k (S_{C_{cr}}) \quad (\text{B.4})$$

cr is a counter used to count the k scores of a specific criteria relative to any other criteria in the network. This equation B.4 is used to sum up all the scores for each criteria involved in the network.

B.1.0.4 Formula for Normalized Weights

Now we normalize weights for each criteria in C using the following formula.

$$NW_C = \frac{S_C}{\sum_{i=1}^m S_{C_i}} \quad (\text{B.5})$$

In above equation B.5, NW_C is the normalized weight for each criteria C equal to the total score S_C of a criteria divided by the sum of all the scores of m number of criteria in a network.

By applying formula B.5 to our example network, we get the normalized weights for both criteria cost and safety as shown below:

$$NW_{Cost} = \frac{S_{Cost}}{S_{Cost} + S_{Safety}}$$

$$NW_{Cost} = \frac{25}{25 + 60.82}$$

$$NW_{Cost} = \frac{25}{85.82}$$

$$NW_{Cost} = 0.291$$

$$NW_{Safety} = \frac{S_{Safety}}{S_{Cost} + S_{Safety}}$$

$$NW_{Safety} = \frac{60.82}{25 + 60.82}$$

$$NW_{Safety} = \frac{60.82}{85.82}$$

$$NW_{Safety} = 0.709$$

These normalized weights for each criteria are then used to feed into the final utility table. These weights determine the relative importance of one criterion to another and rank it based on the experts elicited preferences. For the 2 criteria problem, we get the values for safety and cost and feed them into the Total Score utility node as shown in table B.9.

Nodes	Value
Cost	0.291
Safety	0.709

Table B.9: Total Score Utility

B.1.0.5 Combine Values for Each Criteria

The next step in baseline method is to combine all the values given by each expert in E for each criteria in C . In this 2 criteria example, we combine values for criteria $Cost_i$ and $Safety_j$ given by all five experts. We use geometric mean to aggregate multiple inputs from multiple experts. For aggregating all the scores given by all the experts using the survey questionnaire we have formula B.7. Each expert has her/his own tables of values given to one criteria w.r.t

other as discussed in section B.1.0.2, we find the geometric mean of all the inputs in tables B.2, B.3, B.4, B.5 and B.6, and put the mean value in the respective cell of the resultant table B.10.

The geometric mean of a data set $x_1, x_2, x_3, \dots, x_q$ is given by:

$$\left(\prod_{d=1}^q x_d \right)^{\frac{1}{q}} = \sqrt[q]{x_1 x_2 x_3 \dots x_q} \quad (\text{B.6})$$

Formula:

For integer $i = 1$ to s and

Set a loop for integer $j = 1$ to s (number of states of criteria)

$$U_{ij} = \left(\prod_{E=1}^n X_{E,ij} \right)^{\frac{1}{n}} \quad (\text{B.7})$$

Where s is the total number of states for each criteria (i.e., High, Medium, Low we have 3 states, so $s = 3$, X_E is the value given by expert E in cell of i^{th} row and j^{th} column in the Expert tables B.2, B.3, B.4, B.5 and B.6. n is the total number of experts participating in the survey. U_{ij} represents the cell number in the resultant table B.10 to assign the value calculated by formula B.7.

Cost	Safety		
	High	Medium	Low
High	3.10	2.35	1.15
Medium	3.73	3.95	1.51
Low	4.57	4.18	1.89

Table B.10: Geometric Mean of Values given by all Experts in E

B.1.0.6 Sum Up Values for Each Criteria

Next step is to sum up all the values for each criteria in the network. For this purpose, we use the following equations.

For every pair of criteria C_g and C_h

For integer $i = 1$ to s

$$C_{gi} = \sum_{j=1}^s U_{ij} \quad (\text{B.8})$$

and

$$C_{hi} = \sum_{j=1}^s U_{ji} \quad (\text{B.9})$$

Here C_{gi} represents the value at i^{th} row and j^{th} column in table B.11 for criteria C_g stored row wise while C_{hi} represents the value at j^{th} column and i^{th} row for criteria C_h stored column wise in the same table.

By applying formula B.9 and B.8 we get the total score for each state in both criteria C_g and C_h representing cost and safety respectively.

Cost	Safety			$C_{Cost(i)}$
	High	Medium	Low	
High	3.10	2.35	1.15	6.6
Medium	3.73	3.95	1.51	9.19
Low	4.57	4.18	1.89	10.64
$C_{Safety(i)}$	11.4	10.48	4.55	-

Table B.11: Total Score for each criteria

Next step is to combine all the values for each criteria in C for all the states using the following formula:

$$S_{C_s} = \sum_{l=1}^s S_{C_l} \quad (B.10)$$

Using equation B.10 we sum up every state for each criteria and get a total sum of all the states as well. In next step we normalize those values using formula B.5. Below is the example for further explanation.

For each state s in each criteria C we use this formula to get normalized values.

$$NW_{C_s} = \frac{S_{C_s}}{\sum_{i=1}^m S_{C_{s_i}}} \quad (B.11)$$

Normalize all states of Criteria Cost

Using the equation B.11, we normalize all the states (*High, Medium, Low*) of criteria Cost.

$$NW_{Cost_H} = \frac{S_{Cost_H}}{S_{Cost} + S_{Safety}}$$

$$NW_{Cost_H} = \frac{6.6}{26.43 + 26.43}$$

$$NW_{Cost_H} = \frac{6.6}{52.86}$$

$$NW_{Cost_H} = 0.125$$

$$NW_{Cost_M} = \frac{S_{Cost_M}}{S_{Cost} + S_{Safety}}$$

$$NW_{Cost_M} = \frac{9.19}{26.43 + 26.43}$$

$$NW_{Cost_M} = \frac{9.19}{52.86}$$

$$NW_{Cost_M} = 0.174$$

$$NW_{Cost_L} = \frac{S_{Cost_L}}{S_{Cost} + S_{Safety}}$$

$$NW_{Cost_L} = \frac{10.64}{26.43 + 26.43}$$

$$NW_{Cost_L} = \frac{10.64}{52.86}$$

$$NW_{Cost_L} = 0.2$$

Normalize all states of Criteria Safety

Now using the equation B.11, we normalize all the states (*High, Medium, Low*) of criteria Safety.

$$NW_{Safety_H} = \frac{S_{Safety_H}}{S_{Cost} + S_{Safety}}$$

$$NW_{Safety_H} = \frac{11.4}{26.43 + 26.43}$$

$$NW_{Safety_H} = \frac{11.4}{52.86}$$

$$NW_{Safety_H} = 0.216$$

$$NW_{Safety_M} = \frac{S_{Safety_M}}{S_{Cost} + S_{Safety}}$$

$$NW_{Safety_M} = \frac{10.48}{26.43 + 26.43}$$

$$NW_{Safety_M} = \frac{10.48}{52.86}$$

$$NW_{Safety_M} = 0.198$$

$$NW_{Safety_L} = \frac{S_{Safety_L}}{S_{Cost} + S_{Safety}}$$

$$NW_{Safety_L} = \frac{4.55}{26.43 + 26.43}$$

$$NW_{Safety_L} = \frac{4.55}{52.86}$$

$$NW_{Safety_L} = 0.086$$

Cost	High	Medium	Low
Value	0.125	0.174	0.2

Table B.12: Cost Utility Node

Safety	High	Medium	Low
Value	0.216	0.198	0.086

Table B.13: Safety Utility Node

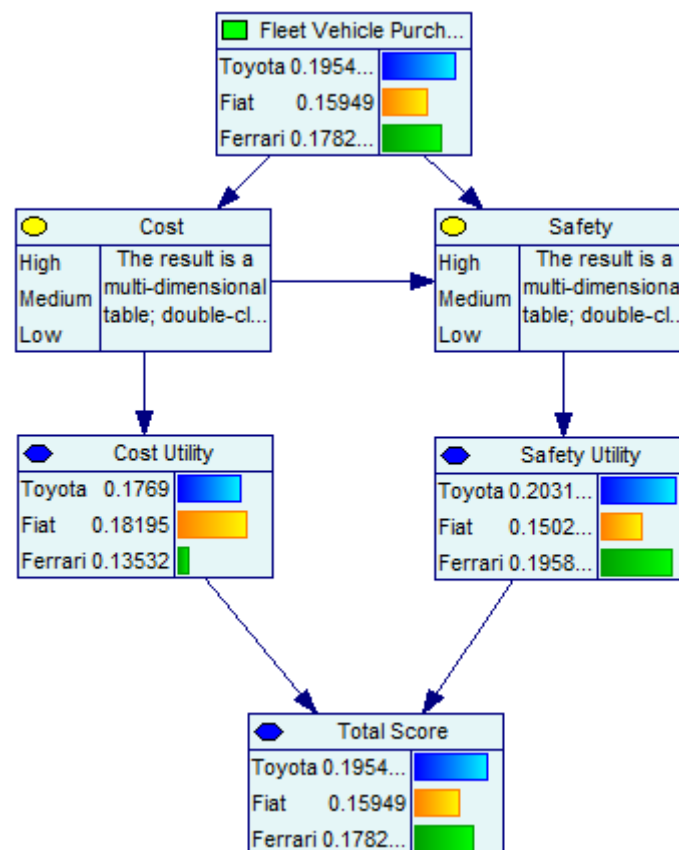


Figure B.2: 2 criteria network after adding the utility values

Modified IDEA Protocol Implementation

Let's have a look at the same 2 criteria sub network (as shown in C.1) of our Fleet Vehicle Purchase problem. We have criteria *Cost* and *Safety*, and a decision has to be made that which criteria is more important than the other. Participants are asked to provide their preferences using the 4 point questions as discussed above. Once we are done with the elicitation process from multiple participants, we get lowest plausible values, highest plausible values, best estimate and confidence. We clean the data using the process explained in section 4.3.1.1 of chapter 5.

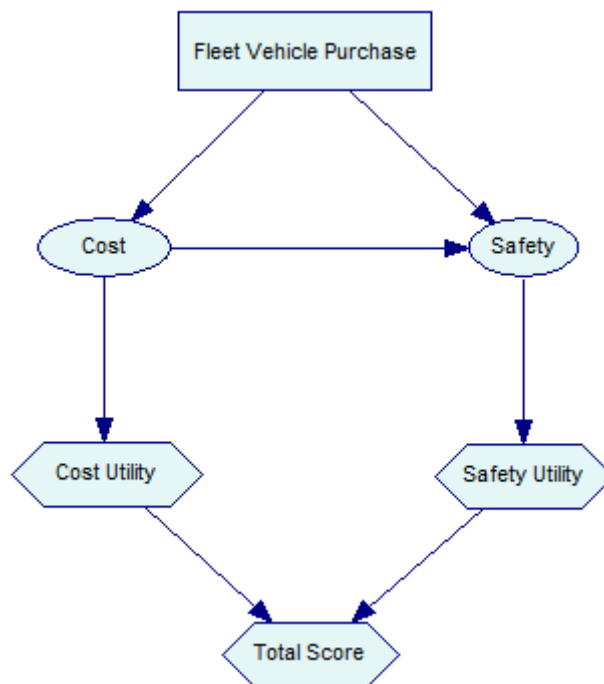


Figure C.1: Car Purchase Example for 2 Criteria

Using the equations 4.6 and 4.7, we standardise the intervals and aggregate the participants' intervals using arithmetic mean. Results drawn from example data for 2 criteria problem are shown in table C.1, including the standardised best score B_S elicited from the partici-

pants.

Criteria	L_S	U_S	B_S
Cost	3.09	4.90	3.7
Safety	3.30	5.61	4.4

Table C.1: Standardised Values

Where there is equal importance E_S of criteria, we standardise the confidence as discussed in algorithm 4.2 and calculate the scores for each criterion using algorithm 4.3. Let's suppose our 2 criteria example, we calculate the mean values of each criteria $S_{C_{Cost,Safety}}$ i.e., Cost and Safety to enter into our network using formula C.1.

$$S_{C_{Cost,Safety}} = \frac{T(L_{S_{Cost,Safety}}) + T(U_{S_{Cost,Safety}})}{2} \quad (C.1)$$

The final values for overall utility to be entered into the BN are shown in table C.2.

Overall Score	Value
Cost	4.44
Safety	4.95

Table C.2: Overall Score using IDEA Based Method

After calculating all the utility functions, we populate the utility nodes in the sub network and get the following results shown in figure C.2.

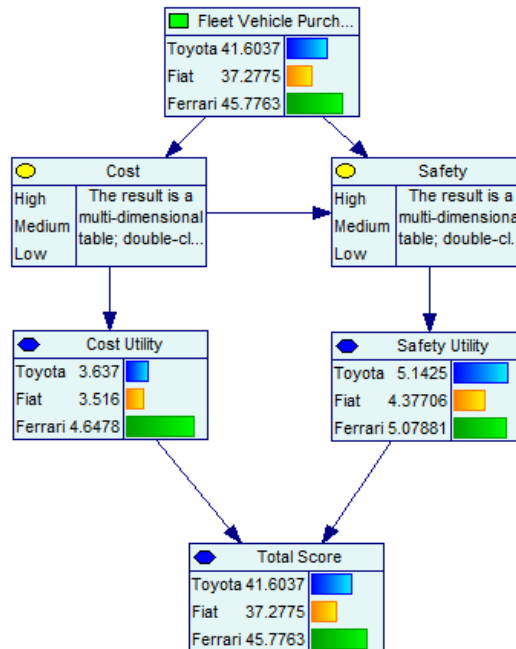


Figure C.2: 2 Criteria Example using IDEA Based Method

AHP Translation into BDN

For each scenario based preference model, we use the mutual information of each criteria with the and add Gaussian noise to generate simulated responses. We calculate the comparison matrix by calculating the geometric mean of pairwise MI values in the simulated responses and adding them to the corresponding cell in the matrix (e.g. using the geometric mean of the Cost vs Safety MIs in the). Once we get the comparison matrix, we calculate the weight for each criterion using the geometric mean of each row representing one criterion.

Using Scenario 5.1, we get the following comparison matrix.

Criteria	Cost	Safety	Customer Satisfaction	Environmental Impact	Geomean	Normalized Weights
Cost	1	1.28	4.97	78.40	4.72	0.50
Safety	0.78	1	3.89	61.29	3.69	0.39
Customer Satisfaction	0.20	0.25	1	15.75	0.94	0.10
Environmental Impact	0.012	0.016	0.063	1	0.060	0.006

Table D.1: Weights Calculated using AHP based on Scenario 5.1

The Normalized Weights in table D.1 are the final weights for each criterion as calculated using AHP method. These weights are the inputs into our BDN and the resultant translated BDN is shown in figure D.1 below.

Table D.2 and table D.3 show the comparison matrix along with the final criteria weights using AHP method. While figure D.2 and figure D.3 are the networks showing the AHP translation into BDNs.

Criteria	Cost	Safety	Customer Satisfaction	Environmental Impact	Geomean	Normalized Weights
Cost	1	73.40	233.89	0.84	10.98	0.45
Safety	0.013	1	3.18	0.011	0.149	0.006
Customer Satisfaction	0.004	0.313	1	0.003	0.04	0.0019
Environmental Impact	1.1177	86.46	275.52	1	12.94	0.536

Table D.2: Weights Calculated using AHP based on Scenario 5.2

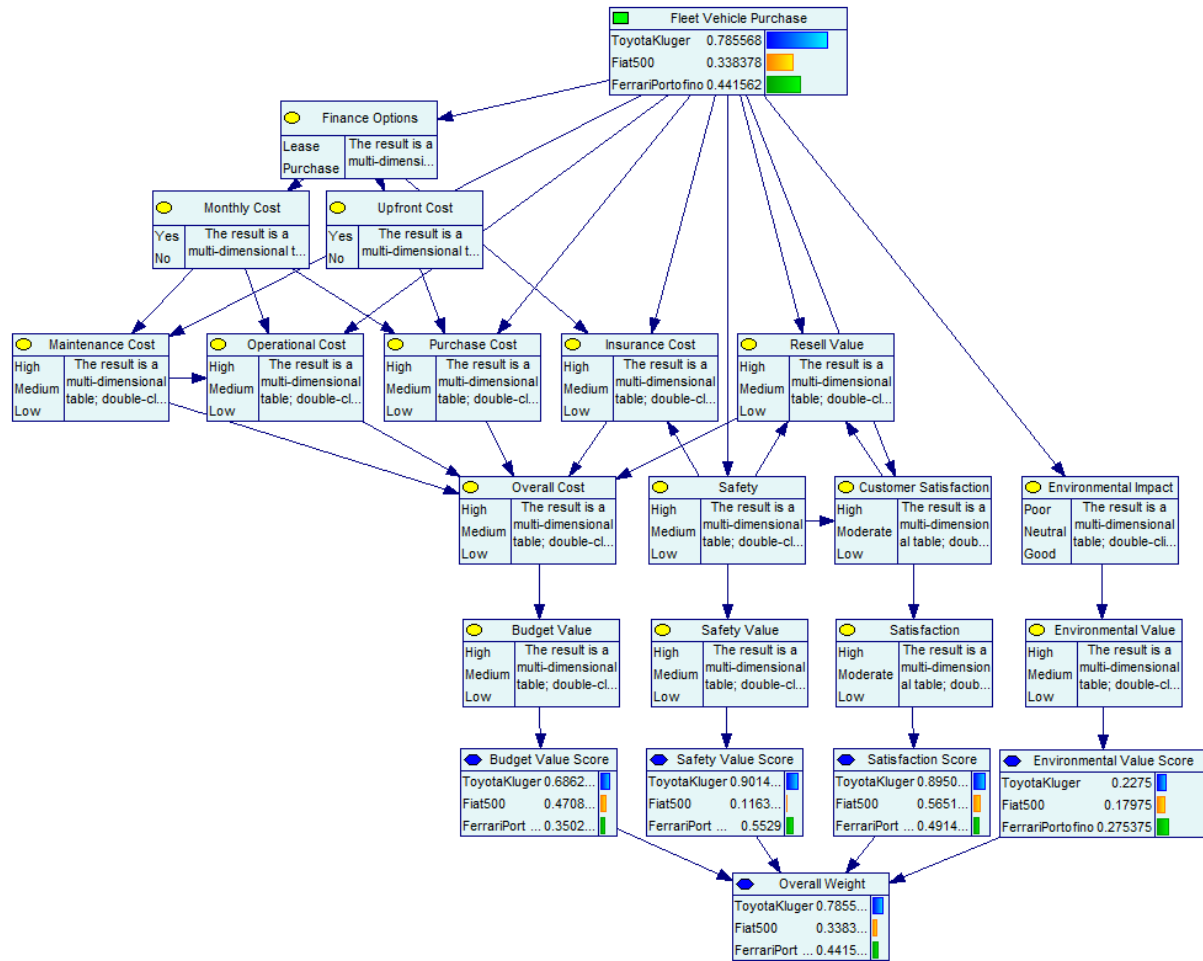


Figure D.1: AHP Translation into BDN Based on Scenario 5.1

Criteria	Cost	Safety	Customer Satisfaction	Environmental Impact	Geomean	Normalized Weights
Cost	1	0.388	6.99	219.16	4.94	0.268
Safety	2.57	1	17.97	563.53	12.70	0.69
Customer Satisfaction	0.14	0.05	1	31.35	0.706	0.038
Environmental Impact	0.0045	0.001	0.031	1	0.022	0.0012

Table D.3: Weights Calculated using AHP based on Scenario 5.3

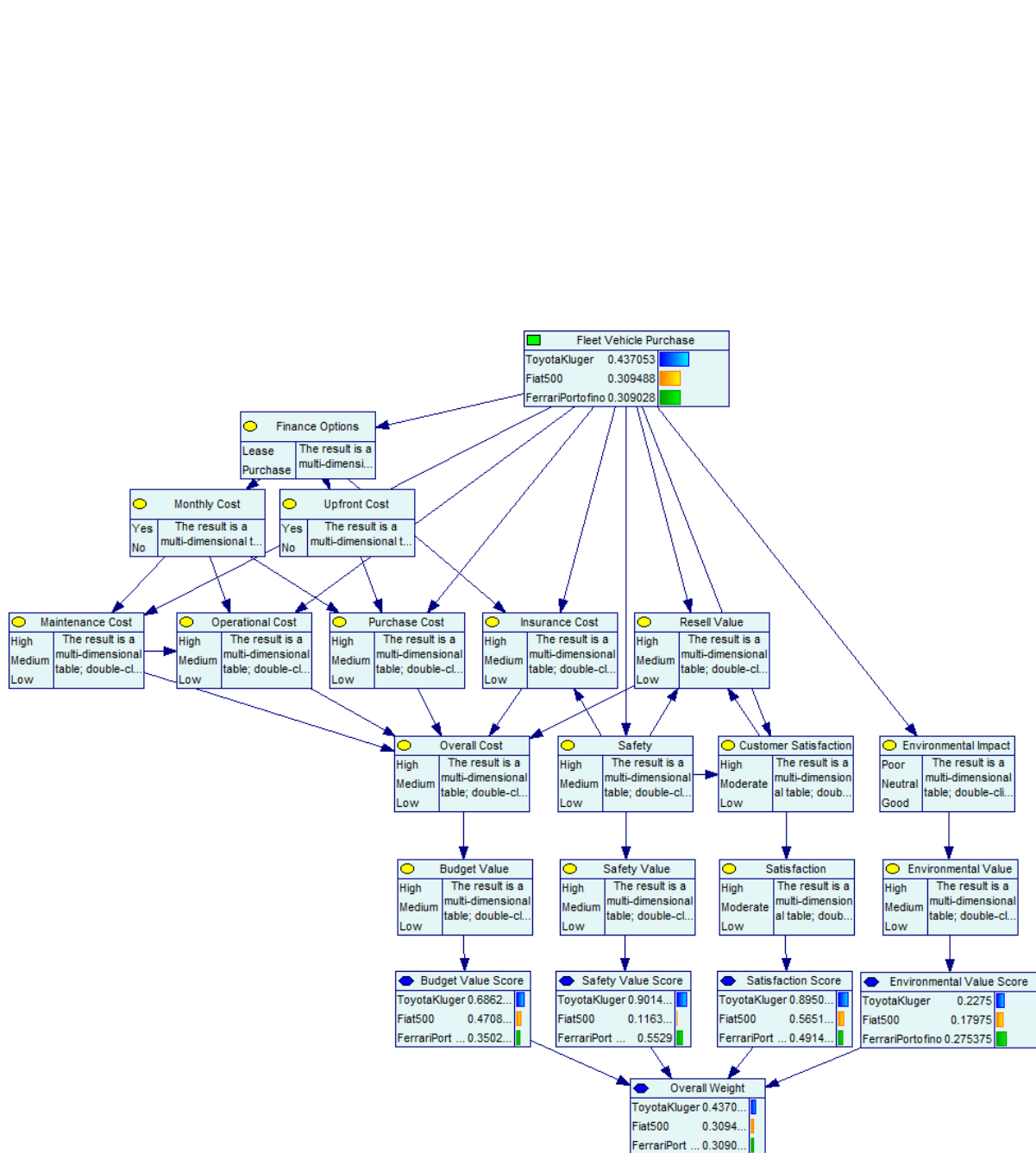


Figure D.2: AHP Translation into BDN Based on Scenario 5.2

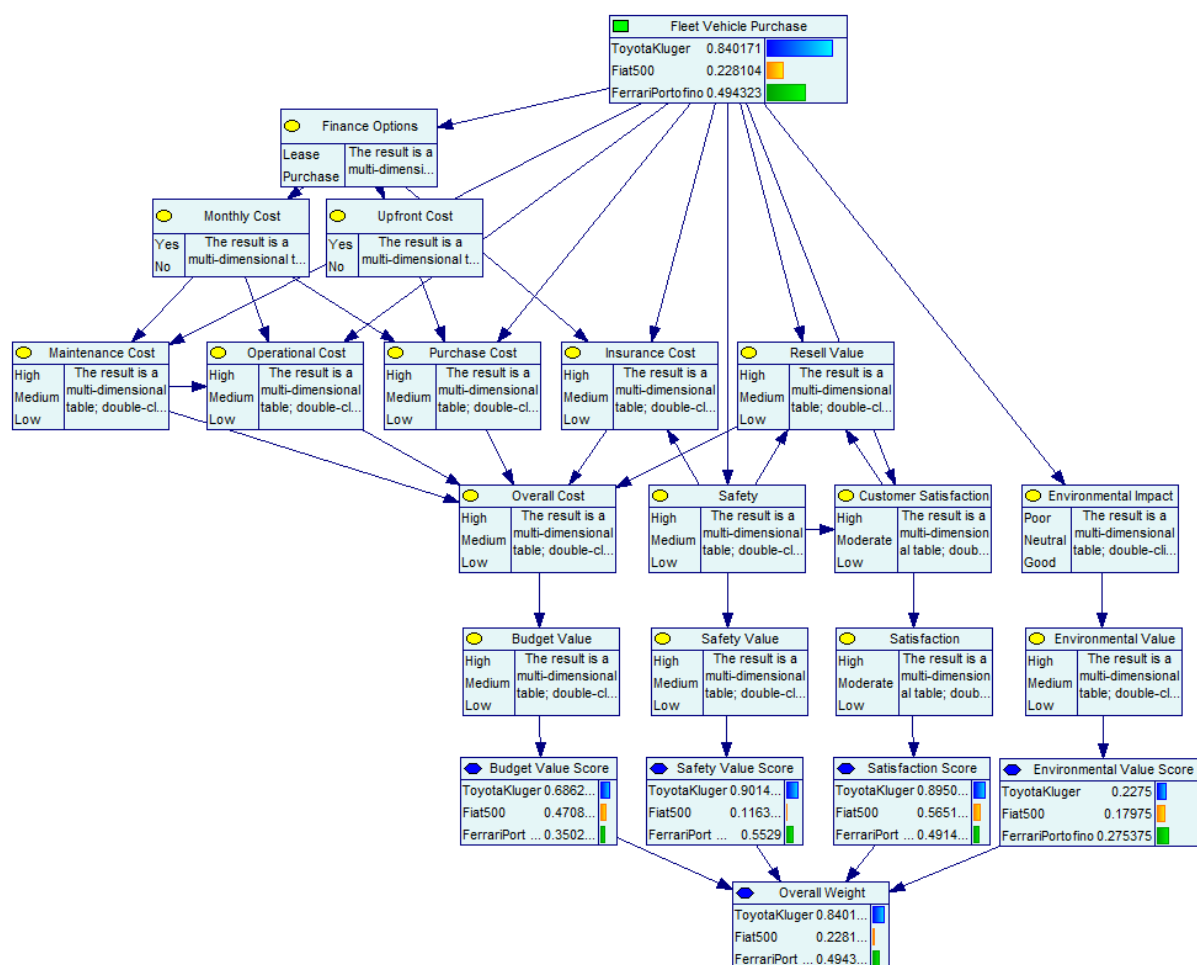


Figure D.3: AHP Translation into BDN Based on Scenario 5.3

Model Based on Scenarios

E.1 Models Based on Preference Scenarios Using Baseline Method

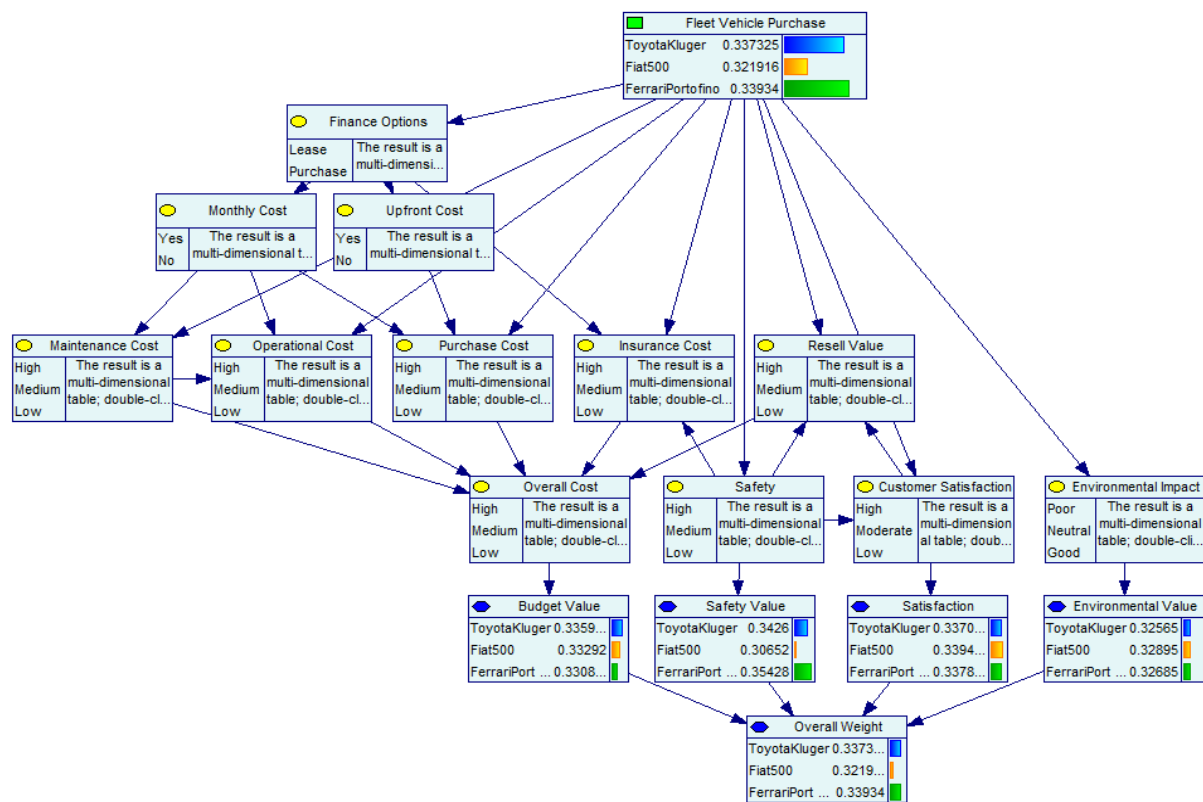


Figure E.1: Model Based on Preference Scenario 5.1 Using Baseline Method

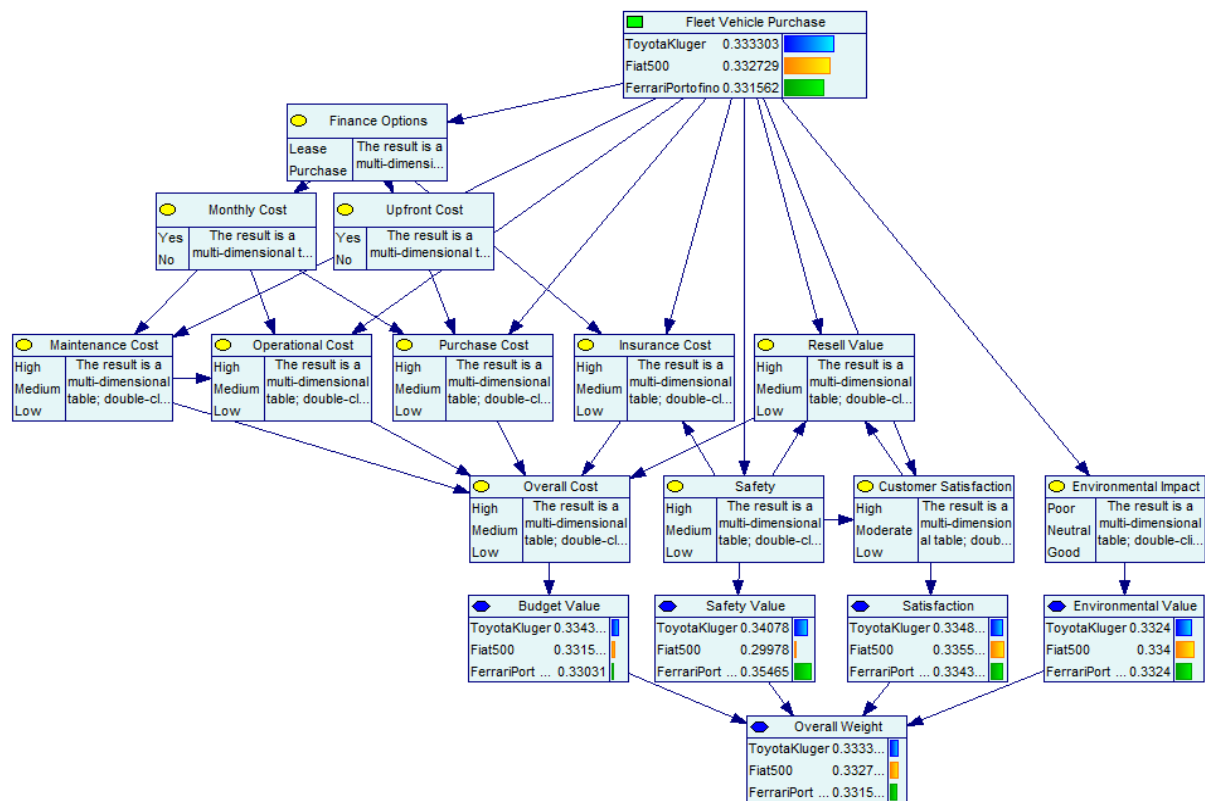


Figure E.2: Model Based on Preference Scenario 5.2 Using Baseline Method

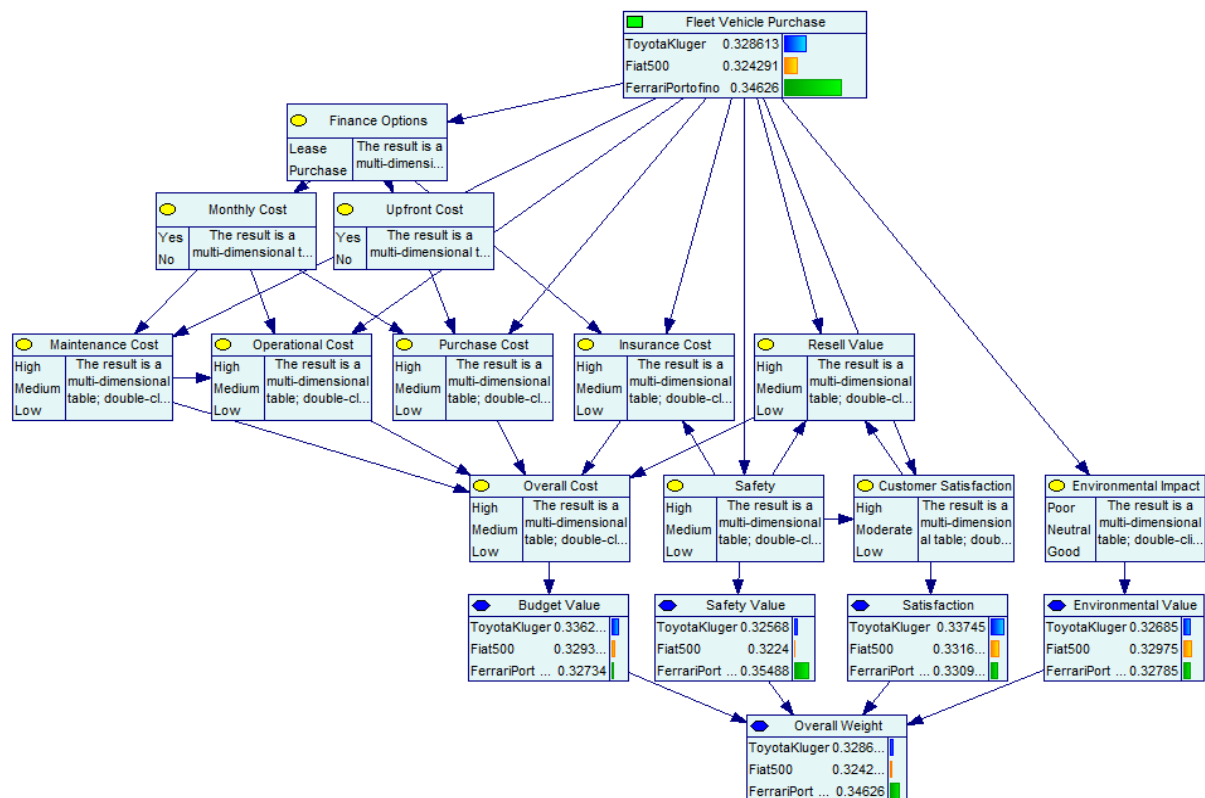


Figure E.3: Model Based on Preference Scenario 5.3 Using Baseline Method

E.2 Models Based on Preference Scenarios Using The 4 Point Method

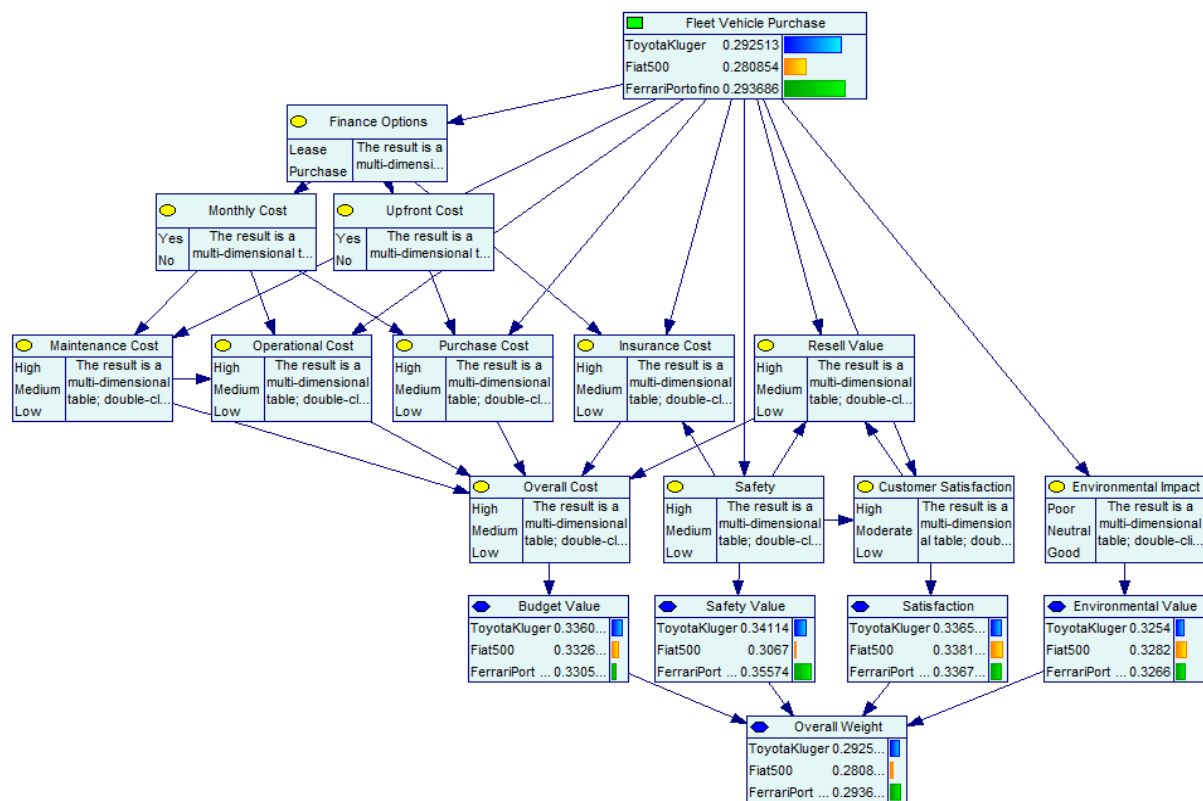


Figure E.4: Model Based on Preference Scenario 5.1 Using 4 Point Method

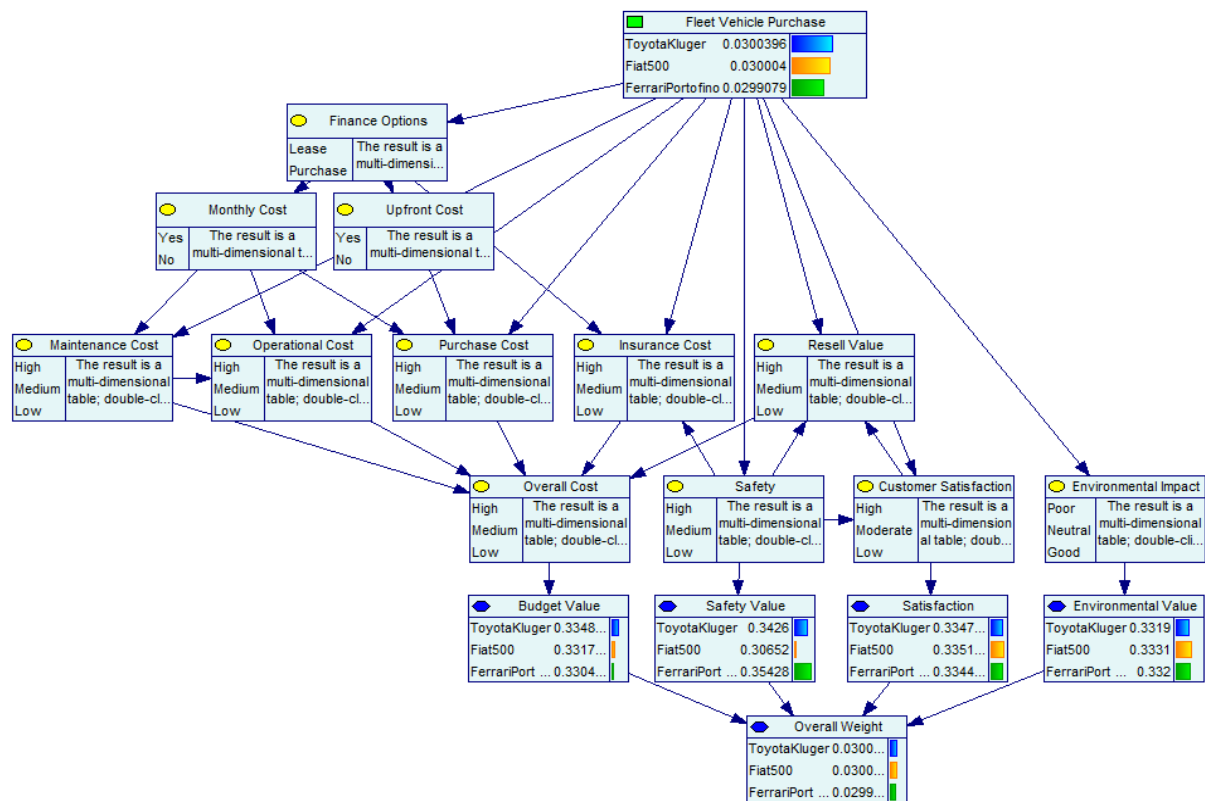


Figure E.5: Model Based on Preference Scenario 5.2 Using 4 Point Method

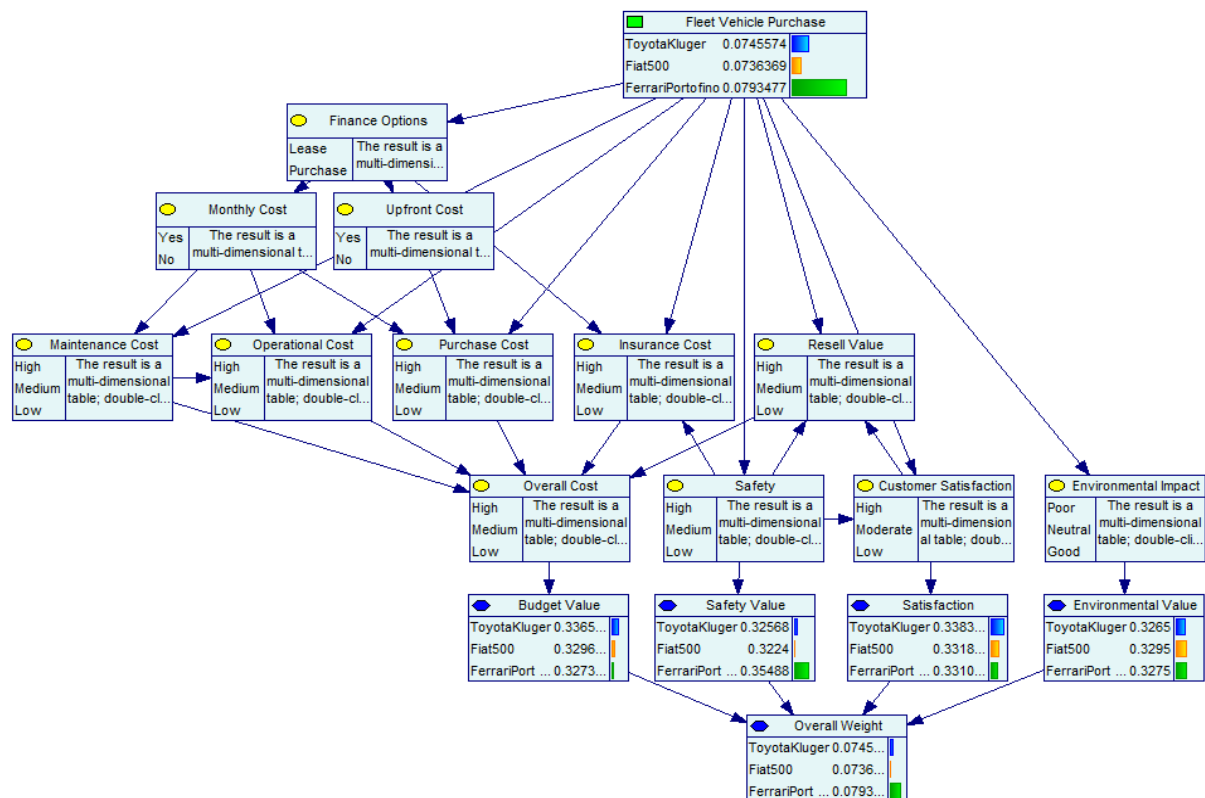


Figure E.6: Model Based on Preference Scenario 5.3 Using 4 Point Method

E.3 Models Based on Preference Scenarios Using AHP Based Method

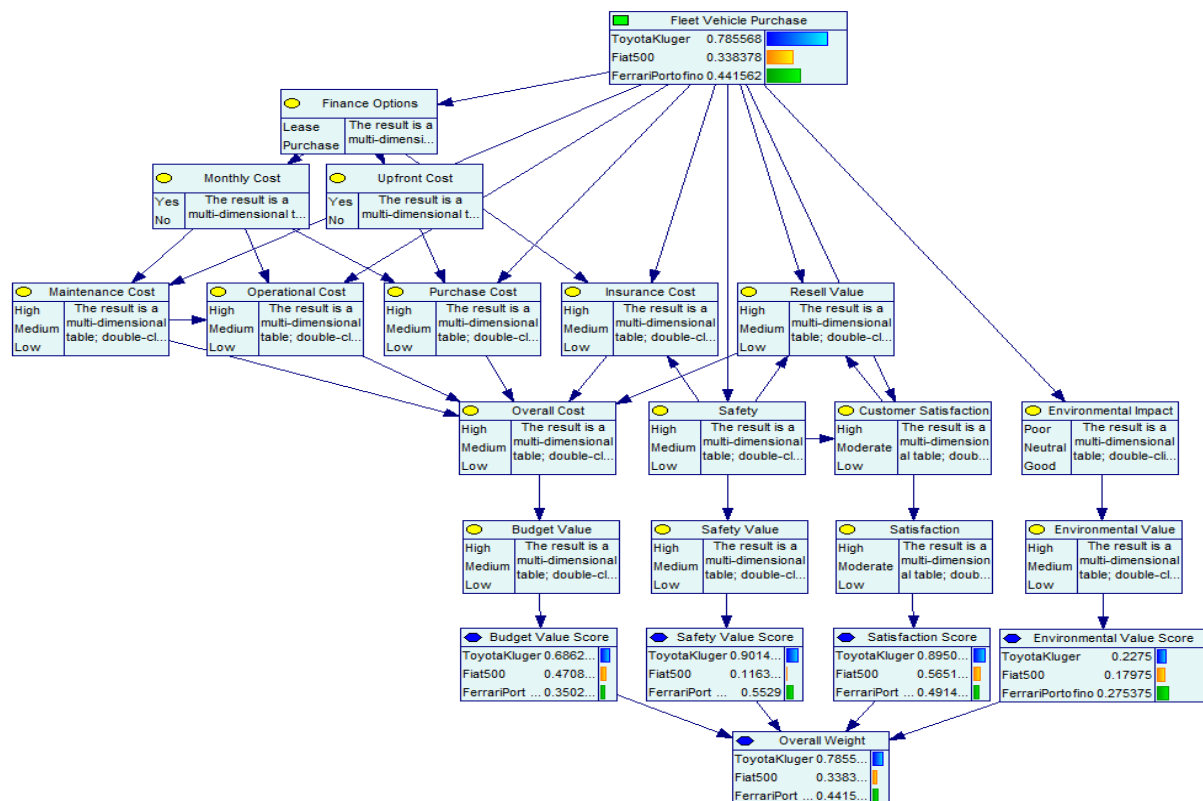


Figure E.7: Model Based on Preference Scenario 5.1 Using AHP Based Method

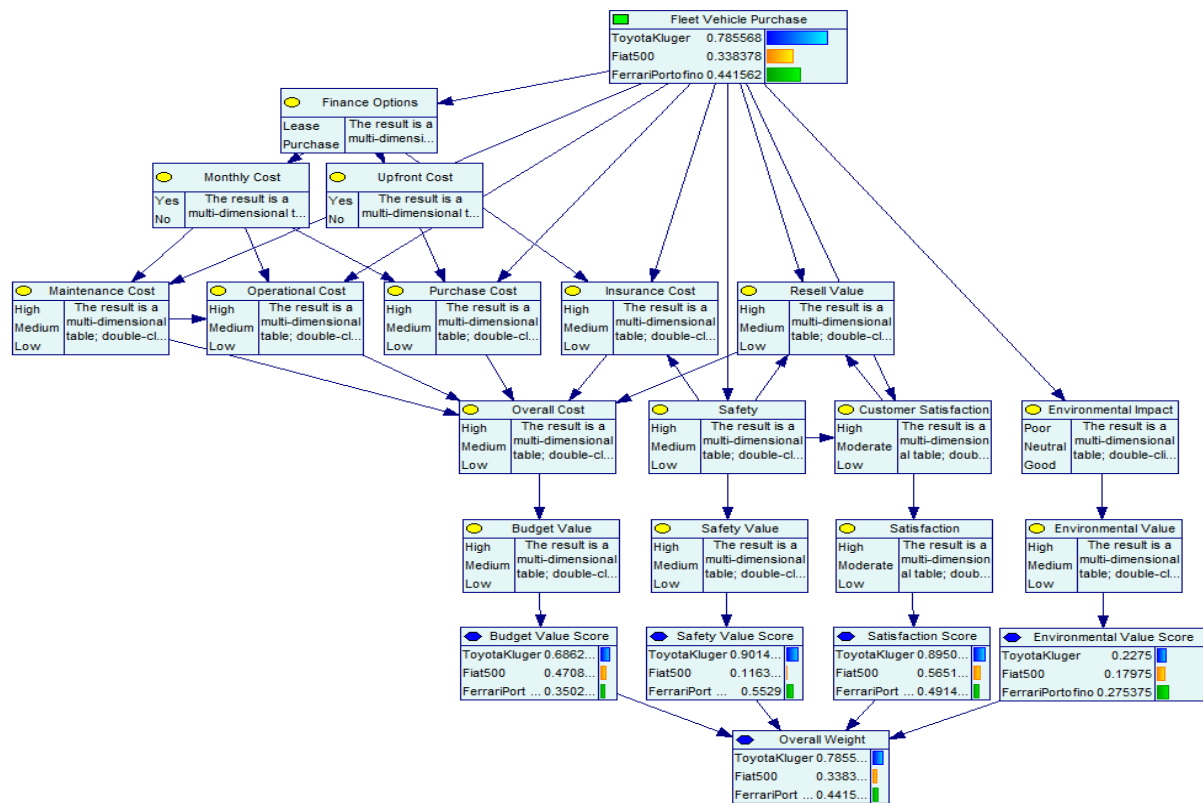


Figure E.8: Model Based on Preference Scenario 5.2 Using AHP Based Method

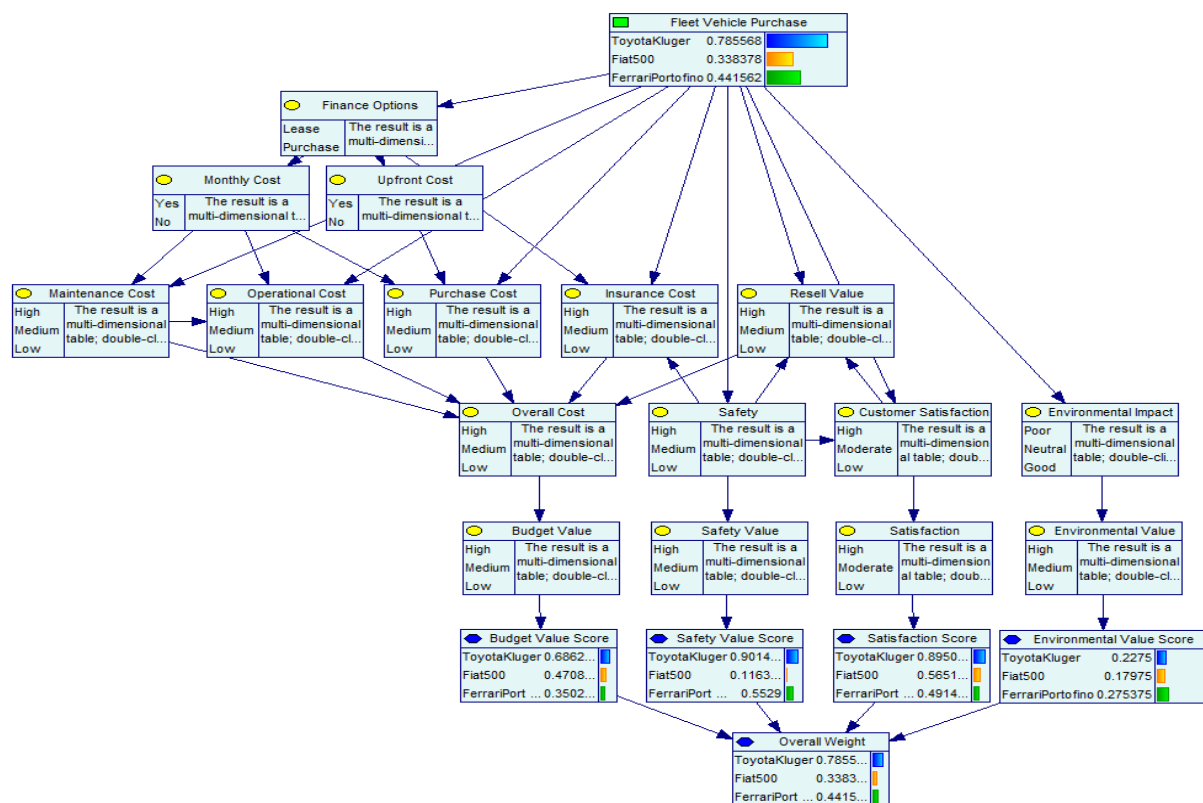


Figure E.9: Model Based on Preference Scenario 5.3 Using AHP Based Method

E.4 Models Based on Preference Scenarios Using CaMML

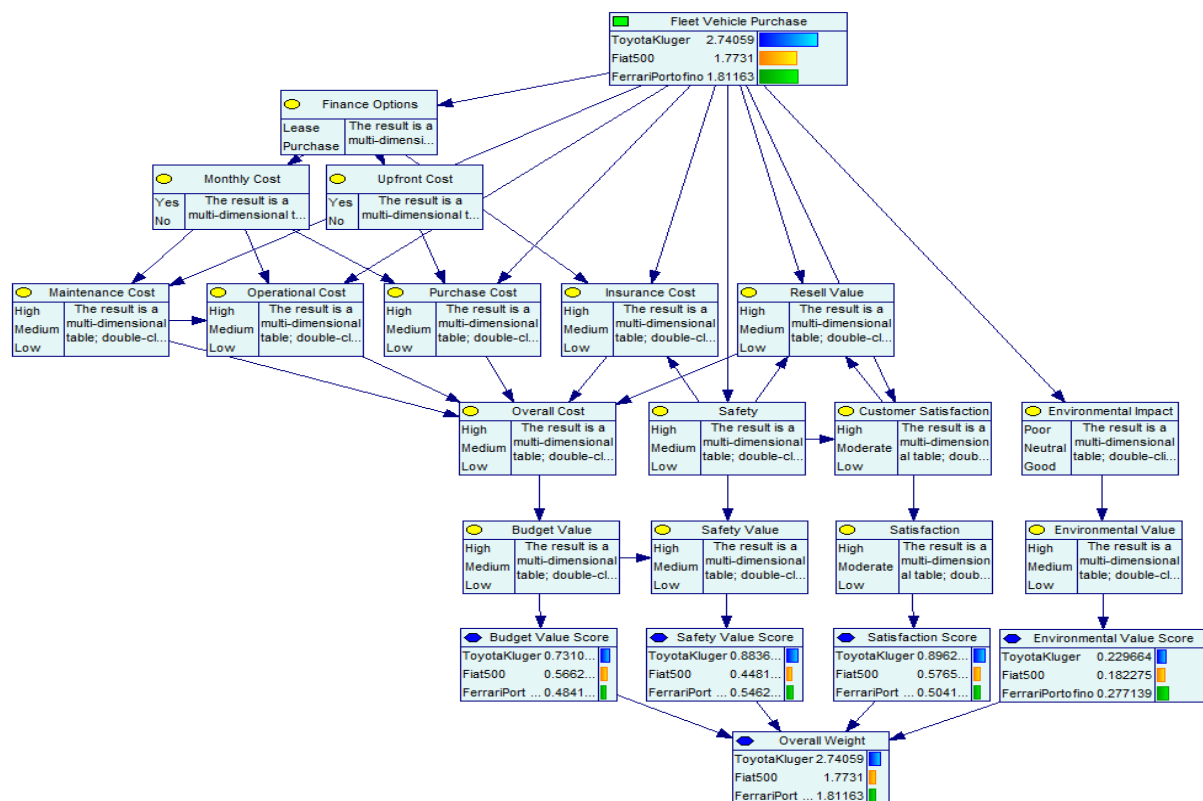
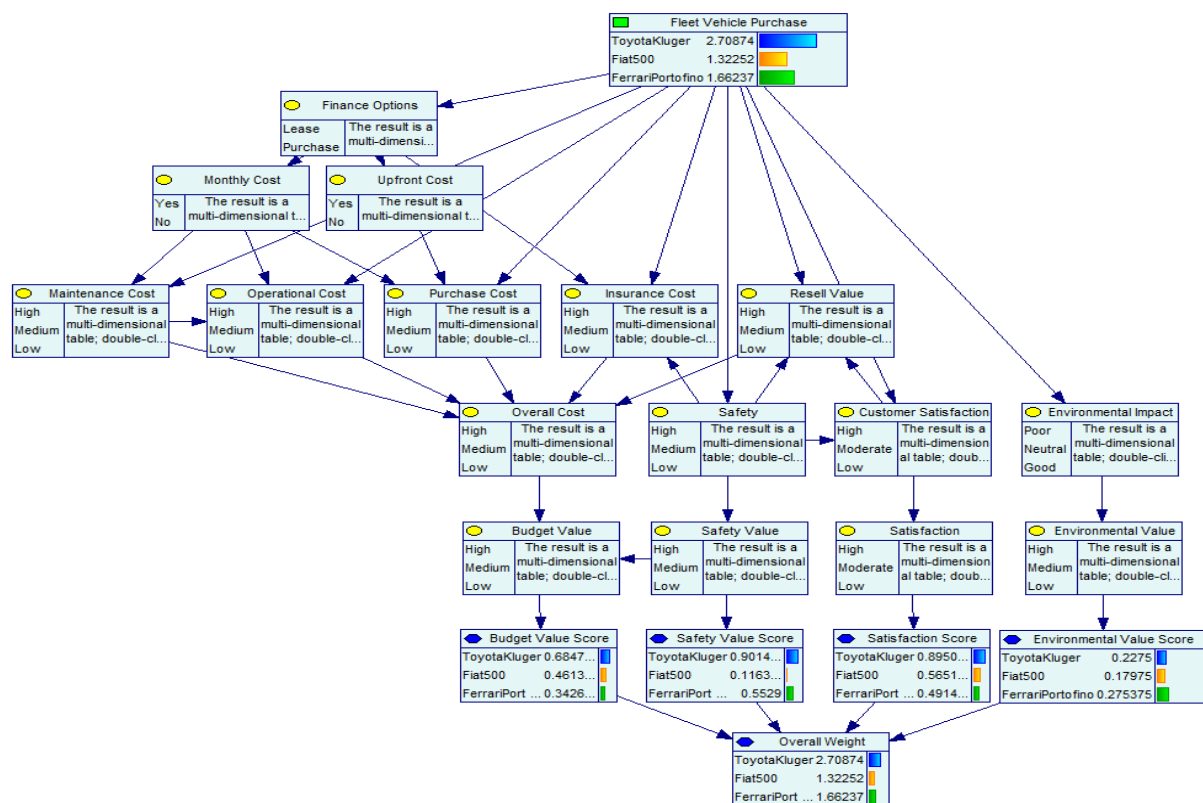
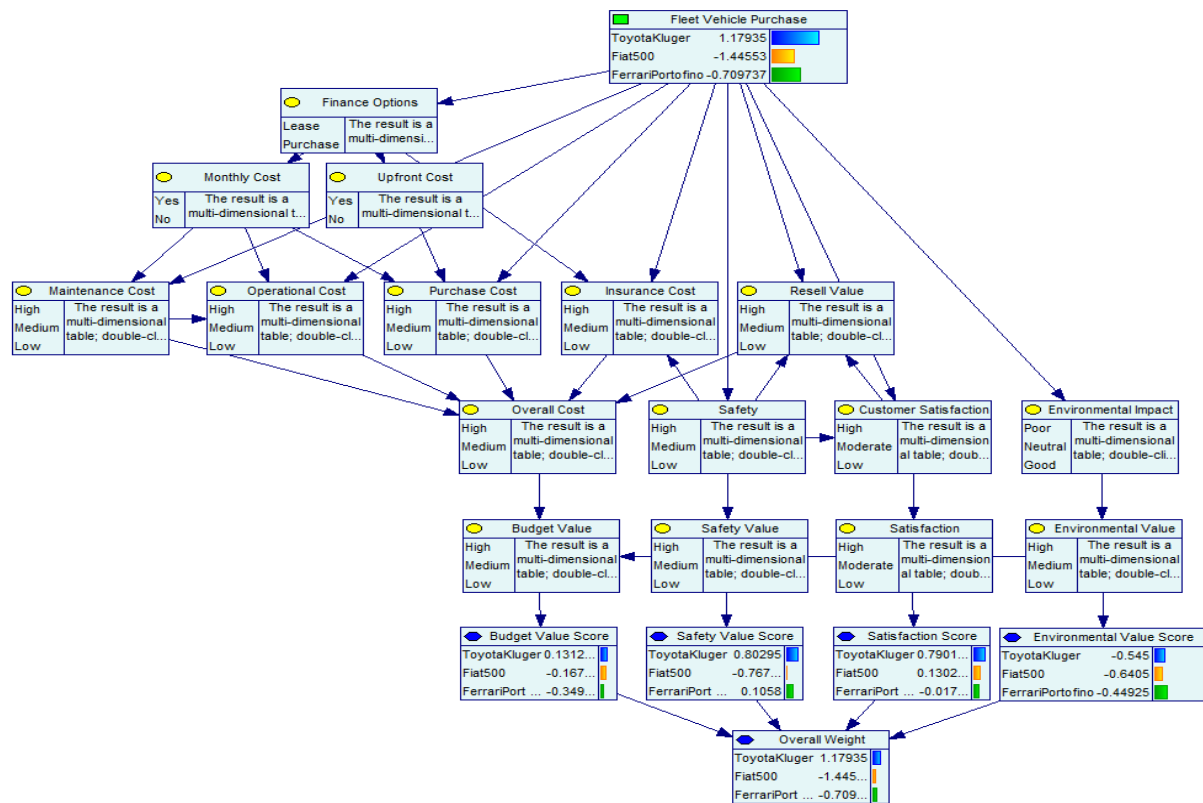


Figure E.10: Model Based on Preference Scenario 5.1 Using CaMML Method



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